SiliconX

Previous Year Questions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

SiliconX

Previous Year Questions

Communications (EC), Electronic Devices and Circuits (EC) and Electromagnetic Fields and Transmission Lines (EC)

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Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Communications (EC), Electronic Devices and Circuits (EC) and Electromagnetic Fields and Transmission Lines (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○ ○ ○ ● ○ ○ Easy ● ● ○ Medium ● ● ● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Subject & Unit Snapshot

Every subject and every unit within it opens with a **Snapshot** box summarizing its real question data:

Questions	the real total for this unit or subject.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks that subject or unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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3 Electromagnetic Fields and Transmission Lines 97

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SUBJECT

1

Communications

Subject Snapshot*				
Stream: EC	**Questions:** 313	**MCQ:** 221	**MSQ:** 6	**NAT:** 86
1M: 107	**2M:** 206	**Descriptive:** 0	**Avg Weightage** ~11M	**Range:** 8M(2020)–14M(2021)

**See preface for how Avg Weightage and Range are calculated.*

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Communications

Unit	Topic	Pages
I.	Random Processes	3
II.	Digital Communications	16
III.	Information Theory	37
IV.	Analog Communications	42

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UNIT

I

Random Processes

Topics

1. Autocorrelation and Power Spectral Density
2. Properties of White Noise
3. Filtering of Random Signals Through LTI Systems

Unit Snapshot*

Stream: EC	**Questions:** 69	**MCQ:** 50	**MSQ:** 1	**NAT:** 18
1M: 23	**2M:** 46	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2021)–8M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows the Normal distribution with zero mean and unity variance.
Y and Z are defined as,

$$Y = X + N \quad \text{and} \quad Z = X^2 + N.$$

Which of the following pairs represents the values of correlation between X and Y and that between X and Z?

A. $\frac{1}{3}$ and 0
B. $\frac{1}{3}$ and $\frac{1}{9}$
C. $\frac{1}{3}$ and $\frac{1}{3}$
D. 1 and 0

Q1.25.1 MCQ 2025 2M

Consider a real-valued random process

$$f(t) = \sum_{n=1}^{N} a_n \, p(t - nT),$$

where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables.

Read the following statements about the random process and choose the correct option.

(i) The mean of the process $f(t)$ is independent of time t.

(ii) The autocorrelation function $E[f(t)f(t+\tau)]$ is independent of time t for all τ.

(Here, $E[\cdot]$ is the expectation operation.)

A. (i) is TRUE and (ii) is FALSE

B. Both (i) and (ii) are TRUE

C. Both (i) and (ii) are FALSE

D. (i) is FALSE and (ii) is TRUE

Q1.24.1 MCQ 2024 1M

A white Gaussian noise $w(t)$ with zero mean and power spectral density $\frac{N_0}{2}$, when applied to a first-order RC low pass filter produces an output $n(t)$. At a particular time $t = t_k$, the variance of the random variable $n(t_k)$ is _____.

A. $\frac{N_0}{4RC}$
B. $\frac{N_0}{2RC}$
C. $\frac{N_0}{RC}$
D. $\frac{2N_0}{RC}$

Q1.24.2 NAT 2024 1M

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval $[0, 1]$. The probability that $X \geq Y$ is _____.

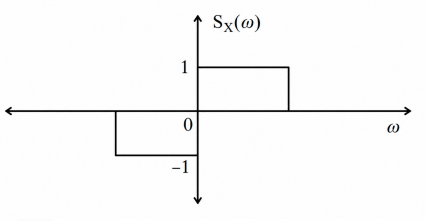
Q1.23.1 MSQ 2023 1M

For a real signal, which of the following is/are valid power spectral density/densities?

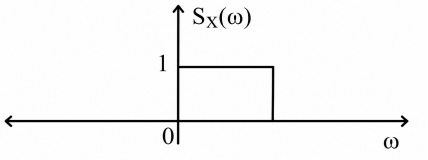
☐ **A.** $S_X(\omega) = \frac{2}{9 + \omega^2}$

☐ **B.** $S_X(\omega) = e^{-\omega^2} \cos^2 \omega$

☐ **C.**



☐ **D.**



Q1.23.2 MCQ 2023 2M

A random variable X, distributed normally as $N(0, 1)$, undergoes the transformation $Y = h(X)$, given in the figure. The form of the probability density function of Y is
(In the options given below, a, b, c are non-zero constants and $g(y)$ is piece-wise continuous function)

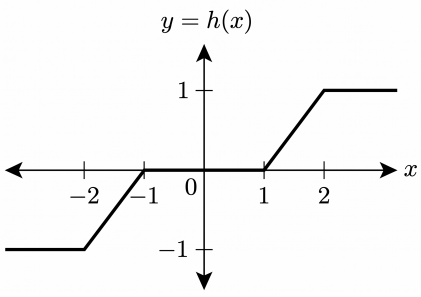


Fig. Q1.23.2

A. $a\delta(y-1) + b\delta(y+1) + g(y)$
B. $a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$
C. $a\delta(y+2) + b\delta(y) + c\delta(y-2) + g(y)$

D. $a\delta(y+2)+b\delta(y-2)+g(y)$

Q1.23.3 NAT 2023 2M

Let $X(t)$ be a white Gaussian noise with power spectral density $\frac{1}{2}$ W/Hz. If $X(t)$ is input to an LTI system with impulse response $e^{-t}u(t)$. The average power of the system output is _____ W *(rounded off to two decimal places).*

Q1.22.1 NAT 2022 2M

Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, $f(x)$, as shown in the figure.

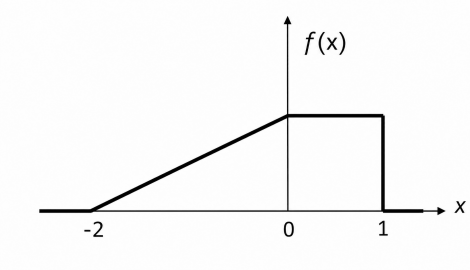


Fig. Q1.22.1

Consider a 1 bit quantizer that maps positive samples to value α and others to value β. If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^*-\beta^*)$ = _____ *(rounded off to two decimal places).*

Q1.21.1 NAT 2021 1M

The autocorrelation function $R_X(\tau)$ of a wide-sense stationary random process $X(t)$ is shown in the figure.

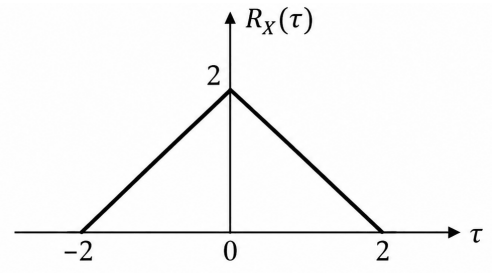


Fig. Q1.21.1

The average power of $X(t)$ is _____

Q1.20.1 NAT 2020 1M

The random variable

$$Y=\int_{-\infty}^{\infty} W(t)\phi(t)\,dt$$

where

$$\phi(t)=\begin{cases}1, & 5\le t\le 7\\ 0, & \text{otherwise}\end{cases}$$

and $W(t)$ is a real white Gaussian noise process with two-sided power spectral density $S_W(f)=3$ W/Hz, for all f. The variance of Y is _____.

Q1.19.1 NAT 2019 2M

Let a random process $Y(t)$ be described as

$$Y(t)=h(t)*X(t)+Z(t),$$

where $X(t)$ is a white noise process with power spectral density $S_X(f)=5$ W/Hz. The filter $h(t)$ has a magnitude response given by

$$|H(f)|=0.5 \quad \text{for } -5\le f\le 5,$$

and zero elsewhere. $Z(t)$ is a stationary random process, uncorrelated with $X(t)$, with power spectral density as shown in the figure. The power in $Y(t)$, in watts, is equal to _____ W (rounded off to two decimal places).

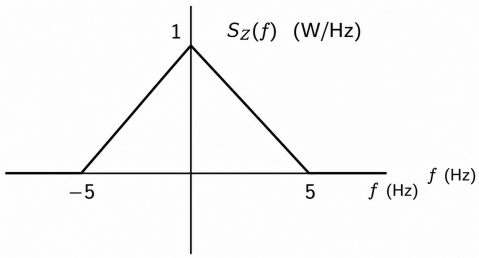


Fig. Q1.19.1

Q1.19.2 NAT 2019 1M

If X and Y are random variables such that

$$E[2X+Y]=0$$

and

$$E[X+2Y]=33,$$

then

$$E[X]+E[Y]=_____.$$

Q1.19.3 NAT 2019 1M

Let Z be an exponential random variable with mean 1. That is, the cumulative distribution function of Z is given by

$$F_Z(x) = \begin{cases} 1 - e^{-x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Then

$$\Pr(Z > 2 \mid Z > 1),$$

rounded off to two decimal places, is equal to _____.

Q1.18.1 MCQ 2018 2M

Consider a white Gaussian noise process $N(t)$ with two-sided power spectral density

$$S_N(f) = 0.5 \text{ W/Hz}$$

as input to a filter with impulse response

$$0.5e^{-t^2/2}$$

(where t is in seconds) resulting in output $Y(t)$. The power in $Y(t)$ in watts is

A. 0.11 **B.** 0.22
C. 0.33 **D.** 0.44

Q1.18.2 NAT 2018 1M

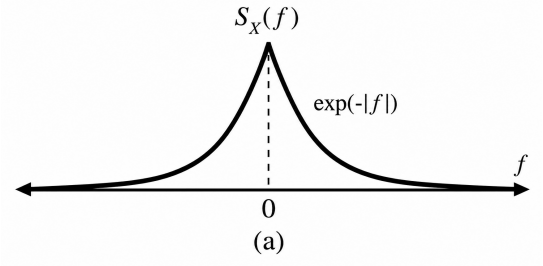
Let X_1, X_2, X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is _____.

Q1.17.1 MCQ 2017 2M

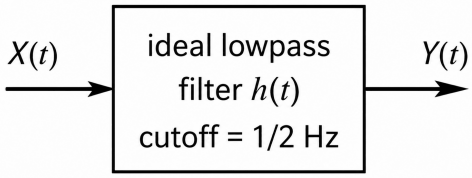
Let $X(t)$ be a wide sense stationary random process with the power spectral density $S_X(f)$ as shown in Figure (a), where f is in Hertz (Hz). The random process $X(t)$ is input to an ideal lowpass filter with the frequency response

$$H(f) = \begin{cases} 1, & |f| \leq \frac{1}{2} \text{ Hz} \\ 0, & |f| > \frac{1}{2} \text{ Hz} \end{cases}$$

as shown in Figure (b). The output of the lowpass filter is $Y(t)$.



(a)



(b)

Let E be the expectation operator and consider the following statements:

1. $E\big(X(t)\big) = E\big(Y(t)\big)$
2. $E\big(X^2(t)\big) = E\big(Y^2(t)\big)$
3. $E\big(Y^2(t)\big) = 2$

Select the correct option:

A. only I is true **B.** only II and III are true
C. only I and II are true **D.** only I and III are true

Q1.17.2 NAT 2017 1M

Consider the random process

$$X(t) = U + Vt,$$

where U is a zero-mean Gaussian random variable and V is a random variable uniformly distributed between 0 and 2. Assume that U and V are statistically independent. The mean value of the random process at $t = 2$ is _____.

Q1.16.1 NAT 2016 2M

An information source generates a binary sequence $\{a_n\}$. a_n can take one of the two possible values -1 and $+1$ with equal probability and are statistically independent and identically distributed. This sequence is precoded to obtain another sequence $\{\beta_n\}$, as

$$\beta_n = a_n + k\, a_{n-3}.$$

The sequence $\{\beta_n\}$ is used to modulate a pulse $g(t)$ to generate the baseband signal

$$X(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT),$$

where

$$g(t) = \begin{cases} 1, & 0 \leq t \leq T \\ 0, & \text{otherwise.} \end{cases}$$

If there is a null at

$$f = \frac{1}{3T}$$

in the power spectral density of $X(t)$, then k is _____.

Q1.16.2 NAT 2016 2M

Consider a random process

$$X(t) = 3V(t) - 8,$$

where $V(t)$ is a zero mean stationary random process with autocorrelation

$$R_V(\tau) = 4e^{-5|\tau|}.$$

The power in $X(t)$ is _____.

Q1.16.3 MCQ 2016 2M

A wide sense stationary random process $X(t)$ passes through the LTI system shown in the figure. If the autocorrelation function of $X(t)$ is $R_X(\tau)$, then the autocorrelation function $R_Y(\tau)$ of the output $Y(t)$ is equal to

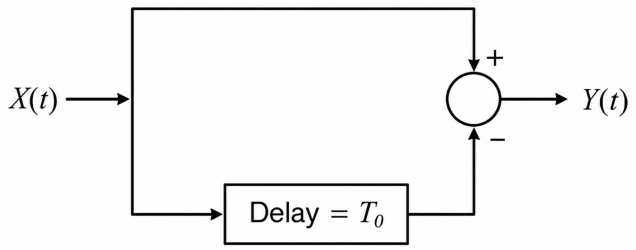


Fig. Q1.16.3

A. $2R_X(\tau) + R_X(\tau - T_0) + R_X(\tau + T_0)$

B. $2R_X(\tau) - R_X(\tau - T_0) - R_X(\tau + T_0)$

C. $2R_X(\tau) + 2R_X(\tau - 2T_0)$

D. $2R_X(\tau) - 2R_X(\tau - 2T_0)$

Q1.15.1 MCQ 2015 2M

A zero mean white Gaussian noise having power spectral density $\frac{N_0}{2}$ is passed through an LTI filter whose impulse response $h(t)$ is shown in the figure. The variance of the filtered noise at $t = 4$ is

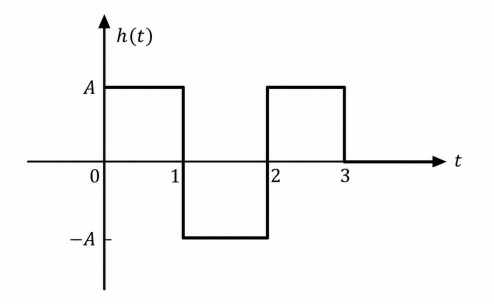


Fig. Q1.15.1

A. $\frac{3}{2}A^2N_0$ **B.** $\frac{3}{4}A^2N_0$

C. A^2N_0 **D.** $\frac{1}{2}A^2N_0$

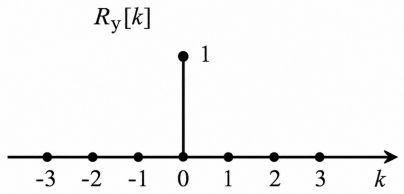
Q1.15.2 MCQ 2015 2M

$\{X_n\}_{n=-\infty}^{\infty}$ is an independent and identically distributed (i.i.d.) random process with X_n equally likely to be +1 or −1. $\{Y_n\}_{n=-\infty}^{\infty}$ is another random process obtained as

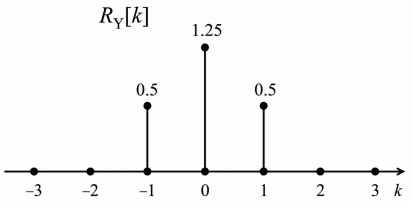
$$Y_n = X_n + 0.5X_{n-1}.$$

The autocorrelation function of $\{Y_n\}_{n=-\infty}^{\infty}$, denoted by $R_Y[k]$, is

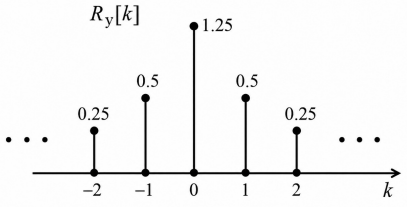
A.



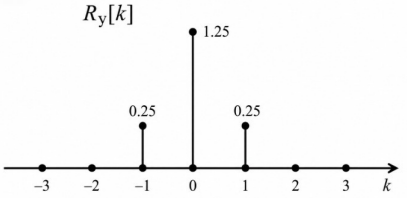
B.



C.



D.



Q1.15.3 MCQ 2015 2M

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables.
If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \geq 1)$ is equal to

A. $pq + (1 - p)(1 - q)$ **B.** pq
C. $p(1 - q)$ **D.** $1 - pq$

Q1.15.4 NAT 2015 2M

A random binary wave $y(t)$ is given by

$$y(t) = \sum_{n=-\infty}^{\infty} X_n \, p(t - nT - \phi)$$

where

$$p(t) = u(t) - u(t - T),$$

$u(t)$ is the unit step function and ϕ is an independent random variable with uniform distribution in $[0, T]$. The sequence $\{X_n\}$ consists of independent and identically distributed binary valued random variables with $P\{X_n = +1\} = P\{X_n = -1\} = 0.5$ for each n.The value of the autocorrelation

$$R_{yy}\left(\frac{3T}{4}\right) \triangleq E\left[y(t)\, y\left(t - \frac{3T}{4}\right)\right]$$

equals _____.

Q1.14.1 MCQ 2014 2M

Consider a random process

$$X(t) = \sqrt{2}\sin(2\pi t + \varphi),$$

where the random phase φ is uniformly distributed in the interval $[0, 2\pi]$. The auto-correlation $E[X(t_1)X(t_2)]$ is

A. $\cos(2\pi(t_1 + t_2))$ **B.** $\sin(2\pi(t_1 - t_2))$
C. $\sin(2\pi(t_1 + t_2))$ **D.** $\cos(2\pi(t_1 - t_2))$

Q1.14.2 NAT 2014 2M

Let $Q(\sqrt{\gamma})$ be the BER of a BPSK system over an AWGN channel with two-sided noise power spectral density $N_0/2$. The parameter γ is a function of bit energy and noise power spectral density.
A system with two independent and identical AWGN channels with noise power spectral density $N_0/2$ is shown in the figure. The BPSK demodulator receives the sum of outputs of both the channels.

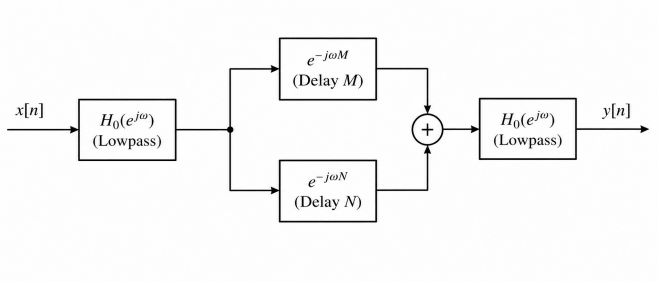


Fig. Q1.14.2

If the BER of this system is $Q(b\sqrt{\gamma})$, then the value of b is _____.

Q1.14.3 NAT 2014 2M

The power spectral density of a real stationary random process $X(t)$ is given by

$$S_X(f) = \begin{cases} \frac{1}{W}, & |f| \leq W \\ 0, & |f| > W \end{cases}$$

The value of the expectation

$$E\left[\pi X(t)\, X\left(t - \frac{1}{4W}\right)\right]$$

is _____.

Q1.14.4 MCQ 2014 2M

Let $X(t)$ be a wide sense stationary (WSS) random process with power spectral density $S_X(f)$. If $Y(t)$ is the process defined as $Y(t) = X(2t - 1)$, the power spectral density $S_Y(f)$ is

A. $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j\pi f}$ **B.** $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j\pi f/2}$
C. $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)$ **D.** $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j2\pi f}$

Q1.14.5 NAT 2014 2M

A real band-limited random process $X(t)$ has two-sided power spectral density

$$S_X(f) = \begin{cases} 10^{-6}(3000 - |f|) \text{ Watts/Hz}, & \text{for } |f| \leq 3 \text{ kHz} \\ 0, & \text{otherwise} \end{cases}$$

where f is the frequency expressed in Hz.The signal $X(t)$ modulates a carrier $\cos(16000\pi t)$ and the resultant signal is passed through an ideal band-pass filter of unity gain with centre frequency of 8 kHz and band-width of 2 kHz. The output power (in Watts) is _____.

Q1.14.6 NAT 2014 2M

Let X_1, X_2, and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $P\{X_1 + X_2 \leq X_3\}$ is _____.

Q1.14.7 MCQ 2014 2M

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2, respectively. The relation which always holds true is

A. $(E[X])^2 > E[X^2]$ **B.** $E[X^2] \geq (E[X])^2$
C. $E[X^2] = (E[X])^2$ **D.** $E[X^2] > (E[X])^2$

Common Data Questions

Answer the question 1.13.1 and question 1.13.2 using the following data. Bits 1 and 0 are transmitted with equal probability.At the receiver, the pdf of the respective received signals for both bits are as shown below.

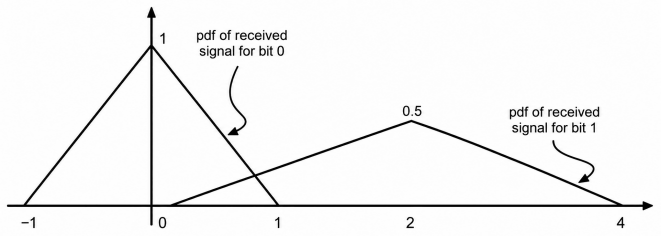


Fig. Q1.14.7

Q1.13.1 MCQ 2013 2M

If the detection threshold is 1, the BER will be

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$
C. $\frac{1}{8}$ **D.** $\frac{1}{16}$

Q1.13.2 MCQ 2013 2M

The optimum threshold to achieve minimum bit error rate (BER) is

A. $\frac{1}{2}$ **B.** $\frac{4}{5}$
C. 1 **D.** $\frac{3}{2}$

Q1.13.3 MCQ 2013 2M

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x

A. $F(x) - G(x) \leq 0$ **B.** $F(x) - G(x) \geq 0$
C. $(F(x) - G(x))\, x \leq 0$ **D.** $(F(x) - G(x))\, x \geq 0$

Q1.13.4 MCQ 2013 2M

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

A. $\frac{4}{9}$ **B.** $\frac{1}{2}$
C. $\frac{2}{3}$ **D.** $\frac{5}{9}$

Q1.12.1 MCQ 2012 1M

The power spectral density of a real process $X(t)$ for positive frequencies is shown below. The values of $E[X^2(t)]$ and $E[|X(t)|]$, respectively, are

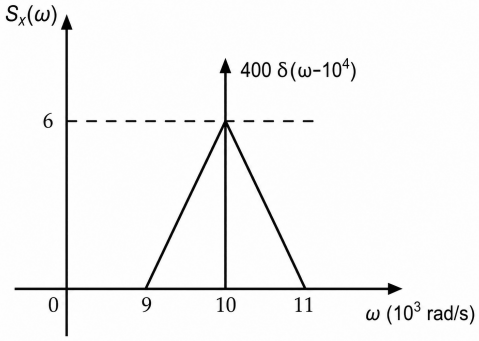


Fig. Q1.12.1

A. $6000/\pi,\ 0$ **B.** $6400/\pi,\ 0$
C. $6400/\pi,\ 20/(\pi\sqrt{2})$ **D.** $6000/\pi,\ 20/(\pi\sqrt{2})$

Q1.12.2 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $1/2$ is

A. $3/4$ **B.** $9/16$
C. $1/4$ **D.** $2/3$

Q1.11.1 MCQ 2011 2M

$X(t)$ is a stationary random process with autocorrelation function

$$R_X(\tau) = \exp(-\pi\tau^2).$$

This process is passed through the system shown below. The power spectral density of the output process $Y(t)$ is

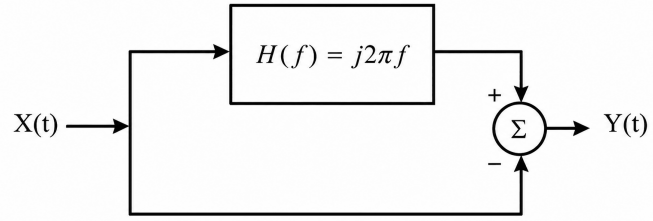


Fig. Q1.11.1

A. $(4\pi^2 f^2 + 1)\exp(-\pi f^2)$ **B.** $(4\pi^2 f^2 - 1)\exp(-\pi f^2)$
C. $(4\pi^2 f^2 + 1)\exp(-\pi f)$ **D.** $(4\pi^2 f^2 - 1)\exp(-\pi f)$

Q1.10.1 MCQ 2010 2M

$X(t)$ is a stationary process with the power spectral density $S_X(f) > 0$ for all f. The process is passed through a system shown below.

Let $S_Y(f)$ be the power spectral density of $Y(t)$. Which one of the following statements is correct?

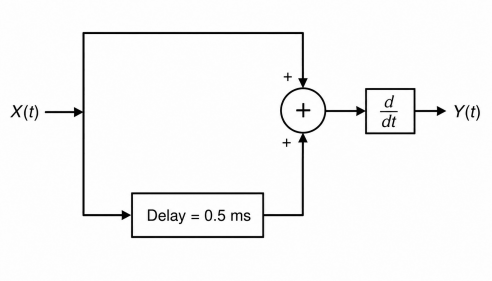


Fig. Q1.10.1

A. $S_Y(f) > 0$ for all f

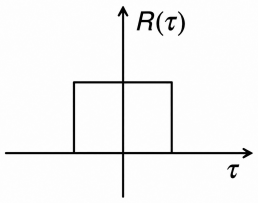
B. $S_Y(f) = 0$ for $|f| > 1$ kHz

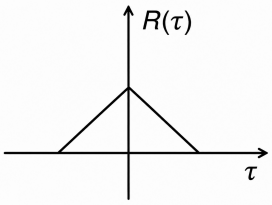
C. $S_Y(f) = 0$ for $f = nf_0$, $f_0 = 2$ kHz, n any integer

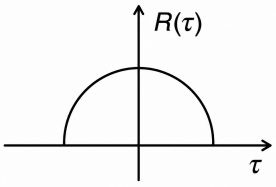
D. $S_Y(f) = 0$ for $f = (2n + 1)f_0$, $f_0 = 1$ kHz, n any integer

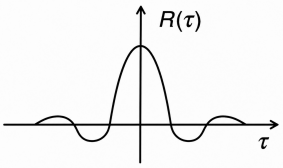
Q1.09.1 MCQ 2009 1M

If the power spectral density of a stationary random process is a sinc-squared function of frequency, the shape of its autocorrelation is _____.

A.


B.


C.


D.


Q1.09.2 MCQ 2009 1M

A white noise process $X(t)$ with two-sided power spectral density

$$S_X(f) = 1 \times 10^{-10} \text{ W/Hz}$$

is input to a filter whose magnitude-squared response $|H(f)|^2$ is triangular, having

$$|H(0)|^2 = 1,$$

and extending from -10 kHz $\le f \le 10$ kHz. The power of the output process $Y(t)$ is

A. 5×10^{-7} W
B. 1×10^{-6} W
C. 2×10^{-6} W
D. 1×10^{-5} W

Q1.08.1 MCQ 2008 2M

Noise with double-sided power spectral density of K over all frequencies is passed through a RC low pass filter with 3 dB cut-off frequency of f_c. The noise power at the filter output is

A. K
B. Kf_c
C. $K\pi f_c$
D. ∞

Q1.08.2 MCQ 2008 1M

The probability density function (PDF) of a random variable X is as shown below.

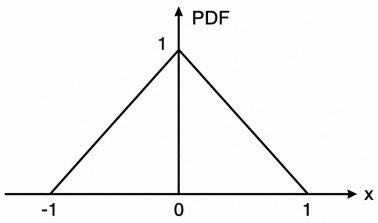
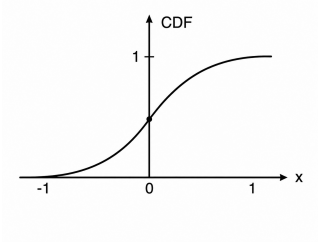
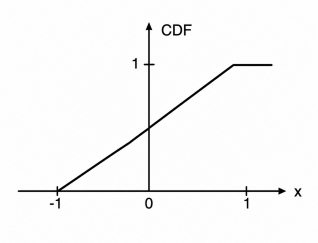
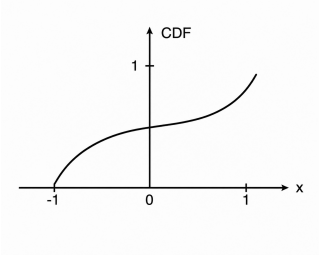


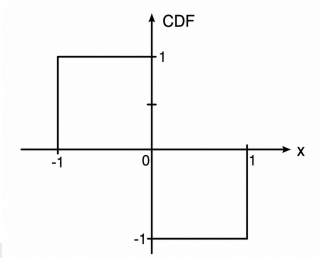
Fig. Q1.08.2

The corresponding cumulative distribution function (CDF) has the form

A.


B.


C.


D.


Q1.07.1 MCQ 2007 1M

If $S(f)$ is the power spectral density of a real, wide-sense stationary random process, then which of the following is ALWAYS true?

A. $S(0) \geq S(f)$
B. $S(f) \geq 0$
C. $S(-f) = -S(f)$
D. $\int_{-\infty}^{\infty} S(f)\, df = 0$

Q1.07.2 MCQ 2007 1M

If $R(\tau)$ is the autocorrelation function of a real, wide-sense stationary random process, then which of the following is NOT true?

A. $R(\tau) = R(-\tau)$
B. $|R(\tau)| \leq R(0)$
C. $R(\tau) = -R(-\tau)$
D. The mean square value of the process is $R(0)$

Q1.06.1 MCQ 2006 2M

A zero-mean white Gaussian noise is passed through an ideal lowpass filter of bandwidth 10 kHz. The output is then uniformly sampled with sampling period $t_s = 0.03$ msec. The samples so obtained would be

A. correlated
B. statistically independent
C. uncorrelated
D. orthogonal

Common Data Questions

Answer the question 1.06.2 and question 1.06.3 using the following data.
The following two questions refer to the generation of a specified Power Spectral Density (PSD) using an LTI system/filter.

Q1.06.2 MCQ 2006 2M

It is desired to generate a stochastic process (as voltage process) with power spectral density

$$S(\omega) = \frac{16}{16 + \omega^2}$$

by driving a Linear-Time-Invariant system by zero mean white noise (as voltage process) with power spectral density being constant equal to 1. The system which can perform the desired task could be

A. first order lowpass R-L filter

B. first order highpass R-C filter

C. tuned L-C filter

D. series R-L-C filter

Q1.06.3 MCQ 2006 2M

The parameters of the system obtained in above question would be

A. first order R-L lowpass filter would have $R = 4\Omega,\ L = 1H$

B. first order R-C highpass filter would have $R = 4\Omega,\ C = 0.25F$

C. tuned L-C filter would have $L = 4H,\ C = 4F$

D. series R-L-C lowpass filter would have $R = 1\Omega,\ L = 4H,\ C = 4F$

Q1.06.4 MCQ 2006 2M

A uniformly distributed random variable X with probability density function

$$f_X(x) = \frac{1}{10}\,(u(x+5) - u(x-5))$$

where $u(\cdot)$ is the unit step function, is passed through a transformation given in the figure below. The probability density function of the transformed random variable Y would be

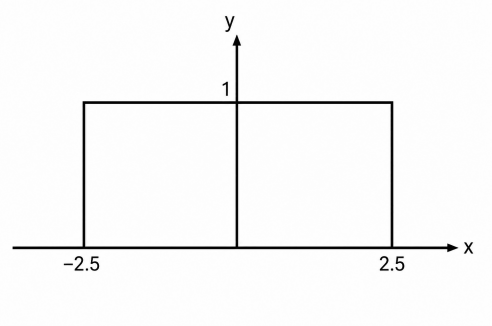


Fig. Q1.06.4

A. $f_Y(y) = \frac{1}{5}\,[u(y+2.5) - u(y-2.25)]$

B. $f_Y(y) = 0.5\,\delta(y) + 0.5\,\delta(y-1)$

C. $f_Y(y) = 0.25\,\delta(y+2.5) + 0.25\,\delta(y-2.5) + 0.5\,\delta(y)$

D. $f_Y(y) = 0.25\,\delta(y+2.5) + 0.25\,\delta(y-2.5) + \frac{1}{10}\,[u(y+2.5) - u(y-2.5)]$

Q1.06.5 MCQ 2006 1M

A probability density function is of the form

$$p(x) = Ke^{-\alpha|x|}, \qquad x \in (-\infty, \infty).$$

The value of K is

A. 0.5 **B.** 1

C. 0.5α **D.** α

Q1.05.1 MCQ 2005 2M

Noise with uniform power spectral density of $N_0 W/\text{Hz}$ is passed through a filter

$$H(\omega) = 2\exp(-j\omega t_d)$$

followed by an ideal low pass filter of bandwidth B Hz. The output noise power in Watts is

A. $2N_0B$ **B.** $4N_0B$

C. $8N_0B$ **D.** $16N_0B$

Q1.05.2 MCQ 2005 2M

An output of a communication channel is a random variable v with the probability density function as shown in figure. The mean square value of v is

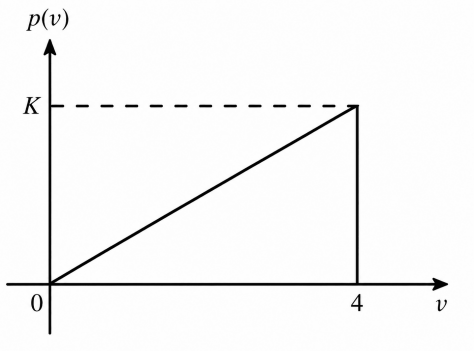


Fig. Q1.05.2

A. 4 **B.** 6

C. 8 **D.** 9

Q1.04.1 MCQ 2004 2M

A random variable X with uniform density in the interval 0 to 1 is quantized as follows:

$$\text{If } 0 \le X \le 0.3, \quad x_q = 0$$
$$\text{If } 0.3 < X \le 1, \quad x_q = 0.7$$

where x_q is the quantized value of X.
The root-mean square value of the quantization noise is

A. 0.573 **B.** 0.198
C. 2.205 **D.** 0.266

Q1.04.2 MCQ 2004 2M

The distribution function $F_X(x)$ of a random variable X is shown in figure below. The probability that $X = 1$ is

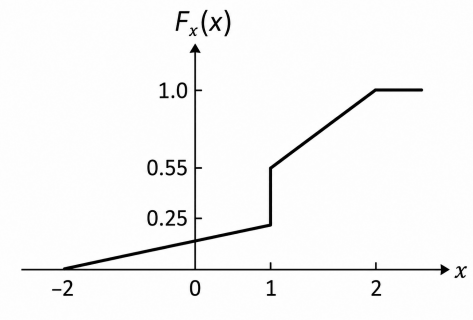


Fig. Q1.04.2

A. zero **B.** 0.25
C. 0.55 **D.** 0.30

Common Data Questions

Answer the question 1.03.1 and question 1.03.2 using the following data.
$X(t)$ is a random process with a constant mean value of 2 and the auto-correlation function

$$R_X(\tau) = 4\left(e^{-0.2|\tau|} + 1\right).$$

Q1.03.1 MCQ 2003 2M

Let Y and Z be the random variables obtained by sampling $X(t)$ at $t = 2$ and $t = 4$ respectively. Let

$$W = Y - Z.$$

The variance of W is

A. 13.36 **B.** 9.36
C. 2.64 **D.** 8.00

Q1.03.2 MCQ 2003 2M

Let X be the Gaussian random variable obtained by sampling the process at $t = t_i$ and let

$$Q(\alpha) = \int_{\alpha}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\, dy.$$

The probability that $X \le 1$ is

A. $1 - Q(0.5)$ **B.** $Q(0.5)$

C. $Q\left(\frac{1}{2\sqrt{2}}\right)$ **D.** $1 - Q\left(\frac{1}{2\sqrt{2}}\right)$

Q1.03.3 MCQ 2003 2M

Let X and Y be two statistically independent random variables uniformly distributed in the ranges $(-1, 1)$ and $(-2, 1)$ respectively. Let $Z = X + Y$. Then the probability that $(Z \le -2)$ is

A. zero **B.** $\frac{1}{6}$

C. $\frac{1}{3}$ **D.** $\frac{1}{12}$

Q1.03.4 MCQ 2003 2M

The noise at the input to an ideal frequency detector is white. The detector is operating above threshold. The power spectral density of the noise at the output is

A. raised-cosine **B.** flat
C. parabolic **D.** Gaussian

Q1.02.1 MCQ 2002 2M

If the variance σ_d^2 of

$$d(n) = x(n) - x(n-1)$$

is one-tenth the variance σ_x^2 of a stationary zero mean discrete time signal $x(n)$, then the normalized autocorrelation function

$$\frac{R_{xx}(k)}{\sigma_x^2}$$

at $k = 1$ is

A. 0.95 **B.** 0.90
C. 0.10 **D.** 0.05

Q1.01.1 MCQ 2001 1M

The PDF of a Gaussian random variable X is given by

$$P_X(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{(x-4)^2}{18}}.$$

The probability of the event $\{X = 4\}$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{3\sqrt{2\pi}}$

C. 0 **D.** $\frac{1}{4}$

Q1.98.1 MCQ 1998 1M

The ACF of a rectangular pulse of duration T is

A. a rectangular pulse of duration T

B. a rectangular pulse of duration $2T$

C. a triangular pulse of duration T

D. a triangular pulse of duration $2T$

Q1.98.2 MCQ 1998 1M

The amplitude spectrum of a Gaussian pulse is

A. uniform **B.** a sine function

C. Gaussian **D.** an impulse function

Q1.98.3 NAT 1998 2M

White noise of two-sided spectral density

$$2 \times 10^{-6} \text{ V}^2/\text{Hz}$$

is applied to a simple R-C low-pass filter whose 3 dB cut-off frequency is 4 kHz. Find the mean squared value of the noise output _____ 10^{-2} V^2.

Q1.97.1 MCQ 1997 1M

A probability density function is given by

$$p(x) = Ke^{-x^2/2}, \qquad -\infty < x < \infty.$$

The value of K should be

A. $\frac{1}{\sqrt{2\pi}}$ **B.** $\sqrt{\frac{2}{\pi}}$

C. $\frac{1}{2\sqrt{\pi}}$ **D.** $\frac{1}{\pi\sqrt{2}}$

Q1.97.2 MCQ 1997 1M

The power spectral density of a deterministic signal is given by

$$\left(\frac{\sin(f)}{f^2}\right)$$

where f is frequency. The autocorrelation function of this signal in the time domain is

A. a rectangular pulse **B.** a delta function

C. a sine pulse **D.** a triangular pulse

Q1.95.1 MCQ 1995 1M

For a narrow band noise with Gaussian quadrature and inphase components, the probability density function of its envelope will be

A. uniform **B.** Gaussian

C. exponential **D.** Rayleigh

Q1.92.1 MCQ 1992 2M

For a random variable X following the probability density function, $p(x)$, shown in figure, the mean and the variance are, respectively.

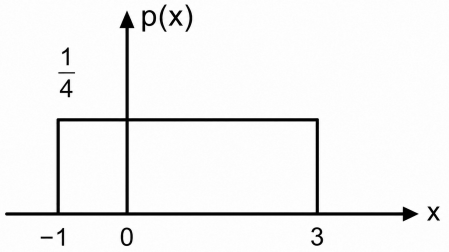


Fig. Q1.92.1

A. 1/2 and 2/3 **B.** 1 and 4/3

C. 1 and 2/3 **D.** 2 and 4/3

Q1.89.1 MCQ 1989 2M

Zero mean Gaussian noise of variance N is applied to a half wave rectifier. The mean squared value of the rectifier output will be:

A. Zero **B.** $N/2$

C. $N/\sqrt{2}$ **D.** N

Q1.87.1 MCQ 1987 2M

The variance of a random variable X is σ_x^2. Then the variance of $-kx$ (where k is a positive constant) is

A. σ_x^2 **B.** $-k\sigma_x^2$

C. $k\sigma_x^2$ **D.** $k^2\sigma_x^2$

Answer Key

Q1.26.1 A	Q1.25.1 A	Q1.24.1 A
Q1.24.2 0.5	Q1.23.1 A,B	Q1.23.2 B
Q1.23.3 0.25	Q1.22.1 1.17	Q1.21.1 2.0
Q1.20.1 6	Q1.19.1 17.40 to 17.60	Q1.19.2 11
Q1.19.3 0.36 to 0.38	Q1.18.1 B	Q1.18.2 0.25
Q1.17.1 A	Q1.17.2 2.0	Q1.16.1 -1.01 to -0.99
Q1.16.2 99 to 101	Q1.16.3 B	Q1.15.1 A
Q1.15.2 B	Q1.15.3 D	Q1.15.4 0.24 to 0.26
Q1.14.1 D	Q1.14.2 1.4 to 1.42	Q1.14.3 3.9 to 4.1
Q1.14.4 C	Q1.14.5 2.4 to 2.6	Q1.14.6 0.15 to 0.18
Q1.14.7 B	Q1.13.1 D	Q1.13.2 B
Q1.13.3 D	Q1.13.4 B	Q1.12.1 B
Q1.12.2 B	Q1.11.1 A	Q1.10.1 D
Q1.09.1 B	Q1.09.2 B	Q1.08.1 C
Q1.08.2 A	Q1.07.1 B	Q1.07.2 C
Q1.06.1 A	Q1.06.2 A	Q1.06.3 A
Q1.06.4 B	Q1.06.5 C	Q1.05.1 B
Q1.05.2 C	Q1.04.1 B	Q1.04.2 D
Q1.03.1 C	Q1.03.2 D	Q1.03.3 D
Q1.03.4 C	Q1.02.1 A	Q1.01.1 C
Q1.98.1 D	Q1.98.2 C	Q1.98.3 2.51
Q1.97.1 A	Q1.97.2 A	Q1.95.1 D
Q1.92.1 B	Q1.89.1 B	Q1.87.1 D

UNIT

II

Digital Communications

Topics

1. PCM and DPCM
2. Digital Modulation - ASK, PSK, FSK, QAM
3. Bandwidth and Inter-Symbol Interference
4. MAP and ML Detection
5. Matched Filter Receiver
6. SNR and BER Analysis
7. Error Correction - Hamming Codes and CRC

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 136	**MCQ:** 98	**MSQ:** 0	**NAT:** 38
1M: 40	**2M:** 96	**Descriptive:** 0	**Avg Weightage** ~5M	**Range:** 1M(2023)–8M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 NAT 2026 1M

A wireless digital transmission scheme is using 16-QAM over an additive white Gaussian noise channel and a maximum-likelihood receiver. Consider the information bit rate from source to be 4×10^6 bits per second.

The minimum transmission bandwidth (in MHz) of the modulated signal necessary for optimum recovery of information at the receiver is _____.

(rounded off to two decimal places)

Q2.26.2 MCQ 2026 1M

Consider a discrete memoryless source with an alphabet of four source symbols.

$s(t)$ is a multi-level $(-1, 0, +1, +2)$ signal representing a long sequence of random symbols from the above source which is generating 10^4 symbols per second.

Which of the following options is the correct value of equivalent Nyquist bandwidth of $s(t)$?

A. 10 kHz **B.** 64 kHz

C. 5 kHz **D.** 20 kHz

Q2.26.3 MCQ 2026 2M

A QPSK modulated signal from an additive white Gaussian noise (AWGN) channel is received with an $E_b/N_0 = 8.4$ dB at the input of a coherent QPSK demodulator. A maximum-likelihood reception method is used in the demodulator.

Assume the complementary error function

$$\text{erfc}(u) \cong \left[\frac{1}{u\sqrt{\pi}}\right] \exp(-u^2).$$

Which is the nearest bit error rate (BER) at the output of the demodulator?

A. 10^{-3} **B.** 10^{-4}

C. 10^{-5} **D.** 10^{-6}

Q2.26.4 NAT 2026 2M

The average bit error rate at the input of a $(7, 4, 1)$ Hamming decoder is 0.10.

The probability that the decoder will fail to decode a received word correctly is _____.

(rounded off to two decimal places)

Q2.25.1 NAT 2025 1M

The generator matrix of a $(6, 3)$ binary linear block code is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

The minimum Hamming distance $d_{\min}$ between codewords equals _____.

(answer in integer)

Q2.25.2 MCQ 2025 2M

A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S. The receiver uses the maximum likelihood decoder to estimate the transmitted symbol S.

Suppose the probability of symbol estimation error P_e is expressed as follows:

$$P_e = \alpha\, P(N > 1),$$

where $P(N > 1)$ denotes the probability that N exceeds 1.What is the value of α?

A. $1/3$ **B.** 1

C. $2/3$ **D.** $4/3$

Q2.24.1 MCQ 2024 1M

A digital communication system transmits through a noiseless bandlimited channel $[-W\ W]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum_n b[n]x(t - nT)$ where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = mT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

A. $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T$

B. $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = \frac{1}{T}$

C. $\sum_{m=-\infty}^{\infty} X(f + mT) = T$

D. $\sum_{m=-\infty}^{\infty} X(f + mT) = \frac{1}{T}$

Q2.24.2 MCQ 2024 2M

A source transmits a symbol s, taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s+w$, where the noise w is zero mean with variance 4 and is independent of s.

Using

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt,$$

the optimum symbol error probability is _____.

A. $\frac{2}{3}Q(2)$ **B.** $\frac{4}{3}Q(1)$

C. $\frac{2}{3}Q(1)$ **D.** $\frac{4}{3}Q(2)$

Q2.24.3 MCQ 2024 2M

The information bit sequence $\{111010101\}$ is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$.

The encoded sequence of bits is _____.

A. $\{1110101011100\}$

B. $\{1110101011101\}$

C. $\{1110101011110\}$

D. $\{1110101010100\}$

Q2.24.4 NAT 2024 2M

Let $X(t) = A\cos(2\pi f_0 t + \theta)$ be a random process, where amplitude A and phase θ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$,the signal to quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

Q2.23.1 NAT 2023 1M

Let the signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input be given as 61.96 dB.

The resolution of the ADC is _____ bits (rounded to the nearest integer).

Q2.22.1 NAT 2022 1M

A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second.

(rounded off to one decimal place)

Q2.22.2 NAT 2022 2M

Consider communication over a memoryless binary symmetric channel using a $(7, 4)$ Hamming code. Each transmitted bit is received correctly with probability $(1-\epsilon)$, and flipped with probability ϵ. For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error.

For $\epsilon = 0.1$, the probability that a transmitted codeword is decoded correctly is _____.

(rounded off to two decimal places)

Q2.22.3 NAT 2022 2M

Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for Y given x_A and x_B are:

$$f_{Y|x_A}(y) = e^{-(y+1)}u(y+1),$$

$$f_{Y|x_B}(y) = e^{(y-1)}\big(1 - u(y-1)\big),$$

where $u(\cdot)$ is the standard unit step function. The probability of symbol error for this system is _____.

(rounded off to two decimal places)

Q2.21.1 NAT 2021 1M

A speech signal, band limited to 4 kHz, is sampled at 1.25 times the Nyquist rate. The speech samples, assumed to be statistically independent and uniformly distributed in the range −5 V to +5 V, are subsequently quantized in an 8-bit uniform quantizer and then transmitted over a voice-grade AWGN telephone channel. If the ratio of transmitted signal power to channel noise power is 26 dB, the minimum channel bandwidth required to ensure reliable transmission of the signal with arbitrarily small probability of transmission error _____ kHz.(rounded off to two decimal places)

Q2.21.2 NAT 2021 1M

A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a delta modulator (DM) operating at a sampling rate of 32 kHz. The minimum step size required to avoid slope overload noise in the DM _____ V.(rounded off to two decimal places)

Q2.21.3 MCQ 2021 2M

A digital transmission system uses a $(7, 4)$ systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-codeword pairs in this code $(m_i; c_i)$, where c_i is the codeword corresponding to the i^{th} message m_i, are known to be $(1100; 0101100)$, $(1110; 0011110)$, $(0110; 1000110)$, then which of the following is a valid codeword in this code?

A. 1101001 **B.** 1011010

C. 0001011 **D.** 0110100

Q2.21.4 NAT 2021 2M

A message signal having peak-to-peak value of 2 V, root mean square value of 0.1 V and bandwidth of 5 kHz is sampled and fed to a pulse code modulation (PCM) system that uses a uniform quantizer. The PCM output is transmitted over a channel that can support a maximum transmission rate of 50 kbps. Assuming that the quantization error is uniformly distributed, the maximum signal to quantization noise ratio that can be obtained by the PCM system _____.(rounded off to two decimal places and calculate answer in linear scale)

Q2.21.5 NAT 2021 2M

A polar non-return to zero (NRZ) waveform, using +2 V and −2 V for representing binary '1' and '0' respectively, is transmitted in the presence of additive zero-mean white Gaussian noise with variance 0.4 V^2. If the *a priori* probability of transmission of a binary '1' is 0.4, the optimum threshold voltage for a maximum *a posteriori* (MAP) receiver _____ V.(rounded off to two decimal places)

Q2.20.1 MCQ 2020 1M

A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous if at least one of its bits is decoded wrongly. The probability that the received block is erroneous is

A. $N(1-\alpha)$ **B.** α^N

C. $1-\alpha^N$ **D.** $1-(1-\alpha)^N$

Q2.20.2 NAT 2020 2M

In a digital communication system, a symbol S randomly chosen from the set $\{s_1, s_2, s_3, s_4\}$ is transmitted. It is given that $s_1 = -3$, $s_2 = -1$,$s_3 = +1$, and $s_4 = +2$. The received symbol is $Y = S + W$. W is a zero-mean unit-variance Gaussian random variable and is independent of S. P_i is the conditional probability of symbol error for the maximum likelihood (ML) decoding when the transmitted symbol $S = s_i$. The index i for which the conditional symbol error probability P_i is the highest is _____.

Q2.19.1 MCQ 2019 1M

A linear Hamming code is used to map 4-bit messages to 7-bit codewords. The encoder mapping is linear. If the message 0001 is mapped to the codeword 0000111, and the message 0011 is mapped to the codeword 1100110, then the message 0010 is mapped to

A. 0010011 **B.** 1100001

C. 1111000 **D.** 1111111

Q2.19.2 MCQ 2019 2M

A single bit, equally likely to be 0 and 1, is to be sent across an additive white Gaussian noise (AWGN) channel with power spectral density $N_0/2$. Binary signaling, with $0 \mapsto p(t)$ and $1 \mapsto q(t)$, is used for the transmission, along with an optimal receiver that minimizes the bit-error probability.

Let $\varphi_1(t), \varphi_2(t)$ form an orthonormal signal set.

If we choose $p(t) = \varphi_1(t)$ and $q(t) = -\varphi_1(t)$, we would obtain a certain bit-error probability P_b.

If we keep $p(t) = \varphi_1(t)$, but take $q(t) = \sqrt{E}\,\varphi_2(t)$, for what value of E would we obtain the same bit-error probability P_b?

A. 0 **B.** 1

C. 2 **D.** 3

Q2.19.3 NAT 2019 2M

A voice signal $m(t)$ is in the frequency range 5 kHz to 15 kHz. The signal is amplitude-modulated to generate an AM signal $f(t) = A(1 + m(t)) \cos 2\pi f_c t$, where $f_c = 600$ kHz. The AM signal $f(t)$ is to be digitized and archived. This is done by first sampling $f(t)$ at 1.2 times the Nyquist frequency, and then quantizing each sample using a 256-level quantizer. Finally, each quantized sample is binary coded using K bits, where K is the minimum number of bits required for the encoding. The rate, in Megabits per second (rounded off to 2 decimal places), of the resulting stream of coded bits is _____ Mbps.

Q2.19.4 NAT 2019 2M

A random variable X takes values -1 and $+1$ with probabilities 0.2 and 0.8, respectively. It is transmitted across a channel which adds noise N, so that the random variable at the channel output is $Y = X + N$. The noise N is independent of X, and is uniformly distributed over the interval $[-2, 2]$. The receiver makes a decision

$$\hat{X} = \begin{cases} -1, & \text{if } Y \le \theta \\ +1, & \text{if } Y > \theta \end{cases}$$

where the threshold $\theta \in [-1, 1]$ is chosen so as to minimize the probability of error $\Pr[\hat{X} \neq X]$. The minimum probability of error, rounded off to 1 decimal place, is _____.

Q2.18.1 NAT 2018 1M

Consider a binary channel code in which each codeword has a fixed length of 5 bits. The Hamming distance between any pair of distinct codewords in this code is at least 2. The maximum number of codewords such a code can contain is _____.

Q2.18.2 NAT 2018 1M

A binary source generates symbols $X \in \{-1, 1\}$ which are transmitted over a noisy channel. The probability of transmitting $X = 1$ is 0.5. Input to the threshold detector is $R = X + N$. The probability density function $f_N(n)$ of the noise N is shown below.

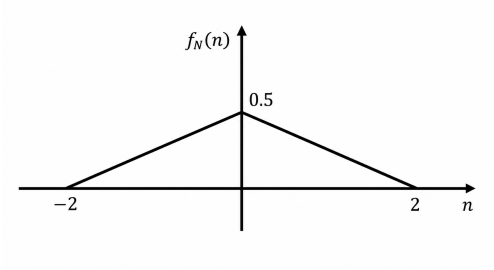


Fig. Q2.18.2

If the detection threshold is zero, then the probability of error (correct to two decimal places) is _____.

Q2.18.3 NAT 2018 2M

A random variable X takes values -0.5 and 0.5 with probabilities $\frac{1}{4}$ and $\frac{3}{4}$, respectively. The noisy observation of X is $Y = X + Z$, where Z has uniform probability density over the interval $(-1, 1)$. X and Z are independent. If the MAP rule based detector outputs $\hat{X}$ as

$$\hat{X} = \begin{cases} -0.5, & Y < \alpha \\ 0.5, & Y \ge \alpha, \end{cases}$$

then the value of α (accurate to two decimal places) is _____.

Q2.17.1 NAT 2017 1M

A sinusoidal message signal is converted to a PCM signal using a uniform quantizer. The required signal-to-quantization noise ratio (SQNR) at the output of the quantizer is 40 dB. The minimum number of bits per sample needed to achieve the desired SQNR is _____.

Q2.17.2 MCQ 2017 1M

Which one of the following statements about differential pulse code modulation (DPCM) is true?

A. The sum of message signal sample with its prediction is quantized

B. The message signal sample is directly quantized, and its prediction is not used

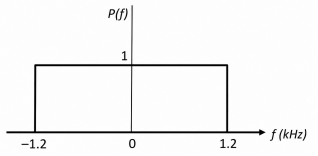
C. The difference of message signal sample and a random signal is quantized

D. The difference of message signal sample with its prediction is quantized

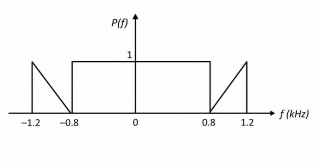
Q2.17.3 MCQ 2017 1M

In a digital communication system, the overall pulse shape $p(t)$ at the receiver before the sampler has the Fourier transform $P(f)$. If the symbols are transmitted at the rate of 2000 symbols per second, for which of the following cases is the inter symbol interference zero?

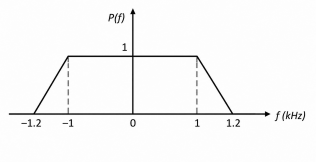
A.



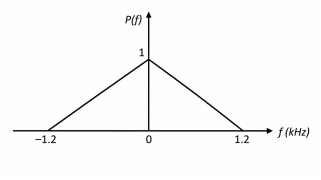
B.



C.



D.



Q2.17.4 MCQ 2017 2M

In binary frequency shift keying (FSK), the given signal waveforms are

$$u_0(t) = 5\cos(20000\pi t);\ 0 \le t \le T,$$

and

$$u_1(t) = 5\cos(22000\pi t);\ 0 \le t \le T,$$

where T is the bit-duration interval and t is in seconds. Both $u_0(t)$ and $u_1(t)$ are zero outside the interval $0 \le t \le T$. With a matched filter (correlator) based receiver, the smallest positive value of T (in milliseconds) required to have $u_0(t)$ and $u_1(t)$ uncorrelated is

A. 0.25 ms **B.** 0.5 ms

C. 0.75 ms **D.** 1.0 ms

Q2.16.1 NAT 2016 1M

Consider binary data transmission at a rate of 56 kbps using baseband binary pulse amplitude modulation (PAM) that is designed to have a raised-cosine spectrum. The transmission bandwidth (in kHz) required for a roll-off factor of 0.25 is _____.

Q2.16.2 NAT 2016 1M

A discrete memoryless source has an alphabet a_1,a_2,a_3, a_4 with corresponding probabilities $\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right\}$.The minimum required average codeword length in bits to represent this source for error-free reconstruction is _____.

Q2.16.3 NAT 2016 1M

A speech signal is sampled at 8 kHz and encoded into PCM format using 8 bits/sample. The PCM data is transmitted through a baseband channel via 4-level PAM. The minimum bandwidth (in kHz) required for transmission is _____.

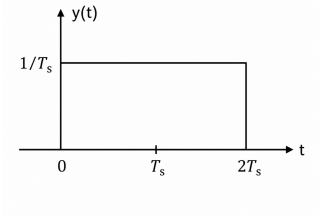
Q2.16.4 MCQ 2016 1M

A binary baseband digital communication system employs the signal

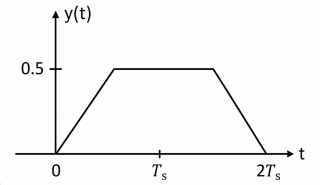
$$p(t) = \begin{cases} \frac{1}{\sqrt{T_s}}, & 0 \le t \le T_s \\ 0, & \text{otherwise} \end{cases}$$

for transmission of bits. The graphical representation of the matched filter output $y(t)$ for this signal will be

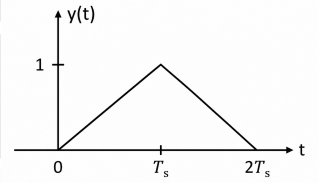
A.



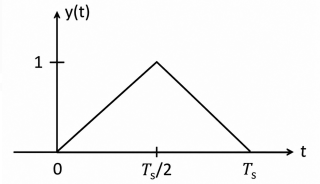
B.



C.



D.



Q2.16.5 NAT 2016 2M

A voice-grade AWGN (additive white Gaussian noise) telephone channel has a bandwidth of 4.0 kHz and two-sided noise power spectral density $\frac{\eta}{2} = 2.5 \times 10^{-5}$ Watt per Hz.If information at the rate of 52 kbps is to be transmitted over this channel with arbitrarily small bit error rate, then the minimum bit-energy E_b (in mJ/bit) necessary is _____.

Q2.16.6 NAT 2016 2M

The bit error probability of a memoryless binary symmetric channel is 10^{-5}. If 10^5 bits are sent over this channel, then the probability that not more than one bit will be in error is _____.

Q2.16.7 NAT 2016 2M

An ideal band-pass channel 500 Hz - 2000 Hz is deployed for communication. A modem is designed to transmit bits at the rate of 4800 bits/s using 16-QAM. The roll-off factor of a pulse with a raised cosine spectrum that utilizes the entire frequency band is _____.

Q2.16.8 NAT 2016 2M

A digital communication system uses a repetition code for channel encoding/decoding. During transmission, each bit is repeated three times (0 is transmitted as 000, and 1 is transmitted as 111). It is assumed that the source puts out symbols independently and with equal probability. The decoder operates as follows: In a block of three received bits, if the number of zeros exceeds the number of ones, the decoder decides in favor of a 0, and if the number of ones exceeds the number of zeros, the decoder decides in favor of a 1. Assuming a binary symmetric channel with crossover probability $p = 0.1$, the average probability of error is _____.

Q2.16.9 MCQ 2016 2M

An analog pulse $s(t)$ is transmitted over an additive white Gaussian noise (AWGN) channel. The received signal is $r(t) = s(t) + n(t)$, where $n(t)$ is additive white Gaussian noise with power spectral density $N_0/2$. The received signal is passed through a filter with impulse response $h(t)$. Let E_s and E_h denote the energies of the pulse $s(t)$ and the filter $h(t)$, respectively. When the signal-to-noise ratio (SNR) is maximized at the output of the filter ($\text{SNR}_{\max}$), which of the following holds?

A. $E_s = E_h;\ \text{SNR}_{\max} = \dfrac{2E_s}{N_0}$

B. $E_s = E_h;\ \text{SNR}_{\max} = \dfrac{E_s}{2N_0}$

C. $E_s > E_h;\ \text{SNR}_{\max} > \dfrac{2E_s}{N_0}$

D. $E_s < E_h;\ \text{SNR}_{\max} = \dfrac{2E_h}{N_0}$

Q2.15.1 NAT 2015 2M

The input X to the Binary Symmetric Channel (BSC) shown in the figure is '1' with probability 0.8. The cross-over probability is $\frac{1}{7}$. If the received bit $Y = 0$, the conditional probability that '1' was transmitted is _____.

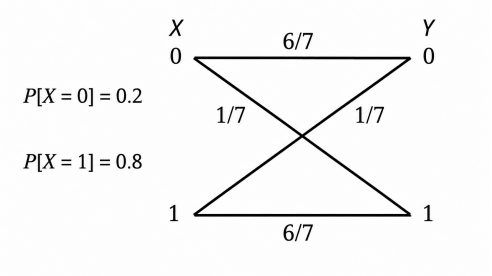


Fig. Q2.15.1

Q2.15.2 MCQ 2015 2M

A source emits bit 0 with probability $\frac{1}{3}$ and bit 1 with probability $\frac{2}{3}$. The emitted bits are communicated to the receiver. The receiver decides for either 0 or 1 based on the received value R. It is given that the conditional density functions of R are as

$$f_{R|0}(r) = \begin{cases} \frac{1}{4}, & -3 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$f_{R|1}(r) = \begin{cases} \frac{1}{6}, & -1 \le x \le 5, \\ 0, & \text{otherwise}. \end{cases}$$

The minimum decision error probability is.

A. 0 **B.** $\frac{1}{12}$

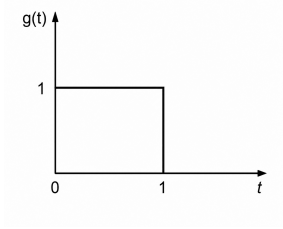
C. $\frac{1}{9}$ **D.** $\frac{1}{6}$

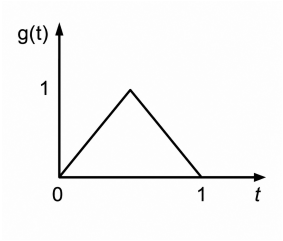
Q2.15.3 NAT 2015 2M

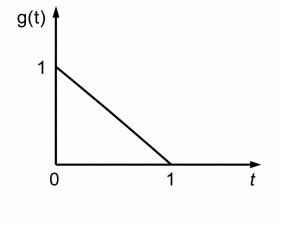
The transmitted signal in a GSM system is of 200 kHz bandwidth and 8 users share a common bandwidth using TDMA. If at a given time 12 users are talking in a cell, the total bandwidth of the signal received by the base station of the cell will be at least (in kHz) _____.

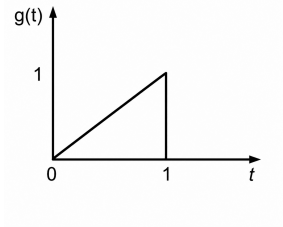
Q2.15.4 MCQ 2015 2M

Consider a binary digital communication system which uses pulses $g(t)$ and $-g(t)$ for transmitting bits over an AWGN channel. If the receiver uses a matched filter, which one of the following pulses will give the minimum probability of bit error?

A. 

B. 

C. 

D. 

Q2.15.5 NAT 2015 1M

A sinusoidal signal of amplitude A is quantized by a uniform quantizer. Assume that the signal utilizes all the representation levels of the quantizer. If the signal to quantization noise ratio is 31.8 dB, the number of levels in the quantizer is _____.

Q2.15.6 MCQ 2015 1M

A sinusoidal signal of 2 kHz frequency is applied to a delta modulator. The sampling rate and step-size Δ of the delta modulator are 20, 000 samples per second and 0.1 V, respectively. To prevent slope overload, the maximum amplitude of the sinusoidal signal (in Volts) is

A. $\frac{1}{2\pi}$ **B.** $\frac{1}{\pi}$

C. $\frac{2}{\pi}$ **D.** π

Q2.15.7 MCQ 2015 1M

The modulation scheme commonly used for transmission from GSM mobile terminals is

A. 4-QAM
B. 16-PSK
C. Walsh-Hadamard orthogonal codes
D. Gaussian Minimum Shift Keying (GMSK)

Q2.14.1 NAT 2014 2M

The input to a 1-bit quantizer is a random variable X with pdf

$$f_X(x) = \begin{cases} 2e^{-2x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

For outputs to be of equal probability, the quantizer threshold should be _____.

Q2.14.2 MCQ 2014 2M

Coherent orthogonal binary FSK modulation is used to transmit two equiprobable symbol waveforms $s_1(t) = \alpha \cos 2\pi f_1 t$ and $s_2(t) = \alpha \cos 2\pi f_2 t$, where $\alpha = 4$ mV. Assume an AWGN channel with two-sided noise power spectral density $\frac{N_0}{2} = 0.5 \times 10^{-12}$ W/Hz. Using an optimal receiver and the relation

$$Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du,$$

the bit error probability for a data rate of 500 kbps is

A. $Q(2)$ **B.** $Q(2\sqrt{2})$

C. $Q(4)$ **D.** $Q(4\sqrt{2})$

Q2.14.3 NAT 2014 2M

In a PCM system, the signal

$$m(t) = \{\sin(100\pi t) + \cos(100\pi t)\} \text{ V}$$

is sampled at the Nyquist rate. The samples are processed by a uniform quantizer with step size 0.75 V. The minimum data rate of the PCM system in bits per second is _____.

Q2.14.4 MCQ 2014 2M

Consider a communication scheme where the binary valued signal X satisfies

$$P\{X = +1\} = 0.75 \quad \text{and} \quad P\{X = -1\} = 0.25.$$

The received signal is

$$Y = X + Z,$$

where Z is a Gaussian random variable with zero mean and variance σ^2. The received signal Y is fed to the threshold detector. The output of the threshold detector $\hat{X}$ is

$$\hat{X} = \begin{cases} +1, & Y > \tau, \\ -1, & Y \le \tau. \end{cases}$$

To achieve a minimum probability of error $P\{\hat{X} \neq X\}$, the threshold τ should be

A. strictly positive
B. zero
C. strictly negative
D. strictly positive, zero, or strictly negative depending on the nonzero value of σ^2

Q2.14.5 NAT 2014 2M ○○○

An M-level PSK modulation scheme is used to transmit independent binary digits over a band-pass channel with bandwidth 100 kHz. The bit rate is 200 kbps and the system characteristic is a raised-cosine spectrum with 100% excess bandwidth. The minimum value of M is _____.

Q2.14.6 MCQ 2014 2M ○○○

Consider a discrete-time channel $Y = X + Z$, where the additive noise Z is signal-dependent. In particular, given the transmitted symbol $X \in \{-a, +a\}$ at any instant, the noise sample Z is chosen independently from a Gaussian distribution with mean βX and unit variance. Assume a threshold detector with zero threshold at the receiver.

When $\beta = 0$, the BER was found to be $Q(a) = 1 \times 10^{-8}$.

$$Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du,$$

and for $v > 1$, use

$$Q(v) \approx e^{-v^2/2}.$$

When $\beta = -0.3$, the BER is closest to

A. 10^{-7} **B.** 10^{-6}
C. 10^{-4} **D.** 10^{-2}

Q2.14.7 NAT 2014 1M ○○○

In a code-division multiple access (CDMA) system with $N = 8$ chips, the maximum number of users who can be assigned mutually orthogonal signature sequences is _____.

Q2.14.8 NAT 2014 1M ○○○

An analog voltage in the range 0 to 8 V is divided in 16 equal intervals for conversion to a 4-bit digital output. The maximum quantization error (in V) is _____.

Q2.14.9 NAT 2014 1M ○○○

A modulated signal is $y(t) = m(t)\cos(40000\pi t)$, where the baseband signal $m(t)$ has frequency components less than 5 kHz only. The minimum required sampling rate (in kHz) at which $y(t)$ should be sampled to recover $m(t)$ is_____.

Q2.13.1 MCQ 2013 1M ○○○

Let $g(t) = e^{-\pi t^2}$, and $h(t)$ is a filter matched to $g(t)$. If $g(t)$ is applied as input to $h(t)$, then the Fourier transform of the output is

A. $e^{-\pi f^2}$ **B.** $e^{-\pi f^2/2}$
C. $e^{-\pi |f|}$ **D.** $e^{-2\pi f^2}$

Q2.13.2 MCQ 2013 1M ○○○

The bit rate of a digital communication system is R kbits/s. The modulation used is 32-QAM. The minimum bandwidth required for ISI-free transmission is

A. $R/10$ Hz **B.** $R/10$ kHz
C. $R/5$ Hz **D.** $R/5$ kHz

Q2.12.1 MCQ 2012 1M ○○○

In a baseband communications link, frequencies upto 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter-symbol interference, the maximum possible signaling rate in symbols per second is

A. 1750 **B.** 2625
C. 4000 **D.** 5250

Q2.12.2 MCQ 2012 2M ○○○

A BPSK scheme operating over an AWGN channel with noise power spectral density of $N_0/2$, uses equiprobable signals

$$s_1(t) = \sqrt{\frac{2E}{T}} \sin(\omega_c t)$$

and

$$s_2(t) = -\sqrt{\frac{2E}{T}}\sin(\omega_c t)$$

over the symbol interval $(0, T)$. If the local oscillator in a coherent receiver is ahead in phase by 45° with respect to the received signal, the probability of error in the resulting system is

A. $Q\left(\sqrt{\frac{2E}{N_0}}\right)$ **B.** $Q\left(\sqrt{\frac{E}{N_0}}\right)$

C. $Q\left(\sqrt{\frac{E}{2N_0}}\right)$ **D.** $Q\left(\sqrt{\frac{E}{4N_0}}\right)$

Q2.12.3 MCQ 2012 2M

A binary symmetric channel (BSC) has a transition probability of $\frac{1}{8}$. If the binary transmit symbol X is such that

$$P(X = 0) = \frac{9}{10},$$

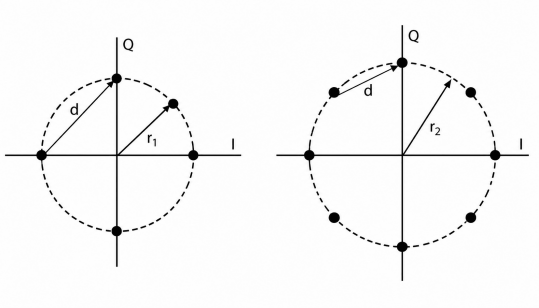
then the probability of error for an optimum receiver is

A. $\frac{7}{80}$ **B.** $\frac{63}{80}$

C. $\frac{9}{10}$ **D.** $\frac{1}{10}$

Common Data Questions

Answer the question 2.11.1 and question 2.11.2 using the following data. A four-phase and an eight-phase signal constellation are shown in the figure below.



Q2.11.1 MCQ 2011 2M

For the constraint that the minimum distance between pairs of signal points be d for both constellations, the radii r_1 and r_2 of the circles are

A. $r_1 = 0.707d,\ r_2 = 2.782d$

B. $r_1 = 0.707d,\ r_2 = 1.932d$

C. $r_1 = 0.707d,\ r_2 = 1.545d$

D. $r_1 = 0.707d,\ r_2 = 1.307d$

Q2.11.2 MCQ 2011 2M

Assuming high SNR and that all signals are equally probable, the additional average transmitted signal energy required by the 8-PSK signal to achieve the same error probability as the 4-PSK signal is

A. 11.90 dB **B.** 8.73 dB

C. 6.79 dB **D.** 5.33 dB

Q2.11.3 MCQ 2011 1M

An analog signal is band-limited to 4 kHz, sampled at the Nyquist rate and the samples are quantized into 4 levels. The quantized levels are assumed to be independent and equally probable. If we transmit two quantized samples per second, the information rate is

A. 1 bit/sec **B.** 2 bits/sec

C. 3 bits/sec **D.** 4 bits/sec

Common Data Questions

Answer the question 2.10.1 and question 2.10.2 using the following data. Consider a baseband binary PAM receiver shown below. The additive channel noise $n(t)$ is white with power spectral density

$$S_N(f) = \frac{N_0}{2} = 10^{-20}\ \text{W/Hz}.$$

The low-pass filter is ideal with unity gain and cutoff frequency 1 MHz. Let Y_k represent the random variable $y(t_k)$.

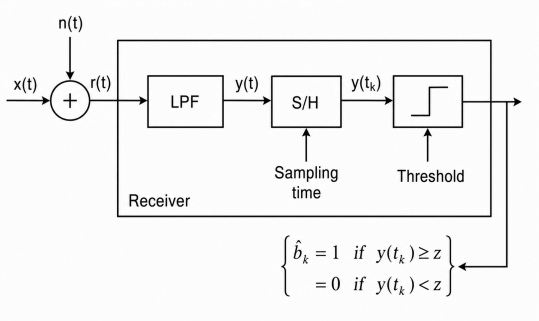
$$Y_k = \begin{cases} N_k, & \text{if transmitted bit } b_k = 0, \\ a + N_k, & \text{if transmitted bit } b_k = 1, \end{cases}$$

where N_k represents the noise sample value. The noise sample has a probability density function

$$p_{N_k}(n) = 0.5\,\alpha e^{-\alpha|n|}.$$

(This has mean zero and variance $2/\alpha^2$.) Assume transmitted bits to be equiprobable and threshold τ is set to

$$\tau = \frac{a}{2} = 10^{-6}\ \text{V}.$$



Q2.10.1 MCQ 2010 2M

The value of the parameter α (in V^{-1}) is

A. 10^{10} **B.** 10^{7}

C. 1.414×10^{-10} **D.** 2×10^{20}

Q2.10.2 MCQ 2010 2M

The probability of bit error is

A. $0.5 \times e^{-0.5}$ **B.** $0.5 \times e^{-5}$

C. $0.5 \times e^{-7}$ **D.** $0.5 \times e^{-10}$

Q2.10.3 MCQ 2010 1M

Consider the pulse shape $s(t)$ as shown. The impulse response $h(t)$ of the filter matched to this pulse is

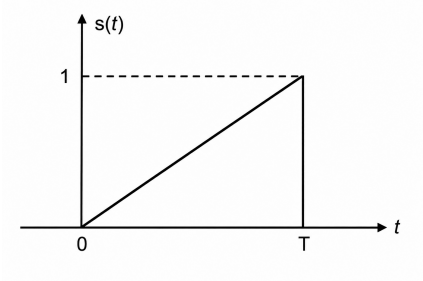
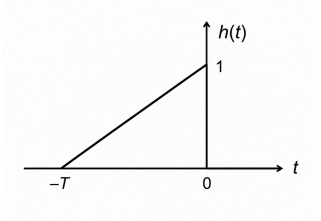
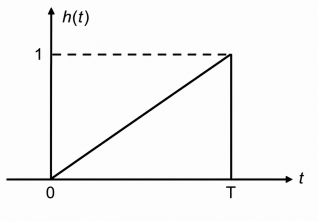


Fig. Q2.10.3

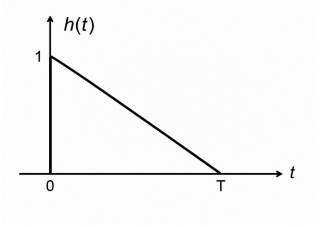
A.



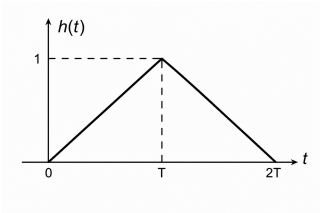
B.



C.



D.



Common Data Questions

Answer the question 2.09.1 and question 2.09.2 using the following data. The amplitude of a random signal is uniformly distributed between −5 V and 5 V.

Q2.09.1 MCQ 2009 2M

If the signal to quantization noise ratio required in uniformly quantizing the signal is 43.5 dB, the step size of the quantization is approximately

A. 0.0333 V **B.** 0.05 V

C. 0.0667 V **D.** 0.10 V

Q2.09.2 MCQ 2009 2M

If the positive values of the signal are uniformly quantized with a step size of 0.05 V, and the negative values are uniformly quantized with a step size of 0.1 V, the resulting signal to quantization noise ratio is approximately

A. 46 dB **B.** 43.8 dB

C. 42 dB **D.** 40 dB

Q2.08.1 MCQ 2008 2M

Four messages band limited to W, W, $2W$ and $3W$ respectively are to be multiplexed using Time Division Multiplexing (TDM). The minimum bandwidth required for transmission of this TDM signal is

A. W **B.** $3W$

C. $6W$ **D.** $7W$

Common Data Questions

Answer the questions 2.08.2 , 2.08.3 , 2.08.4 using the following data.
A speech signal, band limited to 4 kHz and peak voltage

varying between +5 V and −5 V, is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits.

Q2.08.2 MCQ 2008 2M

If the bits 0 and 1 are transmitted using bipolar pulses, the minimum bandwidth required for distortion free transmission is

A. 64 kHz **B.** 32 kHz
C. 8 kHz **D.** 4 kHz

Q2.08.3 MCQ 2008 2M

Assuming the signal to be uniformly distributed between its peak-to-peak value, the signal to noise ratio at the quantizer output is

A. 16 dB **B.** 32 dB
C. 48 dB **D.** 64 dB

Q2.08.4 MCQ 2008 2M

The number of quantization levels required to reduce the quantization noise by a factor of 4 would be

A. 1024 **B.** 512
C. 256 **D.** 64

Q2.07.1 MCQ 2007 2M

In delta modulation, the slope overload distortion can be reduced by

A. decreasing the step size
B. decreasing the granular noise
C. decreasing the sampling rate
D. increasing the step size

Q2.07.2 MCQ 2007 2M

The raised cosine pulse $p(t)$ is used for zero ISI in digital communications. The expression for $p(t)$ with unity roll-off factor is given by

$$p(t) = \frac{\sin 4\pi Wt}{4\pi Wt\left(1 - 16W^2t^2\right)}.$$

The value of $p(t)$ at

$$t = \frac{1}{4W}$$

is

A. −0.5 **B.** 0
C. 0.5 **D.** ∞

Q2.07.3 MCQ 2007 2M

In a GSM system, 8 channels can co-exist in 200 kHz bandwidth using TDMA. A GSM based cellular operator is allocated 5 MHz bandwidth. Assuming a frequency reuse factor of $\frac{1}{5}$, i.e. a five-cell repeat pattern, the maximum number of simultaneous channels that can exist in one cell is

A. 200 **B.** 40
C. 25 **D.** 5

Q2.07.4 MCQ 2007 2M

In a Direct Sequence CDMA system the chip rate is 1.2288×10^6 chips per second. If the processing gain is desired to be AT LEAST 100, the data rate

A. must be less than or equal to 12.288×10^3 bits per sec

B. must be greater than 12.288×10^3 bits per sec

C. must be exactly equal to 12.288×10^3 bits per sec

D. can take any value less than 122.88×10^3 bits per sec

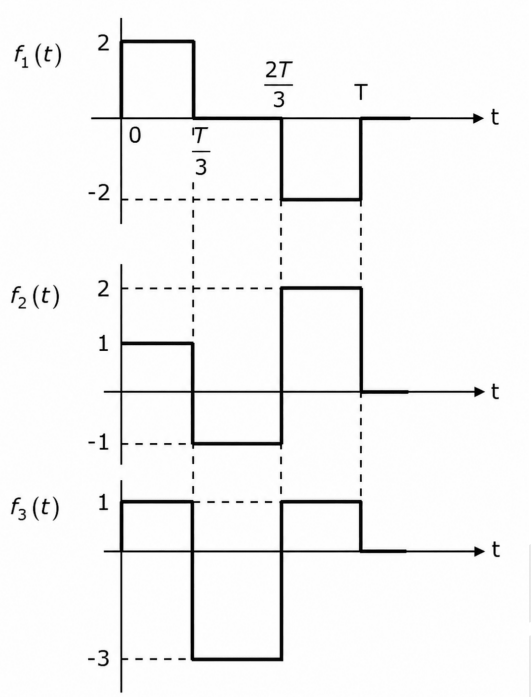
Q2.07.5 MCQ 2007 2M

During transmission over a certain binary communication channel, bit errors occur independently with probability p. The probability of AT MOST one bit in error in a block of n bits is given by

A. p^n **B.** $1 - p^n$
C. $np(1-p)^{n-1} + (1-p)^n$ **D.** $1 - (1-p)^n$

Q2.07.6 MCQ 2007 2M

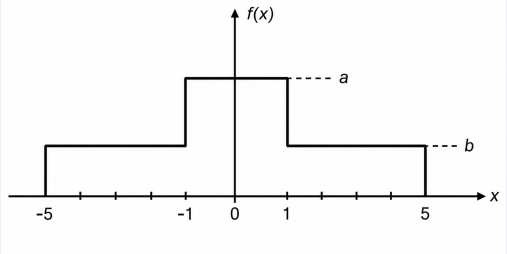
Three functions $f_1(t)$, $f_2(t)$ and $f_3(t)$, which are zero outside the interval $[0, T]$, are shown in the figure. Which of the following statements is correct?



A. $f_1(t)$ and $f_2(t)$ are orthogonal
B. $f_1(t)$ and $f_3(t)$ are orthogonal
C. $f_3(t)$ and $f_3(t)$ are orthogonal
D. $f_1(t)$ and $f_2(t)$ are orthonormal

Common Data Questions

Answer the question 2.07.7 and question 2.07.8 using the following data.
An input to a 6-level quantizer has the probability density function $f(x)$ as shown in the figure. Decision boundaries of the quantizer are chosen so as to maximize the entropy of the quantizer output. It is given that 3 consecutive decision boundaries are -1, 0 and 1.



Q2.07.7 MCQ 2007 2M

The values of a and b are:

A. $a = \frac{1}{6}$ and $b = \frac{1}{12}$
B. $a = \frac{1}{5}$ and $b = \frac{3}{40}$
C. $a = \frac{1}{4}$ and $b = \frac{1}{16}$
D. $a = \frac{1}{3}$ and $b = \frac{1}{24}$

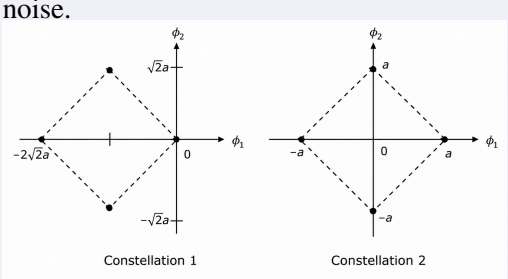
Q2.07.8 MCQ 2007 2M

Assuming that the reconstruction levels of the quantizer are the mid-points of the decision boundaries, the ratio of signal power to quantization noise power is

A. $\frac{152}{9}$
B. $\frac{64}{3}$
C. $\frac{76}{3}$
D. 28

Common Data Questions

Answer the question 2.07.9 and question 2.07.10 using the following data.
Two 4-ary signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis for the two constellations. Assume that the four symbols in both the constellations are equiprobable. Let $\frac{N_0}{2}$ denote the power spectral density of white Gaussian noise.



Q2.07.9 MCQ 2007 2M

The ratio of the average energy of Constellation 1 to the average energy of Constellation 2 is:

A. $4a^2$
B. 4
C. 2
D. 8

Q2.07.10 MCQ 2007 2M

If these constellations are used for digital communications over an AWGN channel, then which of the following statements is true?
A. Probability of symbol error for Constellation 1 is lower
B. Probability of symbol error for Constellation 1 is higher
C. Probability of symbol error is equal for both the constellations
D. The value of N_0 will determine which of the two constellations has a lower probability of symbol error

Q2.06.1 MCQ 2006 2M

The minimum step-size required for a Delta-Modulator operating at 32 K samples/sec to track the signal (here $u(t)$ is the unit-step function)

$$x(t) = 125t\big(u(t) - u(t-1)\big)$$

$$+\big(250 - 125t\big)\big(u(t-1) - u(t-2)\big)$$

so that slope-overload is avoided, would be

A. 2^{-10} **B.** 2^{-8}

C. 2^{-6} **D.** 2^{-4}

Q2.06.2 MCQ 2006 2M

A source generates three symbols with probabilities 0.25, 0.25, 0.50 at a rate of 3000 symbols per second. Assuming independent generation of symbols, the most efficient source encoder would have average bit rate as

A. 6000 bits/sec **B.** 4500 bits/sec

C. 3000 bits/sec **D.** 1500 bits/sec

Common Data Questions

Answer the question 2.06.3 and question 2.06.4 using the following data.

Let

$$g(t) = p(t) * p(t),$$

where $*$ denotes convolution and

$$p(t) = u(t) - u(t-1),$$

with $u(t)$ being the unit step function.

Q2.06.3 MCQ 2006 2M

The impulse response of filter matched to the signal

$$s(t) = g(t) - \delta(t-2) * g(t)$$

is given as:

A. $s(1-t)$ **B.** $-s(1-t)$

C. $-s(t)$ **D.** $s(t)$

Q2.06.4 MCQ 2006 2M

In the following figure the minimum value of the constant "C", which is to be added to $y_1(t)$ such that $y_1(t)$ and $y_2(t)$ are different, is

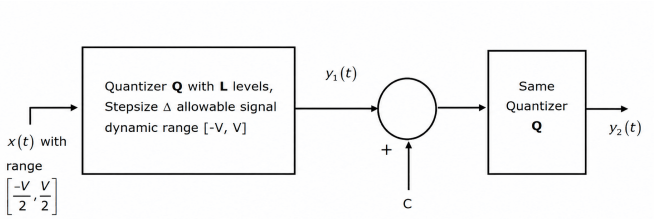


Fig. Q2.06.4

A. Δ **B.** $\dfrac{\Delta}{2}$

C. $\dfrac{\Delta^2}{12}$ **D.** $\dfrac{\Delta}{L}$

Q2.06.5 MCQ 2006 1M

The minimum sampling frequency (in samples/sec) required to reconstruct the following signal from its samples without distortion,

$$x(t) = 5\left(\frac{\sin 2\pi 1000t}{\pi t}\right)^3 + 7\left(\frac{\sin 2\pi 1000t}{\pi t}\right)^2$$

would be

A. 2×10^3 **B.** 4×10^3

C. 6×10^3 **D.** 8×10^3

Q2.05.1 MCQ 2005 2M

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels. The quantization noise power for the quantization region between $-a$ and $+a$ in the figure is

A. $\dfrac{4}{81}$ **B.** $\dfrac{1}{9}$

C. $\dfrac{5}{81}$ **D.** $\dfrac{2}{81}$

Q2.05.2 MCQ 2005 2M

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels. If the probability density function is divided into three regions as shown in the figure, the value of a in the figure is _____.

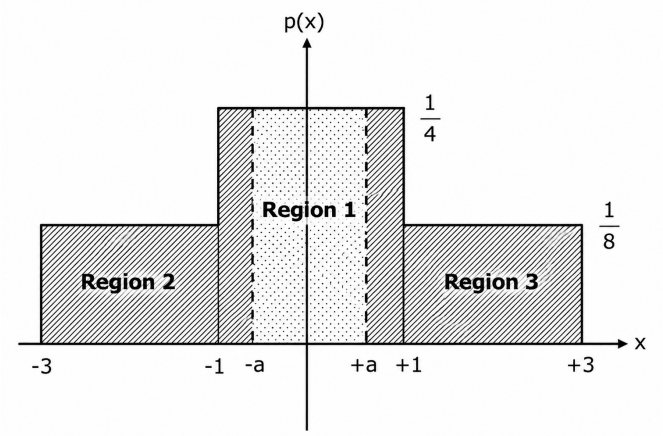
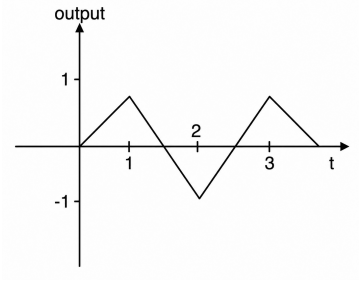
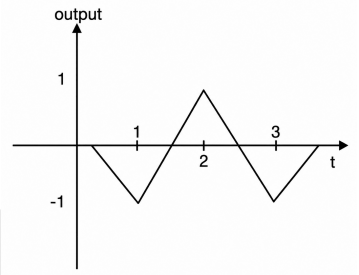


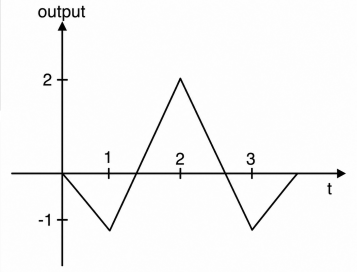
Fig. Q2.05.2

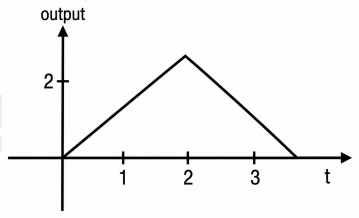
A. $\frac{1}{3}$ **B.** $\frac{2}{3}$

C. $\frac{1}{2}$ **D.** $\frac{1}{4}$

A. 

B. 

C. 

D. 

Q2.05.3 MCQ 2005 2M

A Signal as shown in the figure is applied to a matched filter.Which of the following does represent the output of this matched filter?

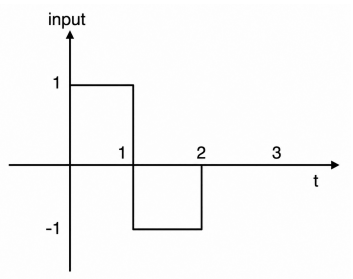


Fig. Q2.05.3

Q2.04.1 MCQ 2004 2M

A source produces binary data at the rate of 10 kbps. The binary symbols are represented as shown in the figure given below.

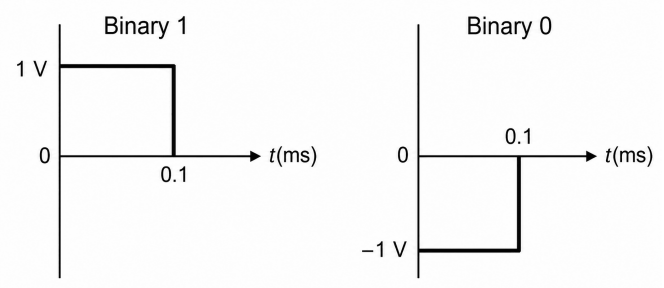


Fig. Q2.04.1

The source output is transmitted using two modulation schemes, namely Binary PSK (BPSK) and Quadrature PSK (QPSK). Let B_1 and B_2 be the bandwidth requirements of BPSK and QPSK respectively. Assume that the bandwidth of the above rectangular pulses is 10 kHz. B_1 and B_2 are

A. $B_1 = 20$ kHz, $B_2 = 20$ kHz

B. $B_1 = 10$ kHz, $B_2 = 20$ kHz

C. $B_1 = 20$ kHz, $B_2 = 10$ kHz

D. $B_1 = 10$ kHz, $B_2 = 10$ kHz

Q2.04.2 MCQ 2004 2M

Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. The bit rate for the multiplexed signal is

A. 115.2 kbps **B.** 28.8 kbps

C. 57.6 kbps **D.** 38.4 kbps

Q2.04.3 MCQ 2004 2M

Consider a binary digital communication system with equally likely 0's and 1's. When binary 0 is transmitted, the detector input can lie between the levels −0.25 V and +0.25 V with equal probability; when binary 1 is transmitted, the voltage at the detector can have any value between 0 and 1 V with equal probability. If the detector has a threshold of 0.2 V (i.e., if the received signal is greater than 0.2 V, the bit is taken as 1), the average bit error probability is

A. 0.15 **B.** 0.2

C. 0.05 **D.** 0.5

Q2.04.4 MCQ 2004 2M

A 1 mW video signal having a bandwidth of 100 MHz is transmitted to a receiver through a cable that has 40 dB loss. If the effective one-side noise spectral density at the receiver is 10^{-20} Watt/Hz, then the signal-to-noise ratio at the receiver is

A. 50 dB **B.** 30 dB

C. 40 dB **D.** 60 dB

Q2.04.5 MCQ 2004 2M

Consider the signal $x(t)$ shown in Fig. Let $h(t)$ denote the impulse response of the filter matched to $x(t)$, with $h(t)$ being non-zero only in the interval 0 to 4 sec. The slope of $h(t)$ in the interval $3 < t < 4$ sec is

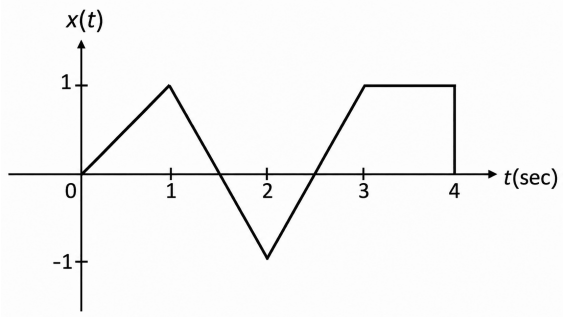


Fig. Q2.04.5

A. $\frac{1}{2}$ sec^{-1} **B.** -1 sec^{-1}

C. $-\frac{1}{2}$ sec^{-1} **D.** 1 sec^{-1}

Q2.04.6 MCQ 2004 2M

In the output of a DM speech encoder, the consecutive pulses are of opposite polarity during time interval $t_1 \leq t \leq t_2$. This indicates that during this interval

A. the input to the modulator is essentially constant

B. the modulator is going through slope overload

C. the accumulator is in saturation

D. the speech signal is being sampled at the Nyquist rate

Q2.04.7 MCQ 2004 2M

In a PCM system, if the code word length is increased from 6 to 8 bits, the signal to quantization noise ratio improves by the factor

A. $\frac{8}{6}$ **B.** 12

C. 16 **D.** 8

Q2.03.1 MCQ 2003 2M

A signal is sampled at 8 kHz and is quantized using an 8-bit uniform quantizer. Assuming SNR_q for a sinusoidal signal, the correct statement for PCM signal with a bit rate of R is

A. $R = 32$ kbps, $\text{SNR}_q = 25.8$ dB

B. $R = 64$ kbps, $\text{SNR}_q = 49.8$ dB

C. $R = 64$ kbps, $\text{SNR}_q = 55.8$ dB

D. $R = 32$ kbps, $\text{SNR}_q = 49.8$ dB

Q2.03.2 MCQ 2003 2M

If S represents the carrier synchronization at the receiver and ρ represents the bandwidth efficiency, then the correct statement for the coherent binary PSK is

A. $\rho = 0.5$, S is required

B. $\rho = 1.0$, S is required

C. $\rho = 0.5$, S is not required
D. $\rho = 1.0$, S is not required

Q2.03.3 MCQ 2003 2M

The input to a linear delta modulator having a step-size $\Delta = 0.628$ is a sine wave with frequency f_m and peak amplitude E_m. If the sampling frequency is $f_s = 40$ kHz,the combination of the sine-wave frequency and the peak amplitude, where slope overload will take place, is
A. $E_m = 0.3$ V, $f_m = 8$ kHz
B. $E_m = 1.5$ V, $f_m = 4$ kHz
C. $E_m = 1.5$ V, $f_m = 2$ kHz
D. $E_m = 0.3$ V, $f_m = 1$ kHz

Q2.03.4 MCQ 2003 2M

If E_b, the energy per bit of a binary digital signal, is 10^{-5} watt-sec and the one-sided power spectral density of the white noise, $N_0 = 10^{-6}$ W/Hz, then the output SNR of the matched filter is
A. 26 dB **B.** 10 dB
C. 20 dB **D.** 13 dB

Q2.03.5 MCQ 2003 2M

A sinusoidal signal with peak-to-peak amplitude of 1.536 V is quantized into 128 levels using a mid-rise uniform quantizer. The quantization-noise power is
A. 0.768 V **B.** 48×10^{-6} V^2
C. 12×10^{-6} V^2 **D.** 3.072 V

Q2.03.6 MCQ 2003 2M

At a given probability of error, binary coherent FSK is inferior to binary coherent PSK by
A. 6 dB **B.** 3 dB
C. 2 dB **D.** 0 dB

Q2.02.1 MCQ 2002 2M

A signal

$$x(t) = 100\cos\left(24\pi \times 10^3\, t\right)$$

is ideally sampled with a sampling period of 50 μsec and then passed through an ideal low-pass filter with cutoff frequency of 15 kHz. Which of the following frequencies is/are present at the filter output?
A. 12 kHz only
B. 8 kHz only
C. 12 kHz and 9 kHz
D. 12 kHz and 8 kHz

Q2.02.2 MCQ 2002 2M

For a bit-rate of 8 Kbps, the best possible values of the transmitted frequencies in a coherent binary FSK system are
A. 16 kHz and 20 kHz
B. 20 kHz and 32 kHz
C. 20 kHz and 40 kHz
D. 32 kHz and 40 kHz

Q2.02.3 MCQ 2002 2M

Consider a sampled signal

$$y(t) = 5 \times 10^{-6}\, x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

where

$$x(t) = 10\cos(8\pi \times 10^3\, t)$$

and

$$T_s = 100\,\mu\text{sec}.$$

When $y(t)$ is passed through an ideal low-pass filter with a cutoff frequency of 5 kHz, the output of the filter is
A. $5 \times 10^{-6}\cos(8\pi \times 10^3\, t)$ **B.** $5 \times 10^{-5}\cos(8\pi \times 10^3\, t)$
C. $5 \times 10^{-1}\cos(8\pi \times 10^3\, t)$ **D.** $10\cos(8\pi \times 10^3\, t)$

Q2.01.1 MCQ 2001 2M

The Nyquist sampling interval, for the signal

$$\text{sinc}(700t) + \text{sinc}(500t)$$

is
A. $\frac{1}{350}$ sec **B.** $\frac{\pi}{350}$ sec
C. $\frac{1}{700}$ sec **D.** $\frac{\pi}{175}$ sec

Q2.01.2 MCQ 2001 2M

A video transmission system transmits 625 picture frames per second. Each frame consists of a 400×400 pixel grid with 64 intensity levels per pixel. The data rate of the system is
A. 16 Mbps **B.** 100 Mbps
C. 600 Mbps **D.** 6.4 Gbps

Q2.00.1 MCQ 2000 2M

In a digital communication system employing Frequency Shift Keying (FSK), the 0 and 1 bit are represented by sine waves of 10 kHz and 25 kHz respectively. These waveforms will be orthogonal for a bit interval of

A. 45 μsec **B.** 200 μsec
C. 50 μsec **D.** 250 μsec

Q2.99.1 MCQ 1999 2M

The peak-to-peak input to an 8-bit PCM encoder is 2 volts. The signal power-to-quantization noise power ratio (in dB) for an input of $0.5\cos(\omega_m t)$ is

A. 47.8 **B.** 49.8
C. 95.6 **D.** 99.6

Q2.99.2 MCQ 1999 2M

The input to a matched filter is given by

$$s(t) = \begin{cases} 10\sin(2\pi \times 10^6 t), & 0 < t < 10^{-4} \text{ sec} \\ 0, & \text{otherwise} \end{cases}$$

The peak amplitude of the filter output is

A. 10 volts **B.** 5 volts
C. 10 millivolts **D.** 5 millivolts

Q2.99.3 MCQ 1999 2M

Four independent messages have bandwidths of 100 Hz, 100 Hz, 200 Hz and 400 Hz respectively. Each is sampled at the Nyquist rate, and the samples are time division multiplexed (TDM) and transmitted. The transmitted sample rate (in Hz) is

A. 1600 **B.** 800
C. 400 **D.** 200

Q2.98.1 MCQ 1998 1M

In a PCM system with uniform quantization, increasing the number of bits from 8 to 9 will reduce the quantization noise power by a factor of

A. 9 **B.** 8
C. 4 **D.** 2

Q2.98.2 MCQ 1998 1M

Quadrature multiplexing is
A. the same as FDM
B. the same as TDM
C. a combination of FDM and TDM
D. quite different from FDM and TDM

Q2.98.3 MCQ 1998 1M

Compression in PCM refers to relative compression of
A. higher signal amplitudes
B. lower signal amplitudes
C. lower signal frequencies
D. higher signal frequencies

Q2.98.4 MCQ 1998 1M

For a given data rate, the bandwidth B_P of a BPSK signal and the bandwidth B_0 of the OOK signal are related as

A. $B_P = \frac{B_0}{4}$ **B.** $B_P = \frac{B_0}{2}$
C. $B_P = B_0$ **D.** $B_P = 2B_0$

Q2.97.1 MCQ 1997 1M

The line code that has zero dc component for pulse transmission of random binary data is
A. Non-return to zero (NRZ)
B. Return to zero (RZ)
C. Alternate Mark Inversion (AMI)
D. None of the above

Q2.97.2 MCQ 1997 1M

A deterministic signal has the power spectrum given in the figure. The minimum sampling rate needed to completely represent this signal is

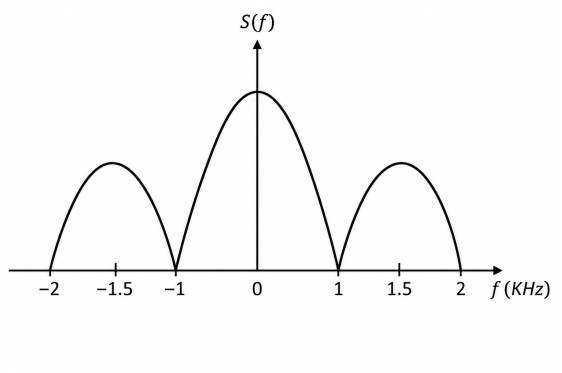


Fig. Q2.97.2

A. 1 kHz **B.** 2 kHz
C. 3 kHz **D.** None of the above

Q2.96.1 MCQ 1996 2M

The number of bits in a binary PCM system is increased from n to $n+1$. As a result, the signal-to-quantization-noise ratio will improve by a factor

A. $\frac{n+1}{n}$ **B.** $2^{\frac{n+1}{n}}$

C. $2^{\frac{2(n+1)}{n}}$ **D.** which is independent of n

Q2.95.1 MCQ 1995 1M

If the number of bits per sample in a PCM system is increased from n to $n+1$, the improvement in signal-to-quantization noise ratio will be

A. 3 dB **B.** 6 dB

C. $2n$ dB **D.** n dB

Q2.95.2 MCQ 1995 1M

The signal-to-quantization noise ratio in an n-bit PCM system

A. depends upon the sampling frequency employed
B. is independent of the value of n
C. increases with increasing value of n
D. decreases with increasing value of n

Q2.94.1 NAT 1994 1M

The bandwidth required for the transmission of a PCM signal increases by a factor of _____ when the number of quantization levels is increased from 4 to 64.

Q2.92.1 MCQ 1992 2M

The bit stream 01001 is differentially encoded using the Delay and Ex-OR scheme for DPSK transmission. Assuming the reference bit as a 1 and assigning phases of 0 and π for 1s and 0s respectively, in the encoded sequence, the transmitted phase sequence becomes

A. $\pi,\ 0,\ \pi,\ \pi,\ 0$ **B.** $0,\ \pi,\ \pi,\ 0,\ 0$

C. $0,\ \pi,\ \pi,\ \pi,\ 0$ **D.** $\pi,\ \pi,\ 0,\ \pi,\ \pi$

Q2.92.2 MCQ 1992 2M

Coherent demodulation of FSK signal can be effected using

A. correlation receiver
B. band-pass filters and envelope detectors
C. matched filter
D. discriminator detection

Q2.92.3 MCQ 1992 2M

Source encoding in a data communication system is done in order to

A. Enhance the information transmission
B. Band-pass filters and envelope rate detectors
C. Conserve the transmitted power
D. Discriminator detection

Q2.91.1 NAT 1991 2M

A binary source has symbol probabilities 0.8 and 0.2. If extension coding (blocks of 4 symbols) is used, the lower and upper bounds on the average code word length are:

(a) lower _____
(b) higher _____

Q2.91.2 NAT 1991 2M

For the signal constellation shown in the figure the type of modulation is

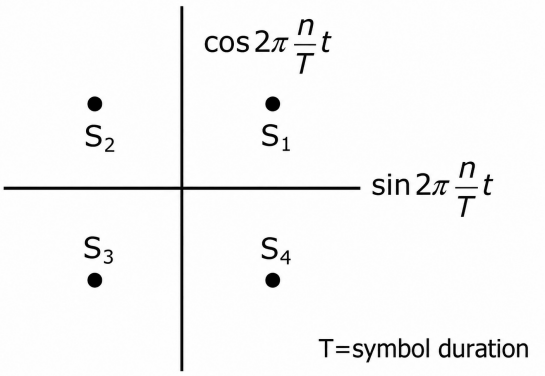


Fig. Q2.91.2

Q2.90.1 MCQ 1990 2M

In a BPSK signal detector, the local oscillator has a fixed phase error of 20°. This phase error deteriorates the output signal-to-noise ratio (SNR) by a factor of

A. $\cos 20°$ **B.** $\cos^2 20°$

C. $\cos 70°$ **D.** $\cos^2 70°$

Q2.89.1 MCQ 1989 2M

In binary data transmission, DPSK is preferred to PSK because

(a) a coherent carrier is not required to be generated at the receiver.
(b) for a given energy per bit, the probability of error is less.

(c) the 180° phase shifts of the carrier are unimportant.
(d) more protection is provided against impulse noise.

Q2.88.1 MCQ 1988 2M

The transfer function of a zero-order hold is

A. $\frac{1-\exp(-Ts)}{s}$ **B.** $\frac{1}{s}$

C. 1 **D.** $\frac{1}{1-\exp(-Ts)}$

Q2.88.2 MCQ 1988 2M

The message bit sequence input to a DPSK modulator is

$$1,\ 1,\ 0,\ 0,\ 1,\ 1.$$

The carrier phase during the reception of the first two message bits is $\pi,\ \pi$. The carrier phase for the remaining four message bits is

A. $\pi,\ \pi,\ 0,\ \pi$ **B.** $0,\ 0,\ \pi,\ \pi$

C. $0,\ \pi,\ \pi,\ \pi$ **D.** $\pi,\ \pi,\ 0,\ 0$

Q2.88.3 MCQ 1988 2M

A signal having uniformly distributed amplitude in the interval $(-V, +V)$ is to be encoded using PCM with uniform quantization. The signal-to-quantization noise ratio is determined by the

A. dynamic range of the signal
B. sampling rate
C. number of quantizing levels
D. power spectrum of signal

Q2.88.4 MCQ 1988 2M

In a digital communication system, transmissions of successive bits through a noisy channel are assumed to be independent events with error probability p. The probability of at most one error in the transmission of an 8-bit sequence is

A. $\frac{7(1-p)}{8}+\frac{p}{8}$ **B.** $(1-p)^8+8p(1-p)^7$

C. $(1-p)^8+(1-p)^7$ **D.** $(1-p)^8+p(1-p)^7$

Q2.87.1 MCQ 1987 2M

Companding in PCM systems leads to improved signal-to-quantization noise ratio. This improvement is for

A. lower frequency components only
B. higher frequency components only
C. lower amplitudes only
D. higher amplitudes only

Answer Key

Q2.26.1 0.95 to 1.05	Q2.26.2 C	Q2.26.3 B
Q2.26.4 0.10 to 0.20	Q2.25.1 3	Q2.25.2 D
Q2.24.1 A(MTA)	Q2.24.2 B	Q2.24.3 A
Q2.24.4 45.00 to 45.30	Q2.23.1 10	Q2.22.1 2.99 to 3.01
Q2.22.2 0.84 to 0.86	Q2.22.3 0.23	Q2.21.1 9.24 to 9.28
Q2.21.2 2.80 to 3.20	Q2.21.3 C	Q2.21.4 30.00 to 34.00

Q2.21.5 0.03 to 0.05	Q2.20.1 D	Q2.20.2 3
Q2.19.1 B	Q2.19.2 D	Q2.19.3 11.80 to 11.82
Q2.19.4 0.1	Q2.18.1 16	Q2.18.2 0.12 to 0.14
Q2.18.3 -0.5 to -0.5	Q2.17.1 7	Q2.17.2 D
Q2.17.3 B	Q2.17.4 B	Q2.16.1 34.5 to 35.5
Q2.16.2 1.74 to 1.76	Q2.16.3 15.9 to 16.1	Q2.16.4 C

Q2.16.5 30.0 to 33.0	Q2.16.6 0.70 to 0.75	Q2.16.7 0.24 to 0.26
Q2.16.8 0.025 to 0.030	Q2.16.9 A	Q2.15.1 0.39 to 0.41
Q2.15.2 D	Q2.15.3 400	Q2.15.4 A
Q2.15.5 32	Q2.15.6 A	Q2.15.7 D
Q2.14.1 0.34 to 0.36	Q2.14.2 C	Q2.14.3 199 to 201
Q2.14.4 C	Q2.14.5 15.9 to 16.1	Q2.14.6 C

Q2.14.7 8	Q2.14.8 0.24 to 0.26	Q2.14.9 9.5 to 10.5
Q2.13.1 D	Q2.13.2 D	Q2.12.1 C
Q2.12.2 B	Q2.12.3 D	Q2.11.1 D
Q2.11.2 D	Q2.11.3 D	Q2.10.1 B
Q2.10.2 D	Q2.10.3 C	Q2.09.1 C
Q2.09.2 C	Q2.08.1 D	Q2.08.2 A
Q2.08.3 C	Q2.08.4 B	Q2.07.1 D
Q2.07.2 C	Q2.07.3 B	Q2.07.4 A
Q2.07.5 C	Q2.07.6 C	Q2.07.7 A
Q2.07.8 D	Q2.07.9 B	Q2.07.10 A
Q2.06.1 B	Q2.06.2 B	Q2.06.3 C
Q2.06.4 B	Q2.06.5 C	Q2.05.1 A
Q2.05.2 B	Q2.05.3 C	Q2.04.1 C
Q2.04.2 C	Q2.04.3 A	Q2.04.4 A
Q2.04.5 B	Q2.04.6 A	Q2.04.7 C
Q2.03.1 B	Q2.03.2 B	Q2.03.3 B
Q2.03.4 D	Q2.03.5 C	Q2.03.6 B
Q2.02.1 D	Q2.02.2 A	Q2.02.3 C
Q2.01.1 C	Q2.01.2 C	Q2.00.1 B
Q2.99.1 43.9 (MTA)	Q2.99.2 D	Q2.99.3 A
Q2.98.1 C	Q2.98.2 D	Q2.98.3 A
Q2.98.4 C	Q2.97.1 C	Q2.97.2 B
Q2.96.1 D	Q2.95.1 B	Q2.95.2 C
Q2.94.1 3	Q2.92.1 C	Q2.92.2 A

Q2.92.3 A	Q2.91.1 2.8 , 3.8	Q2.91.2 QPSK
Q2.90.1 B	Q2.89.1 A	Q2.88.1 A
Q2.88.2 C	Q2.88.3 C	Q2.88.4 B
Q2.87.1 C		

UNIT

III

Information Theory

Topics

1. Entropy and Mutual Information
2. Channel Capacity Theorem

Unit Snapshot*

Stream: EC	**Questions:** 24	**MCQ:** 10	**MSQ:** 2	**NAT:** 12
1M: 8	**2M:** 16	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

Let the relevant bandwidth B of a digital communication system be 1 MHz and

$$kT = -174 \text{ dBm/Hz},$$

where k is Boltzmann's constant and T is the equivalent noise temperature of the receiver.
The received signal power through an additive Gaussian channel is $S = -80$ dBm.
Which of the following statements are true about the Shannon capacity C of the channel?

A. $C = B$ **B.** $C = 2B$
C. $C > 3B$ **D.** $C < B$

Q3.25.1 NAT 2025 2M

X and Y are Bernoulli random variables taking values in $\{0, 1\}$. The joint probability mass function is given by

$$P(X = 0, Y = 0) = 0.06$$

$$P(X = 0, Y = 1) = 0.14$$

$$P(X = 1, Y = 0) = 0.24$$

$$P(X = 1, Y = 1) = 0.56$$

The mutual information $I(X;Y)$ is _____ (rounded off to two decimal places).

Q3.25.2 MSQ 2025 2M

The random variable X takes values in

$$\{-1, 0, 1\}$$

with probabilities

$$P(X = -1) = P(X = 1) = \alpha$$

and

$$P(X = 0) = 1 - 2\alpha, \qquad 0 < \alpha < \frac{1}{2}.$$

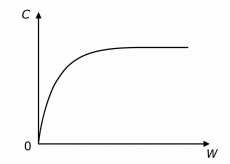
Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α.
Which of the following statements is/are TRUE?

A. $g(0.4) > g(0.3)$ **B.** $g(0.3) > g(0.4)$
C. $g(0.3) > g(0.25)$ **D.** $g(0.25) > g(0.3)$

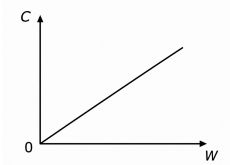
Q3.25.3 MCQ 2025 1M

Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\frac{N_0}{2}$. Let P_{av} denote the average transmit power constraint.
Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?

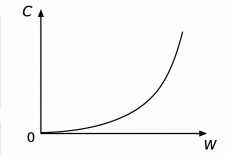
A.



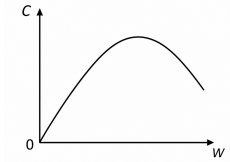
B.



C.



D.



Q3.24.1 NAT 2024 1M

A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____.

Q3.23.1 NAT 2023 2M

The frequency of occurrence of 8 symbols $(a - h)$ is given below:

Symbols	a	b	c	d	e	f	g	h
Frequency of occurrence	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{128}$

Fig. Q3.23.1

A symbol is chosen and it is determined by asking a series of "yes/no" questions which are truthfully answered.
The average number of questions when asked in the most efficient sequence, to determine the chosen symbol, is _____(rounded off to two decimal places).

Q3.22.1 NAT 2022 2M

The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown below.

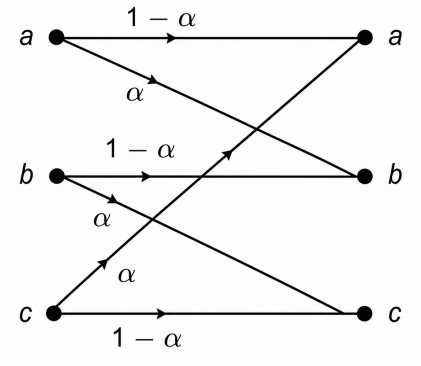


Fig. Q3.22.1

The transition probabilities are as marked.The parameter α lies in the interval $[0.25, 1]$.
The value of α for which the capacity of this channel is maximized is _____(rounded off to two decimal places).

Q3.22.2 MSQ 2022 1M

Let $H(X)$ denote the entropy of a discrete random variable X taking K possible distinct real values.Which of the following statements is/are necessarily true?

A. $H(X) \leq \log_2 K$ bits
B. $H(X) \leq H(2X)$
C. $H(X) \leq H(X^2)$
D. $H(X) \leq H(2^X)$

Q3.17.1 MCQ 2017 2M

Consider a binary memoryless channel characterized by the transition probability diagram shown in the figure.The channel is

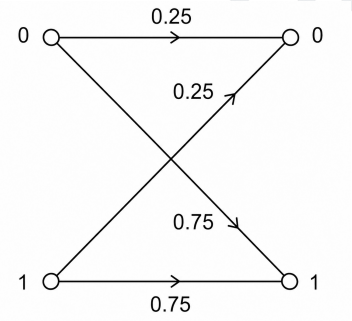
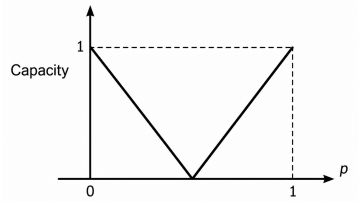


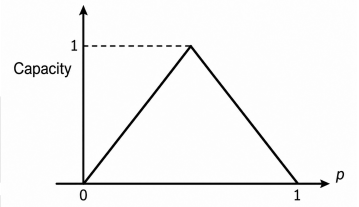
Fig. Q3.17.1

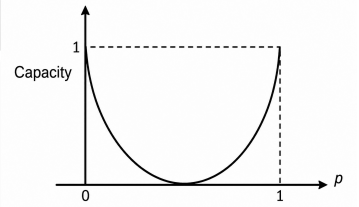
A. Lossless
B. Noiseless
C. Useless
D. Deterministic

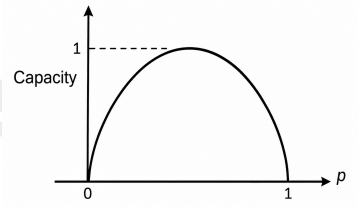
Q3.17.2 MCQ 2017 1M

Which one of the following graphs shows the Shannon capacity (channel capacity) in bits of a memoryless binary symmetric channel with crossover probability p?

A.


B.


C.


D.


Q3.17.3 NAT 2017 1M

Let (X_1, X_2) be independent random variables. X_1 has mean 0 and variance 1, while X_2 has mean 1 and variance 4.
The mutual information $I(X_1; X_2)$ between X_1 and X_2 in bits is _____.

Q3.16.1 NAT 2016 1M

An analog baseband signal, bandlimited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four message symbols that occur independently with probabilities $p_1 = p_4 = 0.125, p_2 = p_3$. The information rate (bits/sec) of the message source is _____.

Q3.16.2 NAT 2016 2M

Consider a discrete memoryless source with alphabet

$$S = \{s_0, s_1, s_2, s_3, \ldots\}$$

and respective probabilities of occurrence

$$P = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots\right\}.$$

The entropy of the source (in bits) is _____.

Q3.16.3 NAT 2016 2M

A binary communication system makes use of the symbols "zero" and "one". There are channel errors. Consider the following events:

x_0 : "zero" is transmitted

x_1 : "one" is transmitted

y_0 : "zero" is received

y_1 : "one" is received

The following probabilities are given:

$$P(x_0) = \frac{1}{2}, \qquad P(y_0|x_0) = \frac{3}{4}, \qquad P(y_0|x_1) = \frac{1}{2}.$$

The information in bits that you obtain when you learn which symbol has been received (while you know that a "zero" has been transmitted) is _____.

Q3.14.1 MCQ 2014 2M

The capacity of a band-limited additive white Gaussian noise (AWGN) channel is given by

$$C = W \log_2\left(1 + \frac{P}{\sigma^2 W}\right) \quad \text{bits/sec},$$

where W is the channel bandwidth, P is the average received power, and σ^2 is the one-sided power spectral density of the AWGN.
For a fixed

$$\frac{P}{\sigma^2} = 1000,$$

the channel capacity (in kbps) with infinite bandwidth ($W \to \infty$) is approximately

A. 1.44 **B.** 1.08
C. 0.72 **D.** 0.36

Q3.14.2 NAT 2014 2M

A fair coin is tossed repeatedly until a "Head" appears for the first time. Let L be the number of tosses required to obtain the first Head. The entropy $H(L)$ in bits is _____.

Q3.14.3 NAT 2014 2M

Consider the Z-channel given in the figure. The input is 0 or 1 with equal probability.If the output is 0, the probability that the input is also 0 equals _____

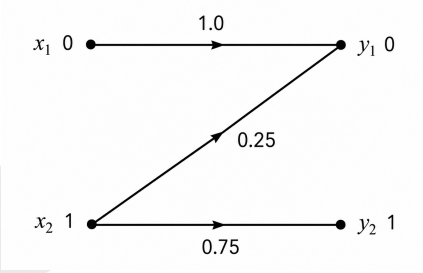


Fig. Q3.14.3

Q3.14.4 NAT 2014 2M

A binary random variable X takes the value of 1 with probability 1/3. X is input to a cascade of 2 independent identical binary symmetric channels (BSCs) each with crossover probability 1/2. The output of BSCs are the random variables Y_1 and Y_2 as shown in the figure.The value of $H(Y_1) + H(Y_2)$ in bits is _____

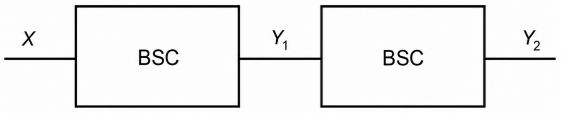


Fig. Q3.14.4

Q3.14.5 NAT 2014 1M

The capacity of a Binary Symmetric Channel (BSC) with cross-over probability 0.5 is _____.

Q3.12.1 MCQ 2012 1M

A source alphabet consists of N symbols with the probability of the first two symbols being the same. A source encoder increases the probability of the first symbol by a small amount ε and decreases that of the second by ε. After encoding, the entropy of the source

A. increases
B. remains the same
C. increases only if $N = 2$
D. decreases

Q3.09.1 MCQ 2009 2M

A communication channel with AWGN operating at a signal-to-noise ratio

$$\text{SNR} \gg 1$$

and bandwidth B has capacity C_1. If the SNR is doubled while keeping the bandwidth constant, the resulting capacity C_2 is given by

A. $C_2 \approx 2C_1$
B. $C_2 \approx C_1 + B$
C. $C_2 \approx C_1 + 2B$
D. $C_2 \approx C_1 + 0.3B$

Q3.08.1 MCQ 2008 2M

A memoryless source emits n symbols, each with probability p. The entropy of the source as a function of n is

A. increases as $\log n$
B. decreases as $\log(1/n)$
C. increases as n
D. increases as $n \log n$

Q3.90.1 MCQ 1990 2M

An image uses 512×512 picture elements. Each picture element can take any one of the 8 distinguishable intensity levels.
The maximum entropy in the above image will be

A. 2097152 bits
B. 786432 bits
C. 648 bits
D. 144 bits

Q3.89.1 MCQ 1989 2M

A source produces 4 symbols with probabilities

$$\frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \frac{1}{8}.$$

For this source, a practical coding scheme has an average codeword length of

$$L = 2 \text{ bits/symbol}.$$

The efficiency of the code is

A. 1
B. $\frac{7}{8}$
C. 0.5
D. 0.25

Answer Key

Q3.26.1 C	**Q3.25.1** 0	**Q3.25.2** B,C
Q3.25.3 A	**Q3.24.1** 4	**Q3.23.1** 1.98
Q3.22.1 1	**Q3.22.2** A,B,D	**Q3.17.1** C
Q3.17.2 C	**Q3.17.3** 0	**Q3.16.1** 360 to 363
Q3.16.2 1.8 to 2.2	**Q3.16.3** 0.8 to 0.82	**Q3.14.1** A
Q3.14.2 1.99 to 2.01	**Q3.14.3** 0.79 to 0.81	**Q3.14.4** 1.9 to 2.1
Q3.14.5 0	**Q3.12.1** D	**Q3.09.1** B
Q3.08.1 A	**Q3.90.1** B	**Q3.89.1** B

UNIT

IV

Analog Communications

Topics

1. Amplitude Modulation and Demodulation
2. Spectra of AM Signals
3. Angle Modulation - FM and PM
4. Demodulation of FM and PM
5. Spectra of FM Signals
6. Superheterodyne Receivers

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 84	**MCQ:** 63	**MSQ:** 3	**NAT:** 18
1M: 36	**2M:** 48	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MSQ 2026 2M

Consider a real, narrowband signal

$$x(t) = A(t)\cos(2\pi f_c t + \theta(t))$$

where the maximum frequency components of $A(t)$ and $\theta(t)$ are f_m and $f_c = 1000 f_m$, respectively.
Which of the following statements is/are correct for

$$-\infty < t < \infty\,?$$

A. $x(t)$ represents a PSK modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
B. $x(t)$ represents an amplitude modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
C. $x(t)$ represents a band-limited Gaussian noise process.
D. $x(t)$ never represents a narrowband FM signal.

Q4.25.1 MSQ 2025 1M

Consider a message signal $m(t)$ which is bandlimited to [-W,W], where W is in Hz.Consider the following two modulation schemes: **Double Sideband Suppressed Carrier (DSB-SC):**

$$f_{\text{DSB}}(t) = A_c\, m(t)\cos(2\pi f_c t)$$

Amplitude Modulation (AM):

$$f_{\text{AM}}(t) = A_c\big(1 + \mu m(t)\big)\cos(2\pi f_c t)$$

where A_c and f_c are the amplitude and frequency of the carrier, respectively, and μ is the modulation index.Consider the following statements:

(i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t.

(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.

Which of the following options is/are correct?

A. (i) is TRUE **B.** (i) is FALSE
C. (ii) is TRUE **D.** (ii) is FALSE

Q4.25.2 MCQ 2025 1M

Consider a frequency-modulated (FM) signal

$$f(t) = A_c\cos\left(2\pi f_c t + 3\sin(2\pi f_1 t) + 4\sin(6\pi f_1 t)\right),$$

where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that

$$f_c > 100 f_1.$$

The peak frequency deviation of the FM signal in Hz is _____

A. $15 f_1$ **B.** $12 f_1$
C. $4 f_1$ **D.** $2 f_1$

Q4.24.1 NAT 2024 1M

An amplitude modulator has output (in Volts)

$$s(t) = A\cos(400\pi t) + B\cos(360\pi t) + B\cos(440\pi t).$$

The carrier power normalized to a $1\,\Omega$ resistance is 50 Watts. The ratio of the total sideband power to the total power is $\frac{1}{9}$. The value of B (in Volts), rounded off to two decimal places, is _____

Q4.23.1 MCQ 2023 2M

Let a frequency modulated (FM) signal be

$$x(t) = A\cos\left(\omega_c t + k_f \int_{-\infty}^{t} m(\lambda)\, d\lambda\right),$$

where $m(t)$ is a message signal of bandwidth W. Let B_T denote the FM bandwidth.The signal is passed through a nonlinear system whose output is

$$y(t) = 2x(t) + 5x^2(t).$$

The minimum value of ω_c required to recover $x(t)$ from $y(t)$ is

A. $B_T + W$ **B.** $\frac{3}{2}B_T$
C. $2B_T + W$ **D.** $\frac{5}{2}B_T$

Q4.22.1 MSQ 2022 2M

Consider an FM broadcast that employs a pre-emphasis filter with frequency response

$$H_{pe}(\omega) = 1 + j\frac{\omega}{\omega_0}$$

where

$$\omega_0 = 10^4 \text{ rad/s}.$$

For the RC network shown below to act as the corresponding de-emphasis filter, the appropriate pair(s) of (R, C) values is/are:

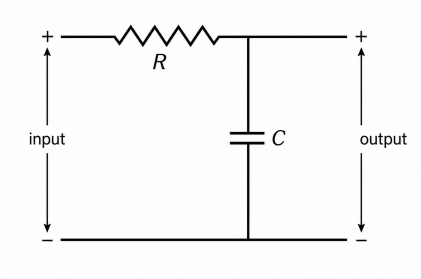


Fig. Q4.22.1

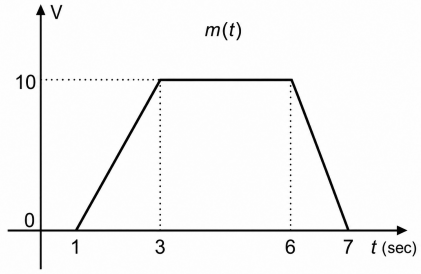


Fig. Q4.20.1

A. $R = 1\,\text{k}\Omega,\ C = 0.1\,\mu\text{F}$ **B.** $R = 2\,\text{k}\Omega,\ C = 1\,\mu\text{F}$

C. $R = 1\,\text{k}\Omega,\ C = 2\,\mu\text{F}$ **D.** $R = 2\,\text{k}\Omega,\ C = 0.5\,\mu\text{F}$

Q4.21.1 NAT 2021 2M

A superheterodyne receiver is tuned to 600 kHz. If the local oscillator feeds a 1000 kHz signal to the mixer, the image frequency (in integer) is _____ kHz.

Q4.21.2 NAT 2021 2M

A sinusoidal message signal having root mean square value of 4 V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2 rad/V. If the carrier signal is

$$c(t) = 2\cos(2\pi \times 10^6 t),$$

the maximum instantaneous frequency of the phase modulated signal (rounded off to one decimal place) is _____ Hz.

Q4.21.3 NAT 2021 1M

Consider a carrier signal which is amplitude modulated by a single-tone sinusoidal message signal with a modulation index of 50%.If the carrier and one of the sidebands are suppressed in the modulated signal, the percentage of power saved (rounded off to one decimal place) is _____.

Q4.20.1 NAT 2020 2M

The phase modulated and frequency modulated signals corresponding to the message signal $m(t)$ are given by

$$S_{PM}(t) = \cos\big(1000\pi t + K_p m(t)\big)$$

and

$$S_{FM}(t) = \cos\left(1000\pi t + K_f \int_{-\infty}^{t} m(\tau)\, d\tau\right),$$

where K_p is the phase deviation constant (rad/V) and K_f is the frequency deviation constant (rad/s/V). If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$are equal, determine

$$\frac{K_p}{K_f}.$$

Q4.20.2 MCQ 2020 2M

For the modulated signal $x(t) = m(t)\cos(2\pi f_c t)$, the message signal is $m(t) = 4\cos(1000\pi t)$, and the carrier frequency is $f_c = 1$ MHz.The signal $x(t)$ is applied to a coherent demodulator consisting of a multiplier and an ideal low-pass filter (LPF). The local oscillator signal is $\cos\big(2\pi(f_c + 40)t\big)$, and the LPF has cut-off frequencies ±510 Hz. Determine the output $y(t)$ of the demodulator.

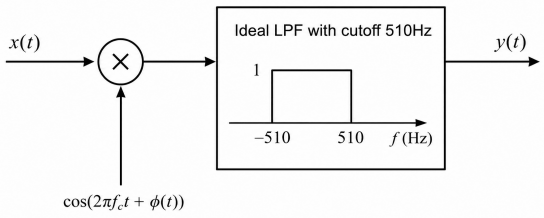


Fig. Q4.20.2

Q4.19.1 NAT 2019 1M

The baseband signal $m(t)$ shown in the figure is phase-modulated to generate the PM signal

$$\phi(t) = 2\cos\big(2\pi f_c t + k\, m(t)\big).$$

The time axis in the figure is in milliseconds. If the carrier frequency is $f_c = 50$ kHz and $k = 10\pi$,then the ratio of the minimum instantaneous frequency (in kHz) to the maximum instantaneous frequency (in kHz) is _____.(rounded off to two decimal places).

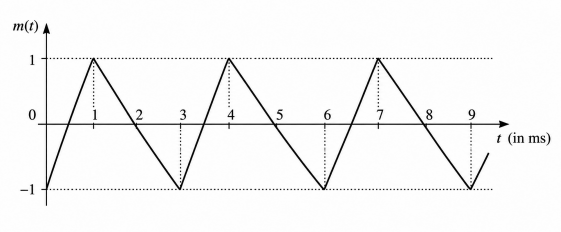


Fig. Q4.19.1

Q4.18.1 MCQ 2018 2M

Let

$$c(t) = A_c \cos(2\pi f_c t)$$

and

$$m(t) = \cos(2\pi f_m t).$$

It is given that

$$f_c \gg 5f_m.$$

The signal

$$v_i(t) = c(t) + m(t)$$

is applied to the input of a non-linear device whose output is

$$v_o(t) = a\, v_i(t) + b\, v_i^2(t),$$

where a and b are positive constants.The output of the non-linear device is passed through an ideal band-pass filter (BPF) having center frequency f_c and bandwidth $3f_m$, producing an amplitude-modulated (AM) wave. If the sideband power of the AM wave is half of the carrier power, then the ratio

$$\frac{a}{b}$$

is

A. 0.25 **B.** 0.5

C. 1 **D.** 2

Q4.18.2 NAT 2018 1M

Consider the following amplitude modulated signal:

$$s(t) = \cos(2000\pi t) + 4\cos(2400\pi t) + \cos(2800\pi t).$$

The ratio (accurate to three decimal places) of the power of the message signal to the power of the carrier signal is _____.

Q4.17.1 NAT 2017 2M

A modulating signal is given by

$$x(t) = 5\sin\left(4\pi \times 10^3 t - 10\pi\cos(2\pi \times 10^3 t)\right)\ \text{V}$$

and is fed to a phase modulator with phase deviation constant $k_p = 5$ rad/V. If the carrier frequency is $f_c = 20$ kHz, the instantaneous frequency (in kHz) at $t = 0.5$ ms is _____

Q4.17.2 NAT 2017 2M

The un-modulated carrier power in an AM transmitter is 5 kW. This carrier is modulated by a sinusoidal modulating signal. The maximum percentage modulation is 50%. If it is reduced to 40%, then the maximum un-modulated carrier power (in kW) that can be used without overloading the transmitter is _____

Q4.16.1 NAT 2016 1M

A superheterodyne receiver operates in the frequency range of 58 MHz – 68 MHz.The intermediate frequency f_{IF} and local oscillator frequency f_{LO} are chosen such that

$$f_{IF} \le f_{LO}.$$

It is required that the image frequencies fall outside the 58 MHz – 68 MHz band. The minimum required f_{IF} (in MHz) is _____

Q4.16.2 NAT 2016 1M

For a superheterodyne receiver, the intermediate frequency is $f_{IF} = 15$ MHz and the local oscillator frequency is $f_{LO} = 3.5$ GHz. If the frequency of the received signal is greater than the local oscillator frequency, then the image frequency (in MHz) is _____

Q4.16.3 NAT 2016 1M

The amplitude of a sinusoidal carrier is modulated by a single-tone sinusoid to obtain the amplitude modulated signal

$$s(t) = 5\cos(1600\pi t) + 20\cos(1800\pi t) + 5\cos(2000\pi t).$$

The value of the modulation index is _____

Q4.16.4 MCQ 2016 1M

A frequency synthesizer consists of a Phase Locked Loop (PLL) and a divide-by-N counter. The divider provides outputs corresponding to

$$\div 2, \quad \div 4, \quad \div 8, \quad \div 16.$$

The synthesizer is excited with an input signal of frequency

$$f_{in} = 5 \text{ kHz}.$$

The free-running frequency of the VCO is 20 kHz. The commutator switch selects the divider outputs repeatedly in the order

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4,$$

corresponding to $N = 2, 4, 8, 16$. The corresponding frequencies synthesized are

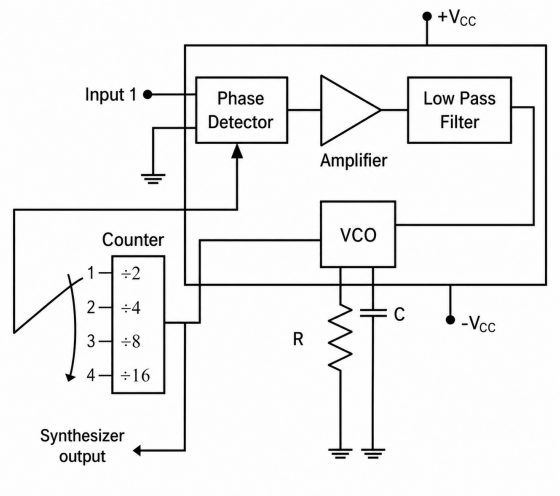


Fig. Q4.16.4

A. 10 kHz, 20 kHz, 40 kHz, 80 kHz
B. 20 kHz, 40 kHz, 80 kHz, 160 kHz
C. 80 kHz, 40 kHz, 20 kHz, 10 kHz
D. 160 kHz, 80 kHz, 40 kHz, 20 kHz

Q4.15.1 MCQ 2015 2M

The complex envelope of the bandpass signal

$$x(t) = -\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)\sin\left(\pi t - \frac{\pi}{4}\right)$$

centered about $f_c = \frac{1}{2}$ Hzis _____

A.

$$\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}$$

B.

$$\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{-j\frac{\pi}{4}}$$

C.

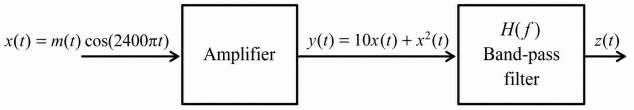
$$\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}$$

D.

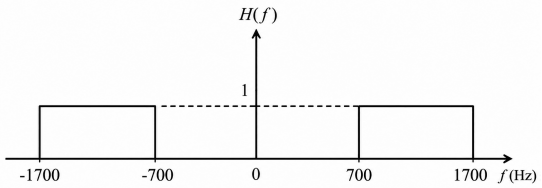
$$\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{-j\frac{\pi}{4}}$$

Q4.15.2 NAT 2015 2M

In the system shown in Figure (a), $m(t)$ is a low-pass signal with bandwidth W Hz. The frequency response of the band-pass filter $H(f)$ is shown in Figure (b). If it is desired that the output signal $z(t) = 10x(t)$, the maximum value of W (in Hz) should be strictly less than _____



(a)



(b)

Q4.15.3 MCQ 2015 1M

A message signal $m(t) = A_m \sin(2\pi f_m t)$ is used to modulate the phase of a carrier $A_c \cos(2\pi f_c t)$ to obtain the phase-modulated signal

$$y(t) = A_c \cos(2\pi f_c t + m(t)).$$

The bandwidth of $y(t)$ _____
A. depends on A_m but not on f_m
B. depends on f_m but not on A_m
C. depends on both A_m and f_m
D. does not depend on A_m or f_m

Q4.15.4 MCQ 2015 1M

Consider the signal

$$s(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t)$$

where $\hat{m}(t)$ denotes the Hilbert transform of $m(t)$, and the bandwidth of $m(t)$ is very small compared to f_c. The signal $s(t)$ is a _____

A. high-pass signal
B. low-pass signal
C. band-pass signal
D. double sideband suppressed carrier signal

Q4.14.1 NAT 2014 2M

If the figure, $M(f)$ is the Fourier transform of the message signal $m(t)$ where $A = 100$ Hz and $B = 40$ Hz. Given $v(t) = \cos(2\pi f_c t)$ and $w(t) = \cos\big(2\pi(f_c + A)t\big)$, where $f_c > A$. The cutoff frequencies of both the filters are f_c.The bandwidth of the signal at the output of the modulator (in Hz) is _____

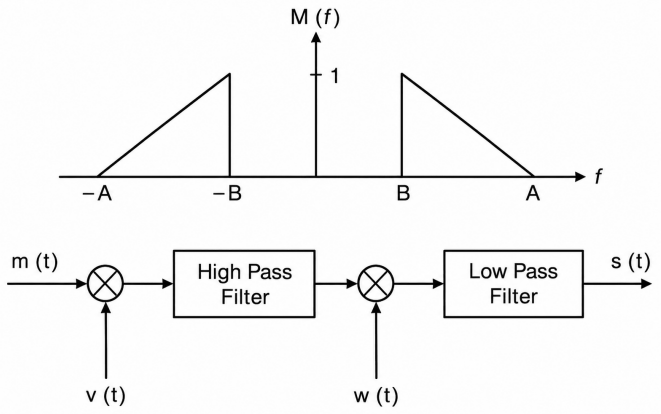


Fig. Q4.14.1

Q4.14.2 NAT 2014 1M

Consider sinusoidal modulation in an AM system. Assuming no overmodulation, the modulation index (μ) when the maximum and minimum values of the envelope, respectively, are 3 V and 1 V, is _____

Q4.14.3 MCQ 2014 1M

Consider a FM signal

$$f(t) = \cos[2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)]\,.$$

The maximum deviation of the instantaneous frequency from the carrier frequency f_c is

A. $\beta_1 f_1 + \beta_2 f_2$ **B.** $\beta_1 f_2 + \beta_2 f_1$
C. $\beta_1 + \beta_2$ **D.** $f_1 + f_2$

Q4.14.4 NAT 2014 1M

In a double side-band (DSB) full carrier AM transmission system, if the modulation index is doubled, then the ratio of total sideband power to the carrier power increases by a factor of _____

Q4.12.1 MCQ 2012 2M

The signal $m(t)$ as shown is applied both to a phase modulator (with k_p as the phase constant) and a frequency modulator (with k_f as the frequency constant) having the same carrier frequency.

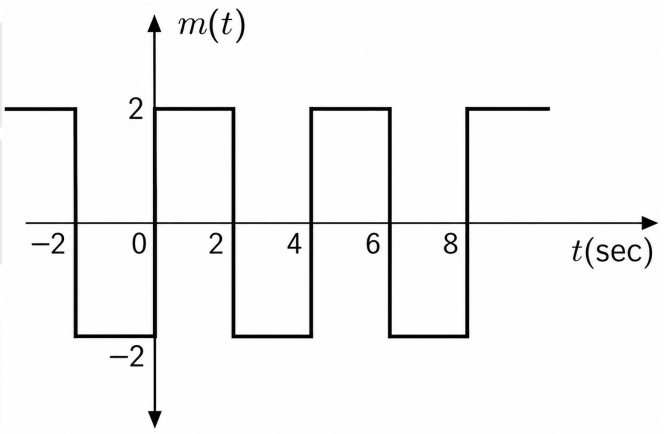


Fig. Q4.12.1

The ratio k_p/k_f (in rad/Hz) for the same maximum phase deviation is

A. 8π **B.** 4π
C. 2π **D.** π

Q4.11.1 MCQ 2011 2M

A message signal $m(t) = \cos 2000\pi t + 4\cos 4000\pi t$ modulates the carrier $c(t) = \cos 2\pi f_c t$ where $f_c = 1$MHz to produce an AM signal. For demodulating the generated AM signal using an envelope detector, the time constant RC of the detector circuit should satisfy

A. 0.5 ms $< RC <$ 1 ms **B.** 1 μs $\ll RC <$ 0.5 ms
C. $RC \ll \mu$s **D.** $RC \gg$ 0.5 ms

Q4.11.2 MCQ 2011 1M

The Column-I lists the attributes and the Column-II lists the modulation systems. Match the attribute to the modulation system that best meets it.

	Column-I		Column-II
P	Power efficient transmission of signals	1	Conventional AM
Q	Most bandwidth efficient transmission of voice signals	2	FM
R	Simplest receiver structure	3	VSB
S	Bandwidth efficient transmission of signals with Significant dc component	4	SSB-SC

Fig. Q4.11.2

A. P-4;Q-2;R-1;S-3 **B.** P-2;Q-4;R-1;S-3
C. P-3;Q-2;R-1;S-4 **D.** P-2;Q-4;R-3;S-1

Q4.10.1 MCQ 2010 1M

Consider an angle modulated signal
$x(t) = 6\cos\left[2\pi \times 10^6 t + 2\sin(8000\pi t) + 4\cos(8000\pi t)\right]$ V
The average power of $x(t)$ is

A. 10 W **B.** 18 W
C. 20 W **D.** 28 W

Q4.10.2 MCQ 2010 1M

Suppose that the modulating signal is $m(t) = 2\cos(2\pi f_m t)$ and the carrier signal is $X_c(t) = A_c\cos(2\pi f_c t)$, which one of the following is a conventional AM signal without over-modulation

A. $x(t) = A_c m(t)\cos(2\pi f_c t)$
B. $x(t) = A_c\,[1 + m(t)]\cos(2\pi f_c t)$
C. $x(t) = A_c\cos(2\pi f_c t) + \frac{A_c}{4}m(t)\cos(2\pi f_c t)$
D. $x(t) = A_c\cos(2\pi f_c t)\cos(2\pi f_c t) + A_c\sin(2\pi f_m t)\sin(2\pi f_c t)$

Q4.09.1 MCQ 2009 2M

A message signal given by

$$m(t) = \left(\frac{1}{2}\right)\cos\omega_1 t - \left(\frac{1}{2}\right)\sin\omega_2 t$$

is amplitude-modulated with a carrier of frequency ω_c to generate

$$s(t) = [1 + m(t)]\cos\omega_c t$$

What is the power efficiency achieved by this modulation scheme?

A. 8.33% **B.** 11.11%
C. 20% **D.** 25%

Q4.09.2 MCQ 2009 1M

For a message signal $m(t) = \cos(2\pi f_m t)$ and carrier of frequency f_c, which of the following represents a single side-band (SSB) signal?

A. $\cos(2\pi f_m t)\cos(2\pi f_c t)$
B. $\cos(2\pi f_c t)$
C. $\cos[2\pi(f_c + f_m)t]$
D. $[1 + \cos(2\pi f_m t)]\cos(2\pi f_c t)$

Q4.08.1 MCQ 2008 2M

The signal

$$\cos\omega_c t + 0.5\cos\omega_m t\,\sin\omega_c t$$

is

A. FM only **B.** AM only
C. both AM and FM **D.** neither AM nor FM

Q4.08.2 MCQ 2008 2M

Consider the frequency modulated signal

$$10\cos\left[2\pi \times 10^5 t + 5\sin(2\pi \times 1500t) + 7.5\sin(2\pi \times 1000t)\right]$$

with carrier frequency of 10^5 Hz. The modulation index is

A. 12.5 **B.** 10
C. 7.5 **D.** 5

Q4.08.3 MCQ 2008 1M

Consider the amplitude modulated (AM) signal

$$A_c\cos\omega_c t + 2\cos\omega_m t\cos\omega_c t.$$

For demodulating the signal using envelope detector, the minimum value of A_c should be

A. 2 **B.** 1
C. 0.5 **D.** 0

Q4.07.1 MCQ 2007 2M

In the following scheme, if the spectrum M(f) of m(t) is as shown, then the spectrum Y(f) of y(t) will be

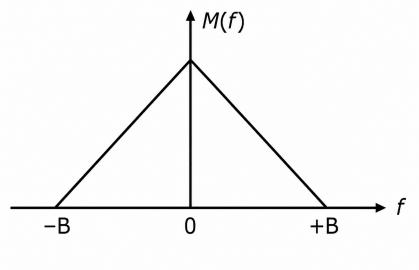


Fig. Q4.07.1

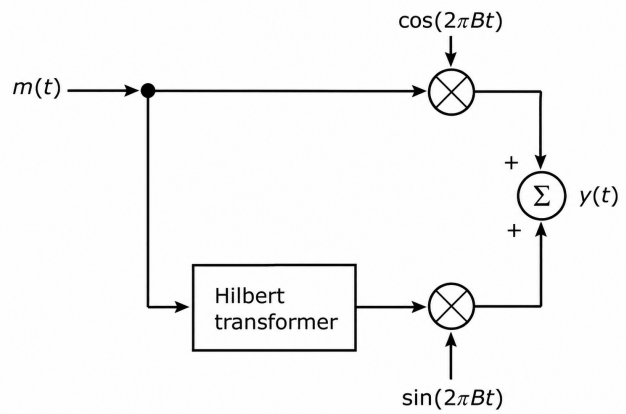
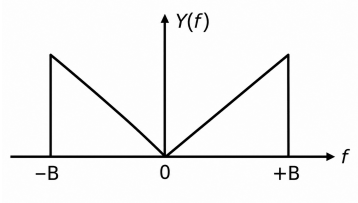
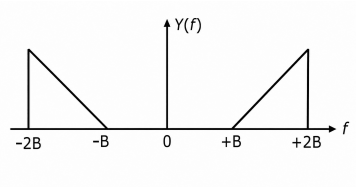
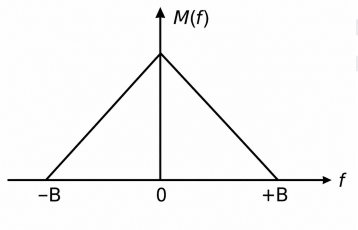
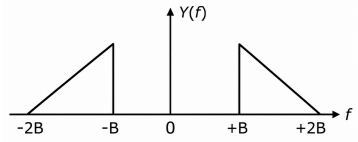


Fig. Q4.07.1

A.


B.


C.


D.


Q4.06.1 MCQ 2006 2M

Consider the following Amplitude Modulated (AM) signal, where $f_m < B$:

$$X_{AM}(t) = 10\,(1 + 0.5 \sin 2\pi f_m t) \cos 2\pi f_c t.$$

The AM signal gets added to a noise with Power Spectral Density $S_n(f)$ given in the figure below. The ratio of average sideband power to mean noise power would be

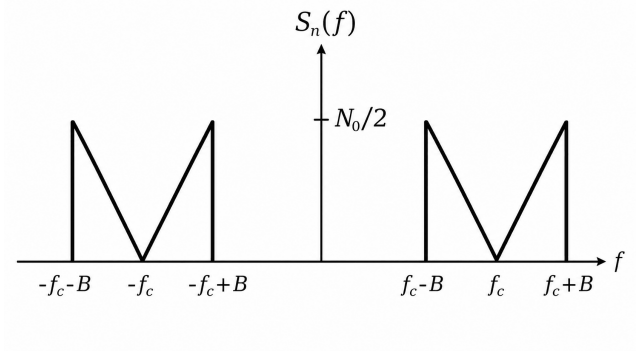


Fig. Q4.06.1

A. $\frac{25}{8N_0B}$ **B.** $\frac{25}{4N_0B}$

C. $\frac{25}{2N_0B}$ **D.** $\frac{25}{N_0B}$

Q4.06.2 MCQ 2006 2M

Consider the following Amplitude Modulated (AM) signal, where $f_m < B$:

$$X_{AM}(t) = 10\,(1 + 0.5 \sin 2\pi f_m t) \cos 2\pi f_c t.$$

The average side-band power for the AM signal given above is

A. 25 **B.** 12.5

C. 6.25 **D.** 3.125

Q4.06.3 MCQ 2006 2M

Let $g(t) = p(t) * p(t)$ where $*$ denotes convolution and $p(t) = u(t) - u(t-1)$ with $u(t)$ being the unit step function.An Amplitude Modulated signal is given as

$$X_{AM}(t) = 100\,(p(t) + 0.5g(t)) \cos \omega_c t$$

in the interval $0 \le t \le 1$. One set of possible values of modulating signal and modulation index would be

A. t, 0.5 **B.** t, 1.0

C. t, 2.0 **D.** t^2, 0.5

Q4.06.4 MCQ 2006 2M

The diagonal clipping in Amplitude Demodulation (using envelope detector) can be avoided if RC time-constant of the envelope detector satisfies the following condition, (here W is message bandwidth and ω_c is carrier frequency both in (rad/sec))

A. $RC < \frac{1}{W}$ **B.** $RC > \frac{1}{W}$

C. $RC < \frac{1}{\omega_c}$ **D.** $RC > \frac{1}{\omega_c}$

Q4.06.5 MCQ 2006 2M

A message signal with bandwidth 10 kHz is Lower-Side Band SSB modulated with carrier frequency $f_{c1} = 10^6$ Hz. The resulting signal is then passed through a Narrow-Band Frequency Modulator with carrier frequency $f_{c2} = 10^9$ Hz. The bandwidth of the output would be

A. 4×10^4 Hz **B.** 2×10^6 Hz

C. 2×10^9 Hz **D.** 2×10^{10} Hz

Q4.05.1 MCQ 2005 2M

A carrier is phase modulated (PM) with frequency deviation of 10 kHz by a single tone frequency of 1 kHz. If the single tone frequency is increased to 2 kHz, assuming that phase deviation remains unchanged, the bandwidth of the PM signal is

A. 21 kHz **B.** 22 kHz

C. 42 kHz **D.** 44 kHz

Q4.05.2 MCQ 2005 2M

A device with input $x(t)$ and output $y(t)$ is characterized by:

$$y(t) = x^2(t).$$

An FM signal with frequency deviation of 90 kHz and modulating signal bandwidth of 5 kHz is applied to this device. The bandwidth of the output signal is

A. 370 kHz **B.** 190 kHz

C. 380 kHz **D.** 95 kHz

Q4.05.3 MCQ 2005 1M

Which of the following analog modulation scheme requires the minimum transmitted power and minimum channel bandwidth ?

A. VSB **B.** $DSB - SC$

C. SSB **D.** AM

Q4.05.4 MCQ 2005 1M

Find the correct match between group 1 and group 2.

Group 1	Group 2
P - $\{1 + km(t)\} A\sin(\omega_c t)$	W - Phase modulation
Q - $km(t)\, A\sin(\omega_c t)$	X - Frequency modulation
R - $A\sin\{\omega_c t + km(t)\}$	Y - Amplitude modulation
S - $A\sin\left(\omega_c t + k\int_{-\infty}^{t} m(\tau)\, d\tau\right)$	Z - DSB-SC modulation

Fig. Q4.05.4

A. P-Z,Q-Y,R-X,S-W **B.** P-W,Q-X,R-Y,S-Z

C. P-X;Q-W;R-Z;S-Y **D.** P-Y;Q-Z;R-W;S-X

Q4.04.1 MCQ 2004 2M

Consider a system shown in the figure. Let $X(f)$ and $Y(f)$ denote the Fourier transforms of $x(t)$ and $y(t)$ respectively. The ideal HPF has the cutoff frequency 10 kHz.

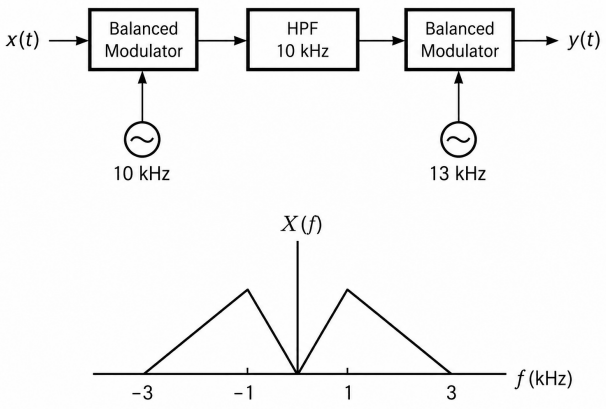


Fig. Q4.04.1

The positive frequencies where $Y(f)$ has spectral peaks are

A. 1 kHz and 24 kHz **B.** 2 kHz and 24 kHz

C. 1 kHz and 14 kHz **D.** 2 kHz and 14 kHz

Q4.04.2 MCQ 2004 2M

Two sinusoidal signals of same amplitude and frequencies 10 kHz and 10.1 kHz are added together. The combined signal is given to an ideal frequency detector. The output of the detector is

A. 0.1 kHz sinusoid **B.** 20.1 kHz sinusoid

C. a linear function of time **D.** a constant

Q4.04.3 MCQ 2004 2M

A 100 MHz carrier of 1 V amplitude and a 1 MHz modulating signal of 1 V amplitude are fed to a balanced modulator. The output of the modulator is passed through an ideal high-pass filter with cut-off frequency of 100 MHz. The output of the filter is added with 100 MHz signal of 1 V amplitude and 90° phase shift as shown in the figure. The envelope of the resultant signal is

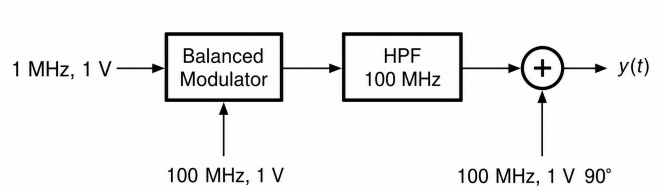


Fig. Q4.04.3

A. constant

B. $\sqrt{1+\sin(2\pi\times10^6 t)}$

C. $\sqrt{\frac{5}{4}-\sin(2\pi\times10^6 t)}$

D. $\sqrt{\frac{5}{4}+\cos(2\pi\times10^6 t)}$

Q4.04.4 MCQ 2004 1M

An AM signal and a narrow-band FM signal with identical carriers, modulating signals and modulation indices of 0.1 are added together. The resultant signal can be closely approximated by

A. broadband FM

B. SSB with carrier

C. DSB-SC

D. SSB without carrier

Q4.04.5 MCQ 2004 1M

An AM signal is detected using an envelope detector. The carrier frequency and modulating signal frequency are 1 MHz and 2 kHz respectively. An appropriate value for the time constant of the envelope detector is

A. 500 μsec

B. 20 μsec

C. 0.2 μsec

D. 1 μsec

Q4.03.1 MCQ 2003 2M

A superheterodyne receiver is to operate in the frequency range 550 kHz – 1650 kHz, with the intermediate frequency of 450 kHz. Let $R = C_{max}/C_{min}$ denote the required capacitance ratio of the local oscillator and I denote the image frequency (in kHz) of the incoming signal. If the receiver is tuned to 700 kHz, then

A. $R = 4.41,\ I = 1600$

B. $R = 2.10,\ I = 1150$

C. $R = 3.0,\ I = 1600$

D. $R = 9.0,\ I = 1150$

Q4.03.2 MCQ 2003 2M

Choose the correct one from among the alternative A,B,C,D after matching an item in Group 1 with most appropriate item in Group 2.

Group 1		Group 2	
P	Ring modulator	1	Clock recovery
Q	VCO	2	Demodulation of FM
R	Foster-Seely discriminator	3	Frequency conversion
S	Mixer	4	Summing the two inputs
		5	Generation of FM
		6	Generation of DSB-SC

Fig. Q4.03.2

A. P-1,Q-3,R-2,S-4

B. P-6,Q-5,R-2,S-3

C. P-6;Q-1;R-3;S-2

D. P-5;Q-6;R-1;S-3

Common Data Questions

Answer the question 4.03.3 and question 4.03.4 using the following data.
Let $m(t) = \cos\left[(4\pi\times10^3)t\right]$ be the message signal and $c(t) = 5\cos\left[(2\pi\times10^6)t\right]$ be the carrier.

Q4.03.3 MCQ 2003 2M

$c(t)$ and $m(t)$ are used to generate an FM signal. If the peak frequency deviation of the generated FM signal is three times the transmission bandwidth of the AM signal, then the coefficient of the term $\cos\left[2\pi(1008\times10^3)t\right]$ in the FM signal (in terms of the Bessel coefficients) is

A. $5J_4(3)$

B. $\frac{5}{2}J_8(3)$

C. $\frac{5}{2}J_8(4)$

D. $5J_4(6)$

Q4.03.4 MCQ 2003 2M

$c(t)$ and $m(t)$ are used to generate an AM signal. The modulation index of the generated AM signal is 0.5. Then the quantity

$$\frac{\text{Total sideband power}}{\text{Carrier power}}$$

is

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$

C. $\frac{1}{3}$ **D.** $\frac{1}{8}$

Q4.03.5 MCQ 2003 2M

A DSB-SC signal is to be generated with a carrier frequency $f_c = 1$ MHz using a non-linear device with the input-output characteristic

$$V_0 = a_0 v_i + a_1 v_i^3$$

where a_0 and a_1 are constants. The output of the non-linear device can be filtered by an appropriate band pass filter.Let

$$v_i = A_c' \cos(2\pi f_c' t) + m(t),$$

where $m(t)$ is the message signal. Then the value of f_c' (in MHz) is

A. 1.0 **B.** 0.333

C. 0.5 **D.** 3.0

Q4.03.6 MCQ 2003 1M

The input to a coherent detector is DSB-SC signal plus noise. The noise at the detector output is

A. the in-phase component **B.** the quadrature-component

C. zero **D.** the envelope

Q4.02.1 MCQ 2002 2M

In the figure,

$$m(t) = \frac{2\sin 2\pi t}{t}, \qquad s(t) = \cos 200\pi t$$

and

$$n(t) = \frac{\sin 199\pi t}{t}.$$

The output $y(t)$ will be

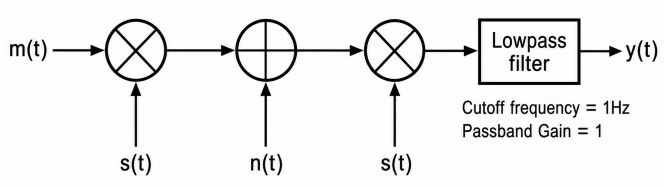


Fig. Q4.02.1

A. $\frac{\sin 2\pi t}{t}$

B. $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t}\cos 3\pi t$

C. $\frac{\sin 2\pi t}{t} + \frac{\sin 0.5\pi t}{t}\cos 1.5\pi t$

D. $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t}\cos 0.75\pi t$

Q4.02.2 MCQ 2002 2M

An angle modulated signal is given by

$$s(t) = \cos 2\pi \left(2 \times 10^6 t + 30\sin 150t + 40\cos 150t\right)$$

The maximum frequency and phase deviations of $s(t)$ are

A. 10.5 kHz, 140π rad **B.** 6 kHz, 80π rad

C. 10.5 kHz, 100π rad **D.** 7.5 kHz, 100π rad

Q4.02.3 MCQ 2002 1M

A 1 MHz sinusoidal carrier is amplitude modulated by a symmetrical square wave of period 100μsec. Which of the following frequencies will NOT be present in the modulated signal?

A. 990 kHz **B.** 1010 kHz

C. 1020 kHz **D.** 1030 kHz

Q4.00.1 MCQ 2000 2M

In an FM system, a carrier of 100MHz is modulated by a sinusoidal signal of 5kHz. The bandwidth by Carson's approximation is 1MHz.If $y(t) = (\text{modulated waveform})^3$, then by using Carson's approximation, the bandwidth of $y(t)$ around 300MHz and the spacing of spectral components are, respectively.

A. 3 MHz, 5 kHz **B.** 1 MHz, 15 kHz

C. 3 MHz, 15 kHz **D.** 1 MHz, 5 kHz

Q4.00.2 MCQ 2000 2M

A message $m(t)$ band limited to the frequency f_m has a power of P_m. The power of the output signal in the figure is

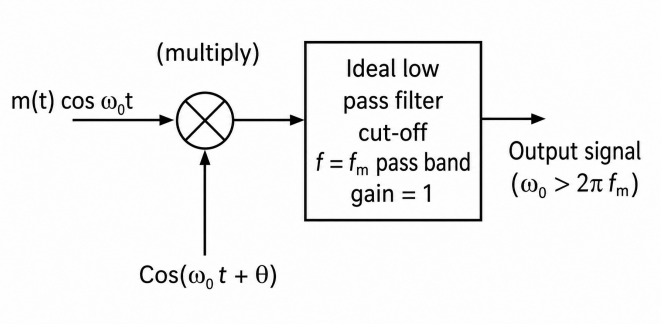


Fig. Q4.00.2

A. $\frac{P_m \cos\theta}{2}$ **B.** $\frac{P_m}{4}$

C. $\frac{P_m \sin^2\theta}{4}$ **D.** $\frac{P_m \cos^2\theta}{4}$

Q4.00.3 MCQ 2000 1M

The amplitude modulated wave form

$$s(t) = A_c\,[1 + K_a m(t)]\cos\omega_c t$$

is fed to an ideal envelope detector. The maximum magnitude of $K_a m(t)$ is greater than 1. Which of the following could be the detector output?

A. $A_c m(t)$ **B.** $A_c^2\,[1 + K_a m(t)]^2$

C. $|A_c\,[1 + K_a m(t)]|$ **D.** $A_c\,[1 + K_a m(t)]^2$

Q4.99.1 MCQ 1999 1M

A modulated signal is given by,

$$s(t) = m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t),$$

where the base band signal $m_1(t)$ and $m_2(t)$ have bandwidths of 10 kHz and 15 kHz, respectively. The bandwidth of the modulated signal, in kHz, is

A. 10 kHz **B.** 15 kHz

C. 25 kHz **D.** 30 kHz

Q4.98.1 MCQ 1998 1M

A DSB-SC signal is generated using the carrier $\cos(\omega_c t+\theta)$ and modulating signal $x(t)$. The envelope of the DSB-SC signal is

A. $x(t)$ **B.** $|x(t)|$

C. only positive portion of $x(t)$ **D.** $x(t)\cos\theta$

Q4.98.2 MCQ 1998 1M

The image channel selectivity of super heterodyne receiver depends upon

A. IF amplifiers only

B. RF and IF amplifiers only

C. Pre selector, RF and IF amplifiers

D. Pre selector and RF amplifiers only

Q4.96.1 MCQ 1996 1M

The image channel rejection in a super heterodyne reciever comes from

A. IF stages only

B. RF stages only

C. detector and RF stages only

D. detector, RF and IF stages

Q4.96.2 MCQ 1996 2M

An FM signal with a modulation index 9 is applied to a frequency tripler. The modulation index in the output signal will be

A. 0 **B.** 3

C. 9 **D.** 27

Q4.95.1 MCQ 1995 1M

A PAM signal can be detected by using

A. an ADC **B.** an integrator

C. a band pass filter **D.** a high pass filter

Q4.95.2 MCQ 1995 1M

A PLL can be used to demodulate

A. PAM signals **B.** PCM signals

C. FM signals **D.** DSB-SC signals

Q4.95.3 MCQ 1995 1M

The image(second) channel selectivity of a super heterodyne communication receiver is determined by

A. antenna and pre selector

B. the pre selector and RF amplifier

C. the pre selector and IF amplifier

D. the RF and IF amplifier

Q4.94.1 MCQ 1994 2M

Match the following

List I	List II
a) SSB	1) Envelope detector
b) AM	2) Integrate and dump
c) BPSK	3) Hilbert transform
	4) Ratio detector
	5) PLL

Fig. Q4.94.1

Q4.94.2 MCQ 1994 1M

$v(t) = 5\left[\cos(10^6\pi t) - \sin(10^3\pi t) \times \sin(10^6\pi t)\right]$ represents

A. DSB suppressed carrier signal

B. AM signal

C. SSB upper sideband signal

D. Narrow band FM signal

Q4.94.3 NAT 1994 1M

A 10 MHz carrier is frequency modulated by a sinusoidal signal of 500Hz, the maximum frequency deviation being 50 kHz. The bandwidth required (in kHz) as given by the Carson's rule is

Q4.93.1 MCQ 1993 2M

Which of the following demodulator(s) can be used for demodulating the signal

$$x(t) = 5(1 + 2\cos 200\pi t)\cos 20000\pi t$$

A. Envelope demodulator

B. Square-law demodulator

C. Synchronous demodulator

D. None of the above

Q4.93.2 NAT 1993 2M

A super-heterodyne radio receiver with an intermediate frequency of 455 KHz is tuned to a station operating at 1200 KHz. The associated image frequency (in kHz) is

Q4.92.1 MCQ 1992 2M

The maximum power efficiency of an AM modulator is

A. 25% **B.** 50%

C. 75% **D.** 100%

Q4.90.1 MCQ 1990 2M

In commercial TV transmission in India, picture and speech signals are modulated respectively as:

A. VSB and VSB **B.** VSB and SSB

C. VSB and FM **D.** FM and VSB

Q4.89.1 MCQ 1989 2M

A signal $x(t) = 2\cos(\pi \cdot 10^4 t)$ volts is applied to an FM modulator with the sensitivity constant of 10 kHz/volt. Then the modulation index of the FM wave is

A. 4 **B.** 2

C. $\frac{4}{\pi}$ **D.** $\frac{2}{\pi}$

Q4.89.2 MCQ 1989 2M

Which of the following schemes suffer(s) from the threshold effect?

A. AM detection using envelope detection

B. AM detection using synchronous detection

C. FM detection using a discriminator

D. SSB detection with synchronous detection

Q4.87.1 MCQ 1987 2M

A carrier $A_c \cos \omega_c t$ is frequency modulated by a signal $E_m \cos \omega_m t$. The modulation index is m_f. The expression for the resulting FM signal is:

A. $A_c \cos\left[\omega_c t + m_f \sin \omega_m t\right]$

B. $A_c \cos\left[\omega_c t + m_f \cos \omega_m t\right]$

C. $A_c \cos\left[\omega_c t + 2\pi m_f \cos \omega_m t\right]$

D. $A_c \cos\left[\omega_c t + \frac{2\pi m_f}{\omega_m} E_m \cos \omega_m t\right]$

Q4.87.2 MCQ 1987 2M

In a radar receiver the antenna is connected to the receiver through a waveguide. Placing the preamplifier on the antenna side of the waveguide rather than on the receiver side leads to:

A. A reduction in the overall noise figure.

B. A reduction in interference.

C. An improvement in selectivity characteristics.

D. An improvement in directional characteristics.

Answer Key

Q4.26.1 A, B, C	Q4.25.1 A,D	Q4.25.2 A
Q4.24.1 2.49 to 2.51	Q4.23.1 B	Q4.22.1 A
Q4.21.1 1400	Q4.21.2 1011310 to 1011320	Q4.21.3 94.2 to 94.6
Q4.20.1 2	Q4.20.2 B	Q4.19.1 0.74 to 0.76
Q4.18.1 D	Q4.18.2 0.25	Q4.17.1 70
Q4.17.2 5.19 to 5.23	Q4.16.1 4.9 to 5.1	Q4.16.2 3485

Q4.16.3 0.49 to 0.51	Q4.16.4 A	Q4.15.1 C
Q4.15.2 349 to 351	Q4.15.3 C	Q4.15.4 C
Q4.14.1 59.9 to 60.1	Q4.14.2 0.45 to 0.55	Q4.14.3 A
Q4.14.4 3.94 to 4.05	Q4.12.1 B	Q4.11.1 B
Q4.11.2 B	Q4.10.1 B	Q4.10.2 C
Q4.09.1 C	Q4.09.2 C	Q4.08.1 C

Q4.08.2 B	Q4.08.3 A	Q4.07.1 A
Q4.06.1 B	Q4.06.2 C	Q4.06.3 A
Q4.06.4 A	Q4.06.5 B	Q4.05.1 D
Q4.05.2 A	Q4.05.3 C	Q4.05.4 D
Q4.04.1 B	Q4.04.2 D	Q4.04.3 C
Q4.04.4 B	Q4.04.5 B	Q4.03.1 A

Q4.03.2 B	Q4.03.3 D	Q4.03.4 D
Q4.03.5 C	Q4.03.6 A	Q4.02.1 C
Q4.02.2 D	Q4.02.3 C	Q4.00.1 A
Q4.00.2 D	Q4.00.3 C	Q4.99.1 D
Q4.98.1 B	Q4.98.2 D	Q4.96.1 B
Q4.96.2 D	Q4.95.1 B	Q4.95.2 C
Q4.95.3 B	Q4.94.1 A-3,B-1,C-2	Q4.94.2 D
Q4.94.3 101	Q4.93.1 C	Q4.93.2 2110
Q4.92.1 B	Q4.90.1 C	Q4.89.1 A
Q4.89.2 C	Q4.87.1 A	Q4.87.2 A

SUBJECT

2

Electronic Devices and Circuits

Subject Snapshot*				
Stream: EC	**Questions:** 208	**MCQ:** 145	**MSQ:** 10	**NAT:** 53
1M: 118	**2M:** 90	**Descriptive:** 0	**Avg Weightage** ~8M	**Range:** 1M(2013)–10M(2024)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Electronic Devices and Circuits

Unit	Topic	Pages
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II.	PN Junctions	71
III.	BJTs	81
IV.	MOSCap and MOSFETs	86
V.	Optoelectronic Devices	94

PrepFusion

UNIT

I

Basic Semiconductor Physics

Topics

1. Physics and Fundamentals
2. Carrier Transport Phenomena
3. Semiconductor Under Non-Equilibrium
4. Fermi Dirac Distribution and Energy Band Diagrams

Unit Snapshot*

Stream: EC	**Questions:** 75	**MCQ:** 55	**MSQ:** 4	**NAT:** 16
1M: 46	**2M:** 29	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2020)–11M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 2M

Consider that the concentration of electrons in a semiconductor bar varies linearly from 2×10^{17} cm^{-3} at $x = 1$ μm to 1×10^{16} cm^{-3} at $x = 4$ μm along the x-direction. Assume that the concentration of electrons is not varying along other directions (that is along y- and z-directions). [Given: the mobility of electron is 1400 cm^{2}V^{-1}s^{-1}, thermal voltage is 25 mV and electronic charge is 1.6×10^{-19} C.] The density of electron diffusion current (in A/mm^2) is ___. (rounded off to two decimal places)

Q1.25.1 MSQ 2025 1M

Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

- ☐ **A.** Arsenic
- ☐ **B.** Boron
- ☐ **C.** Gallium
- ☐ **D.** Phosphorous

Q1.25.2 MCQ 2025 2M

The intrinsic carrier concentration of a semiconductor is 2.5×10^{16} /m^3 at 300 K. If the electron and hole mobilities are 0.15 m^2/Vs and 0.05 m^2/Vs, respectively, then the intrinsic resistivity of the semiconductor (in kΩ.m) at 300 K is _____.
(Charge of an electron $e = 1.6 \times 10^{-19}$ C.)

A. 1.65 **B.** 1.25 **C.** 0.85 **D.** 1.95

Q1.25.3 MCQ 2025 2M

The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is 0.38 m^2/Vs. The electron diffusivity D_n at 300 K (in cm^2/s, rounded off to the nearest integer) is _____.
(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

A. 26 **B.** 98 **C.** 38 **D.** 10

Q1.24.1 MSQ 2024 1M

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?

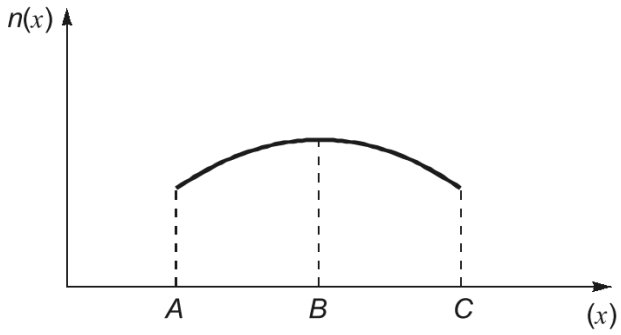


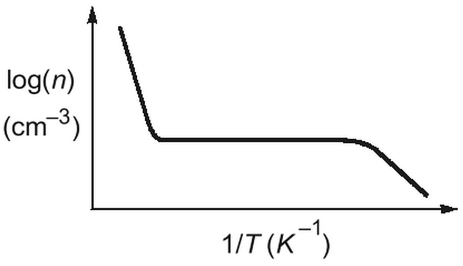
Fig. Q1.24.1

- ☐ **A.** For x between B and C, the electron diffusion current is directed from C to B.
- ☐ **B.** For x between B and A. the electric field is directed from A to B.
- ☐ **C.** For x between B and A, the electron drift current is directed from B to A.
- ☐ **D.** For x between B and C, the electric field is directed from B to C .

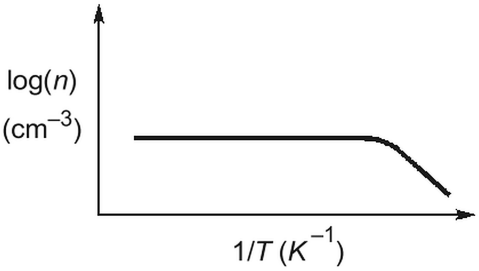
Q1.24.2 MCQ 2024 1M

For non-degenerately doped n-type silicon, which one of the following plots represent the temperature (T) dependence of free electron concentration (n)?

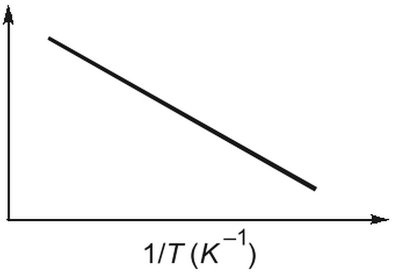
A.



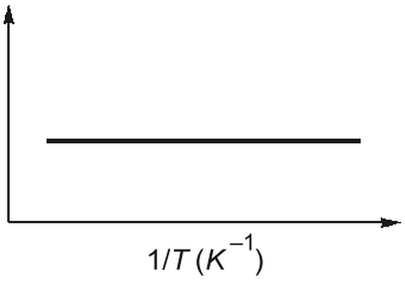
B.



C.



D.



Q1.24.3 NAT 2024 2M

A non-degenerate n-type semiconductor has 5% neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV. The difference between E_C and E_D (in eV, rounded off to two decimal places) is _____.

Q1.23.1 MCQ 2023 1M

In a semiconductor, if the Fermi energy level lies in the conduction band, then the semiconductor is known as

A. degenerate n-type. **B.** degenerate p-type.

C. non-degenerate n-type. **D.** non-degenerate p-type.

Q1.23.2 MCQ 2023 1M

For an intrinsic semiconductor at temperature $T = 0$ K, which of the following statement is true?

A. All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.

B. All energy states in the valence band are empty of electrons and all energy states in the conduction band are filled with electrons.

C. All energy states in the valence and conduction band are filled with holes.

D. All energy states in the valence and conduction band are filled with electrons.

Q1.23.3 NAT 2023 2M

In an extrinsic semiconductor, the hole concentration is given to be $1.5n_i$ where n_i is the intrinsic carrier concentration of 1×10^{10} cm^{-3}. The ratio of electron to hole mobility for equal hole and electron drift current is given as _____ (*rounded off to two decimal places*).

Q1.23.4 NAT 2023 2M

In a semiconductor device, the Fermi-energy level is 0.35 eV above the valence band energy. The effective density of states in the valence band at $T = 300$ K is 1×10^{19} cm^{-3}. The thermal equilibrium hole concentration in silicon at 400 K is _____ $\times 10^{13}$ cm^{-3} (*rounded off to two decimal places*).

Given kT at 300 K is 0.026 eV.

Q1.22.1 MSQ 2022 2M

Select the correct statement(s) regarding semiconductor devices:

☐ **A.** Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.

☐ **B.** Collector region is generally more heavily doped than base region in a *BJT*.

☐ **C.** Total current is spatially constant in a two terminal electronic device in dark under steady-state condition.

☐ **D.** Mobility of electrons always increases with temperature in silicon beyond 300 K.

Q1.22.2 MCQ 2022 1M

Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3}. Electron and hole diffusion lengths are 2 μm and 1 μm, respectively.

The left side of the bar ($x = 0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x = 0$ because of the laser. The steady state electron density at $x = 0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x = 2$ μm, is _____.

A. 0.37×10^{14} cm^{-3} **B.** 0.63×10^{13} cm^{-3}

C. 3.7×10^{14} cm^{-3} **D.** 10^{3} cm^{-3}

Q1.22.3 MCQ 2022 1M ○○○

In a non-degenerate bulk semiconductor with electron density $n = 10^{16}\ cm^{-3}$, the value of $E_C - E_{F_n} = 200\ meV$, where E_C and E_{F_n} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is $10^{10}\ cm^{-3}$. For $n = 0.5 \times 10^{16}\ cm^{-3}$, the closest approximation of the value of $(E_C - E_{F_n})$, among the given options, is ____

A. 226 meV **B.** 174 meV

C. 218 meV **D.** 182 meV

Q1.22.4 NAT 2022 2M ○○○

A p-type semiconductor with zero electric field is under illumination (low level injection) in steady-state condition. Excess minority carrier density is zero at $x = \pm 2l_n$, where $l_n = 10^{-4}$ cm is the diffusion length of electrons. Assume electronic charge, $q = -1.6 \times 10^{-19}$ C. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at $x = \pm 2l_n$ is ___ mA/cm^2 (Rounded off to two decimal places).

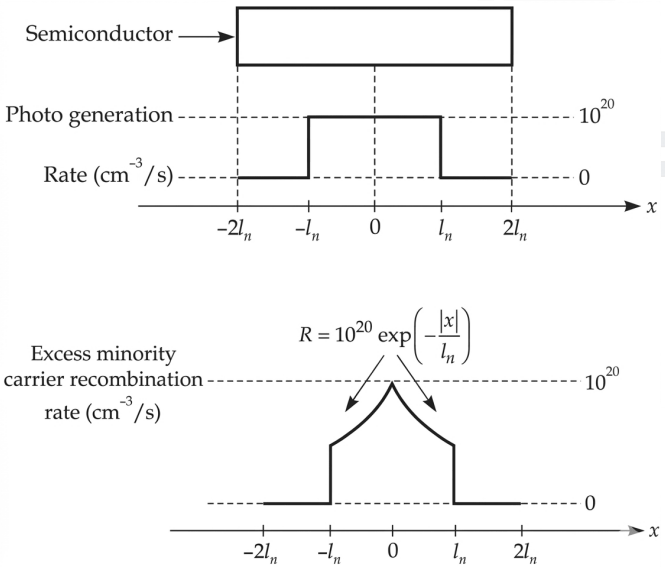


Fig. Q1.22.4

Q1.21.1 MCQ 2021 1M ○○○

A bar of silicon is doped with boron concentration of 10^{16} cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron hole pairs are generated throughout the volume of the bar at the rate of 10^{20} cm^{-3} s^{-1}. If the recombination lifetime is 100 μs, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then the approximate product of steady-state electron and hole concentrations due to this light exposure is,

A. 10^{32} cm^{-6} **B.** 2×10^{20} cm^{-6}

C. 2×10^{32} cm^{-6} **D.** 10^{20} cm^{-6}

Q1.21.2 MCQ 2021 1M ○○○

The energy band diagram of a p-type semiconductor bar of length L under equilibrium condition (i.e.. the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valence band at the two edges of the bar is Δ.

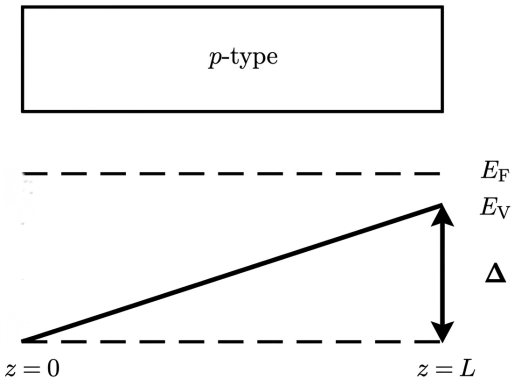


Fig. Q1.21.2

If the charge of an electron is q, then the magnitude of the electric field developed inside this semiconductor bar is

A. $\frac{\Delta}{qL}$ **B.** $\frac{2\Delta}{qL}$ **C.** $\frac{\Delta}{2qL}$ **D.** $\frac{3\Delta}{2qL}$

Q1.20.1 MCQ 2020 1M ○○○

A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV. The intrinsic Fermi level is shifted from mid-bandgap energy level by

A. 13.45 meV **B.** 18.02 meV

C. 26.90 meV **D.** 9.01 meV

Q1.17.1 NAT 2017 2M

The dependence of drift velocity of electrons on electric field in a semiconductor is shown below. The semiconductor has a uniform electron concentration of $n = 1 \times 10^{16}$ cm^{-3} and electronic charge $q = 1.6 \times 10^{-19}$ C. If a bias of 5 V is applied across a 1 μm region of this semiconductor, the resulting current density in this region, (in kA/cm^2), is ____

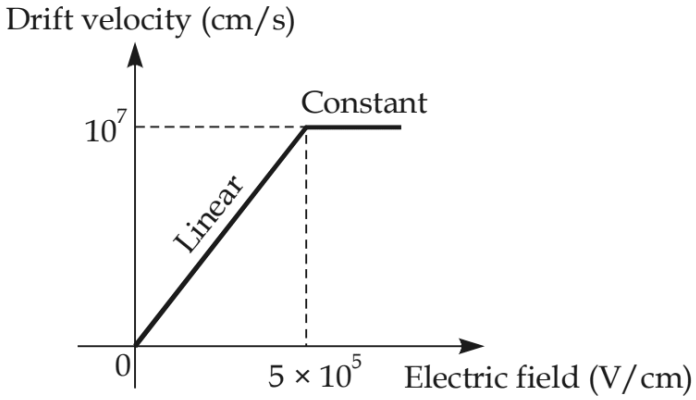


Fig. Q1.17.1

Q1.17.2 NAT 2017 2M

As shown a uniformly doped Silicon (Si) bar of length $L = 0.1$ μm with a donor concentration $N_D = 10^{16}$ cm^{-3} is illuminated at $x = 0$ such that electron and hole pairs are generated at the rate of $G_L = G_{L0}(1 - \frac{x}{L})$, $0 \le x \le L$, where $G_{L0} = 10^{17}$ cm$^{-3}s^{-1}$. Hole lifetime is $10^{-4}s$, electronic charge $q = 1.6 \times 10^{-19}C$, hole diffusion coefficient $D_P = 100$ cm$^2/s$ and low level injection condition prevails. Assuming a linearly decaying steady state excess hole concentration that goes to 0 at $x = L$, the magnitude of the diffusion current density at $x = L/2$, in A/cm^2, is _____.

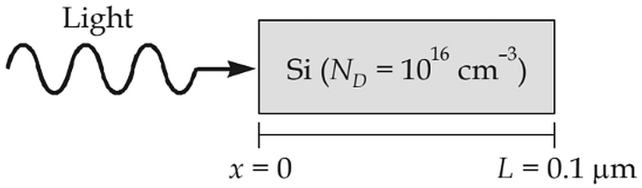


Fig. Q1.17.2

Q1.17.3 MCQ 2017 1M

A bar of Gallium Arsenide ($GaAs$) is doped with Silicon such that the Silicon atoms occupy Gallium and Arsenic sites in the $GaAs$ crystal. Which one of the following statement is true?

A. Silicon atoms act as p-type dopants in Arsenic sites and n-type dopants in Gallium sites

B. Silicon atoms act as n-type dopants in Arsenic sites and p-type dopants in Gallium sites

C. Silicon atoms act as p-type dopants in Arsenic as well as Gallium sites

D. Silicon atoms act as n-type dopants in Arsenic as well as Gallium sites

Q1.16.1 MCQ 2016 2M

Consider a silicon sample at $T = 300$ K, with a uniform donor density $N_d = 5 \times 10^{16}$ cm^{-3} illuminated uniformly such that the optical generation rate is $G_{opt} = 1.5 \times 10^{20}$ cm$^{-3}s^{-1}$ throughout the sample. The incident radiation is turned off at $t = 0$. Assume low-level injection to be valid and ignore surface effects. The carrier lifetimes are $\tau_{p0} = 0.1\mu s$ and $\tau_{n0} = 0.5\mu s$

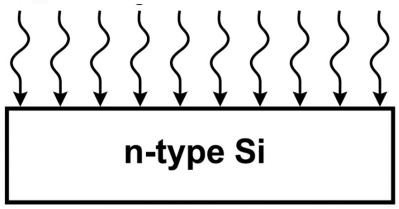


Fig. Q1.16.1

The hole concentration at $t = 0$ and the hole concentration at $t = 0.3$ μs, respectively, are

A. 1.5×10^{13} cm^{-3} and 7.47×10^{11} cm^{-3}

B. 1.5×10^{13} cm^{-3} and 8.23×10^{11} cm^{-3}

C. 7.5×10^{13} cm^{-3} and 3.73×10^{11} cm^{-3}

D. 7.5×10^{13} cm^{-3} and 4.12×10^{11} cm^{-3}

Q1.16.2 NAT 2016 2M

The figure below shows the doping distribution in a p-type semiconductor in log scale. The magnitude of the electric field (in kV/cm) in the semiconductor due to non uniform doping is _____

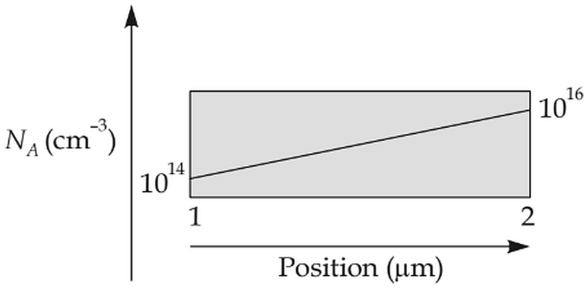


Fig. Q1.16.2

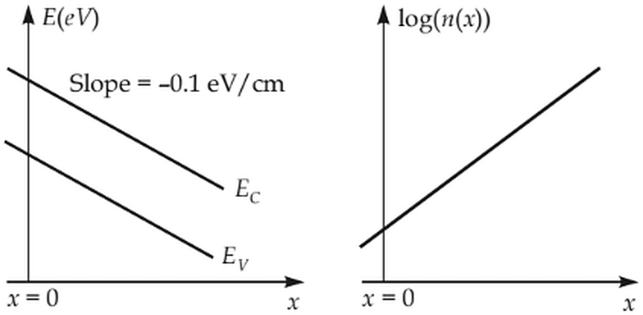


Fig. Q1.15.1

Q1.16.3 MCQ 2016 1M

A small percentage of impurity is added to an intrinsic semiconductor at 300 K. Which one of the following statements is true for the energy band diagram shown in the following figure?

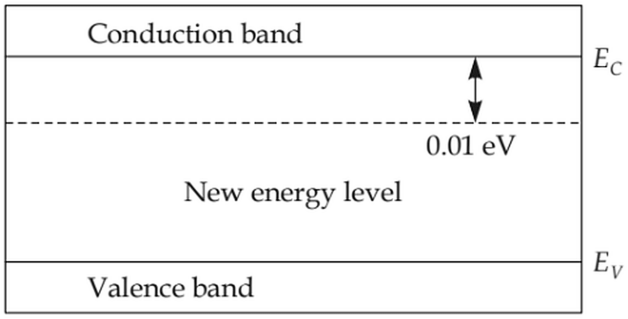


Fig. Q1.16.3

A. Intrinsic semiconductor doped with pentavalent atoms to form n-type semiconductor

B. Intrinsic semiconductor doped with trivalent atoms to form n-type semiconductor

C. Intrinsic semiconductor doped with pentavalent atoms to form p-type semiconductor

D. Intrinsic semiconductor doped with trivalent atoms to form p-type semiconductor

Q1.15.1 MCQ 2015 2M

The energy band diagram and the electron density profile $n(x)$ in a semiconductor are shown in the figures.

Assume that $n(x) = 10^{15} exp(\frac{q\alpha x}{kT})$ cm^{-3}, with $\alpha = 0.1$ V/cm and x expressed in cm. Given $\frac{kT}{q} = 0.026$ V, $D_n = 36$ cm$^2 s^{-1}$, and $\frac{D}{\mu} = \frac{kT}{q}$.

The electron current density (in A/cm^2) at $x = 0$ is

A. -4.4×10^{-2} **B.** -2.2×10^{-2}

C. 0 **D.** 2.2×10^{-2}

Q1.15.2 NAT 2015 2M

A dc voltage of 10 V is applied across an n-type silicon bar having a rectangular cross-section and a length of 1 cm as shown in figure. The donor doping concentration N_D and the mobility of electrons μ_n are 10^{16} cm^{-3} and 1000 cm^2 V^{-1} s^{-1}, respectively. The average time (in μs) taken by the electrons to move from one end of the bar to other end is _____.

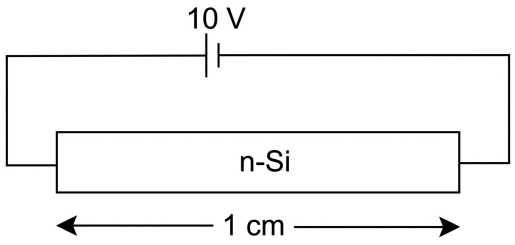


Fig. Q1.15.2

Q1.15.3 NAT 2015 1M

A piece of silicon is doped uniformly with phosphorous with a doping concentration of 10^{16} /cm^3. The expected value of mobility versus doping concentration for silicon assuming full dopant ionization is shown below. The charge of an electron is 1.6×10^{-19}C. The conductivity (in S cm^{-1}) of the silicon sample at 300 K is _____.

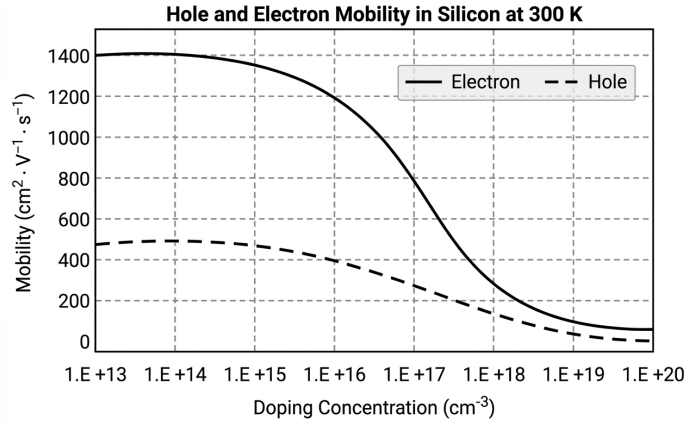


Fig. Q1.15.3

Q1.15.4 NAT 2015 1M

An n-type silicon sample is uniformly illuminated with light which generates 10^{20} electron-hole pairs per cm^3 per second. The minority carrier lifetime in the sample is 1 μs. In the steady state, the hole concentration in the sample is approximately 10^x, where x is an integer. The value of x is _____.

Q1.15.5 NAT 2015 1M

A silicon sample is uniformly doped with donor type impurities with a concentration of 10^{16} / cm^3. The electron and hole mobilities in the sample are 1200 $cm^2/V-s$ and 400 $cm^2/V-s$ respectively. Assume complete ionization of impurities. The charge of an electron is 1.6×10^{-19} C. The resistivity of the sample (in Ω-cm) is _____.

Q1.14.1 MCQ 2014 2M

An N -type semiconductor having uniform doping is biased as shown in the figure.

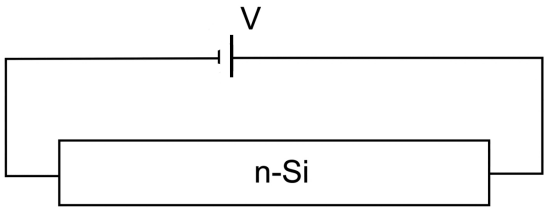
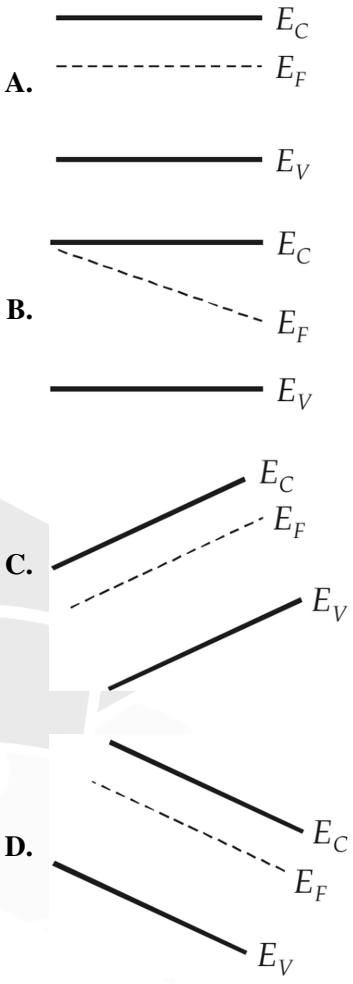


Fig. Q1.14.1

If E_C is the lowest energy level of the conduction band, E_V is the highest energy level of the valence band and E_F is the Fermi level, which one of the following represents the energy band diagram for the biased N-type semiconductor ?



Q1.14.2 NAT 2014 2M

Consider a silicon sample doped with $N_D = 1 \times 10^{15}$ /cm^3 donor atoms. Assume that the intrinsic carrier concentration $n_i = 1.5 \times 10^{10}$ /cm^3. If the sample is additionally doped with $N_A = 1 \times 10^{18}$ /cm^3 acceptor atoms, the approximate number of electrons/cm^3 in the sample, at T = 300 K, will be _____.

Q1.14.3 NAT 2014 1M

The cut-off wavelength (in μ m) of light that can be used for intrinsic excitation of a semiconductor material of bandgap $E_g = 1.1$ eV is _____.

Q1.14.4 MCQ 2014 1M

In the figure, $ln(\rho_i)$ is plotted as a function of $1/T$, where ρ_i is the intrinsic resistivity of silicon, T is the temperature, and the plot is almost linear. The slope of the line can be used to estimate

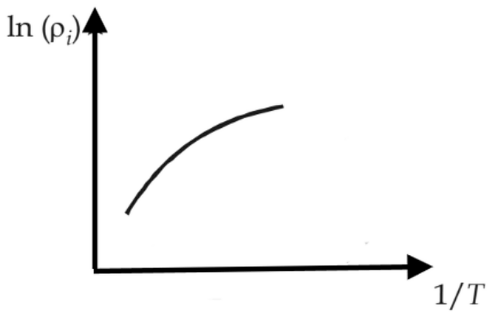


Fig. Q1.14.4

A. band gap energy of silicon (E_g)

B. sum of electron and hole mobility in silicon ($\mu_n + \mu_p$)

C. reciprocal of the sum of electron and hole mobility in silicon $(\mu_n + \mu_p)^{-1}$

D. intrinsic carrier concentration of silicon (n_i)

Q1.14.5 MCQ 2014 1M

At T = 300 K, the band gap and the intrinsic carrier concentration of GaAs are 1.42 eV and $10^6\ cm^{-3}$, respectively. In order to generate electron hole pairs in GaAs, which one of the wavelength (λ_C) ranges of incident radiation, is most suitable ? (Given that : Plank's constant is 6.62×10^{-34}J-s, velocity of light is $3\times10^{10}\ cm/s$ and charge of electron is 1.6×10^{-19}C)

A. $0.42\mu m < \lambda_c < 0.87\mu m$

B. $0.87\mu m < \lambda_c < 1.42\mu m$

C. $1.42\mu m < \lambda_c < 1.62\mu m$

D. $1.62\mu m < \lambda_c < 6.62\mu m$

Q1.14.6 NAT 2014 1M

At T = 300 K, the hole mobility of a semiconductor $\mu_P = 500\ cm^2/Vs$ and $\frac{KT}{q} = 26\ mV$.

The hole diffusion constant D_P in cm^2/s is ____.

Q1.14.7 MCQ 2014 1M

A thin P-type silicon sample is uniformly illuminated with light which generates excess carriers. The recombination rate is directly proportional to

A. the minority carrier mobility

B. the minority carrier recombination lifetime

C. the majority carrier concentration

D. the excess minority carrier concentration

Q1.14.8 NAT 2014 2M

Assume electronic charge $q = 1.6\times10^{-19}\ C$, $kT/q = 25mV$ and electron mobility $\mu_n = 1000\ cm^2/Vs$. If the concentration gradient of electrons injected into a P-type silicon sample is $1\times10^{21}\ /cm^4$, the magnitude of electron diffusion current density (in A/cm^2) is ____.

Q1.14.9 MCQ 2014 1M

A silicon bar is doped with donor impurities $N_D = 2.25\times10^{15}\ atom/cm^3$. Given the intrinsic carrier concentration of silicon at T = 300 K is $n_i = 1.5\times10^{10}\ cm^{-3}$. Assuming complete impurity ionization, the equilibrium electron and hole concentrations are

A. $n_0 = 1.5\times10^{16}\ cm^{-3}, p_0 = 1.5\times10^{5}\ cm^{-3}$

B. $n_0 = 1.5\times10^{10}\ cm^{-3}, p_0 = 1.5\times10^{15}\ cm^{-3}$

C. $n_0 = 2.25\times10^{15}\ cm^{-3}, p_0 = 1.5\times10^{10}\ cm^{-3}$

D. $n_0 = 2.25\times10^{15}\ cm^{-3}, p_0 = 1\times10^{5}\ cm^{-3}$

Q1.14.10 MCQ 2014 2M

The doping concentrations on the p-side and n-side of a silicon diode are $1\times10^{16}\ cm^{-3}$ and $1\times10^{17}\ cm^{-3}$, respectively. A forward bias of 0.3 V is applied to the diode. At T=300K, the intrinsic carrier concentration of silicon $n_i = 1.5\times10^{10}\ cm^{-3}$ and $\frac{KT}{q} = 26\ mV$. The electron concentration at the edge of the depletion region on the p-side is

A. $2.3\times10^{9}\ cm^{-3}$ **B.** $1\times10^{16}\ cm^{-3}$

C. $1\times10^{17}\ cm^{-3}$ **D.** $2.25\times10^{9}\ cm^{-3}$

Q1.11.1 MCQ 2011 1M

Drift current in the semiconductors depends upon

A. only the electric field

B. only the carrier concentration gradient

C. both the electric field and the carrier concentration

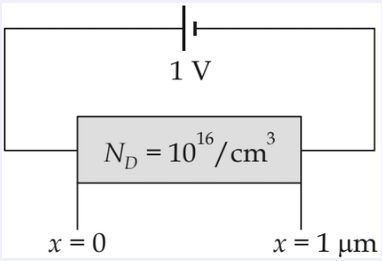
D. both the electric field and the carrier concentration gradient

Common Data Questions

Answer the next two questions using the following data.

The silicon sample with unit cross-sectional areas shown below is in thermal equilibrium. The following information is given:

T = 300 K, electronic charge = 1.6×10^{-19} C, thermal voltage = 26 mV and electron mobility = 1350 cm^2/V-s.



Q1.10.1 MCQ 2010 2M

The magnitude of the electric field at $x = 0.5\ \mu$m is

A. 1 kV/cm **B.** 5 kV/cm

C. 10 kV/cm **D.** 26 kV/cm

Q1.10.2 MCQ 2010 2M

The magnitude of the electron drift current density at $x = 0.5\ \mu$m is

A. 2.16×10^4 A/cm^2 **B.** 1.08×10^4 A/cm^2

C. 4.32×10^3 A/cm^2 **D.** 6.48×10^2 A/cm^2

Q1.09.1 MCQ 2009 1M

The ratio of the mobility to the diffusion coefficient in a semiconductor has the units

A. V^{-1} **B.** $cm.V^1$ **C.** $V.cm^{-1}$ **D.** $V.s$

Q1.09.2 MCQ 2009 1M

In an n-type silicon crystal at room temperature, which of the following can have a concentration of $4 \times 10^{19}\ cm^{-3}$?

A. Silicon atoms **B.** Holes

C. Dopant atoms **D.** Valence electrons

Q1.08.1 MCQ 2008 1M

Silicon is doped with boron to a concentration of 4×10^{17} atoms/ cm^3. Assume the intrinsic carrier concentration of silicon to be $1.5 \times 10^{10}/cm^3$ and the value of kT/q to be 25 mV at 300 K. Compared to undopped silicon, the fermi level of doped silicon

A. goes down by 0.31 eV

B. goes up by 0.13 eV

C. goes down by 0.427 eV

D. goes up by 0.427 eV

Q1.08.2 MCQ 2008 1M

Which of the following is true?

A. A silicon wafer heavily doped with boron is a p^+ substrate

B. A silicon wafer lightly doped with boron is a p^+ substrate

C. A silicon wafer heavily doped with arsenic is a p^+ substrate

D. A silicon wafer lightly doped with arsenic is a p^+ substrate

Q1.07.1 MCQ 2007 1M

The electron and hole concentrations in an intrinsic semiconductor are n_i per cm^3 at 300 K. Now, if acceptor impurities are introduced with a concentration of N_A per cm^3 (where $N_A >> n_i$), the electron concentration per cm^3 at 300 K will be

A. n_i **B.** $n_i + N_A$ **C.** $N_A - n_i$ **D.** $\frac{n_i^2}{N_A}$

Q1.06.1 MCQ 2006 1M

A heavily doped n-type semiconductor has the following data:

Hole-electron mobility ratio : 0.4

Doping concentration : 4.2×10^8 atoms/m^3

Intrinsic concentration : 1.5×10^4 atoms/m^3

The ratio of conductance of the n-type semiconductor to that of the intrinsic semiconductor of same material and at same temperature is given by

A. 0.00005 **B.** 2000 **C.** 10000 **D.** 20000

Q1.06.2 MCQ 2006 1M

The majority carriers in an n-type semiconductor have an average drift velocity v in a direction perpendicular to a uniform magnetic field B. The electric field E induced due to Hall effect acts in the direction

A. $v \times B$ **B.** $B \times v$

C. along v **D.** opposite to v

Q1.06.3 MCQ 2006 1M

Under low level injection assumption, the injected minority carrier current for an extrinsic semiconductor is essentially the

A. Diffusion current **B.** Drift current

C. Recombination current **D.** Induced current

Q1.06.4 MCQ 2006 1M

The concentration of minority carriers in an extrinsic semiconductor under equilibrium is

A. Directly proportional to doping concentration

B. Inversely proportional to the doping concentration

C. Directly proportional to the intrinsic concentration

D. Inversely proportional to the intrinsic concentration

Q1.05.1 MCQ 2005 1M

A Silicon sample A is doped with 10^{18} atoms/cm^3 of boron. Another sample B of identical dimension is doped with 10^{18} atoms/cm^3 phosphorus. The ratio of electron to hole mobility is 3. The ratio of conductivity of the sample A to B is

A. 3 **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{2}$

Q1.05.2 MCQ 2005 1M

The primary reason for the widespread use of Silicon in semiconductor device technology is

A. abundance of Silicon on the surface of the Earth

B. larger bandgap of Silicon in comparison to Germanium

C. favorable properties of Silicon dioxide (SiO2)

D. lower melting point

Q1.05.3 MCQ 2005 1M

The bandgap of Silicon at room temperature is

A. 1.3 eV **B.** 0.7 eV **C.** 1.1 eV **D.** 1.4 eV

Q1.04.1 MCQ 2004 1M

The longest wavelength that can be absorbed by silicon, which has the bandgap of 1.12 eV, is 1.1 μm. If the longest wavelength that can be absorbed by another material is 0.87 μm, then bandgap of this material is

A. 1.416 eV **B.** 0.886 eV

C. 0.854 eV **D.** 0.706 eV

Q1.04.2 MCQ 2004 1M

The resistivity of a uniformly doped n-type silicon sample is 0.5 Ωcm. If the electron mobility (μ_n) is 1250 cm^2/Vsec and the charge of an electron is 1.6×10^{-19} Coulomb, the donor impurity concentration (N_D) in the sample is

A. $2 \times 10^{16}/cm^3$ **B.** $1 \times 10^{16}/cm^3$

C. $2.5 \times 10^{15}/cm^3$ **D.** $5 \times 10^{15}/cm^3$

Q1.04.3 MCQ 2004 1M

The impurity commonly used for realizing the base region of a silicon n - p - n transistor is

A. Gallium **B.** Indium

C. Boron **D.** Phosphorus

Q1.03.1 MCQ 2003 1M

The electron concentration in a sample of uniformly doped n-type silicon at 300 K varies linearly from $10^{17}/cm^3$ at $x = 0$ to $6 \times 10^{16}/cm^3$ at $x = 2\mu m$. Assume a situation that electrons are supplied to keep this concentration gradient constant with time. If electronic charge is 1.6×10^{-19} coulomb and the diffusion constant $D_n = 35\ cm^2/s$, the current density in the silicon, if no electric field is present, is

A. zero **B.** $-112A/cm^2$

C. $+1120A/cm^2$ **D.** $-1120A/cm^2$

Q1.03.2 MCQ 2003 1M

An n-type silicon bar 0.1 cm long and 100 μm^2 in cross-sectional area has a majority carrier concentration of $5 \times 10^{20}/m^3$ and the carrier mobility is 0.13 $m^2/V-s$ at 300 K. If the charge of an electron is 1.6×10^{-19} coulomb, then the resistance of the bar is

A. 10^6 Ohm **B.** 10^4 Ohm

C. 10^{-1} Ohm **D.** 10^{-4} Ohm

Q1.02.1 MCQ 2002 1M

The intrinsic carrier concentration of silicon sample at 300 K is $1.5 \times 10^{16}/m^3$. If after doping, the number of majority carriers is $5 \times 10^{20}/m^3$, the minority carrier density is

A. $4.50 \times 10^{11}/m^3$ **B.** $3.333 \times 10^4/m^3$

C. $5.00 \times 10^{20}/m^3$ **D.** $3.00 \times 10^{-5}/m^3$

Q1.98.1 MCQ 1998 1M

The units of $\frac{q}{kT}$ are

A. V **B.** V^{-1} **C.** J **D.** J/K

Q1.95.1 MCQ 1995 1M

A small concentration of minority carriers is injected into a homogeneous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carries a distance of 1 cm in 20 μsec. The mobility (in cm^2/V-sec) will be

A. 1, 000 **B.** 2, 000 **C.** 5, 000 **D.** 500, 000

Q1.95.2 MCQ 1995 1M

In a p-type Si simple the hole concentration is $2.25 \times 10^{15}/cm^3$. The intrinsic carrier concentration is $1.5 \times 10^{10}/cm^3$ the electron concentration is

A. zero **B.** $10^{10}/cm^3$

C. $10^5/cm^3$ **D.** $1.5 \times 10^{25}/cm^3$

Q1.95.3 MCQ 1995 1M

The probability that an electron in a metal occupies the Fermi level, at any temperature ($T > 0$ K)

A. 0 **B.** 1 **C.** 0.5 **D.** 1.0

Q1.95.4 MCQ 1995 1M

The drift velocity of electrons, in silicon

A. is proportional to the electric field for all values of electric field.

B. is independent of the electric field.

C. increases at low values of electric field and decreases at high values of electric field exhibiting negative differential resistance.

D. increases linearly with electric field at low values of electric field and gradually saturates at higher values of electric field.

Q1.94.1 MCQ 1994 1M

A p-type silicon sample has a higher conductivity compared to an n-type silicon sample having the same dopant concentration. (True/False) _____

Q1.92.1 MCQ 1992 2M

A semiconductor is irradiated with light such that carriers are uniformly generated throughout its volume. The semiconductor is n-type with $N_D = 10^{19}/cm^3$. If the excess electron concentration in the steady-state is $\Delta n = 10^{15}/cm^3$ and if $\tau_p = 10\,\mu$sec. (minority carriers life time) the generation rate due to irradiation

A. is 10^{20} e-h pairs/cm^3/s

B. is 10^{24} e-h pairs/cm^3/s

C. is 10^{10} e-h pairs/cm^3/s

D. cannot be determined, the given data is insufficient

Q1.91.1 MCQ 1991 2M

A silicon sample is uniformly doped with 10^{16} phosphorus atoms/cm^3 and 2×10^{16} boron atoms/cm^3. If all the dopants are fully ionized, the material is

A. n-type with carrier concentration of $10^{16}/cm^3$.

B. p-type with carrier concentration of $10^{16}/cm^3$.

C. p-type with carrier of $2 \times 10^{16}/cm^3$.

D. n-type with a carrier concentration of $2 \times 10^{16}/cm^3$.

Q1.90.1 MSQ 1990 2M

Under high electric fields, in a semiconductor with increasing electric field,

A. the mobility of charge carriers decreases.

B. the mobility of charge carriers increases.

C. the velocity of the charge carriers saturates.

D. the velocity of the charge carriers increases.

Q1.89.1 MCQ 1989 2M

The concentration of ionized acceptors and donors in a semiconductor are N_A, N_D respectively. If $N_A > N_D$ and n_i is the intrinsic concentration, the position of the Fermi level with respect to the intrinsic level depends on

A. $N_A - N_D$ **B.** $N_A + N_D$ **C.** $\frac{N_A N_D}{n_i^2}$ **D.** n_i

Q1.89.2 MCQ 1989 2M ○○○

Due to illumination by light, the electron and hole concentrations in a heavily doped N type semiconductor increases by Δn and Δp respectively if n_i is the intrinsic concentration then,

A. $\Delta n < \Delta p$ **B.** $\Delta n > \Delta p$

C. $\Delta n = \Delta p$ **D.** $\Delta n \times \Delta p = n_i^2$

Q1.87.1 MCQ 1987 2M ○○○

Consider two energy levels: E_1, E ev above the Fermi level and E_2, E ev below the Fermi level. P_1 and P_2 are respectively the probability of E_1 being occupied by an electron and E_2 being empty. Then,

A. $P_1 > P_2$ **B.** $P_1 = P_2$

C. $P_1 < P_2$ **D.** P_1 & P_2 depend on number of free electrons

Q1.87.2 MCQ 1987 2M ○○○

In an intrinsic semiconductor the free electron concentration depends on

A. effective mass of electrons only

B. effective mass of holes only

C. temperature of the semiconductor

D. width of the forbidden energy band of the semiconductor

Q1.87.3 MCQ 1987 2M ○○○

According to the Einstein relation, for any semiconductor the ratio of diffusion constant to mobility of carriers

A. depends upon the temperature of the semiconductor.

B. depends upon the type of the semiconductor.

C. varies with life time of the semiconductor.

D. is a universal constant.

Q1.87.4 MCQ 1987 2M ○○○

Direct band gap semiconductors

A. exhibit short carrier life time and they are used for fabricating BJT's.

B. exhibit long carrier life time and they are used for fabricating BJT's.

C. exhibit short carrier life time and they are used for fabricating lasers.

D. exhibit long carrier life time and they are used for fabricating lasers.

Answer Key

Q1.26.1 -36.50 to -34.50 OR 34.50 to 36.50	Q1.25.1 A,D	Q1.25.2 B
Q1.25.3 B	Q1.24.1 A, C, D	Q1.24.2 A
Q1.24.3 0.17 to 0.19	Q1.23.1 A	Q1.23.2 A
Q1.23.3 2.20 to 2.30	Q1.23.4 60.00 to 70.00	Q1.22.1 A, C
Q1.22.2 A	Q1.22.3 C	Q1.22.4 0.57 to 0.61
Q1.21.1 C	Q1.21.2 A	Q1.20.1 D
Q1.17.1 1.5 to 1.7	Q1.17.2 15.9 to 16.1	Q1.17.3 A
Q1.16.1 A	Q1.16.2 1.1 to 1.25	Q1.16.3 A
Q1.15.1 C	Q1.15.2 95 to 105	Q1.15.3 1.8 to 2
Q1.15.4 14	Q1.15.5 0.50 to 0.54	Q1.14.1 D
Q1.14.2 224.9 to 225.1	Q1.14.3 1.12 to 1.14	Q1.14.4 A
Q1.14.5 A	Q1.14.6 12.9 to 13.1	Q1.14.7 D
Q1.14.8 3990 to 4010	Q1.14.9 D	Q1.14.10 A
Q1.11.1 C	Q1.10.1 C	Q1.10.2 A
Q1.09.1 A	Q1.09.2 C	Q1.08.1 C
Q1.08.2 A	Q1.07.1 D	Q1.06.1 D
Q1.06.2 B	Q1.06.3 A	Q1.06.4 B
Q1.05.1 B	Q1.05.2 A	Q1.05.3 C

Q1.04.1 A	Q1.04.2 B	Q1.04.3 C
Q1.03.1 D	Q1.03.2 A	Q1.02.1 A
Q1.98.1 B	Q1.95.1 C	Q1.95.2 C
Q1.95.3 C	Q1.95.4 D	Q1.94.1 False
Q1.92.1 A	Q1.91.1 B	Q1.90.1 A,C
Q1.89.1 A	Q1.89.2 C	Q1.87.1 B
Q1.87.2 C	Q1.87.3 A	Q1.87.4 C

PrepFusion

UNIT

II

PN Junctions

Topics

1. Basic Semiconductor Junctions
2. PN Junction Under Equilibrium
3. PN Junction Under Biasing

Unit Snapshot*

Stream: EC	**Questions:** 56	**MCQ:** 42	**MSQ:** 2	**NAT:** 12
1M: 30	**2M:** 26	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 3M(2013)–10M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

Consider carrier transport in a Zener diode in the breakdown region. Which is the dominant transport mechanism for current flow in this case?

A. Drift **B.** Diffusion

C. Tunneling **D.** Ballistic transport

Q2.26.2 MSQ 2026 1M

Consider a p-n junction diode when it is forward biased with 2 V. Which of the following is/are the correct magnitude(s) of the energy difference between quasi Fermi-levels, E_{fn} in the n-side and E_{fp} in the p-side?

- ☐ **A.** 2 eV
- ☐ **B.** 1 eV
- ☐ **C.** 2 V
- ☐ **D.** 1 V

Q2.25.1 NAT 2025 2M

An ideal p-n junction germanium diode has a reverse saturation current of 10 μA at 300 K. The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____.

(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

Q2.21.1 NAT 2021 2M

A silicon P-N junction is shown in the figure. The doping in the P region is $5 \times 10^{16}\ cm^{-3}$ and doping in the N region is $10 \times 10^{16}\ cm^{-3}$. The parameters given are

Built-in voltage $(\Phi_{bi}) = 0.8V$

Electron charge $(q) = 1.6 \times 10^{-19}C$

Vacuum permittivity $(\epsilon_0) = 8.85 \times 10^{-12}\ F/m$

Relative permittivity of silicon $(\epsilon_{Si}) = 12$

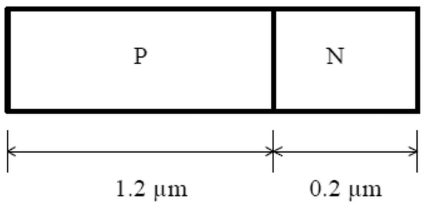


Fig. Q2.21.1

The magnitude of reverse bias voltage that would completely deplete one of the two regions (P or N) prior to the other (rounded off to one decimal place) is _____ V.

Q2.20.1 MCQ 2020 1M

Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is U_{max}. If the electron and the hole capture cross-section are equal, which one of the following is False?

A. With all other parameters unchanged, U_{max} decreases if the intrinsic carrier density is reduced.

B. U_{max} occurs at the edges of the depletion region in the device.

C. U_{max} depends exponentially on the applied bias.

D. With all other parameters unchanged, U_{max} increases if the thermal velocity of the carriers increases.

Q2.20.2 MCQ 2020 2M

A one-sided abrupt pn junction diode has a depletion capacitance C_D of 50 pF at a reverse bias of 0.2 V. The plot of $1/C_D^2$ versus the applied voltage V for this diode is a straight line as shown in the figure below. The slope of the plot is ___ $\times 10^{20}\ F^{-2}\ V^{-1}$.

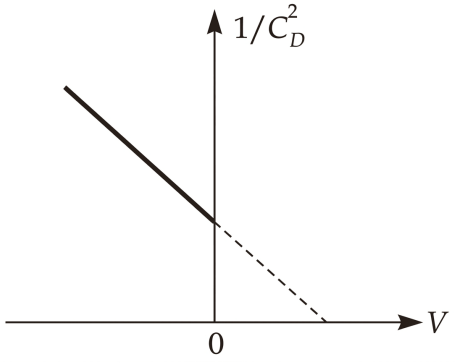


Fig. Q2.20.2

A. -5.7 **B.** -3.8 **C.** -1.2 **D.** -0.4

Q2.19.1 NAT 2019 2M

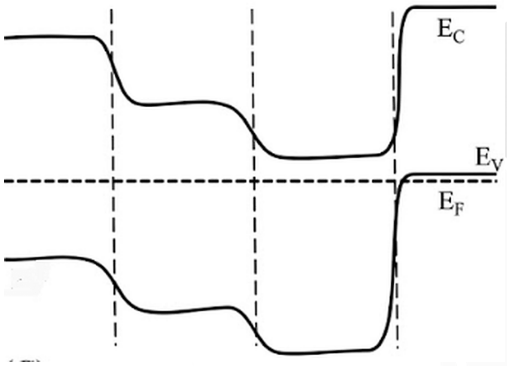
In an ideal pn junction with an ideality factor of 1 at $T = 300\ K$, the magnitude of the reverse-bias voltage required to reach 75% of its reverse saturation current, rounded off to 2 decimal places, is _____ mV. [$k = 1.38 \times 10^{-23}\ JK^{-1}$, $h = 6.625 \times 10^{-34}\ Js$, $q = 1.602 \times 10^{-19}\ C$]

Q2.19.2 MCQ 2019 1M

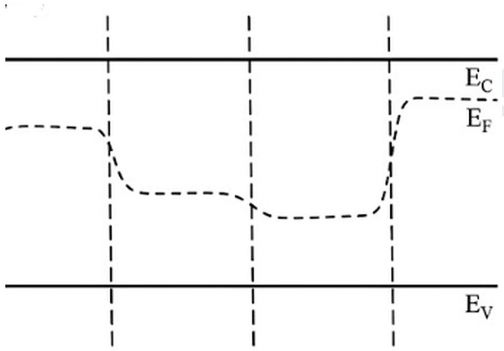
Which one of the following options describes correctly the equilibrium band diagram at $T = 300\ K$ of a Silicon $pnp^{+}p^{++}$ configuration shown in the figure?

Fig. Q2.19.2

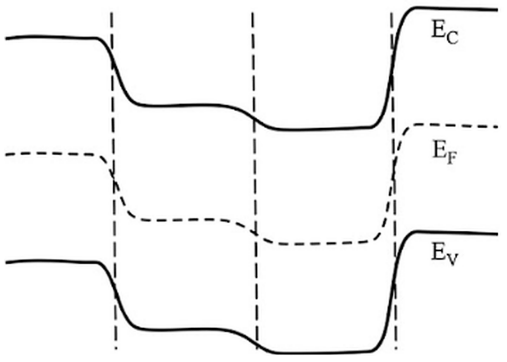
A.



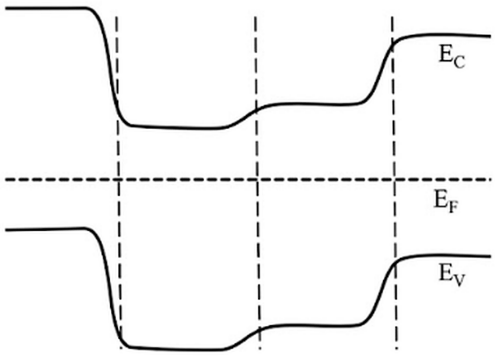
B.



C.



D.



Q2.18.1 NAT 2018 2M

A junction is made between p^{-} Si with doping density $N_{A1} = 10^{15} cm^{-3}$ and p Si with doping density $N_{A2} = 10^{17} cm^{-3}$. Given: Boltzmann constant $k = 1.38 \times 10^{-23} JK^{-1}$, electronic charge $q = 1.6 \times 10^{-19} C$. Assume 100% acceptor ionization. At room temperature ($T = 300K$), the magnitude of the built-in potential (in volts, correct to two decimal places) across this junction will be ____.

Q2.18.2 NAT 2018 1M

A p-n step junction diode with a contact potential of $0.65V$ has a depletion width of $1\mu m$ at equilibrium. The forward voltage (in volts, correct to two decimal places) at which this width reduces to $0.6\mu m$ is ____.

Q2.18.3 MCQ 2018 1M

In a p-n junction diode at equilibrium, which one of the following statements is NOT TRUE?

A. The hole and electron diffusion current components are in the same direction.

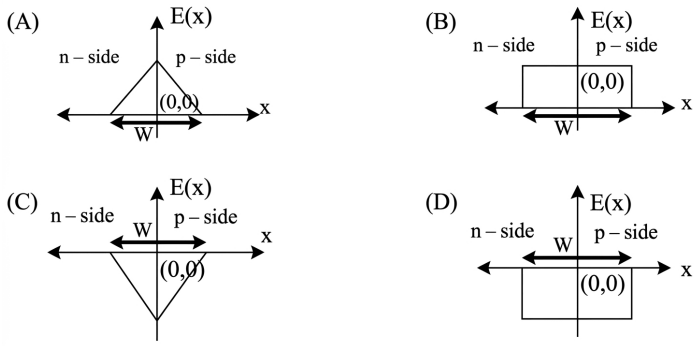
B. The hole and electron drift current components are in the same direction.

C. On an average, holes and electrons drift in opposite direction.

D. On an average, electrons drift and diffuse in the same direction.

Q2.17.1 MCQ 2017 2M

An abrupt pn junction (located at $x = 0$) is uniformly doped on both p and n sides. The width of the depletion region is W and the electric field variation in the x-direction is $E(x)$. Which of the following figures represents the electric field profile near the pn junction?



A. A **B.** B **C.** C **D.** D

Q2.17.2 NAT 2017 2M

As shown, two Silicon (Si) abrupt p-n junction diodes are fabricated with uniform donor doping concentration of $N_{D1} = 10^{14}cm^{-3}$ and $N_{D2} = 10^{16}cm^{-3}$ in the n-regions of the diodes, and uniform acceptor doping concentration of $N_{A1} = 10^{14}cm^{-3}$ and $N_{A2} = 10^{16}cm^{-3}$ in the p regions of the diodes, respectively. Assuming that the reverse bias voltage is >> built-in potentials of the diodes, the ratio $\frac{C_2}{C_1}$ of their reverse bias capacitances for the same applied reverse bias, is ____.

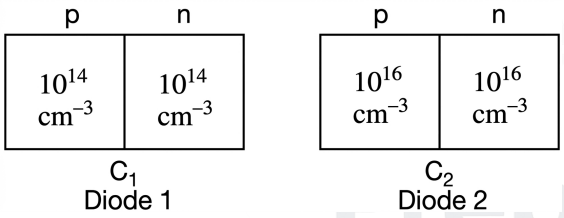


Fig. Q2.17.2

Q2.17.3 MCQ 2017 1M

An $n^+ - n$ Silicon device is fabricated with uniform and non-degenerate donor doping concentrations of $N_{D1} = 1 \times 10^{18}cm^{-3}$ and $N_{D2} = 1 \times 10^{15}cm^{-3}$ corresponding to the n^+ and n regions respectively. At the operational temperature T, assume complete impurity ionization, $kT/q = 25mV$, and intrinsic carrier concentration to be $n_i = 1 \times 10^{10}cm^{-3}$. What is the magnitude of the built-in potential of this device?

A. 0.748V **B.** 0.460V **C.** 0.288V **D.** 0.173V

Q2.16.1 MCQ 2016 1M

The I-V characteristics of three types of diodes at the room temperature, made of semiconductors X, Y and Z, are shown in the figure. Assume that the diodes are uniformly doped and identical in all respects except their materials. If E_{gX}, E_{gY} and E_{gZ} are the band gaps of X, Y and Z, respectively, then

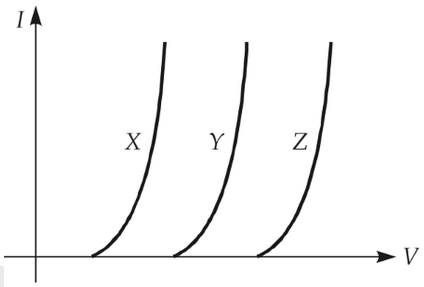


Fig. Q2.16.1

A. $E_{gX} > E_{gY} > E_{gZ}$

B. $E_{gX} = E_{gY} = E_{gZ}$

C. $E_{gX} < E_{gY} < E_{gZ}$

D. no relationship among these band gaps exists.

Q2.16.2 NAT 2016 2M

Consider a region of silicon devoid of electrons and holes, with an ionized donor density of $N_d^+ = 10^{17}cm^{-3}$. The electric field at x= 0 is $0V/cm$ and the electric field at x=L is $50kV/cm$ in the positive x direction. Assume that the electric field is zero in the y and z directions at all points.

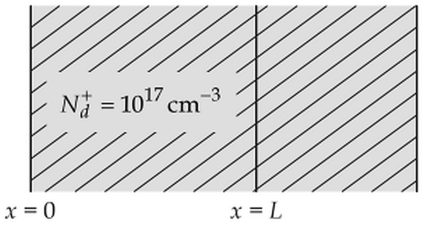


Fig. Q2.16.2

Given $q = 1.6 \times 10^{-19}coulomb$, $\epsilon_0 = 8.85 \times 10^{-14}F/cm$, $\epsilon_r = 11.7$ for silicon, the value of L in nm is _____.

Q2.16.3 MCQ 2016 2M

Consider avalanche breakdown in a silicon p^+n junction. The n-region is uniformly doped with a donor density N_D. Assume that breakdown occurs when the magnitude of the electric field at any point in the device becomes equal to the critical field E_{crit}. Assume E_{crit} to be independent of N_D. If the built-in voltage of the p^+n junction is much smaller than the breakdown voltage, V_{BR}, the relationship between V_{BR} and N_D is

A. $V_{BR} \times \sqrt{N_D} = constant$

B. $N_D \times \sqrt{V_{BR}} = constant$

C. $N_D \times V_{BR} = constant$

D. $N_D / V_{BR} = constant$

Q2.16.4 NAT 2016 2M

Consider a silicon p-n junction with a uniform acceptor doping concentration of $10^{17} cm^{-3}$ on the p-side and a uniform donor doping concentration of $10^{16} cm^{-3}$ on the n-side. No external voltage is applied to the diode. Given: $kT/q = 26mV$, $n_i = 1.5 \times 10^{10} cm^{-3}$, $\varepsilon_{Si} = 12\varepsilon_0$, $\varepsilon_0 = 8.85 \times 10^{-14} F/m$, and $q = 1.6 \times 10^{-19} C$. The charge per unit junction area ($nC\ cm^{-2}$) in the depletion region on the p-side is ____.

Q2.15.1 MCQ 2015 2M

The electric field profile in the depletion region of a p-n junction in equilibrium is shown in the figure. Which one of the following statements is NOT TRUE?

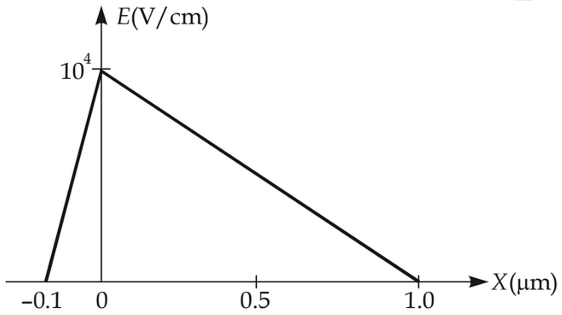


Fig. Q2.15.1

A. The left side of the junction is n-type and the right side is p-type

B. Both the n-type and p-type depletion regions are uniformly doped

C. The potential difference across the depletion region is 700 mV

D. If the p-type region has a doping concentration of 10^{15} cm^{-3}, then the doping concentration in the n-type region will be 10^{16} cm^{-3}

Q2.15.2 NAT 2015 2M

For a silicon diode with long P and N regions, the accepter and donor impurity concentrations are 1×10^{17} cm^{-3} and 1×10^{15} cm^{-3}, respectively. The lifetimes of electrons in P region and holes in N region are both 100 μs. The electron and hole diffusion coefficients are 49 cm^2/s and 36 cm^2/s, respectively. Assume $kT/q = 26$ mV, the intrinsic carrier concentration is 1×10^{10} cm^{-3}, and $q = 1.6 \times 10^{-19}$ C. When a forward voltage of 208 mV is applied across the diode, the hole current density (in nA/cm^2) injected from P region to N region is ____.

Q2.15.3 NAT 2015 2M

The built-in potential of an abrupt p-n junction is 0.75 V. If its junction capacitance (C_J) at a reverse bias (V_R) of 1.25 V is 5 pF, the value of C_J (in pF) when $V_R = 7.25$ V is ____.

Q2.15.4 MCQ 2015 1M

A region of negative differential resistance is observed in the current voltage characteristics of a silicon PN junction if

A. both the P-region and the N-region are heavily doped

B. the N-region is heavily doped compared to the P-region

C. the P-region is heavily doped compared to the N-region

D. an intrinsic silicon region is inserted between the P-region and the N-region

Q2.14.1 MCQ 2014 2M

The donor and accepter impurities in an abrupt junction silicon diode are 1×10^{16} cm^{-3} and 5×10^{18} cm^{-3}, respectively. Assume that the intrinsic carrier concentration in silicon $n_i = 1.5 \times 10^{10}$ cm^{-3} at 300 K, $\frac{kT}{q} = 26$ mV and the permittivity of silicon $\varepsilon_{si} = 1.04 \times 10^{-12}$ F/cm. The built-in potential and the depletion width of the diode under thermal equilibrium conditions, respectively, are

A. 0.7 V and 1×10^{-4} cm

B. 0.86 V and 1×10^{-4} cm

C. 0.7 V and 3.3×10^{-5} cm

D. 0.86 V and 3.3×10^{-5} cm

Q2.14.2 NAT 2014 2M ◯◯◯

Consider an abrupt PN junction (at $T = 300\,\text{K}$) shown in the figure. The depletion region width X_n on the N-side of the junction is $0.2\,\mu\text{m}$ and the permittivity of silicon (ε_{si}) is $1.044 \times 10^{-12}\,\text{F/cm}$. At the junction, the approximate value of the peak electric field (in kV/cm) is ____.

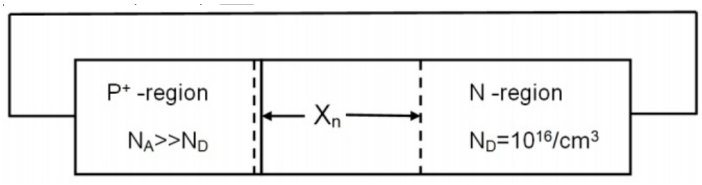


Fig. Q2.14.2

Q2.14.3 MCQ 2014 2M ◯◯◯

When a silicon diode having a doping concentration of $N_A = 9 \times 10^{16}\ cm^{-3}$ on p -side and $N_D = 1 \times 10^{16}\ cm^{-3}$ on n-side is reverse biased, the total depletion width is found to be $3\ \mu$m. Given that the permittivity of silicon is 1.04×10^{-12} F/cm, the depletion width on the p-side and the maximum electric field in the depletion region, respectively, are

A. $2.7\mu m$ and $2.3 * 10^5\ V/cm$

B. $0.3\mu m$ and $4.15 * 10^5\ V/cm$

C. $0.3\mu m$ and $0.42 * 10^5\ V/cm$

D. $2.1\mu m$ and $0.42 * 10^5\ V/cm$

Q2.13.1 MCQ 2013 1M ◯◯◯

In a forward biased pn junction diode, the sequence of events that best describes the mechanism of current flow is

A. injection, and subsequent diffusion and recombination of minority carriers

B. injection, and subsequent drift and generation of minority carriers

C. extraction, and subsequent diffusion and generation of minority carriers

D. extraction, and subsequent drift and recombination of minority carriers

Q2.11.1 MCQ 2011 1M ◯◯◯

A Zener diode, when used in voltage stabilization circuits, is biased in

A. reverse bias region below the breakdown voltage

B. reverse breakdown region

C. forward bias region

D. forward bias constant current mode

Q2.11.2 MCQ 2011 1M ◯◯◯

A silicon PN junction is forward biased with a constant current at room temperature. When the temperature is increased by 10° C, the forward bias voltage across the PN junction

A. increases by 60 mV **B.** decreases by 60 mV

C. increases by 25 mV **D.** decreases by 25 mV

Q2.10.1 MCQ 2010 1M ◯◯◯

Compared to a p-n junction with $N_A = N_D = 10^{14}/\text{cm}^3$, which one of the following statements is TRUE for a p-n junction with $N_A = N_D = 10^{20}/\text{cm}^3$?

A. Reverse breakdown voltage is lower and depletion capacitance is lower

B. Reverse breakdown voltage is higher and depletion capacitance is lower

C. Reverse breakdown voltage is lower and depletion capacitance is higher

D. Reverse breakdown voltage is higher and depletion capacitance is higher

Common Data Questions

Answer the next two questions using the following data.

Consider a silicon p-n junction at room temperature having the following parameters:
Doping on the n-side = $1 \times 10^{17}\ \text{cm}^{-3}$
Depletion width on the n-side = $0.1\ \mu$m
Depletion width on the p-side = $1.0\ \mu$m
Intrinsic carrier concentration = $1.4 \times 10^{10}\ \text{cm}^{-3}$
Thermal voltage = 26 mV
Permittivity of free space = $8.85 \times 10^{-14}\ \text{Fcm}^{-1}$
Dielectric constant of silicon = 12

Q2.09.1 MCQ 2009 1M

The peak electric field in the device is

A. 0.15 MVcm^{-1}, directed from p-region to n-region

B. 0.15 MVcm^{-1}, directed from n-region to p-region

C. 1.80 MVcm^{-1}, directed from p-region to n-region

D. 1.80 MVcm^{-1}, directed from n-region to p-region

Q2.09.2 MCQ 2009 1M

The built-in potential of the junction

A. is 0.70 V

B. is 0.76 V

C. is 0.82 V

D. Cannot be estimated from the data given

Q2.08.1 MCQ 2008 1M

Consider the following assertions.
S1 : For Zener effect to occur, a very abrupt junction is required.
S2 : For quantum tunneling to occur, a very narrow energy barrier is required.
Which of the following is correct ?

A. Only $S2$ is true

B. $S1$ and $S2$ are both true but $S2$ is not a reason for $S1$

C. $S1$ and $S2$ are both true but $S2$ is a reason for $S1$

D. Both $S1$ and $S2$ are false

Q2.08.2 MCQ 2008 1M

Which of the following is NOT associated with a p-n junction ?

A. Junction Capacitance

B. Charge Storage Capacitance

C. Depletion Capacitance

D. Channel Length Modulation

Q2.07.1 MCQ 2007 1M

A p^+n junction has a built-in potential of 0.8 V. The depletion layer width a reverse bias of 1.2 V is 2 μm . For a reverse bias of 7.2 V, the depletion layer width will be

A. 4 μm **B.** 4.9 μm **C.** 8 μm **D.** 12 μm

Q2.07.2 MCQ 2007 1M

In a p^+n junction diode under reverse bias the magnitude of electric field is maximum at

A. the edge of the depletion region on the p-side

B. the edge of the depletion region on the n-side

C. the p^+n junction

D. the centre of the depletion region on the n-side

Q2.05.1 MCQ 2005 1M

A Silicon PN junction diode under reverse bias has depletion region of width 10 μm. The relative permittivity of Silicon, $\varepsilon_r = 11.7$ and the permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m . The depletion capacitance of the diode per square meter is

A. 100 μF **B.** 10 μF **C.** 1 μF **D.** 20 μF

Q2.05.2 MCQ 2005 1M

A Silicon PN junction at a temperature of 20°C has a reverse saturation current of 10 pico-Amperes (pA). The reverse saturation current at 40°C for the same bias is approximately

A. 30pA **B.** 40pA **C.** 50pA **D.** 60pA

Q2.04.1 MCQ 2004 1M

Consider an abrupt p-n junction. Let V_{bi} be the built-in potential of this junction and V_R be the applied reverse bias. If the junction capacitance (C_1) is 1 pF for $V_{bi} + V_R = 1$ V, then for $V_{bi} + V_R = 4$ V, C_j will be

A. 4 pF **B.** 2 pF **C.** 0.25 pF **D.** 0.5 pF

Q2.04.2 MCQ 2004 1M

In an abrupt p-n junction, the doping concentrations on the p-side and n-side are $N_A = 9 \times 10^{16}/\text{cm}^3$ and $N_D = 1 \times 10^{16}/\text{cm}^3$ respectively. The p-n junction is reverse biased and the total depletion width is 3 μm. The depletion width on the p-side is

A. 2.7 μm **B.** 0.3 μm **C.** 2.25 μm **D.** 0.75 μm

Q2.03.1 MCQ 2003 1M

At 300 K, for a diode current of 1 mA, a certain germanium diode requires a forward bias of 0.1435 V, whereas a certain silicon diode requires a forward bias of 0.718 V. Under the conditions state above, the closest approximation of the ratio of reverse saturation current in germanium diode to that in silicon diode is

A. 1 **B.** 5

C. 4×10^3 **D.** 8×10^3

Q2.02.1 MCQ 2002 1M

In Figure shown below, a silicon is carrying a constant current of 1 mA. When the temperature of the diode is 20°C, V_D is found to be 700 mV. If the temperature rises to 40°C, V_D becomes approximately equal to

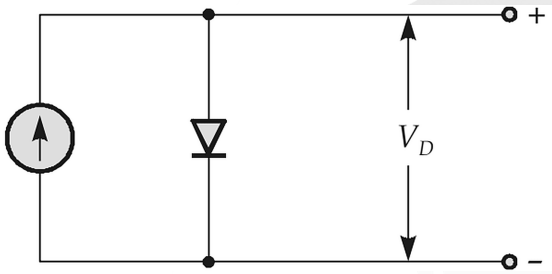


Fig. Q2.02.1

A. 740 mV **B.** 660 mV **C.** 680 mV **D.** 700 mV

Q2.98.1 MCQ 1998 2M

A pn-junction in series with a 100 Ω resistor, is forwarded biased. So that a current of 100 mA flows. If the voltage across this combination is instantaneously reversed at $t = 0$ current through diodes is approximately given by

A. 0 mA **B.** −100 mA **C.** 100 mA **D.** 50 mA

Q2.98.2 MCQ 1998 1M

The static characteristic of an adequately forward biased p-n junction is a straight line, if the plot is of

A. $\log I$ vs $\log V$ **B.** $\log I$ vs V

C. I vs $\log V$ **D.** I vs V

Q2.95.1 MCQ 1995 1M

The depletion capacitance, C_J of an abruptly p-n junction with constant doping on either side varies with R.B. V_R as,

A. $C_J \propto V_R$ **B.** $C_J \propto V_R^{-1}$

C. $C_J \propto V_R^{-1/2}$ **D.** $C_J \propto V_R^{-1/3}$

Q2.95.2 MCQ 1995 1M

A Zener diode works on the principle of

A. tunneling of charge carriers across the junction.

B. thermionic emission.

C. diffusion of charge carriers across the junction.

D. hopping of charge carriers across the junction.

Q2.95.3 MCQ 1995 1M

The diffusion potential across a p-n junction

A. decreases with increasing doping concentration.

B. increases with decreasing bandgap.

C. does not depend on doping concentrations.

D. increases with increase in doping concentration.

Q2.93.1 MSQ 1993 2M

The built-in potential (diffusion potential) in a p-n junction

A. is equal to the difference in the Fermi-level of the two sides, expressed in volts.

B. increases with the increase in the doping levels of the two sides.

C. increases with the increase in temperature.

D. is equal to the average of the Fermi-level of the two sides.

Q2.91.1 NAT 1991 2M

Referring to the below figure the switch 'S' is in position 1 initially and steady-state condition exist from time $t = 0$ to $t = t_0$, the switch is suddenly thrown into position 2. The current 'I' through the 10 kΩ resistor as a function of time t, from $t = 0$ is (Give the sketch showing the magnitudes of the current at $t = 0$, $t = t_0$ and $t = \infty$).

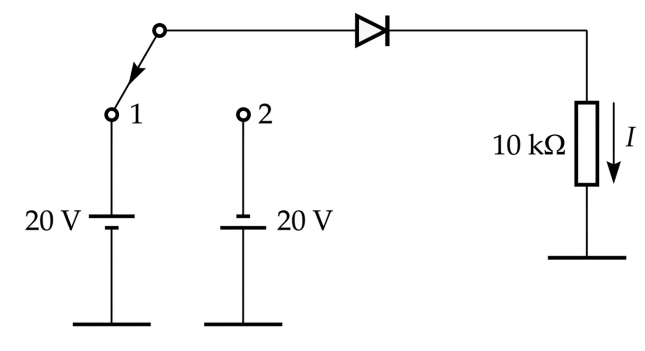


Fig. Q2.91.1

Q2.91.2 MCQ 1991 2M

The small signal capacitance of an abrupt $P^+ n$ junction is 1 nF/cm^2 at zero bias. If the built-in voltage is 1 volt, the capacitance at a reverse bias voltage of 99 volts in nF/cm^2 is

A. 10 **B.** 0.1 **C.** 0.01 **D.** 100

Q2.90.1 MCQ 1990 2M

In a uniformly doped abrupt p-n junction the doping level of the n-side is four (4) times the doping level of the p-side the ratio of the depletion layer width of n-side verses p-side is

A. 0.25 **B.** 0.5 **C.** 1.0 **D.** 2.0

Q2.90.2 MCQ 1990 1M

In a junction diode

A. the depletion capacitance increases with increase in the reverse bias.

B. the depletion capacitance decreases with increase in the reverse bias.

C. the depletion capacitance increases with increase in the forward bias.

D. the depletion capacitance is much higher than the depletion capacitance when it is forward biased.

Q2.89.1 MCQ 1989 2M

The switching speed of P^+N junction (having a heavily doped P region) depends primarily on

A. the mobility of minority carriers in the P^+ region.

B. the lifetime of minority carriers in the P^+ region.

C. the mobility of majority carriers in the N region.

D. the lifetime of majority carriers in the N region.

Q2.89.2 MCQ 1989 2M

In a Zener diode

A. only the P region is heavily doped.

B. only the N region is heavily doped.

C. both P and N regions are heavily doped.

D. both P and P regions are lightly doped.

Q2.88.1 MCQ 1988 2M

In the circuit shown below the current voltage relationship when D_1 and D_2 are identical is given by (Assume Ge diodes)

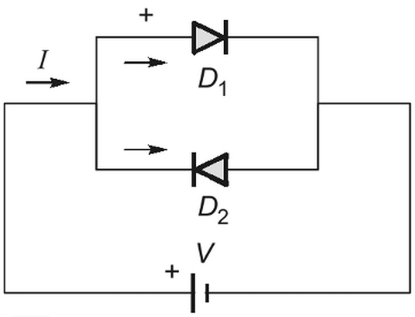


Fig. Q2.88.1

A. $V = \frac{KT}{q} \sinh\left(\frac{I}{2}\right)$

B. $V = \frac{KT}{q} \ln\left(\frac{I}{I_o}\right)$

C. $V = \frac{KT}{q} \sinh^{-1}\left(\frac{I}{2}\right)$

D. $V = \frac{KT}{q}[\exp(-I) - 1]$

Q2.88.2 MCQ 1988 2M

For a pn-junction match the type of breakdown with phenomenon:

List - I:

1. Avalanche breakdown
2. Zener breakdown
3. Punch through

List - II:

A. Collision of carriers with crystal ions

B. Early effect

C. Rupture of covalent bond due to strong electric field.

A. 1-B, 2-A, 3-C **B.** 1-C, 2-A, 3-B
C. 1-A, 2-B, 3-C **D.** 1-A, 2-C, 3-B

Q2.87.1 MCQ 1987 2M ◯◯◯

The diffusion capacitance of a *p*-*n* junction

A. decreases with increasing current and increasing temperature.

B. decreases with decreasing current and increasing temperature.

C. increases with increasing current and increasing temperature.

D. does not depend on current and temperature.

Q2.15.1 C	Q2.15.2 28 to 30	Q2.15.3 2.4 to 2.6
Q2.15.4 A	Q2.14.1 D	Q2.14.2 -32 to -30
Q2.14.3 B	Q2.13.1 A	Q2.11.1 B
Q2.11.2 D	Q2.10.1 C	Q2.09.1 B
Q2.09.2 B	Q2.08.1 C	Q2.08.2 D
Q2.07.1 A	Q2.07.2 C	Q2.05.1 B
Q2.05.2 B	Q2.04.1 D	Q2.04.2 B
Q2.03.1 C	Q2.02.1 B	Q2.98.1 B
Q2.98.2 B	Q2.95.1 C	Q2.95.2 A
Q2.95.3 D	Q2.93.1 A, B	Q2.91.1 *
Q2.91.2 B	Q2.90.1 A	Q2.90.2 B
Q2.89.1 MTA	Q2.89.2 C	Q2.88.1 B
Q2.88.2 D	Q2.87.1 B	

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q2.26.1 C	Q2.26.2 A	Q2.25.1 0.23 to 0.24
Q2.21.1 8.1 to 8.4	Q2.20.1 B	Q2.20.2 MTA
Q2.19.1 34 to 38	Q2.19.2 A	Q2.18.1 0.11 to 0.13
Q2.18.2 0.4 to 0.43	Q2.18.3 D	Q2.17.1 A
Q2.17.2 10	Q2.17.3 D	Q2.16.1 C
Q2.16.2 30 to 34	Q2.16.3 C	Q2.16.4 -5 to -4.6

UNIT

III

BJTs

Topics

1. Introduction to BJTs and Mode of Operations
2. Ideal and Non-Ideal Characteristics of BJTs

Unit Snapshot*

Stream: EC	**Questions:** 22	**MCQ:** 14	**MSQ:** 2	**NAT:** 6
1M: 13	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2015)–3M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MSQ 2026 1M

Figure shows the output characteristics of two different Bipolar Junction Transistors (BJT), BJT 1 with magnitude of Early voltage $|V_{A1}|$, and BJT 2 with magnitude of Early voltage $|V_{A2}|$. Which of the following options is/are correct regarding the Early voltages?

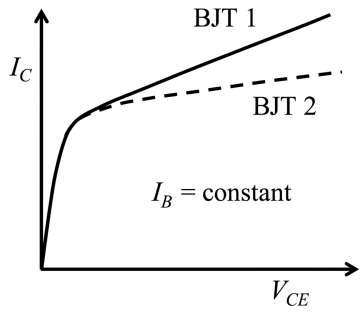


Fig. Q3.26.1

- **A.** $|V_{A1}| > |V_{A2}|$
- **B.** $|V_{A1}|$ is infinitely large
- **C.** $|V_{A1}| < |V_{A2}|$
- **D.** $|V_{A2}|$ is finite

Q3.24.1 MSQ 2024 2M

Which of the following statements is/are true for a BJT with respect to its DC current gain β?

- **A.** Under high-level injection condition in forward active mode, β will decrease with increase in the magnitude of collector current.
- **B.** Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, β will decrease with increase in the magnitude of collector current.
- **C.** β will be lower when the BJT is in saturation region compared to when it is in active region.
- **D.** A higher value of β will lead To a lower value of the collector-to-emitter breakdown voltage.

Q3.17.1 MCQ 2017 1M

An npn bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base - collector junction is increased, then

- **A.** the effective base width increases and common - emitter current gain increases
- **B.** the effective base width increases and common - emitter current gain decreases
- **C.** the effective base width decreases and common - emitter current gain increases
- **D.** the effective base width decreases and common - emitter current gain decreases

Q3.17.2 MCQ 2017 1M

For a narrow base PNP BJT, the excess minority carrier concentration (Δn_E for emitter, Δp_{E0} for base, Δn_c for collector) normalized to equilibrium minority carrier concentration (n_{E0} for emmitter, p_{B0} for base, n_{C0} for collector) in the quasi-neutral emitter, base and collector regions are shown below. Which one of the following biasing modes is the transistor operating in ?

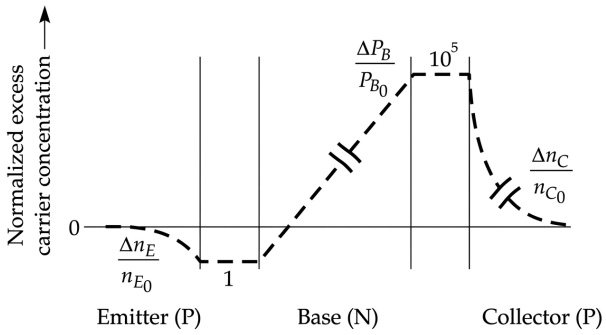


Fig. Q3.17.2

A. Forward active **B.** Saturation
C. Inverse active **D.** Cutoff

Q3.16.1 NAT 2016 2M

The injected excess electron concentration profile in the base region of an npn BJT, biased in the active region, is linear, as shown in the figure. If the area of the emitter-base junction is 0.001 cm^2, $\mu_n = 800\ cm^2/(V-s)$ in the base region and depletion layer widths are negligible, then the collector current I_C (in mA) at room temperature is ____
(Given: thermal voltage $V_T = 26$ mV at room temperature, electronic charge $q = 1.6 \times 10^{-19}$ C)

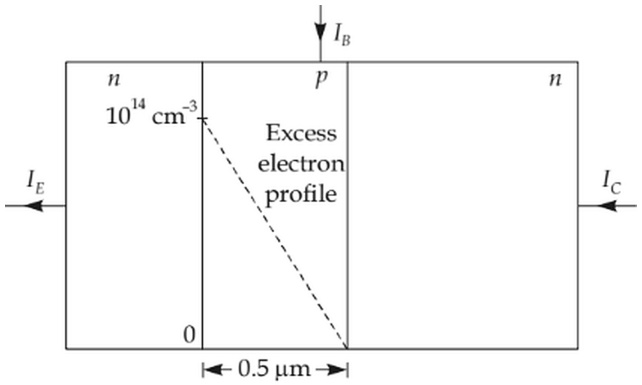


Fig. Q3.16.1

Q3.15.1 NAT 2015 2M

An npn BJT having reverse saturation current $I_S = 10^{-15}$ A is biased in the forward active region with $V_{BE} = 700$ mV. The thermal voltage (V_T) is 25 mV and the current gain (β) may vary from 50 to 150 due to manufacturing variations. The maximum emitter current (in μA) is ____.

Q3.15.2 MCQ 2015 1M

If the base width in a bipolar junction transistor is doubled, which one of the following statements will be TRUE?

A. Current gain will increase

B. Unity gain frequency will increase.

C. Emitter-base junction capacitance will increase

D. Early Voltage will increase

Q3.14.1 NAT 2014 2M

Consider two BJTs biased at the same collector current with area $A_1 = 0.2\,\mu\text{m} \times 0.2\,\mu\text{m}$ and $A_2 = 300\,\mu\text{m} \times 300\,\mu\text{m}$. Assuming that all other device parameters are identical, kT/q = 26 mV, the intrinsic carrier concentration is 1×10^{10} cm^{-3}, and $q = 1.6 \times 10^{-19}$ C, the difference between the base-emitter voltages (in mV) of the two BJTs (i.e., $V_{BE1} - V_{BE2}$) is ____.

Q3.14.2 NAT 2014 2M

A BJT is biased in forward active mode. Assume $V_{BE} = 0.7$ V, kT/q = 25mV and reverse saturation current $I_S = 10^{-13}$ mA. The transconductance of the BJT (in mA/V) is ____.

Q3.11.1 MCQ 2011 1M

For a BJT the common base current gain $\alpha = 0.98$ and the collector base junction reverse bias saturation current $I_{CO} = 0.6\,\mu$A. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B = 20\,\mu$A. The collector current I_C for this mode of operation is

A. 0.98 mA **B.** 0.99 mA **C.** 1.0 mA **D.** 1.01 mA

Q3.10.1 MCQ 2010 1M

In a uniformly doped BJT, assume that N_E, N_B and N_C are the emitter, base and collector doping in atoms/cm^3, respectively. If the emitter injection efficiency of the BJT is close to unity, which one of the following condition is TRUE

A. $N_E = N_B = N_C$

B. $N_E \gg N_B$ and $N_B > N_C$

C. $N_E = N_B$ and $N_B < N_C$

D. $N_E < N_B < N_C$

Q3.06.1 MCQ 2006 1M

The phenomenon known as "Early Effect" in a bipolar transistor refers to a reduction of the effective base-width caused by

A. Electron - hole recombination at the base

B. The reverse biasing of the base - collector junction

C. The forward biasing of emitter-base junction

D. The early removal of stored base charge during saturation-to-cut off switching

Q3.04.1 MCQ 2004 1M

The neutral base width of a bipolar transistor, biased in the active region, is 0.5 μm. The maximum electron concentration and the diffusion constant in the base are $10^{14}/cm^3$ and $D_n = 25\ cm^2/sec$ respectively. Assuming negligible recombination in the base, the collector current density is (the electron charge is 1.6×10^{-19} - Coulomb)

A. $800 A/cm^2$ **B.** $8 A/cm^2$

C. $200 A/cm^2$ **D.** $2 A/cm^2$

Q3.04.2 MCQ 2004 1M

Consider the following statements S1 and S2. S1 : The β of a bipolar transistor reduces if the base width is increased. S2 : The β of a bipolar transistor increases if the dopoing concentration in the base is increased. Which remarks of the following is correct ?

A. S1 is FALSE and S2 is TRUE

B. Both S1 and S2 are TRUE

C. Both S1 and S2 are FALSE

D. S1 is TRUE and S2 is FALSE

Q3.04.3 MCQ 2004 1M

If for a silicon npn transistor, the base-to-emitter voltage (V_{BE}) is 0.7 V and the collector-to-base voltage (V_{CB}) is 0.2 V, then the transistor is operating in the

A. normal active mode **B.** saturation mode

C. inverse active mode **D.** cutoff mode

Q3.99.1 MCQ 1999 1M

The early-effect in a bipolar junction transistor is caused by

A. fast-turn-on

B. fast-turn-off

C. large collector-base reverse bias

D. large emitter-base forward bias

Q3.97.1 MCQ 1997 2M

In a bipolar transistor at room temperature, if the emitter current is doubled, the voltage across its base-emitter junction

A. doubles **B.** halves

C. increases by about 20 mV **D.** decreases by about 20 mV

Q3.96.1 MCQ 1996 2M

If a transistor is operating with both of its junctions forward biased, but with the collector base forward bias greater than the emitter-base forward bias, then it is operating in the

A. forward active mode. **B.** reverse saturation mode.

C. reverse active mode. **D.** forward saturation mode.

Q3.95.1 MCQ 1995 1M

A BJT is said to be operating in the saturation region if

A. both the junctions are reverse biased.

B. base-emitter junction is reverse biased and base-collector junction is forward biased.

C. base-emitter junction is forward biased and base-collector junction is reverse biased.

D. both the junctions are forward biased.

Q3.95.2 MCQ 1995 1M

The breakdown voltage of a transistor with its base open is BV_{CEO} and that with emitter open is BV_{CBO}, then

A. $BV_{CEO} = BV_{CBO}$

B. $BV_{CEO} > BV_{CBO}$

C. $BV_{CEO} < BV_{CBO}$

D. BV_{CEO} is not related to BV_{CBO}

Q3.94.1 NAT 1994 2M

Match the following:

List-I:

A. The current gain of a BJT will be increased.

B. The current gain of a BJT will be reduced.

C. The break-down voltage of a BJT will be reduced.

List-II:

1. The collector doping concentration is increased.
2. The base width is reduced.
3. The emitter doping concentration to base doping concentration ratio is reduced.
4. The base doping concentration is increased keeping the ratio of the emitter doping concentration to base doping concentration constant.
5. The collector doping concentration is reduced.

Q3.92.1 NAT 1992 2M ◯ ◯ ◯

In a transistor having finite β, forward bias across the base emitter junction is kept constant and the reverse bias across the collector base junction is increased. Neglecting the leakage across the collector base junction and the depletion region generations current, the base current will ____ (increase/decrease/remains constant).

Answer Key

Q3.26.1 C, D	**Q3.24.1** A, C, D	**Q3.17.1** C
Q3.17.2 C	**Q3.16.1** 6.55 to 6.75	**Q3.15.1** 1465 to 1485
Q3.15.2 D	**Q3.14.1** 378 to 381	**Q3.14.2** 5.7 to 5.9
Q3.11.1 D	**Q3.10.1** B	**Q3.06.1** B
Q3.04.1 B	**Q3.04.2** D	**Q3.04.3** A
Q3.99.1 C	**Q3.97.1** C	**Q3.96.1** B
Q3.95.1 D	**Q3.95.2** C	**Q3.94.1** A-2, B-3, C-1
Q3.92.1 decrease		

PrepFusion

UNIT

IV

MOSCap and MOSFETs

Topics

1. Different Modes of Operations of MOSCAPs
2. MOS Physics

Unit Snapshot*

Stream: EC	**Questions:** 43	**MCQ:** 29	**MSQ:** 1	**NAT:** 13
1M: 25	**2M:** 18	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2019)–7M(2021)

**See preface for how Avg Weightage and Range are calculated.*

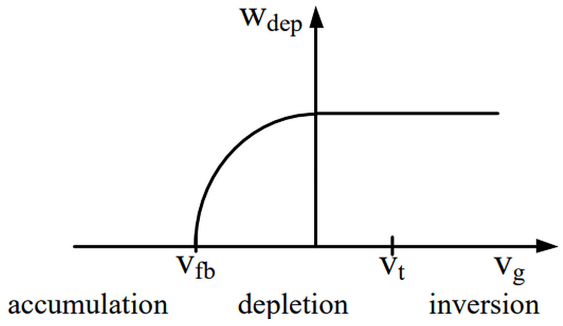
Q4.24.1 NAT 2024 2M

Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm^2 at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is 8.85×10^{-14} F/cm. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

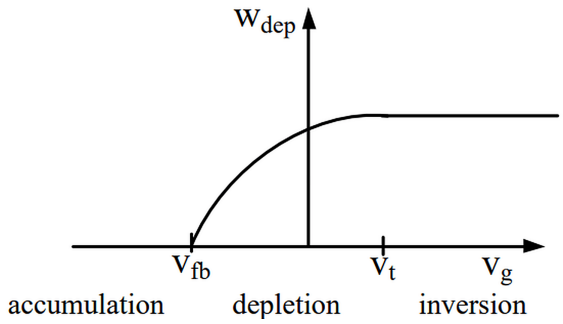
Q4.23.1 MCQ 2023 1M

For a MOS capacitor, V_{fb} and V_t are the flat-band voltage and the threshold voltage, respectively. The variation of the depletion width (W_{dep}) for varying gate voltage (V_g) is best represented by

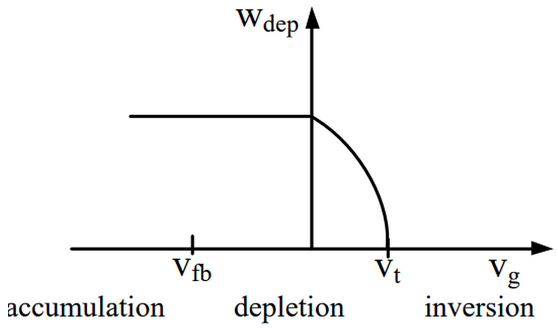
A.



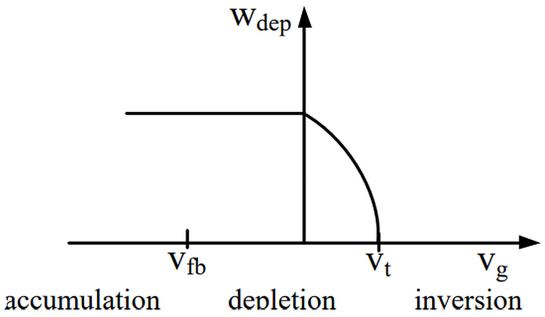
B.



C.



D.



Q4.22.1 NAT 2022 1M

An ideal MOS capacitor (p-type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with $V_G = 2$ V. The corresponding inversion charge density Q_{IN} is 2.2 μC/cm^2. Assume oxide capacitance per unit area as $C_{OX} = 1.7$ μF/cm^2. For $V_G = 4$ V, the value of Q_{IN} is ____ μC/cm^2 (rounded off to one decimal place).

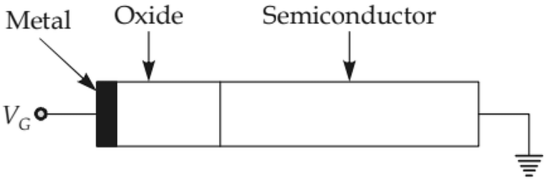


Fig. Q4.22.1

Q4.21.1 MCQ 2021 2M

For an n-channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity $\left(\frac{\partial V_T}{\partial |V_{BS}|}\right)$ is found to be 50 mV/V at a substrate voltage $|V_{BS}| = 2$ V, where V_T is the threshold voltage of the MOSFET. Assume that, $|V_{BS}| \gg 2\Phi_B$, where $q\Phi_B$ is the separation between the Fermi energy level E_F and the intrinsic level E_i in the bulk.

Parameters given are: Electron charge (q) = 1.6×10^{-19} C , Vacuum permittivity (ε_0) = 8.85×10^{-12} F/m , Relative permittivity of silicon (ε_{Si}) = 12 , Relative permittivity of oxide (ε_{ox}) = 4

The doping concentration of the substrate is

A. 7.37×10^{15} cm^{-3} **B.** 4.37×10^{15} cm^{-3}

C. 2.37×10^{15} cm^{-3} **D.** 9.37×10^{15} cm^{-3}

Q4.20.1 MCQ 2020 2M

The band diagram of a p-type semiconductor with a band-gap of 1 eV is shown. Using this semiconductor, a MOS capacitor having V_{TH} of −0.16 V, C'_{ox} of 100 nF/cm^2 and a metal work function of 3.87 eV is fabricated. There is no charge within the oxide. If the voltage across the capacitor

is V_{TH}, the magnitude of depletion charge per unit area (in C/cm^2) is

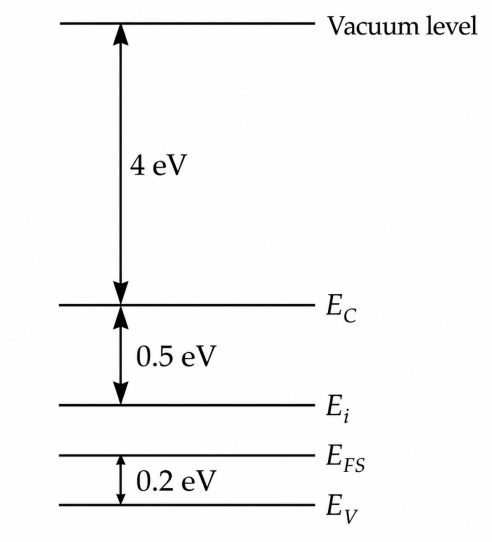


Fig. Q4.20.1

A. 1.7×10^{-8} **B.** 0.52×10^{-8}

C. 1.41×10^{-8} **D.** 0.93×10^{-8}

Q4.19.1 MCQ 2019 1M

The figure shows the high-frequency C-V curve of a MOS capacitor (at T = 300 K) with $\Phi_{ms} = 0$ V and no oxide charges. The flat-band, inversion, and accumulation conditions are represented, respectively, by the points

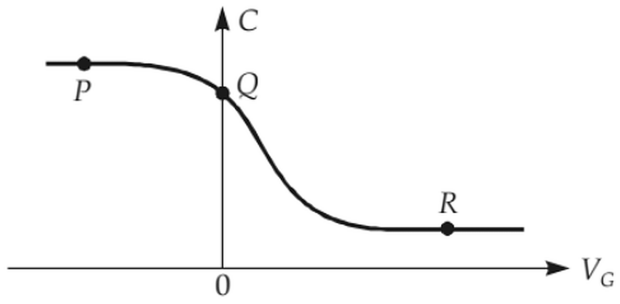


Fig. Q4.19.1

A. P, Q, R **B.** Q, R, P

C. R, P, Q **D.** Q, P, R

Q4.17.1 MCQ 2017 2M

Two n-channel MOSFETs, T1 and T2, are identical in all respects except that the width of T2 is double that of T1. Both the transistors are biased in the saturation region of operation, but the gate overdrive voltage ($V_{GS} - V_{TH}$) of T2 is double that of T1, where V_{GS} and V_{TH} are the gate-to-source voltage and threshold voltage of the transistors, respectively. If the drain current and transconductance of T1 are I_{D1} and g_{m1} respectively, the corresponding values of these two parameters for T2 are

A. $8I_{D1}$ and $2g_{m1}$ **B.** $8I_{D1}$ and $4g_{m1}$

C. $4I_{D1}$ and $4g_{m1}$ **D.** $4I_{D1}$ and $2g_{m1}$

Q4.17.2 NAT 2017 2M

A MOS capacitor is fabricated on p-type Si (silicon) where the metal work function is 4.1 eV and electron affinity of Si is 4.0 eV. $E_C - E_F = 0.9$ eV, where E_C and E_F are the conduction band minimum and the Fermi energy levels of Si, respectively. Oxide $\epsilon_r = 3.9$, $\epsilon_o = 8.85 \times 10^{-14}$ F/cm, oxide thickness $t_{ox} = 0.1\,\mu m$ and electronic charge $q = 1.6 \times 10^{-19}$ C. If the measured flat band voltage of the capacitor is -1V, then the magnitude of the fixed charge at the oxide-semiconductor interface, in nC/cm^2, is ____.

Q4.16.1 NAT 2016 2M

Figures I and II show two MOS capacitors of unit area. The capacitor in Figure I has insulator materials X (of thickness $t_1 = 1$ nm and dielectric constant $\epsilon_1 = 4$) and Y (of thickness $t_2 = 3$ nm and dielectric constant $\epsilon_2 = 20$). The capacitor in Figure II has only insulator material X of thickness t_{Eq}. If the capacitors are of equal capacitance, then the value of t_{Eq} (in nm) is ____

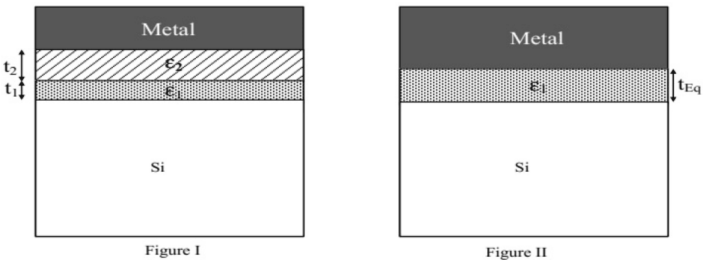


Fig. Q4.16.1

Q4.16.2 MCQ 2016 1M

The figure shows the band diagram of a Metal Oxide Semiconductor (MOS). The surface region of this MOS is in

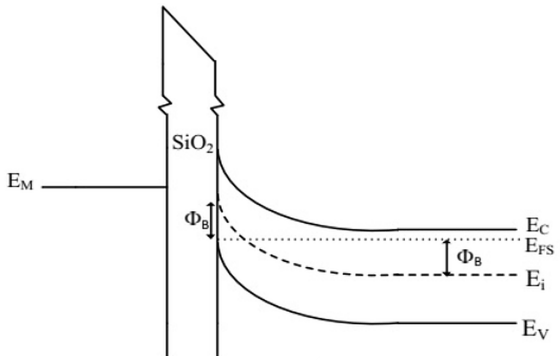


Fig. Q4.16.2

A. inversion **B.** accumulation

C. depletion **D.** flat band

Q4.16.3 MCQ 2016 2M

A voltage V_G is applied across a MOS capacitor with metal gate and p-type silicon substrate at T=300 K. The inversion carrier density (in number of carriers per unit area) for $V_G = 0.8$ V is 2×10^{11} cm^{-2}. For $V_G = 1.3$ V, the inversion carrier density is 4×10^{11} cm^{-2}. What is the value of the inversion carrier density for $V_G = 1.8$ V?

A. 4.5×10^{11} cm^{-2} **B.** 6.0×10^{11} cm^{-2}

C. 7.2×10^{11} cm^{-2} **D.** 8.4×10^{11} cm^{-2}

Q4.16.4 MCQ 2016 1M

A long-channel NMOS transistor is biased in the linear region with $V_{DS} = 50$ mV and is used as a resistance. Which one of the following statements is NOT correct?

A. If the device width W is increased, the resistance decreases.

B. If the threshold voltage is reduced, the resistance decreases.

C. If the device length L is increased, the resistance increases.

D. If V_{GS} is increased, the resistance increases.

Q4.16.5 NAT 2016 2M

Consider an n-channel metal oxide semiconductor field effect transistor (MOSFET) with a gate-tosource voltage of 1.8 V. Assume that $\frac{W}{L} = 4$, $\mu_N C_{ox} = 70 \times 10^{-6}$ A V^{-2}, the threshold voltage is 0.3 V, and the channel length modulation parameter is 0.09 V^{-1}. In the saturation region, the drain conductance (in micro seimens) is ____

Q4.16.6 MCQ 2016 1M

Consider the following statements for a metal oxide semiconductor field effect transistor (MOSFET):

- **P**: As channel length reduces, OFF-state current increases.
- **Q**: As channel length reduces, output resistance increases.
- **R**: As channel length reduces, threshold voltage remains constant.
- **S**: As channel length reduces, ON current increases.

Which of the above statements are INCORRECT?

A. P and Q **B.** P and S

C. Q and R **D.** R and S

Q4.15.1 NAT 2015 2M

The current in an enhancement mode NMOS transistor biased in saturation mode was measured to be 1 mA at a drain-source voltage of 5 V. When the drain-source voltage was increased to 6 V while keeping gate-source voltage same, the drain current increased to 1.02 mA. Assume that drain to source saturation voltage is much smaller than the applied drain-source voltage. The channel length modulation parameter λ (in V^{-1}) is ____.

Q4.15.2 MCQ 2015 2M

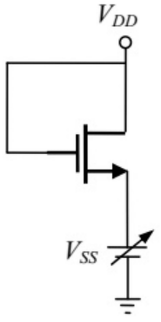
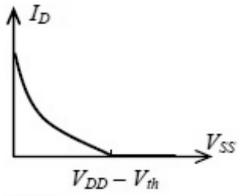
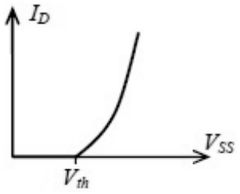
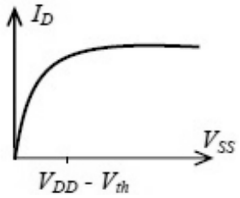


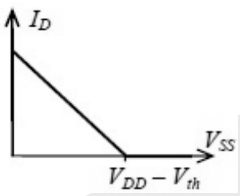
Fig. Q4.15.2

For the NMOSFET in the circuit shown, the threshold voltage is V_{th} , where $V_{th} > 0$. The source voltage V_{SS} is varied from 0 to V_{DD}. Neglecting the channel length modulation, the drain current I_D as a function of V_{SS} is represented by

A.


B.


C.


D.


Q4.15.3 NAT 2015 2M

In a MOS capacitor with an oxide layer thickness of 10 nm, the maximum depletion layer thickness is 100 nm. The permittivities of the semiconductor and the oxide layer are ε_s and ε_{ox} respectively. Assuming $\varepsilon_s/\varepsilon_{ox} = 3$, the ratio of the maximum capacitance to the minimum capacitance of this MOS capacitor is ____.

Q4.15.4 NAT 2015 2M

A MOSFET in saturation has a drain current of 1 mA for $V_{DS} = 0.5$ V. If the channel length modulation coefficient is 0.05 V^{-1}, the output resistance (in $k\Omega$) of the MOSFET is ____.

Q4.14.1 NAT 2014 2M

An ideal MOS capacitor has boron doping-concentration of 10^{15} cm^{-3} in the substrate. When a gate voltage is applied, a depletion region of width 0.5 μm is formed with a surface (channel) potential of 0.2 V. Given $\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm and the relative permittivities of silicon and silicon dioxide are 12 and 4, respectively, the peak electric field (in V/μm) in the oxide region is ____.

Q4.14.2 NAT 2014 2M

The slope of the I_D vs. V_{GS} curve of an n-channel MOSFET in linear regime is $10^{-3}\,\Omega^{-1}$ at $V_{DS} = 0.1$V. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs. V_{GS} curve (in $\sqrt{\text{A}}$/V) under saturation regime is approximately ____.

Q4.14.3 NAT 2014 2M

A depletion type N-channel MOSFET is biased in its linear region for use as a voltage controlled resistor. Assume threshold voltage $V_{TH} = -0.5$ V, $V_{GS} = 2.0$ V, $V_{DS} = 5$ V, $W/L = 100$, $C_{OX} = 10^{-8}$ F/cm^2 and $\mu_n = 800$ cm^2/V – s. The value of the resistance of the voltage controlled resistor (in Ω) is ____.

Q4.14.4 MCQ 2014 1M

If fixed positive charges are present in the gate oxide of an n-channel enhancement type MOSFET, it will lead to

A. a decrease in the threshold voltage

B. channel length modulation

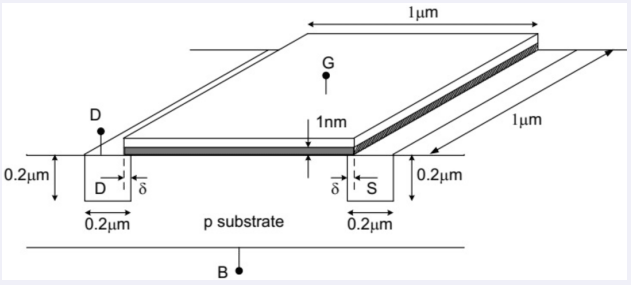
C. an increase in substrate leakage current

D. an increase in accumulation capacitance

Common Data Questions

Answer the next two questions using the following data.

In the three dimensional view of a silicon n-channel MOS transistor shown below, $\delta = 20$ nm. The transistor is of width 1 μm. The depletion width formed at every p-n junction is 10 nm. The relative permittivities of Si and SiO_2, respectively, are 11.7 and 3.9, and $\varepsilon_0 = 8.9 \times 10^{-12}$ F/m.



Q4.12.1 MCQ 2012 1M

The source-body junction capacitance is approximately

A. 2 fF **B.** 7 fF **C.** 2 pF **D.** 7 pF

Q4.12.2 MCQ 2012 1M

The gate-source overlap capacitance is approximately

A. 0.7 fF **B.** 0.7 pF **C.** 0.35 fF **D.** 0.24 pF

Q4.12.3 MCQ 2012 1M

The source of a silicon ($n_i = 10^{10}$ per cm^3) n-channel MOS transistor has an area of 1 sq μm and a depth of 1 μm. If the dopant density in the source is $10^{19}/cm^3$, the number of holes in the source region with the above volume is approximately

A. 10^7 **B.** 100 **C.** 10 **D.** 0

Q4.10.1 MCQ 2010 1M

At room temperature, a possible value for the mobility of electrons in the inversion layer of a silicon n-channel MOSFET is

A. $150\,cm^2/V-s$ **B.** $1350\,cm^2/V-s$

C. $1800\,cm^2/V-s$ **D.** $3600\,cm^2/V-s$

Q4.09.1 MCQ 2009 1M

Consider the following two statements about the internal conditions in a n-channel MOSFET operating in the active region.

S1 : The inversion charge decreases from source to drain. S2 : The channel potential increases from source to drain.

Which of the following is correct?

A. Only S2 is true

B. Both S1 and S2 are false

C. Both S1 and S2 are true, but S2 is not a reason for S1

D. Both S1 and S2 are true, and S2 is a reason for S1

Q4.08.1 MCQ 2008 1M

The drain current of MOSFET in saturation is given by $I_D = K(V_{GS} - V_T)^2$ where K is a constant. The magnitude of the transconductance g_m is

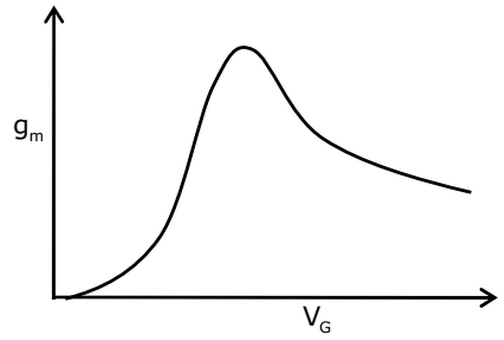
A. $\dfrac{K(V_{GS}-V_T)^2}{V_{DS}}$ **B.** $2K(V_{GS}-V_T)$

C. $\dfrac{I_d}{V_{GS}-V_{DS}}$ **D.** $\dfrac{K(V_{GS}-V_T)^2}{V_{GS}}$

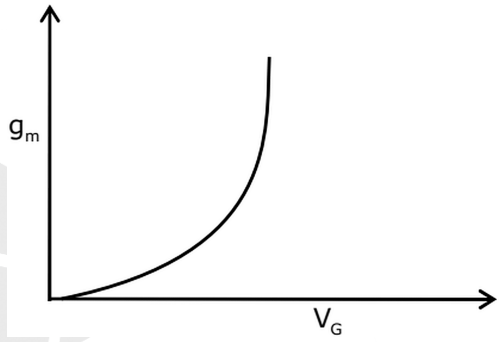
Q4.08.2 MCQ 2008 1M

The measured trans conductance g_m of an NMOS transistor operating in the linear region is plotted against the gate voltage V_G at a constant drain voltage V_D. Which of the following figures represents the expected dependence of g_m on V_G?

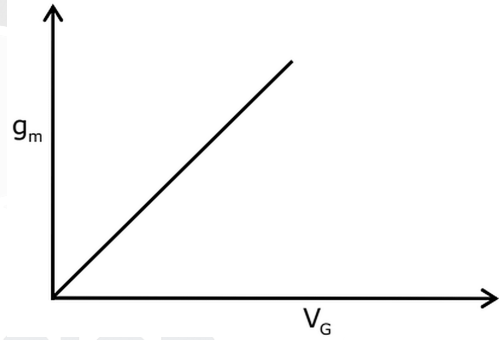
A.



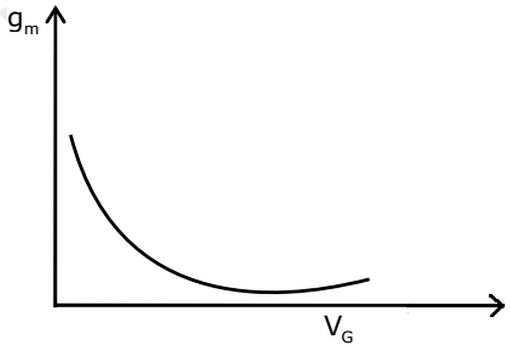
B.



C.



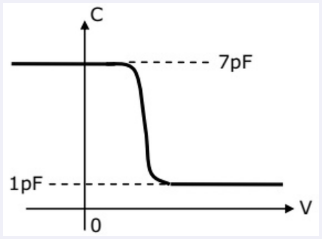
D.



Common Data Questions

Answer the next three questions using the following data.

The figure shows the high-frequency capacitance-voltage (C-V) characteristics of a Metal/SiO_2/silicon (MOS) capacitor having an area of $1 \times 10^{-4}\,cm^2$. Assume that the permittivities ($\varepsilon_0\varepsilon_r$) of silicon and SiO_2 are 1×10^{-12} F/cm and 3.5×10^{-13} F/cm respectively.



Q4.07.1 MCQ 2007 1M

Consider the following statements about the C-V characteristics plot : S1 : The MOS capacitor has as n-type substrate S2 : If positive charges are introduced in the oxide, the C-V polt will shift to the left. Then which of the following is true?

A. Both S1 and S2 are true

B. S1 is true and S2 is false

C. S1 is false and S2 is true

D. Both S1 and S2 are false

Q4.07.2 MCQ 2007 1M

The maximum depletion layer width in silicon is

A. 0.143 μm **B.** 0.857 μm **C.** 1 μm **D.** 1.143 μm

Q4.07.3 MCQ 2007 1M

The gate oxide thickness in the MOS capacitor is

A. 50 nm **B.** 350 nm **C.** 143 nm **D.** 1 nm

Q4.05.1 MCQ 2005 1M

A MOS capacitor made using p type substrate is in the accumulation mode. The dominant charge in the channel is due to the presence of

A. holes **B.** electrons

C. positively charged ions **D.** negatively charged ions

Q4.04.1 MCQ 2004 1M

Consider the following statements S1 and S2.

S1 : The threshold voltage (V_T) of MOS capacitor decreases with increase in gate oxide thickness.

S2 : The threshold voltage (V_T) of a MOS capacitor decreases with increase in substrate doping concentration. Which Marks of the following is correct ?

A. S1 is FALSE and S2 is TRUE

B. Both S1 and S2 are TRUE

C. S1 is TRUE and S2 is FALSE

D. Both S1 and S2 are FALSE

Q4.03.1 MCQ 2003 1M

For an n - channel enhancement type MOSFET, if the source is connected at a higher potential than that of the bulk (i.e. $V_{SB} > 0$), the threshold voltage V_T of the MOSFET will

A. remain unchanged **B.** decrease

C. change polarity **D.** increase

Q4.01.1 MCQ 2001 1M

The effective channel length of a MOSFET in saturation decreases with increase in

A. gate voltage **B.** drain voltage

C. source voltage **D.** body voltage

Q4.01.2 MCQ 2001 1M

MOSFET can be used as a

A. current controlled capacitor

B. voltage controlled capacitor

C. current controlled inductor

D. voltage controlled inductor

Q4.94.1 NAT 1994 1M

Channel current is reduced on application of a more positive voltage to the GATE of the depletion mode n-channel MOSFET.

(True/False)

Q4.94.2 MCQ 1994 1M

The threshold voltage of an n-channel MOSFET can be increased by

A. increasing the channel carrier concentration.

B. reducing the channel carrier concentration.

C. reducing the GATE oxide thickness.

D. reducing the channel length.

Q4.94.3 NAT 1994 1M

The transit time of a current carriers through the channel of an FET decides its ____ characteristic.

Q4.90.1 MSQ 1990 2M

Which of the following effects can be caused by a rise in the temperature?

A. Increase in MOSFET current (I_{DS})

B. Increase in BJT current (I_C)

C. Decrease in MOSFET current (I_{DS})

D. Decrease in BJT current (I_C)

Q4.89.1 MCQ 1989 2M

In a MOSFET, the polarity of the inversion layer is the same as that of the

A. charge on the gate electrode

B. minority carries in the drain

C. majority carriers in the substrate

D. majority carriers in the source

Q4.88.1 MCQ 1988 2M

In MOSFET devices the N-channel type is better than the P-channel type in the following respects

A. it has better noise immunity.

B. it is faster.

C. it is TTL compatible.

D. it has better drive capability.

Answer Key

Q4.24.1 4.10 to 4.50	Q4.23.1 B	Q4.22.1 5.5 to 5.7
Q4.21.1 A	Q4.20.1 A	Q4.19.1 B
Q4.17.1 B	Q4.17.2 6.85 to 6.95	Q4.16.1 1.55 to 1.65
Q4.16.2 A	Q4.16.3 B	Q4.16.4 D
Q4.16.5 28 to 29	Q4.16.6 C	Q4.15.1 0.018 to 0.026
Q4.15.2 A	Q4.15.3 4.3 to 4.4	Q4.15.4 19 to 21
Q4.14.1 2.3 to 2.5	Q4.14.2 0.06 to 0.08	Q4.14.3 499 to 501
Q4.14.4 A	Q4.12.1 B	Q4.12.2 A
Q4.12.3 D	Q4.10.1 A	Q4.09.1 D
Q4.08.1 B	Q4.08.2 A	Q4.07.1 C
Q4.07.2 B	Q4.07.3 A	Q4.05.1 A
Q4.04.1 D	Q4.03.1 D	Q4.01.1 B
Q4.01.2 B	Q4.94.1 False	Q4.94.2 B
Q4.94.3 switching	Q4.90.1 B , C	Q4.89.1 D
Q4.88.1 B		

UNIT

V

Optoelectronic Devices

Topics

1. Solar Cell
2. Photodiode
3. LEDs

Unit Snapshot*

Stream: EC	**Questions:** 12	**MCQ:** 5	**MSQ:** 1	**NAT:** 6
1M: 4	**2M:** 8	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 0M(2016)–4M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MSQ 2026 2M ○○○

Consider an LED based on a direct bandgap semiconductor material with energy bandgap 1.3 eV. Given: Planck's constant, $h = 6.63 \times 10^{-34}$ J s and speed of light in free space is 3×10^8 m s^{-1}. In which of the following wavelength ranges the LED will NOT emit?

- ☐ **A.** 1410 ± 20 nm
- ☐ **B.** 1090 ± 20 nm
- ☐ **C.** 950 ± 20 nm
- ☐ **D.** 510 ± 20 nm

Q5.24.1 NAT 2024 2M ○○○

The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, rounded off to two decimal places) is _____.

Q5.20.1 MCQ 2020 2M ○○○

A pn junction solar cell of area 1.0 cm^2, illuminated uniformly with 100 $mWcm^{-2}$, has the following parameters: Efficiency = 15%, open circuit voltage = 0.7V, fill factor = 0.8, and thickness = 200μm. The charge of an electron is $1.6 \times 10^{-19}C$. The average optical generation rate (in $cm^{-3}s^{-1}$) is

A. 0.84×10^{19} **B.** 5.57×10^{19}

C. 1.04×10^{19} **D.** 83.6×10^{19}

Q5.19.1 NAT 2019 2M ○○○

A Germanium sample of dimensions 1cmx1cm is illuminated with a 20 mW, 600 nm laser light source as shown in the figure.

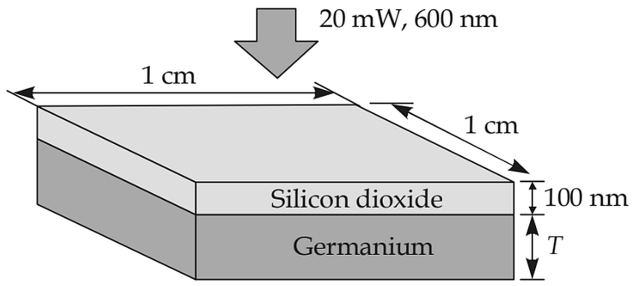


Fig. Q5.19.1

The illuminated sample surface has a 100 nm of lossless Silicon dioxide layer that reflects one-fourth of the incident light. From the remaining light, one-third of the power is reflected from the Silicon dioxide-Germanium interface, one-third is absorbed in the Germanium layer, and one-third is transmitted through the other side of the sample. If the absorption coefficient of Germanium at 600 nm is $3 \times 10^4 cm^{-1}$ and the bandgap is 0.66 eV, the thickness of the Germanium layer, rounded off to 3 decimal places, is ____ μm.

Q5.19.2 MCQ 2019 2M ○○○

The quantum efficiency (η) and responsivity (R) at a wavelength λ (in μm) in a p-i-n photodetector are related by

A. $R = \dfrac{\eta\lambda}{1.24}$ **B.** $R = \dfrac{\lambda}{1.24\eta}$

C. $R = \dfrac{1.24\lambda}{\eta}$ **D.** $R = \dfrac{1.24}{\eta\lambda}$

Q5.18.1 NAT 2018 2M ○○○

A solar cell of area 1.0cm^2 , operating at 1.0 sun intensity, has a short circuit current of 20mA, and an open circuit voltage of 0.65V. Assuming room temperature operation and thermal equivalent voltage of 26mV, the open circuit voltage (in volts, correct to two decimal places) at 0.2 sun intensity is ____ .

Q5.18.2 MCQ 2018 2M ○○○

Red (R), Green (G) and Blue (B) Light Emitting Diodes (LEDs) were fabricated using p-n junctions of three different inorganic semiconductors having different band-gaps. The builtin voltages of red, green and blue diodes are V_R, V_G, V_B respectively. Assume donor and acceptor doping to be the same (N_A and N_D respectively) in the p and n sides of all the three diodes. Which one of the following relationships about the built-in voltages is TRUE?

A. $V_R > V_G > V_B$ **B.** $V_R < V_G < V_B$

C. $V_R = V_G = V_B$ **D.** $V_R > V_G < V_B$

Q5.17.1 NAT 2017 2M ○○○

For a particular intensity of incident light on a silicon pn junction solar cell, the photocurrent density J_L is 2.5mA/cm^2 and the open-circuit voltage V_{OC} is 0.451V. Consider thermal voltage V_T to be 25mV. If the intensity of the incident light is increased by 20 times,assuming that the temperature remains unchanged, V_{OC} (in volts) will be ____ .

Q5.16.1 NAT 2016 1M

The figure shows the I-V characteristics of a solar cell illuminated uniformly with solar light of power $100 mW/cm^2$. The solar cell has an area of $3 cm^2$ and a fill factor of 0.7. The maximum efficiency (in %) of the device is ____ .

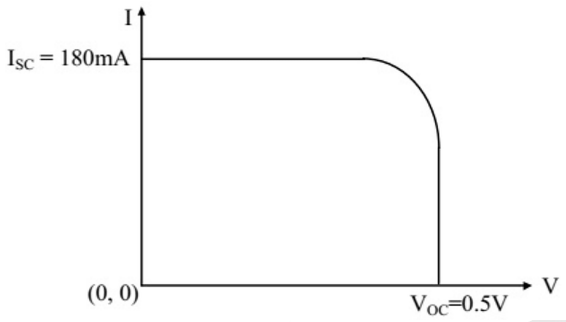


Fig. Q5.16.1

Q5.14.1 NAT 2014 1M

When the optical power incident on a photodiode is 10 μW and the responsivity is 0.8 A/W, the photocurrent generated (in μA) is ____.

Q5.07.1 MCQ 2007 1M

Group I lists three types of p-n junction diodes. Match each device in Group I with one of the option in Group II to indicate the bias condition of the device in its normal mode of operation.

Group I	Group II
(P) Zener Diode	(1) Forward bias
(Q) Solar cell	(2) Reverse bias
(R) LASER diode	
(S) Avalanche Photodiode	

A. P - 1, Q - 2, R - 1 , S - 2

B. P - 2, Q - 1, R - 1 , S - 2

C. P - 2, Q - 2, R - 1 , S - 2

D. P - 2, Q - 1, R - 2 , S - 2

Q5.03.1 MCQ 2003 1M

A particular green LED emits light of wavelength 5490 $\mathring{A}$. The energy bandgap of the semiconductor material used there is (Planck's constant = $6.26 \times 10^{-34}\ Js$).

A. 2.26 eV **B.** 1.98 eV **C.** 1.17 eV **D.** 0.74 eV

Answer Key

Q5.26.1 A, B, D	Q5.24.1 4.00 to 4.26	Q5.20.1 A
Q5.19.1 0.23 to 0.232	Q5.19.2 A	Q5.18.1 0.59 to 0.63
Q5.18.2 B	Q5.17.1 0.51 to 0.62	Q5.16.1 20.5 to 21.5
Q5.14.1 8 to 8	Q5.07.1 B	Q5.03.1 A

SUBJECT

3

Electromagnetic Fields and Transmission Lines

Subject Snapshot*

Stream: EC	**Questions:** 257	**MCQ:** 195	**MSQ:** 14	**NAT:** 48
1M: 98	**2M:** 159	**Descriptive:** 0	**Avg Weightage** ~8M	**Range:** 5M(2013)–11M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Electromagnetic Fields and Transmission Lines

Unit **Topic** **Pages**

I. Coordinate Geometry & Maxwell's Equation - Basics 99
II. EM Waves 105
III. Transmission Lines & Smith Chart 116
IV. Rectangular & Circular Waveguides 130
V. Antenna & Antenna Array 136

PrepFusion

UNIT

I

Coordinate Geometry & Maxwell's Equation - Basics

Topics

1. Basics
2. Maxwell's Equation for Static & Time Varying Fields

Unit Snapshot*

Stream: EC	**Questions:** 32	**MCQ:** 23	**MSQ:** 1	**NAT:** 8
1M: 9	**2M:** 23	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 5M(2019)–13M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ 2025 2M

A square metal sheet of 4 m × 4 m is placed on the x-y plane as shown in the figure below.

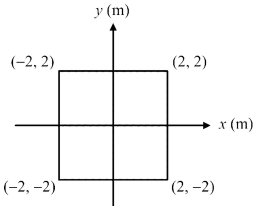


Fig. Q1.25.1

If the surface charge density (in μC/m^2) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in μC, rounded off to the nearest integer) on the sheet is _____.

A. 16 **B.** 85 **C.** 64 **D.** 256

Q1.25.2 MCQ 2025 2M

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m.
The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____.

A. 0.46 **B.** 1.82 **C.** 0.58 **D.** 1.12

Q1.25.3 NAT 2025 2M

Two resistors are connected in a circuit loop of area 5 m^2, as shown in the figure below. The circuit loop is placed on the x-y plane.

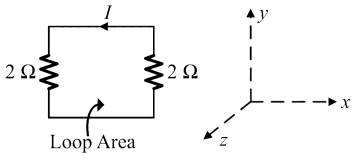


Fig. Q1.25.3

When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z-axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is _____.

Q1.24.1 MCQ 2024 2M

Consider the Earth to be a perfect sphere of radius R. Then the surface area of the region, enclosed by the 60°N latitude circle, that contains the north pole in its interior is _____.

A. $(2 - \sqrt{3})\pi R^2$ **B.** $\dfrac{(\sqrt{2} - 1)\pi R^2}{2}$

C. $\dfrac{2\pi R^2}{3}$ **D.** $\dfrac{(2 + \sqrt{3})\pi R^2}{8\sqrt{2}}$

Q1.24.2 MCQ 2024 1M

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

A. Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.

B. Only $\vec{F}_1$ is an electrostatic field.

C. Only $\vec{F}_2$ is an electrostatic field.

D. Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

Q1.22.1 NAT 2022 2M

In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by

$$\vec{D}(x, y, z) = (x^3\vec{i} + y^3\vec{j} + xy^2\vec{k})\ \text{C/m}^2,$$

where $\vec{i}$, $\vec{j}$, $\vec{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at (±0.5 m, ±0.5 m, ±0.5 m). The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Q1.22.2 MCQ 2022 1M

In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2\sin(t + \pi/2)$. The displacement current (in Amperes) through the capacitor is _____.

A. $2\sin(t)$

B. $2\sin(t + \pi)$

C. $2\sin(t + \pi/2)$

D. 0

Q1.21.1 NAT 2021 2M

For a vector field $\mathbf{D} = \rho \cos^2 \varphi \mathbf{a}_\rho + z^2 \sin^2 \varphi \mathbf{a}_\varphi$ in a cylindrical coordinate system (ρ, φ, z) with unit vectors $\mathbf{a}_\rho$, $\mathbf{a}_\varphi$ and $\mathbf{a}_z$, the net flux of $\mathbf{D}$ leaving the closed surface of the cylinder $(\rho = 3, 0 \le z \le 2)$ (*rounded off to two decimal places*) is _____.

Q1.21.2 NAT 2021 1M

Consider the vector field $\mathbf{F} = \mathbf{a}_x(4y - c_1 z) + \mathbf{a}_y(4x + 2z) + \mathbf{a}_z(2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$. If the field $\mathbf{F}$ is irrotational (conservative), then the constant c_1 (*in integer*) is _____.

Q1.19.1 MCQ 2019 1M

In the table shown, **List-I** and **List-II** respectively, contain terms appearing on the left-hand side and the right-hand side of Maxwell's equations (in their standard form). Match the left-hand side withe the corresponding right-hand side.

List-I	List-II
1. $\nabla \cdot \vec{D}$	**P.** 0
2. $\nabla \times \vec{E}$	**Q.** ρ
3. $\nabla \cdot \vec{B}$	**R.** $-\dfrac{\partial \vec{B}}{\partial t}$
4. $\nabla \times \vec{H}$	**S.** $\vec{J} + \dfrac{\partial \vec{D}}{\partial t}$

A. 1 - R, 2 - Q, 3 - S, 4 - P

B. 1 - Q, 2 - S, 3 - P, 4 - R

C. 1 - P, 2 - R, 3 - Q, 4 - S

D. 1 - Q, 2 - R, 3 - P, 4 - S

Q1.16.1 MCQ 2016 2M

The parallel-plate capacitor shown in the figure has movable plates. The capacitor is charged so that the energy stored in it is E when the plate separation is d. The capacitor is then isolated electrically and the plates are moved such that the plate separation becomes $2d$.

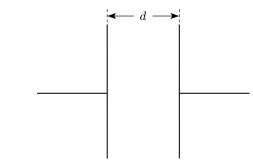


Fig. Q1.16.1

At this new plate separation, what is the energy stored in the capacitor, neglecting fringing effect?

A. $2E$ **B.** $\sqrt{2}E$

C. E **D.** $E/2$

Q1.16.2 MCQ 2016 2M

The current density in a medium is given by

$$\vec{J} = \frac{400 \sin \theta}{2\pi(r^2 + 4)} \hat{a}_r, \ \text{Am}^{-2}$$

The total current and the average current density flowing through the portion of a spherical surface, $r =$ 0.8 m, $\dfrac{\pi}{12} \le \theta \le \dfrac{\pi}{4}$, $0 \le \phi \le 2\pi$ are given, respectively by

A. 15.09 A, 12.86 Am^{-2}

B. 18.73 A, 13.65 Am^{-2}

C. 12.86 A, 9.23 Am^{-2}

D. 10.28 A, 7.56 Am^{-2}

Q1.16.3 NAT 2016 1M

Concentric spherical shells of radii 2 m, 4 m and 8 m carry uniform surface charge densities of 20 nC/m^2, -4 nC/m^2 and ρ_s, respectively. The value of ρ_s (nC/m^2) required to ensure that the electric flux density $\vec{D} = \vec{0}$ at radius 10 m is _____.

Q1.15.1 NAT 2015 2M

A vector field $\vec{D} = 2\rho^2 \hat{a}_\rho + z\hat{a}_z$ exists inside a cylindrical region enclosed by the surfaces $\rho = 1$, $z = 0$ and $z = 5$. Let S be the surface bounding this cylindrical region. The surface integral of this field on S $\oiint_S \vec{D} \cdot d\vec{s}$ is _____.

Q1.15.2 NAT 2015 1M

In a source free region in vacuum, if the electrostatic potential $\phi = 2x^2 + y^2 + cz^2$, the value of constant c must be _____.

Q1.14.1 MCQ 2014 2M

If $\vec{E} = -(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$ is the electric field in a source free region, a valid expression for the electrostatic potential is

A. $xy^3 - yz^2$ **B.** $2xy^3 - xyz^2$

C. $y^3 + xyz^2$ **D.** $2xy^3 - 3xyz^2$

Q1.14.2 MCQ 2014 2M

A region shown below contains a perfect conducting half-space and air. The surface current $\vec{K}_s$ on the surface of the perfect conductor is $\vec{K}_s = \hat{x}2$ amperes per meter. The tangential $\vec{H}$ field in the air just above the perfect conductor is

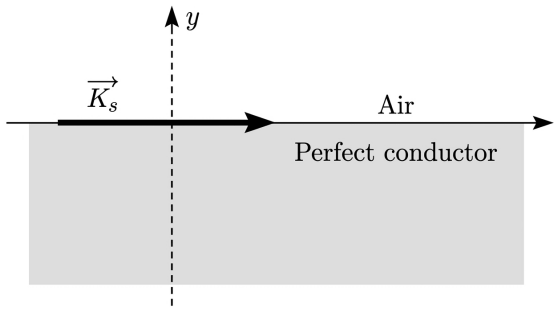


Fig. Q1.14.2

A. $(\hat{x} + \hat{z})$ amperes per meter

B. $\hat{x}2$ amperes per meter

C. $-\hat{z}2$ amperes per meter

D. $\hat{z}2$ amperes per meter

Q1.11.1 MCQ 2011 2M

A current sheet $\vec{J} = 10\hat{u}_y$ A/m lies on the dielectric interface $x = 0$ between two dielectric media with $\varepsilon_{r1} = 1, \mu_{r1} = 1$ in Region-1 ($x < 0$) and $\varepsilon_{r2} = 2, \mu_{r2} = 2$ in Region-2 ($x > 0$). If the magnetic field in Region-1 at $x = 0^-$ is $\vec{H}_1 = 3\hat{u}_x + 30\hat{u}_y$ A/m, the magnetic field in Region-2 at $x = 0^+$ is

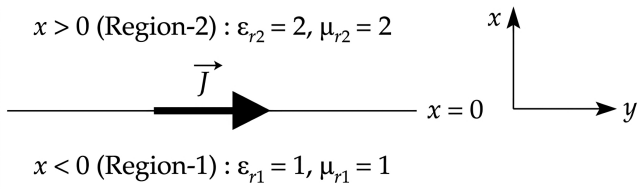


Fig. Q1.11.1

A. $\vec{H}_2 = 1.5\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m

B. $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m

C. $\vec{H}_2 = 1.5\hat{u}_x + 40\hat{u}_y$ A/m

D. $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y + 10\hat{u}_z$ A/m

Q1.09.1 MCQ 2009 2M

If a vector field $\vec{V}$ is related to another vector field $\vec{A}$ through $\vec{V} = \nabla \times \vec{A}$, which of the following is true?

Note: C and S_C refer to any closed contour and any surface whose boundary is C.

A. $\oint_C \vec{V} \cdot \vec{dl} = \iint_{S_C} \vec{A} \cdot \vec{ds}$

B. $\oint_C \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{ds}$

C. $\oint_C \nabla \times \vec{V} \cdot \vec{dl} = \iint_{S_C} \nabla \times \vec{A} \cdot \vec{ds}$

D. $\oint_C \nabla \times \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{ds}$

Q1.08.1 MCQ 2008 2M

For static electric and magnetic fields in an inhomogeneous source free medium, which of the following represents the correct form of two of Maxwell's equations?

A. $\nabla \cdot \vec{E} = 0$, $\nabla \times \vec{B} = 0$

B. $\nabla \cdot \vec{E} = 0$, $\nabla \cdot \vec{B} = 0$

C. $\nabla \times \vec{E} = 0$, $\nabla \times \vec{B} = 0$

D. $\nabla \times \vec{E} = 0$, $\nabla \cdot \vec{B} = 0$

Q1.07.1 MCQ 2007 1M

If C is a closed curve enclosing a surface S, then the magnetic field intensity $\vec{H}$, the current density $\vec{J}$ and the electric flux density $\vec{D}$ are related by

A. $\iint_S \vec{H} \cdot \vec{ds} = \oint_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{dl}$

B. $\int_C \vec{H} \cdot \vec{dl} = \oiint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{ds}$

C. $\oiint_S \vec{H} \cdot \vec{ds} = \int_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{dl}$

D. $\oint_C \vec{H} \cdot \vec{dl} = \iint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{ds}$

Q1.06.1 MCQ 2006 2M

A medium is divided into regions I and II about $x = 0$ plane, as shown in the figure below. An electromagnetic wave with electric field $\vec{E}_1 = 4\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ is incident normally on the interface from region-I. The electric field $\vec{E}_2$ in region-II at the interface is

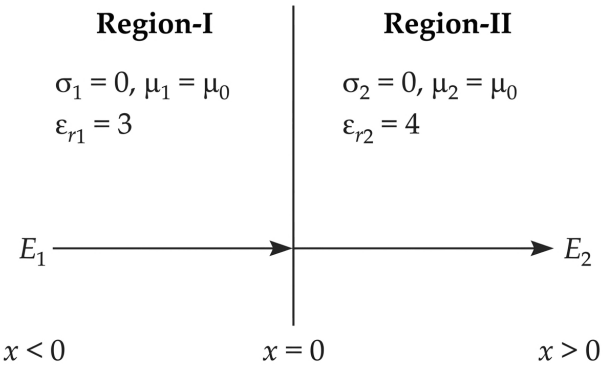


Fig. Q1.06.1

A. $\vec{E}_2 = \vec{E}_1$ **B.** $4\hat{a}_x + 0.75\hat{a}_y - 1.25\hat{a}_z$

C. $3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ **D.** $-3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$

Q1.03.1 MCQ 2003 2M

Medium 1 has the electrical permittivity $\epsilon_1 = 1.5\epsilon_0$ Farad/m and occupies the region to the left of $x = 0$ plane. Medium 2 has the electrical permittivity $\epsilon_2 = 2.5\epsilon_0$ Farad/m and occupies the region to the right of $x = 0$ plane. If E_1 in medium 1 is $E_1(2u_x - 3u_y + 1u_z)$ Volt/m, then E_2 in medium 2 is

A. $(2.0u_x - 7.5u_y + 2.5u_z)$ Volt/m

B. $(2.0u_x - 2.0u_y + 0.6u_z)$ Volt/m

C. $(1.2u_x - 3.0u_y + 1.0u_z)$ Volt/m

D. $(1.2u_x - 2.0u_y + 0.6u_z)$ Volt/m

Q1.03.2 MCQ 2003 2M

If the electric field intensity is given by

$$\vec{E} = (x\hat{u}_x + y\hat{u}_y + z\hat{u}_z) \text{ Volt/m}$$

The potential difference between $X(2, 0, 0)$ and $Y(1, 2, 3)$ is

A. +1 Volt **B.** -1 Volt

C. +5 Volt **D.** +6 Volt

Q1.03.3 MCQ 2003 1M

The unit of $\nabla \times \vec{H}$ is

A. Ampere **B.** Ampere/meter

C. Ampere/meter^2 **D.** Ampere-meter

Q1.02.1 MCQ 2002 2M

The electric field on the surface of a perfect conductor is 2 V/m. The conductor is immersed in water with $\epsilon = 80\epsilon_0$. The surface charge density on the conductor is

A. 0 C/m^2 **B.** 2 C/m^2

C. 1.8×10^{-11} C/m^2 **D.** 1.41×10^{-9} C/m^2

Q1.96.1 MCQ 1996 1M

A metal sphere with 1 m radius and a surface charge density of 10 C/m^2 is enclosed in a cube of 10 m side. The total outward electric displacement flux normal to the surface of the cube is

A. 40π Coulombs **B.** 10π Coulombs

C. 5π Coulombs **D.** None of the above

Q1.93.1 NAT 1993 2M

Given, $\vec{V} = x\cos^2 y\,\hat{i} + x^2 e^z\,\hat{j} + z\sin^2 y\,\hat{k}$ and S the surface of unit cube with one corner at the origin and edges parallel to the coordinate axis, the value of the integral $\iint_C \vec{V} \cdot \hat{n}\, dS$ is _____.

Q1.90.1 MCQ 1990 2M

Which of the following field equations indicate that the free magnetic charge do not exist?

A. $\vec{H} = \frac{1}{\mu}\nabla \times \vec{A}$ **B.** $\vec{H} = \oint \frac{Id\vec{l} \times \vec{R}}{4\pi R^2}$

C. $\nabla \cdot \vec{H} = 0$ **D.** $\nabla \times \vec{H} = \vec{J}$

Q1.88.1 MCQ 1988 2M

Vector potential is a vector

A. whose curl is equal to the magnetic flux density.

B. whose curl is equal to the electric field intensity.

C. whose divergence is equal to the electric potential.

D. which is equal to the vector product $\vec{E} \times \vec{H}$.

Q1.88.2 MSQ 1988 2M

On either side of a charge free interface between two media

☐ **A.** the normal components of the electric field are equal.

☐ **B.** the tangential components of the electric field are equal.

☐ **C.** the normal components of the electric flux density are equal.

☐ **D.** the tangential components of the electric flux density are equal.

Q1.87.1 MCQ | 1987 | 2M | ◯◯◯

An electrostatic field is said to be conservative when:

A. The divergence of the field is equal to zero.

B. The curl of the field is equal to zero.

C. The curl of the field is equal to $-\frac{\partial E}{\partial t^2}$.

D. The Laplacian of the field is equal to $\mu\varepsilon\frac{\partial^2 E}{\partial t^2}$.

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q1.25.1 C	**Q1.25.2** A	**Q1.25.3** 0.62 to 0.63
Q1.24.1 A	**Q1.24.2** B	**Q1.22.1** 0.50
Q1.22.2 C	**Q1.21.1** 56.55	**Q1.21.2** 0
Q1.19.1 D	**Q1.16.1** A	**Q1.16.2** MTA
Q1.16.3 -0.28 to -0.22	**Q1.15.1** 78.5 to 78.6	**Q1.15.2** -3
Q1.14.1 D	**Q1.14.2** D	**Q1.11.1** A
Q1.09.1 B	**Q1.08.1** D	**Q1.07.1** D
Q1.06.1 C	**Q1.03.1** C	**Q1.03.2** C
Q1.03.3 C	**Q1.02.1** D	**Q1.96.1** A
Q1.93.1 1	**Q1.90.1** C	**Q1.88.1** A
Q1.88.2 B,C	**Q1.87.1** B	

UNIT

II

EM Waves

Topics

1. EM Wave Propagation in Different Medium
2. Poynting Vector & Poynting Theorem
3. Boundary Conditions
4. Normal & Oblige Incidence of EM Wave
5. Wave Polarization

Unit Snapshot*

Stream: EC	**Questions:** 71	**MCQ:** 60	**MSQ:** 4	**NAT:** 7
1M: 31	**2M:** 40	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 3M(2023)–12M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

The electric field of a monochromatic plane wave travelling in a lossless isotropic and homogeneous medium is given by

$$\vec{E}(z,t) = E_0[\hat{x}\cos(\omega t - kz) + \hat{y}\sin(\omega t - kz)]$$

in a right-handed orthogonal co-ordinate system. Which of the following is the correct polarization of the electromagnetic wave?

A. Right-handed circularly polarized

B. Left-handed circularly polarized

C. Linearly polarized

D. Linearly polarized with $-45°$ angle to $\hat{x}$

Q2.26.2 MSQ 2026 1M

Which option(s) represents/ represent the dielectric loss tangent of a substrate?

☐ **A.** Ratio of the real to imaginary parts of the total displacement current

☐ **B.** $(\omega\epsilon'' + \sigma)/(\omega\epsilon')$

☐ **C.** Ratio of the electric susceptibility to permittivity

☐ **D.** Ratio of the polarization vector $\vec{P}$ to the displacement vector $\vec{D}$

Q2.24.1 MCQ 2024 2M

A uniform plane wave with electric field $\vec{E}(x) = A_y\hat{a}_y e^{-j\frac{2\pi x}{3}}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____.

A. -6 **B.** $-\frac{3}{2}$

C. -3 **D.** $-\frac{3}{4}$

Q2.23.1 MSQ 2023 2M

The electric field of a plane electromagnetic wave is $E = a_x C_{1x}\cos(\omega t - \beta z) + a_y C_{1y}\cos(\omega t - \beta z + \theta)$ V/m Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?

☐ **A.** $C_{1x} = 1, C_{1y} = 1, \theta = \pi/4$

☐ **B.** $C_{1x} = 2, C_{1y} = 1, \theta = \pi/2$

☐ **C.** $C_{1x} = 1, C_{1y} = 2, \theta = 3\pi/2$

☐ **D.** $C_{1x} = 2, C_{1y} = 1, \theta = 3\pi/4$

Q2.21.1 MCQ 2021 1M

Consider a rectangular coordinate system (x, y, z) with unit vectors a_x, a_y and a_z. A plane wave traveling in the region $z \geq 0$ with electric field vector, $E = 10\cos(2\times10^8 t + \beta z)a_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity, $\varepsilon = \varepsilon_0$, permeability, $\mu = 4\mu_0$, where $\varepsilon_0 = 8.85\times10^{-12}$ F/m and $\mu_0 = 4\pi\times10^{-7}$ H/m). The value of the reflection coefficient is,

A. $\frac{3}{5}$ **B.** $\frac{2}{3}$

C. $\frac{2}{5}$ **D.** $\frac{1}{3}$

Q2.20.1 NAT 2020 2M

The magnetic field of a uniform plane wave in vacuum is given by

$$\vec{H}(x,y,z,t) = (\hat{a}_x + 2\hat{a}_y + b\hat{a}_z)\cos(\omega t + 3x - y - z)$$

The value of b is _____.

Q2.18.1 MCQ 2018 2M

The distance (in m) a wave has to propagate in a medium having a skin depth of 0.1 m so that the amplitude of the wave attenuates by 20 dB, is

A. 0.12 **B.** 0.23

C. 0.46 **D.** 2.3

Q2.18.2 NAT 2018 2M ○○○

A uniform plane wave travelling in free space and having the electric field,

$$\vec{E} = (\sqrt{2}\hat{a}_x - \hat{a}_z)\cos[6\sqrt{3}\pi \times 10^8 t - 2\pi(x + \sqrt{2}z)] \text{ V/m}$$

is incident on a dielectric medium (relative permittivity > 1, relative permeability $= 1$) as shown in the figure and there is no reflected wave.

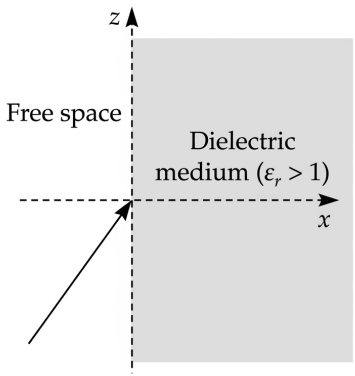


Fig. Q2.18.2

The relative permittivity (correct to two decimal places) of the dielectric medium is _____.

Q2.17.1 NAT 2017 2M ○○○

The permittivity of water at optical frequencies is $1.75\varepsilon_0$. It is found that an isotropic light source at a distance 'd' under water forms an illuminated circular area of radius 5 m as shown in the figure. The critical angle is θ_c.

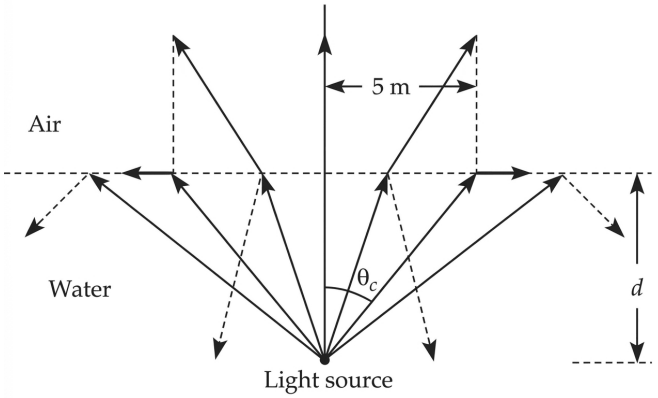


Fig. Q2.17.1

The value of 'd' (in meter) is _____.

Q2.17.2 MCQ 2017 2M ○○○

The expression for an electric field in free space is,

$$\vec{E} = E_0(\hat{x} + \hat{y} + j2\hat{z})e^{-j(\omega t - kx + ky)}$$

where, x, y, z represent the spatial coordinates, t represents time, and ω, k are constants. The electric field

A. does not represent a plane wave.

B. represents a circularly polarized plane wave propagating normal to the z-axis.

C. represents an elliptically polarized plane wave propagating along the x-y plane.

D. represents a linearly polarized plane wave.

Q2.16.1 MCQ 2016 1M ○○○

Faraday's law of electromagnetic induction is mathematically described by which one of the following equations?

A. $\nabla \cdot \vec{B} = 0$

B. $\nabla \cdot \vec{D} = \rho_v$

C. $\nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$

D. $\nabla \times \vec{H} = \sigma\vec{E} + \dfrac{\partial \vec{D}}{\partial t}$

Q2.16.2 MCQ 2016 1M ○○○

If a right-handed circularly polarized wave is incident normally on a plane perfect conductor, then the reflected wave will be

A. right-handed circularly polarized

B. left-handed circularly polarized

C. elliptically polarized with a tilt angle of 45°

D. horizontally polarized

Q2.16.3 MCQ 2016 1M ○○○

Let the electric field vector of a plane electromagnetic wave propagating in a homogenous medium be expressed as,

$$E = \hat{x}E_x e^{-j(\omega t - \beta z)}$$

where the propagation constant β is function of the angular frequency ω. Assume that, $\beta(\omega)$ and E_x are known and are real. From the information available, which one of the following cannot be determined?

A. The type of polarization of the wave.

B. The group velocity of the wave.

C. The phase velocity of the wave.

D. The power flux through the $z = 0$ plane.

Q2.15.1 MCQ 2015 2M

The electric field of a plane wave propagating in a lossless non-magnetic medium is given by the following expression:

$$\vec{E}(z,t) = \hat{a}_x 5\cos(2\pi \times 10^9 t + \beta z)$$
$$+\hat{a}_y 3\cos\left(2\pi \times 10^9 t + \beta z - \frac{\pi}{2}\right)$$

The polarization of the wave is

A. right-handed circular **B.** left-handed elliptical
C. right-handed elliptical **D.** linear

Q2.15.2 MCQ 2015 1M

The electric field of a uniform plane electromagnetic wave is

$$\vec{E} = (\hat{a}_x + j2\hat{a}_y)\exp[j(2\pi \times 10^7 t - 0.2z)]$$

The polarization of the wave is

A. right-handed circular **B.** right-handed elliptical
C. left-handed circular **D.** left-handed elliptical

Q2.15.3 NAT 2015 1M

The electric field intensity of a plane wave travelling in free space is given by the following expression:

$$\vec{E}(x,t) = \hat{a}_y 24\pi\cos(\omega t - k_o x)\ \text{V/m}$$

In this field, consider a square area 10 cm × 10 cm on a plane $x + y = 1$. The total time averaged power (in mW) passing through the square area is _____.

Q2.15.4 NAT 2015 2M

Consider a uniform plane wave with amplitude (E_0) of 10 V/m and 1.1 GHz frequency travelling an air, and incident normally on a dielectric medium with complex relative permittivity (ε_r) and permeability (μ_r) as shown in figure.

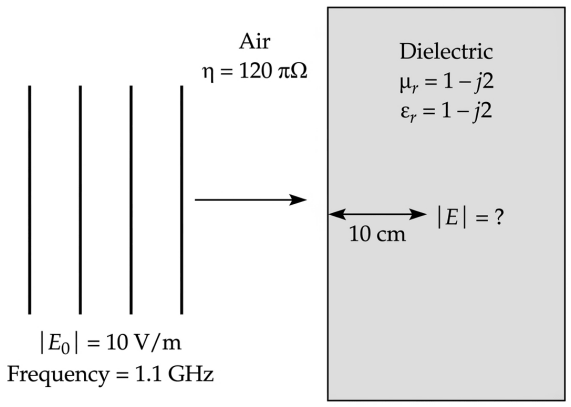


Fig. Q2.15.4

The magnitude of the transmitted electric field component (in V/m) after it has travelled a distance of 10 cm inside the dielectric region is _____.

Q2.15.5 NAT 2015 1M

The electric field component of a plane wave travelling in lossless dielectric medium is given by

$$\vec{E}(z,t) = \hat{a}_y 2\cos\left(10^8 t - \frac{z}{\sqrt{2}}\right)\ \text{V/m}$$

The wavelength (in m) for the wave is _____.

Q2.14.1 NAT 2014 2M

Assume that a plane wave in air with an electric field $\vec{E} = 10\cos(\omega t - 3x - \sqrt{3}z)\hat{a}_y$ V/m is incident on a non-magnetic dielectric slab or relative permittivity 3 which covers the region $z > 0$. The angle of transmission in the dielectric slab is _____ degrees.

Q2.14.2 MCQ 2014 2M

If the electric field of a plane wave is

$$\vec{E}(z,t) = \hat{x}3\cos(\omega t - kz + 30°) - \hat{y}4\sin(\omega t - kz + 45°)$$

mV/m

The polarization state of the plane wave is

A. left elliptical **B.** left circular
C. right elliptical **D.** right circular

Q2.14.3 MCQ 2014 1M

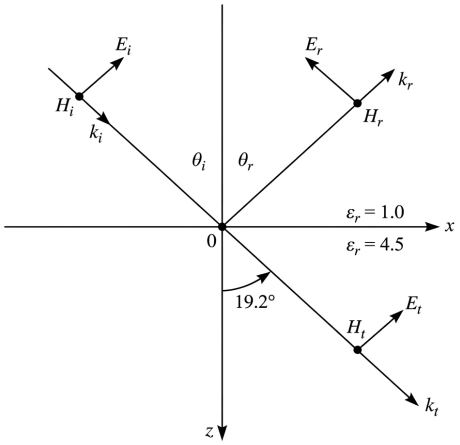
Which one of the following field patterns represents a TEM wave travelling in the positive x-direction?

A. $E = +8\hat{y}, H = -4\hat{z}$ **B.** $E = -2\hat{y}, H = -3\hat{z}$
C. $E = +2\hat{z}, H = +2\hat{y}$ **D.** $E = -3\hat{y}, H = +4\hat{z}$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions (Q2.13.1 and Q2.13.2): A monochromatic plane wave of wavelength $\lambda = 600\ \mu m$ is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$ and $\vec{E}_t$ denote incident, reflected, and transmitted electric field vectors associated with the wave.



Q2.13.1 MCQ 2013 2M

A monochromatic plane wave of wavelength $\lambda = 600\ \mu m$ is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$ and $\vec{E}_t$ denote the incident, reflected and transmitted electric field vectors associated with the wave.

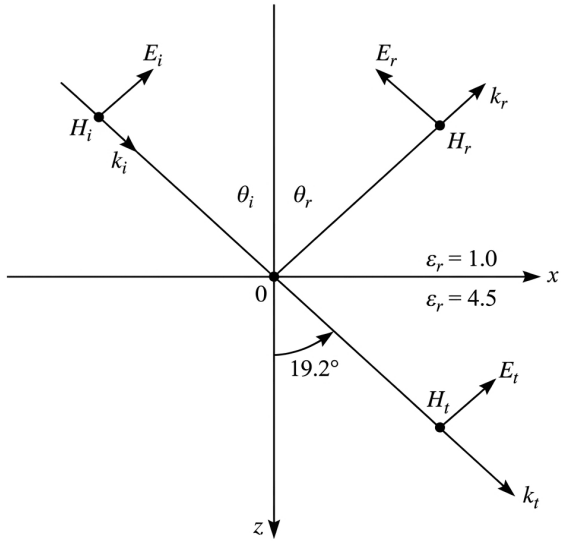


Fig. Q2.13.1

The angle of incidence θ_i and the expression for $\vec{E}_i$ are

A. 60° and $\frac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}}$ V/m

B. 45° and $\frac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 z}{3}}$ V/m

C. 45° and $\frac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}}$ V/m

D. 60° and $\frac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 z}{3}}$ V/m

Q2.13.2 MCQ 2013 2M

The expression for $\vec{E}_r$ is

A. $-0.23\frac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x-z)}{3\sqrt{2}}}$ V/m

B. $-\frac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{j\frac{\pi \times 10^4 z}{3}}$ V/m

C. $0.44\frac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x-z)}{3}}$ V/m

D. $\frac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3}}$ V/m

Q2.12.1 MCQ 2012 1M

A plane wave propagating in air with

$$\vec{E} = (8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z)e^{j(\omega t + 3x - 4y)}\ \text{V/m}$$

is incident on a perfectly conducting slab positioned at $x \le 0$. The $\vec{E}$ field of the reflected waves is

A. $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m

B. $(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m

C. $(8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m

D. $(8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m

Q2.12.2 MCQ 2012 1M

The electric field a uniform plane electromagnetic wave in free space, along the positive x-direction, is given by

$$\vec{E} = 10(\hat{a}_y + j\hat{a}_z)e^{-j25x}$$

The frequency and polarization of the wave, respectively are

A. 1.2 GHz and left circular

B. 4 GHz and left circular

C. 1.2 GHz and right circular

D. 4 Hz and right circular

Q2.11.1 MCQ 2011 2M

The electric and magnetic fields for a TEM wave of frequency 14 GHz in a homogeneous medium of relative permittivity ε_r and relative permeability $\mu_r = 1$ are given by

$$\vec{E} = E_p e^{j(\omega t - 280\pi y)} \hat{u}_z \text{ V/m}$$

$$\vec{H} = 3e^{j(\omega t - 280\pi y)} \hat{u}_x \text{ V/m}$$

Assuming the speed of light in free space to be 3×10^8 m/s, intrinsic impedance of free space to be 120π, the relative permittivity ε_r of the medium and the electric field amplitude E_p are

A. $\varepsilon_r = 3, E_p = 120\pi$ **B.** $\varepsilon_r = 3, E_p = 360\pi$

C. $\varepsilon_r = 9, E_p = 360\pi$ **D.** $\varepsilon_r = 9, E_p = 120\pi$

Q2.10.1 MCQ 2010 2M

A plane wave having the electric field component, $\vec{E}_i = 24\cos(3 \times 10^8 t - \beta y)\hat{a}_z$ V/m and travelling in free space is incident normally on a lossless medium with $\mu = \mu_0$ and $\epsilon = 9\epsilon_0$ which occupies the region $y \geq 0$. The reflected magnetic field component is given by

A. $\frac{1}{10\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

B. $\frac{1}{20\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

C. $-\frac{1}{20\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

D. $-\frac{1}{10\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

Q2.10.2 MCQ 2010 1M

The electric field component of a time harmonic plane EM wave travelling in a non-magnetic lossless dielectric medium has an amplitude of 1 V/m. If the relative permittivity of the medium is 4, the magnitude of the time average power density vector (in W/m^2) is

A. $\frac{1}{30\pi}$ **B.** $\frac{1}{60\pi}$

C. $\frac{1}{120\pi}$ **D.** $\frac{1}{240\pi}$

Q2.08.1 MCQ 2008 2M

A uniform plane wave in the free space is normally incident on an infinitely thick dielectric slab (dielectric constant $\varepsilon_r = 9$). The magnitude of the reflection coefficient is

A. 0 **B.** 0.3

C. 0.5 **D.** 0.8

Q2.07.1 MCQ 2007 1M

A plane wave of wavelength λ is travelling in a direction making an angle 30° with positive x-axis and 90° with positive y-axis. The $\vec{E}$ field of the plane wave can be represented as (E_0 is a constant)

A. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\sqrt{3}\pi}{\lambda}x - \frac{\pi}{\lambda}z\right)}$

B. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x - \frac{\sqrt{3}\pi}{\lambda}z\right)}$

C. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t + \frac{\sqrt{3}\pi}{\lambda}x + \frac{\pi}{\lambda}z\right)}$

D. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x + \frac{\sqrt{3}\pi}{\lambda}z\right)}$

Q2.07.2 MCQ 2007 2M

A Right Circularly Polarized (RCP) plane wave is incident at an angle of 60° to the normal, on an air-dielectric interface. If the reflected wave is linearly polarized, the relative dielectric constant ε_{r2} is

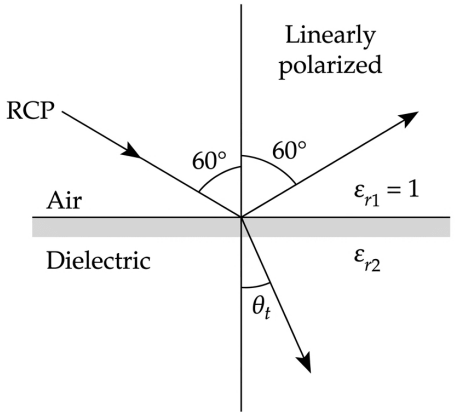


Fig. Q2.07.2

A. $\sqrt{2}$ **B.** $\sqrt{3}$

C. 2 **D.** 3

Q2.07.3 MCQ 2007 2M

The $\vec{H}$ field (in A/m) of a plane wave propagating in free space is given by

$$\vec{H} = \hat{x}\frac{5\sqrt{3}}{\eta_0}\cos(\omega t - \beta z) + \hat{y}\frac{5}{\eta_0}\sin\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

The time average power flow density in watts is

A. $\frac{\eta_0}{100}$ **B.** $\frac{100}{\eta_0}$

C. $50\eta_0^2$ **D.** $\frac{50}{\eta_0}$

Q2.06.1 MCQ 2006 2M

When a plane wave travelling in free-space is incident normally on a medium having $\epsilon_r = 4.0$, the fraction of power transmitted into the medium is given by

A. $\frac{8}{9}$ **B.** $\frac{1}{2}$

C. $\frac{1}{3}$ **D.** $\frac{5}{6}$

Q2.06.2 MCQ 2006 2M

A medium of relative permittivity $\epsilon_{r2} = 2$ forms an interface with free-space. A point source of electromagnetic energy is located in the medium at a depth of 1 meter from the interface. Due to the total internal reflection, the transmitted beam has a circular cross-section over the interface. The area of the beam cross-section at the interface is given by

A. 2π m^2 **B.** π^2 m^2

C. $\frac{\pi}{2}$ m^2 **D.** π m^2

Q2.06.3 MCQ 2006 1M

The electric field on an electromagnetic wave propagating in the positive z-direction is given by

$$E = \hat{a}_x \sin(\omega t - \beta z) + \hat{a}_y \sin\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

The wave is

A. linearly polarized in the z-direction.

B. elliptically polarized.

C. left-hand circularly polarized.

D. right-hand circularly polarized.

Q2.05.1 MCQ 2005 1M

Refractive index of glass is 1.5. Find the wavelength of a beam of light with frequency of 10^{14} Hz in glass. Assume velocity of light is 3×10^8 m/s in vacuum.

A. 3 μm **B.** 3 mm

C. 2 μm **D.** 1 mm

Q2.05.2 MCQ 2005 1M

The magnetic field intensity vector of a plane wave is given by

$$\vec{H}(x,t) = 10\sin(50000t + 0.004x + 30)\hat{a}_y$$

where, $\hat{a}_y$ denotes the unit vector in y-direction. The wave is propagating with a phase velocity

A. 5×10^4 m/s **B.** 3×10^8 m/s

C. 1.25×10^7 m/s **D.** 3×10^6 m/s

Q2.04.1 MCQ 2004 2M

If,

$$\vec{E} = (\hat{a}_x + j\hat{a}_y)\, e^{jkz - j\omega t} \text{ and}$$

$$\vec{H} = \left(\frac{k}{\omega\mu}\right)(\hat{a}_y + j\hat{a}_x)\, e^{jkz - j\omega t}$$

the time averaged pointing vector is

A. null vector **B.** $\left(\frac{k}{\omega\mu}\right)\hat{a}_z$

C. $\left(\frac{2k}{\omega\mu}\right)\hat{a}_z$ **D.** $\left(\frac{k}{2\omega\mu}\right)\hat{a}_z$

Q2.04.2 MCQ 2004 2M

A parallel plate air-filled capacitor has plate area of 10^{-4} m^2 and plate separation of 10^{-3} m. It is connected to a 0.5 V, 3.6 GHz source. The magnitude of the displacement current is ($\varepsilon_0 = 1/36\pi \times 10^{-9}$ F/m)

A. 10 mA **B.** 100 mA

C. 10 A **D.** 1.59 mA

Q2.04.3 MCQ 2004 2M

A plane electromagnetic wave propagating in free space is incident normally on a large slab of lossless, non-magnetic, dielectric material with $\epsilon > \epsilon_0$. Maxima and minima are observed when the electric field is measured in front of the slab. The maximum electric field is found to be 5 times the minimum field. The intrinsic impedance of the medium should be

A. 120 $\pi\Omega$ **B.** 60 $\pi\Omega$

C. 600 $\pi\Omega$ **D.** 24 $\pi\Omega$

Q2.03.1 MCQ 2003 2M

If the electric field intensity associated with a uniform plane electromagnetic wave travelling in a perfect dielectric medium is given by

$$E(z,t) = 10\cos(2\pi \times 10^7 t - 0.1\pi z) \text{ Volt/m}$$

The velocity of the travelling wave is

A. 3.00×10^8 m/sec **B.** 2.00×10^8 m/sec

C. 6.28×10^7 m/sec **D.** 2.00×10^7 m/sec

Q2.03.2 MCQ 2003 2M

A uniform plane wave travelling in air is incident on the plane boundary between air and another dielectric medium with $\varepsilon_r = 4$. The reflection coefficient for electric field at normal incidence, is

A. zero **B.** $0.5\angle 180°$

C. $0.333\angle 0°$ **D.** $0.333\angle 180°$

Q2.03.3 MCQ 2003 1M

The depth of penetration of electromagnetic wave in a medium having conductivity σ at a frequency of 1 MHz is 25 cm. The depth of penetration at a frequency of 4 MHz will be

A. 6.25 cm **B.** 12.50 cm

C. 50.00 cm **D.** 100.00 cm

Q2.02.1 MCQ 2002 2M

Distilled water at 25°C is characterized by

$$\sigma = 1.7 \times 10^{-4} \text{ ℧/m and } \epsilon = 78\epsilon_0$$

at a frequency of 3 GHz. Its loss tangent $\tan\delta$ is

A. 1.3×10^{-5} **B.** 1.3×10^{-3}

C. $1.7 \times 10^{-4}/78$ **D.** $1.7 \times 10^{-4}/(78\ \epsilon_0)$

Q2.02.2 MCQ 2002 2M

A plane wave is characterized by

$$\vec{E} = \left(0.5\hat{x} + \hat{y}e^{j\pi/2}\right) e^{j\omega t - jkz}$$

The wave is

A. linearly polarized **B.** circularly polarized

C. elliptically polarized **D.** unpolarized

Q2.01.1 MCQ 2001 2M

A material has conductivity of 10^{-2} ℧/m and a relative permittivity of 4. The frequency at which the conduction current in the medium is equal to the displacement current is

A. 45 MHz **B.** 90 MHz

C. 450 MHz **D.** 900 MHz

Q2.01.2 MCQ 2001 1M

If a plane electromagnetic wave satisfies the equation $\frac{\partial^2 E_x}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 E_x}{\partial t^2}$, the wave propagates in the

A. x-direction

B. z-direction

C. y-direction

D. xy-plane at an angle of 45° between the x and z-directions

Q2.00.1 MCQ 2000 2M

Two coaxial cables 1 and 2 are filled with different dielectric constants ϵ_{r1} and ϵ_{r2} respectively. The ratio of wavelengths in the two cables, (λ_1/λ_2) is

A. $\sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}}$ **B.** $\sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}$

C. $\frac{\epsilon_{r1}}{\epsilon_{r2}}$ **D.** $\frac{\epsilon_{r2}}{\epsilon_{r1}}$

Q2.00.2 MCQ 2000 2M

A uniform plane wave in air impinges at 45° angle on a lossless dielectric material with dielectric constant ϵ_r. The transmitted wave propagates in a 30° direction with respect to the normal. The value of ϵ_r is

A. 1.5 **B.** $\sqrt{1.5}$

C. 2 **D.** $\sqrt{2}$

Q2.00.3 MCQ 2000 1M

A TEM wave is incident normally upon a perfect conductor. The E and H fields at the boundary will be respectively

A. minimum and minimum **B.** maximum and maximum

C. minimum and maximum **D.** maximum and minimum

Q2.99.1 MCQ 1999 2M

A plane wave propagation through a medium [$\epsilon_r = 8, \mu_r = 2$ and $\sigma = 0$] has its electric field given by

$$\vec{E} = 0.5\sin(10^8 t - \beta z)\ \text{V/m}$$

The wave impedance, (in Ω) is

A. 377 **B.** $198.5\angle 180°$

C. $182.9\angle 14°$ **D.** 188.5

Q2.99.2 MCQ 1999 1M

Identify which one of the following will not satisfy the wave equation?

A. $50e^{j(\omega t - 3z)}$ **B.** $\sin[\omega(10z + 5t)]$

C. $\cos(y^2 + 5t)$ **D.** $\sin(x)\cos(t)$

Q2.98.1 MCQ 1998 1M

The Maxwell's equation, $\nabla \times \vec{H} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t}$ is based on

A. Ampere's law **B.** Gauss's law

C. Faraday's law **D.** Coulomb's law

Q2.98.2 MCQ 1998 1M

A loop is rotating about the y-axis in a magnetic field $\vec{B} = B_0\cos(\omega t + \phi)\hat{a}_x$ T. The voltage in the loop is

A. zero

B. due to rotation only

C. due to transformer action only

D. due to both rotation and transformer action

Q2.98.3 MCQ 1998 1M

The time averaged pointing vector, (in W/m^2), for a wave with $\vec{E} = 24e^{j(\omega t + \beta z)}\hat{a}_y$ V/m in free space is

A. $-\dfrac{2.4}{\pi}\hat{a}_z$ **B.** $\dfrac{2.4}{\pi}\hat{a}_z$

C. $\dfrac{4.8}{\pi}\hat{a}_z$ **D.** $-\dfrac{4.8}{\pi}\hat{a}_z$

Q2.98.4 MCQ 1998 1M

The polarization of a wave with electric field vector $\vec{E} = E_o e^{j(\omega t - \beta z)}(\hat{a}_x + \hat{a}_y)$ is

A. linear **B.** elliptical

C. left hand circular **D.** right hand circular

Q2.98.5 MCQ 1998 1M

The depth of penetration of a wave in a lossy dielectric increases with increasing

A. conductivity **B.** permeability

C. wavelength **D.** permittivity

Q2.98.6 MCQ 1998 1M

The wavelength of a wave with propagation constant $(0.1\pi + j0.2\pi)$ m^{-1} is

A. $\dfrac{2}{\sqrt{0.05}}$ m **B.** 10 m

C. 20 m **D.** 30 m

Q2.98.7 MCQ 1998 1M

The intrinsic impedance of copper at high frequency is

A. purely resistive

B. purely inductive

C. complex with a capacitive component

D. complex with an inductive component

Q2.96.1 MCQ 1996 2M

Some unknown material has a conductivity of 10^6 ℧/m and a permeability of $4\pi \times 10^{-7}$ H/m. The skin depth for the material at 1 GHz is

A. 15.9 μm **B.** 20.9 μm

C. 25.9 μm **D.** 30.9 μm

Q2.96.2 MCQ 1996 2M

A uniform plane wave in air is normally incident on an infinitely thick slab. If the refractive index of the glass slab is 1.5, then the percentage of the incident power that is reflected from the air glass interface is

A. 0% **B.** 4%

C. 20% **D.** 100%

Q2.95.1 MCQ 1995 1M

Copper behaves as a

A. conductor always

B. conductor or dielectric depending on the applied electric field strength

C. conductor or dielectric depending on the frequency

D. conductor or dielectric depending on the electric current density

Q2.95.2 MCQ 1995 1M

The intrinsic impedance of a lossy dielectric medium is given by

A. $\frac{j\omega\mu}{\sigma}$ **B.** $\frac{j\omega\epsilon}{\mu}$

C. $\sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$ **D.** $\sqrt{\frac{\mu}{\epsilon}}$

Q2.94.1 MCQ 1994 1M

A plane electromagnetic wave travelling along the $+z$-direction, has its electric field given by $E_x = 2\cos(\omega t)$ and $E_y = 2\cos(\omega t + 90^\circ)$ The wave is

A. linearly polarized **B.** right circularly polarized

C. left circularly polarized **D.** elliptically polarized

Q2.93.1 MSQ 1993 2M

A plane wave is incident normally on a perfect conductor as shown in figure. Here E_x^i, H_y^i and $\vec{p}^i$ are electric field, magnetic field and pointing vector, respectively, for the incident wave. The reflected waver should have

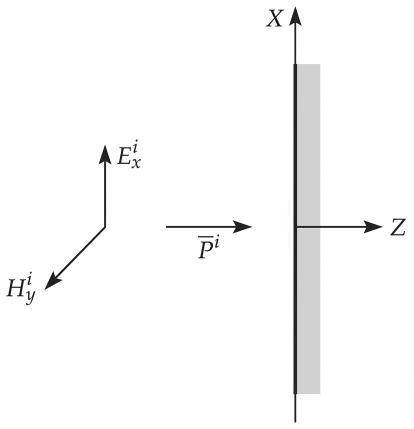


Fig. Q2.93.1

A. $E_x^r = -E_x^i$ **B.** $H_y^r = -H_y^i$

C. $\vec{p}^r = -\vec{p}^i$ **D.** $E_x^r = E_x^i$

Q2.93.2 MCQ 1993 2M

A material is described by the following electrical parameters at a frequency of 10 GHz, $\sigma = 10^6$ $\mho$/m, $\mu = \mu_0$ and $\epsilon/\epsilon_0 = 10$. The material at this frequency is considered to be $\left(\epsilon_0 = \frac{1}{36\pi} \times 10^{-9}\ \text{F/m}\right)$.

A. a good conductor

B. a good dielectric

C. neither a good conductor, nor a good dielectric

D. a good magnetic material

Q2.91.1 MSQ 1991 2M

The electric field component of a uniform plane electromagnetic wave propagation in the Y-direction in a lossless medium will satisfy the equation

☐ **A.** $\frac{\partial^2 E_y}{\partial y^2} = \mu\epsilon\frac{\partial^2 E_y}{\partial t^2}$

☐ **B.** $\frac{\partial^2 E_y}{\partial x^2} = \mu\epsilon\frac{\partial^2 E_y}{\partial t^2}$

☐ **C.** $\frac{\partial^2 E_x}{\partial y^2} = \mu\epsilon\frac{\partial^2 E_x}{\partial t^2}$

☐ **D.** $\frac{\sqrt{E_x^2 + E_z^2}}{\sqrt{H_x^2 + H_z^2}} = \sqrt{\frac{\mu}{\epsilon}}$

Q2.90.1 MCQ 1990 2M

The incoming solar radiation at a place on the surface of the earth is 1.2 kW/m^2. The amplitude of the electric field corresponding to this incident power is nearly equal to

A. 80 mV/m **B.** 2.5 V/m

C. 30 V/m **D.** 950 V/m

Q2.89.1 MCQ 1989 2M

The skin depth of copper at a frequency of 3 GHz is 1 micro (10^{-6} meter). At 12 GHz, for a non magnetic conductor whose conductivity is 1/9 times that of copper, the skin depth would be

A. $\sqrt{9\times 4}$ microns **B.** $\sqrt{\frac{9}{4}}$ microns

C. $\sqrt{\frac{4}{9}}$ microns **D.** $\frac{1}{\sqrt{9\times 4}}$ microns

Q2.88.1 MCQ 1988 2M

In a good conductor the phase relation between the tangential components of electric field E_t and the magnetic field H_t is as follows:

A. E_t and H_t are in phase.

B. E_t and H_t are out of phase.

C. H_t leads E_t by 90°.

D. E_t leads H_t by 45°.

Q2.87.1 MCQ 1987 2M

For an electromagnetic wave incident from one medium to a second medium, total reflection takes place when

A. the angle of incidence is equal to the Brewster angle with E field perpendicular to the plane of incidence.

B. the angle of incidence is equal to the Brewster angle with E field parallel to the plane of incidence.

C. the angle of incidence is equal to the critical angle with the wave moving from the denser medium to a rarer medium.

D. the angle of incidence is equal to the critical angle with the wave moving from a rarer medium to a rarer medium.

Answer Key

Q2.26.1 A	Q2.26.2 A,B	Q2.24.1 D
Q2.23.1 A,B,D	Q2.21.1 D	Q2.20.1 1
Q2.18.1 B	Q2.18.2 2	Q2.17.1 4.33
Q2.17.2 C	Q2.16.1 C	Q2.16.2 B
Q2.16.3 D	Q2.15.1 B	Q2.15.2 D
Q2.15.3 53.0 to 53.6	Q2.15.4 0.1	Q2.15.5 8.85 to 8.90
Q2.14.1 30	Q2.14.2 A	Q2.14.3 B
Q2.13.1 C	Q2.13.2 A	Q2.12.1 C
Q2.12.2 A	Q2.11.1 D	Q2.10.1 A
Q2.10.2 C	Q2.08.1 C	Q2.07.1 A
Q2.07.2 D	Q2.07.3 D	Q2.06.1 A
Q2.06.2 D	Q2.06.3 C	Q2.05.1 C
Q2.05.2 C	Q2.04.1 A	Q2.04.2 A
Q2.04.3 D	Q2.03.1 B	Q2.03.2 D
Q2.03.3 B	Q2.02.1 A	Q2.02.2 C
Q2.01.1 A	Q2.01.2 B	Q2.00.1 B
Q2.00.2 C	Q2.00.3 C	Q2.99.1 D
Q2.99.2 C	Q2.98.1 A	Q2.98.2 D
Q2.98.3 A	Q2.98.4 A	Q2.98.5 D
Q2.98.6 B	Q2.98.7 D	Q2.96.1 A
Q2.96.2 B	Q2.95.1 A	Q2.95.2 C
Q2.94.1 C	Q2.93.1 A,C	Q2.93.2 A
Q2.91.1 C,D	Q2.90.1 D	Q2.89.1 B
Q2.88.1 D	Q2.87.1 C	

UNIT

III

Transmission Lines & Smith Chart

Topics

1. Types of Transmission Lines
2. Impedance Calculations
3. Standing Wave Patterns
4. DC Analysis
5. Power Transfer
6. Scattering Parameters
7. Smith Chart

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 73	**MCQ:** 53	**MSQ:** 3	**NAT:** 17
1M: 28	**2M:** 45	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2013)–10M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

A complex load (in Ω) is represented as $\Gamma_L = 0.5\angle 30°$ on the Smith chart. A co-axial cable with a characteristic impedance of 50 Ω is connected to the load. The new input impedance of the load now moves to a diametrically opposite point on the same Γ circle on the Smith chart. Which option is the nearest input impedance of the cable connected load (in Ω)?

A. 20.7 - j5.1 **B.** 17.7 - j11.8

C. 97.5 - j65.0 **D.** 97.5 + j65.0

Q3.26.2 NAT 2026 1M

For a lossless passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at −3 dB. For a lossy passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at −4 dB. The percentage of power dissipated in the lossy network at the intersection frequency is _____ (rounded off to two decimal places)

Q3.25.1 NAT 2025 2M

A 50 Ω lossless transmission line is terminated with a load Z_L of $(50 - j75)$ Ω. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

Q3.24.1 NAT 2024 2M

A lossless transmission line with characteristic impedance $Z_0 = 50$ Ω is terminated with an unknown load. The magnitude of the reflection coefficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____.

Q3.24.2 MCQ 2024 2M

Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances z_{inA}, z_{inB}, z_{inC} and z_{inD} (indicated in the figure) are _____.

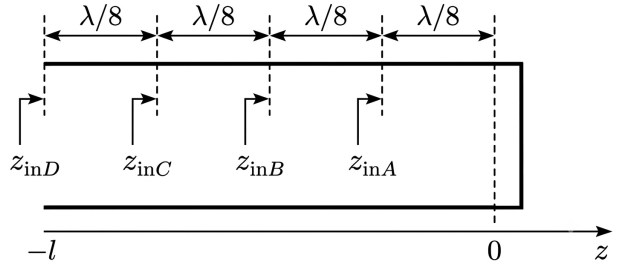


Fig. Q3.24.2

A. $z_{inA} = +0.4j\ \Omega$, $z_{inB} = \infty$, $z_{inC} = -0.4j\ \Omega$, $z_{inD} = 0$

B. $z_{inA} = -1j\ \Omega$, $z_{inB} = 0$, $z_{inC} = +1j\ \Omega$, $z_{inD} = \infty$

C. $z_{inA} = \infty$, $z_{inB} = +0.4j\ \Omega$, $z_{inC} = 0$, $z_{inD} = +0.4j\ \Omega$

D. $z_{inA} = +1j\ \Omega$, $z_{inB} = \infty$, $z_{inC} = -1j\ \Omega$, $z_{inD} = 0$

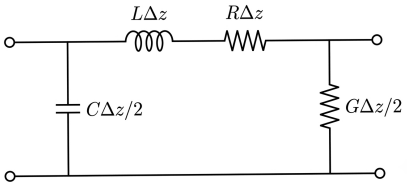
Q3.23.1 NAT 2023 2M

A transparent dielectric coating is applied to glass ($\epsilon_r = 4, \mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75\mu$m). The minimum thickness of the dielectric coating, in μm, that can be used is _____ (rounded off to two decimal places).

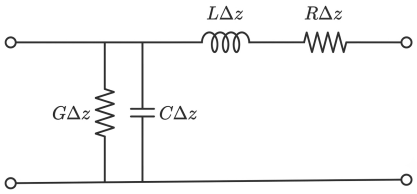
Q3.23.2 MSQ 2023 2M

The following circuit(s) representing a lumped element equivalent of an infinitesimal section of a transmission line is/are

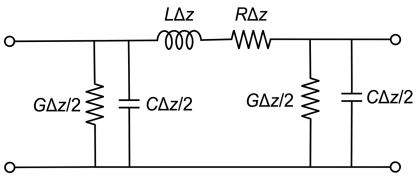
☐ **A.**



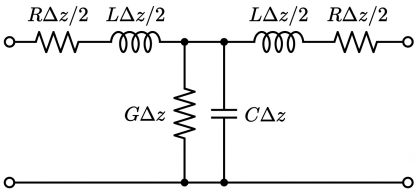
☐ **B.**



☐ **C.**



☐ **D.**



Q3.23.3 MSQ 2023 2M

The standing wave ratio on a 50 Ω lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m, respectively, are

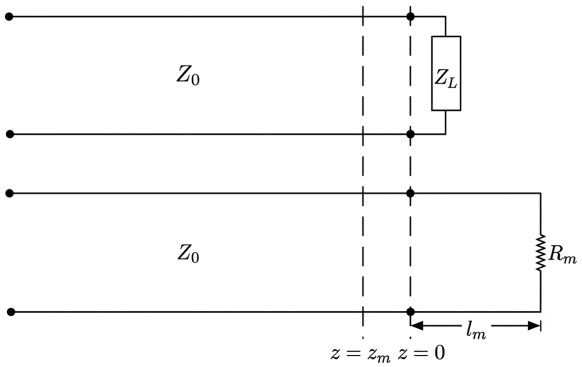


Fig. Q3.23.3

- **A.** $R_m = 100\ \Omega, l_m = 20$ cm
- **B.** $R_m = 25\ \Omega, l_m = 20$ cm
- **C.** $R_m = 100\ \Omega, l_m = 5$ cm
- **D.** $R_m = 25\ \Omega, l_m = 5$ cm

Q3.23.4 MCQ 2023 2M

The S-parameters of a two port network are given as

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

with reference to Z_0. Two lossless transmission line sections of electrical lengths $\theta_1 = \beta l_1$ and $\theta_2 = \beta l_2$ are added to the input and output ports for measurement purposes, respectively. The S-parameters $[S']$ of the resultant two port network are

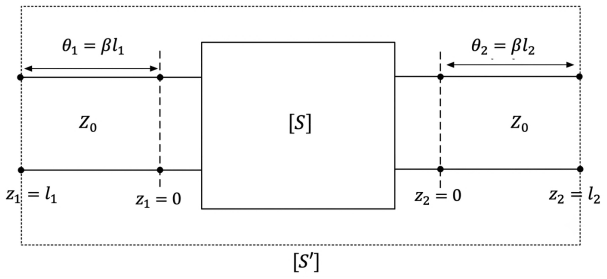


Fig. Q3.23.4

- **A.** $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$
- **B.** $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
- **C.** $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
- **D.** $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

Q3.21.1 MCQ 2021 2M

The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50\ \Omega$ with a load impedance Z_L. A quarter wave line having a characteristic impedance $Z_1 = 75\ \Omega$ is connected to Z_L. Two stubs having characteristic impedance of 75 Ω each are connected to this quarter wave line. One is a short-circuited (S.C.) stub of length of 0.25λ connected across PQ and the other one is an open-circuit (O.C.) stub of length 0.5λ connected across RS.

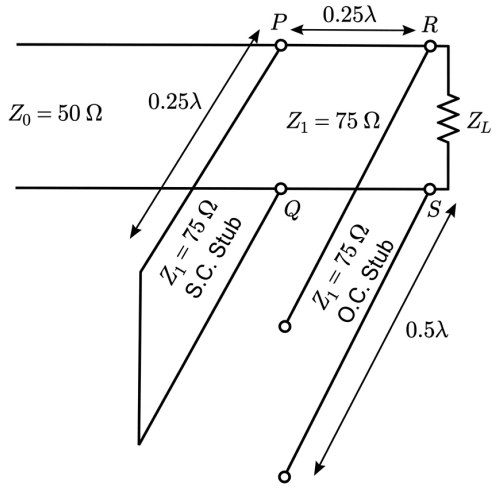


Fig. Q3.21.1

The impedance matching is achieved when the real part of Z_L is

A. 50.0 Ω

B. 33.3 Ω

C. 75.0 Ω

D. 112.5 Ω

Q3.20.1 MCQ 2020 1M

The impedances $Z = jX$ for all X in the range $(-\infty, \infty)$, map to the Smith chart as,

A. a circle of radius 0.5 with center at $(0.5, 0)$.

B. a point at the center of the chart.

C. a line passing through the center of the chart.

D. a circle of radius 1 with center at $(0, 0)$.

Q3.20.2 NAT 2020 1M

A transmission line of length $3\lambda/4$ and having a characteristic impedance of 50 Ω is terminated with a load of 400 Ω. The impedance (rounded off to two decimal places) seen at the input end of the transmission line is _____ Ω.

Q3.18.1 NAT 2018 1M

A lossy transmission line has resistance per unit length $R = 0.05\ \Omega/\text{m}$. The line is distortionless and has characteristic impedance of 50 Ω. The attenuation constant (in Np/m correct to three decimal places) of the line is _____.

Q3.18.2 MCQ 2018 1M

The points P, Q and R shows on the Smith chart (normalized impedance chart) in the following figure represent:

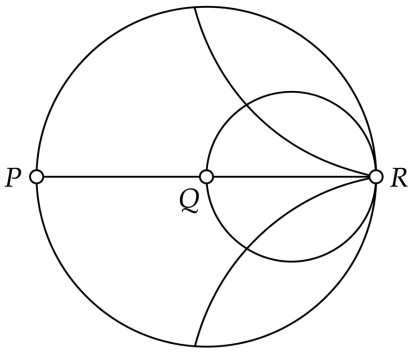


Fig. Q3.18.2

A. P : Open-circuit, Q : Short-circuit, R : Matched load

B. P : Open-circuit, Q : Matched load, R : Short-circuit

C. P : Short circuit, Q : Matched load, R : Open-circuit

D. P : Short circuit, Q : Open-circuit, R : Matched load

Q3.17.1 NAT 2017 2M

An optical fibre is kept along the $\hat{z}$-direction. The refractive indices for the electric fields along $\hat{x}$ and $\hat{y}$-directions in the fibre are $n_x = 1.5000$ and $n_y = 1.5001$, respectively ($n_x \neq n_y$ due to the imperfection in the fibre cross-section). The free space wavelength of a light wave propagating in the fibre is 1.5 μm. If the light wave is circularly polarized at the input of the fibre, the minimum propagation distance after which is becomes linearly polarized (in centimeters), is _____.

Q3.17.2 NAT 2017 1M

A two-wire transmission line terminates in a television set. The VSWR measured on the line is 5.8. The percentage of power that is reflected from the television set is _____.

Q3.17.3 NAT 2017 1M

The voltage of an electromagnetic wave propagating in a coaxial cable with uniform characteristic impedance is $V(l) = e^{-\gamma l + j\omega t}$ Volts, where l is the distance along the length of the cable in meters, $\gamma = (0.1 + j40)\ \text{m}^{-1}$ is the complex propagation constant, and $\omega = 2\pi \times 10^6$ rad/s is the angular frequency. The absolute value of the attenuation in the cable (in dB/meter) is _____.

Q3.16.1 NAT 2016 1M

Light from free space is incident at an angle θ_i to the normal of the face of a step-index large core optical fibre. The core and cladding refractive indices are $n_1 = 1.5$ and $n_2 = 1.4$, respectively.

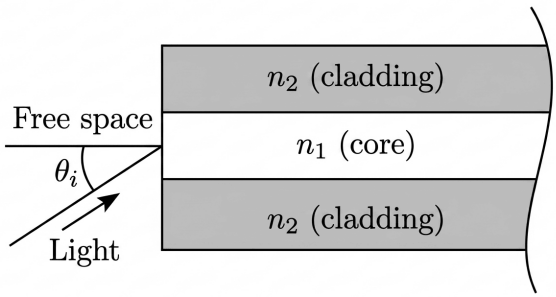


Fig. Q3.16.1

The maximum value of θ_i (in degrees) for which the incident light will be guided in the core of the fibre is _____.

Q3.16.2 NAT 2016 2M

A microwave circuit consisting of lossless transmission lines T_1 and T_2 is shown in the figure. The plot shows the magnitude of the input reflection coefficient $|\Gamma|$ as a function f. The phase velocity of the signal in the transmission lines is 2×10^8 m/s.

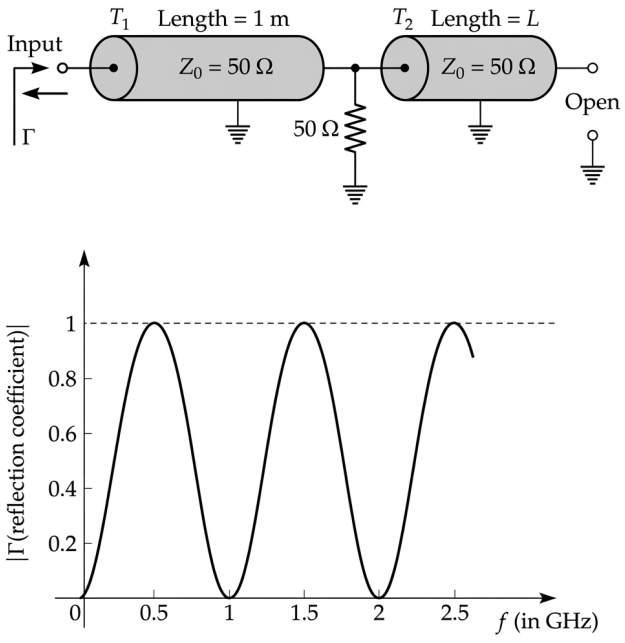


Fig. Q3.16.2

The length L (in meters) of T_2 is _____.

Q3.16.3 MCQ 2016 2M

A lossless microstrip transmission line consists of a trace of width w. It is drawn over a practically infinite ground plane and is separated by a dielectric slab of thickness t and relative permittivity $\varepsilon_r > 1$. The inductance per unit length and the characteristic impedance of this line are L and Z_0, respectively.

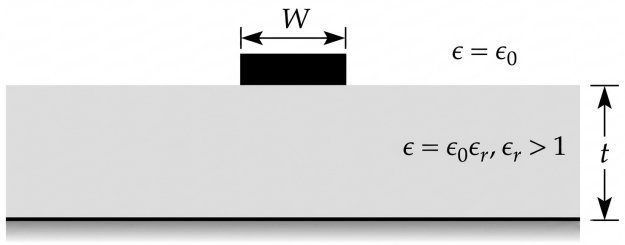


Fig. Q3.16.3

Which one of the following inequalities is always satisfied?

A. $Z_0 > \sqrt{\dfrac{Lt}{\varepsilon_0 \varepsilon_r w}}$ **B.** $Z_0 < \sqrt{\dfrac{Lt}{\varepsilon_0 \varepsilon_r w}}$

C. $Z_0 > \sqrt{\dfrac{Lw}{\varepsilon_0 \varepsilon_r t}}$ **D.** $Z_0 < \sqrt{\dfrac{Lw}{\varepsilon_0 \varepsilon_r t}}$

Q3.16.4 MCQ 2016 1M

The propagation constant of a lossy transmission line is $(2 + j5)\ \text{m}^{-1}$ and its characteristic impedance is $(50 + j0)\ \Omega$ at $\omega = 10^6$ rad s^{-1}. The value of the line constant L, C, R, G are respectively

A. $L = 200\,\mu$H/m, $C = 0.1\,\mu$F/m, $R = 50\,\Omega$/m, $G = 0.02$ S/m

B. $L = 250\,\mu$H/m, $C = 0.1\,\mu$F/m, $R = 100\,\Omega$/m, $G = 0.04$ S/m

C. $L = 200\,\mu$H/m, $C = 0.2\,\mu$F/m, $R = 100\,\Omega$/m, $G = 0.02$ S/m

D. $L = 250\,\mu$H/m, $C = 0.2\,\mu$F/m, $R = 50\,\Omega$/m, $G = 0.04$ S/m

Q3.15.1 NAT 2015 2M

A 200 m long transmission line having parameters shown in the figure is terminated into a load R_L. The line is connected to a 400 V source having source R_s through a switch which is closed at $t = 0$. The transient response of the circuit at the input of the line ($z = 0$) is also drawn in the figure.

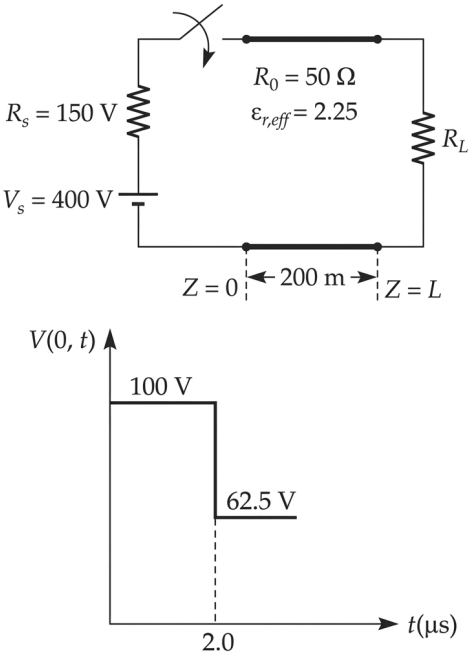


Fig. Q3.15.1

The value of R_L (in Ω) is _____.

Q3.15.2 NAT 2015 2M

A coaxial capacitor of inner radius 1 mm and outer radius 5 mm has a capacitance a per unit length of 172 pF/m. If the ratio of outer radius to inner is doubled, the capacitance per unit length (in pF/m) is _____.

Q3.15.3 MCQ 2015 2M

Consider the 3 m long lossless air-filled transmission line shown in the figure. It has a characteristic impedance of $120\pi\ \Omega$, is terminated by a short-circuit, and is excited with a frequency of 37.5 MHz. What is the nature of the input impedance (Z_{in})?

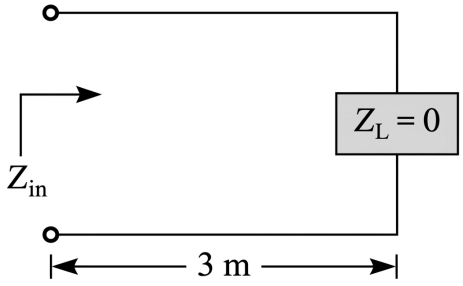


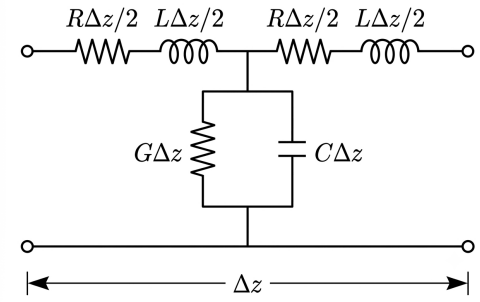
Fig. Q3.15.3

A. Open
B. Short
C. Inductive
D. Capacitive

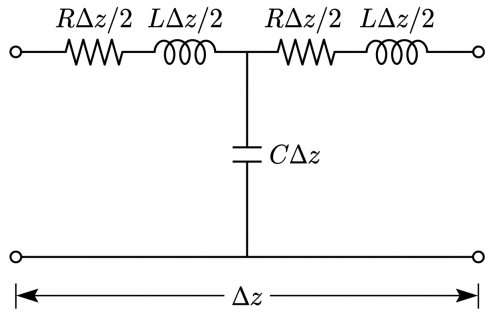
Q3.15.4 MCQ 2015 1M

A co-axial cable is made of two brass conductors. The spacing between the conductors is filled with Teflon ($\epsilon_r = 2.1, \tan\delta = 0$). Which of the following circuits can represent the lumped element model of a small piece of this cable having length Δz?

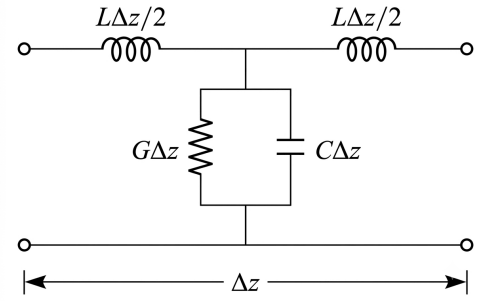
A.



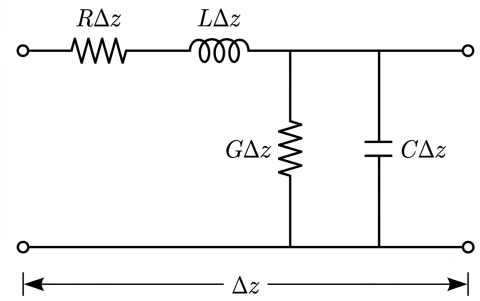
B.



C.



D.



Q3.14.1 MCQ 2014 1M

A two-port network has scattering parameters given by

$$[S] = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$$

If the port-2 of the two-port is short circuited, the s_{11} parameter for the resultant one-port network is

A. $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 + s_{22}}$

B. $\dfrac{s_{11} + s_{11}s_{22} - s_{12}s_{21}}{1 + s_{22}}$

C. $\dfrac{s_{11} + s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$

D. $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$

Q3.14.2 MCQ 2014 1M

In the following figure, the transmitter T_x sends a wideband modulated RF signal via a coaxial cable to the receiver R_x. The output impedance Z_T of T_x, the characteristic impedance Z_0 of the cable and the input impedance Z_R of R_x are all real.

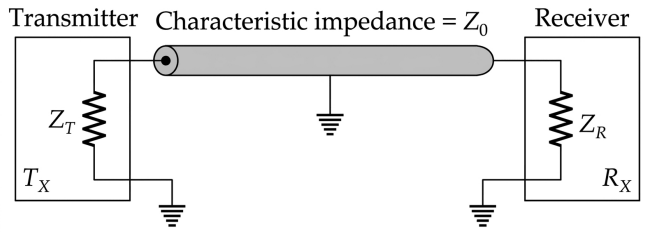


Fig. Q3.14.2

Which one of the following statements is true about the distortion of the received signal due to impedance mismatch?

A. The signal gets distorted if $Z_R \neq Z_0$ irrespective of the value of Z_T.

B. The signal gets distorted if $Z_T \neq Z_0$, irrespective of the value of Z_R.

C. Signal distortion implies impedance mismatch at both ends, $Z_T \neq Z_0$ and $Z_R \neq Z_0$.

D. Impedance mismatches do not result in signal distortion but reduce power transfer efficiency.

Q3.14.3 NAT 2014 2M

In the transmission line shown, the impedance Z_{in} (in Ω) between node A and the ground is _____.

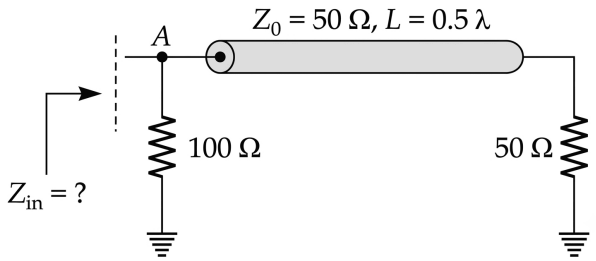


Fig. Q3.14.3

Q3.14.4 NAT 2014 1M

To maximize power transfer, a lossless transmission line is to be matched to a resistive load impedance via a $\lambda/4$ transformer as shown.

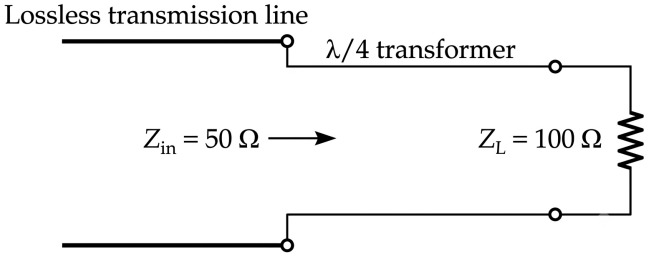


Fig. Q3.14.4

The characteristic impedance (in Ω) of the $\lambda/4$ transformer is _____.

Q3.14.5 NAT 2014 2M

The input impedance of $\lambda/8$ section of a lossless transmission line of characteristic impedance 50 Ω is found to be real when the other end is terminated by a load $Z_L = (R + jX)$ Ω. If X is 30 Ω, the value of R (in Ω) is _____.

Q3.14.6 MCQ 2014 2M

For a parallel plate transmission line, let v be the speed of propagation and z be the characteristic impedance. Neglecting fringe effects, a reduction of the spacing between the plates by a factor of two results in

A. having of v and no change in z.

B. no changes in v and having of z.

C. no change in both v and z.

D. halving of both v and z.

Q3.13.1 MCQ 2013 1M

The return loss of a device is found to be 20 dB. The voltage standing wave ratio (VSWR) and magnitude of reflection coefficient are respectively

A. 1.22 and 0.1 **B.** 0.81 and 0.1

C. -1.22 and 0.1 **D.** 2.44 and 0.2

Q3.12.1 MCQ 2012 2M

A transmission line with a characteristic impedance of 100 Ω is used to match a 50 Ω section to a 200 Ω section. If the matching is to be done both at 429 MHz and 1 GHz, the length of the transmission line can be approximately

A. 82.5 cm **B.** 1.05 m

C. 1.58 m **D.** 1.75 m

Q3.12.2 MCQ 2012 1M

A coaxial cable with an inner diameter of 1 mm and outer diameter of 2.4 mm is filled with a dielectric of relative permittivity 10.89. Given,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \ \varepsilon_0 = \frac{10^{-9}}{36\pi} \text{ F/m},$$

the characteristic impedance of the cable as

A. 330 Ω **B.** 100 Ω

C. 143.3 Ω **D.** 43.4 Ω

Q3.11.1 MCQ 2011 2M

A transmission line of characteristic impedance 50 Ω is terminated in a load impedance Z_L. The VSWR of the line is measured as 5 and the first of the voltage maxima in the line is observed at a distance of $\lambda/4$ from the load. The value of Z_L is

A. 10 Ω **B.** 250 Ω

C. (19.23 + j46.15) Ω **D.** (19.23 - j46.15) Ω

Q3.11.2 MCQ 2011 1M

A transmission line of characteristic impedance 50 Ω is terminated by a 50 Ω load. When excited by a sinusoidal voltage source at 10 GHz, the phase difference between two points spaced 2 mm apart on the line is found to be $\pi/4$ radians. The phase velocity of the wave along the line is

A. 0.8×10^8 m/s **B.** 1.2×10^8 m/s

C. 1.6×10^8 m/s **D.** 3×10^8 m/s

Q3.10.1 MCQ 2010 1M

If the scattering matrix $[S]$ of a two-port network is

$$[S] = \begin{bmatrix} 0.2\angle 0° & 0.9\angle 90° \\ 0.9\angle 90° & 0.1\angle 90° \end{bmatrix}$$

then the network is

A. lossless and reciprocal

B. lossless but not reciprocal

C. not lossless but reciprocal

D. neither lossless nor reciprocal

Q3.10.2 MCQ 2010 1M

In the circuit shown, all the transmission line sections are lossless. The Voltage Standing Wave Ratio (VSWR) on the 60 Ω line is

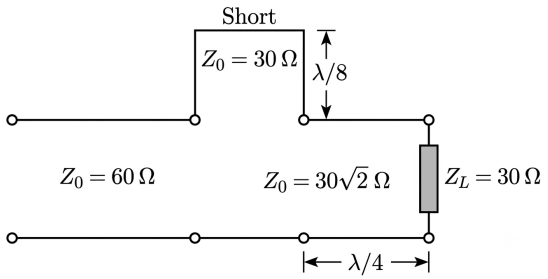


Fig. Q3.10.2

A. 1.00 **B.** 1.64 **C.** 2.50 **D.** 3.00

Q3.10.3 MCQ 2010 1M

A transmission line has a characteristic impedance of 50 Ω and a resistance of 0.1 Ω/m. If the line is distortionless, the attenuation constant (in Np/m) is

A. 500 **B.** 5

C. 0.014 **D.** 0.002

Q3.09.1 MCQ 2009 2M

A transmission line terminates in two branches, each of length $\lambda/4$, as shown. The branches are terminated by 50 Ω loads. The lines are lossless and have the characteristic impedances shown. Determine the impedance Z_i as seen by the source.

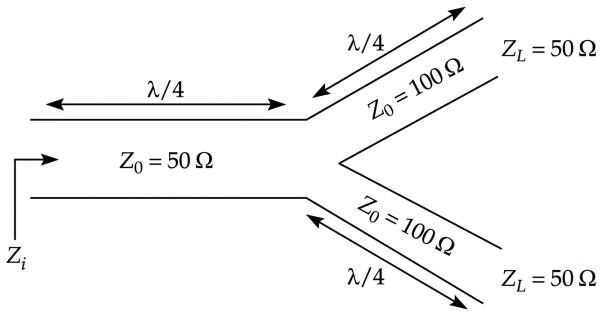


Fig. Q3.09.1

A. 200 Ω **B.** 100 Ω

C. 50 Ω **D.** 25 Ω

Q3.08.1 MCQ 2008 2M

One end of a lossless transmission line having the characteristic impedance of 75 Ω and length of 1 cm is short-circuited. At 3 GHz, the input impedance at the other end of the transmission line is

A. 0 **B.** resistive

C. capacitive **D.** inductive

Q3.08.2 MCQ 2008 2M

In the design of a single mode step index optical fiber close to upper cut-off, the single-mode operation is not preserved if

A. radius as well as operating wavelength are halved.

B. radius as well as operating wavelength are doubled.

C. radius is halved and operating wavelength is doubled.

D. radius is doubled and operating wavelength is halved.

Q3.07.1 MCQ 2007 2M

A load of 50 Ω is connected in shunt in a 2-wire transmission line of $Z_0 = 50\ \Omega$ as shown in the figure. The 2-port scattering parameter matrix (S-matrix) of the shunt element is

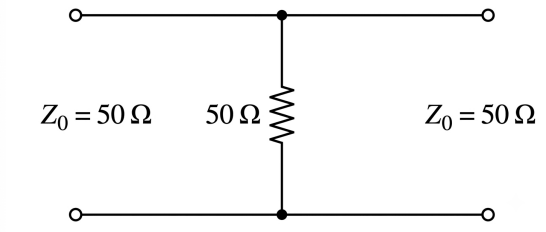


Fig. Q3.07.1

A. $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$ **B.** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$ **D.** $\begin{bmatrix} \frac{1}{4} & -\frac{3}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix}$

Q3.07.2 MCQ 2007 2M

The parallel branches of a 2-wire transmission line are terminated in 100 Ω and 200 Ω resistors as shown in the figure. The characteristic impedance of the line is $Z_0 = 50\ \Omega$ and each section has a length of $\lambda/4$. The voltage reflection coefficient Γ at the input is

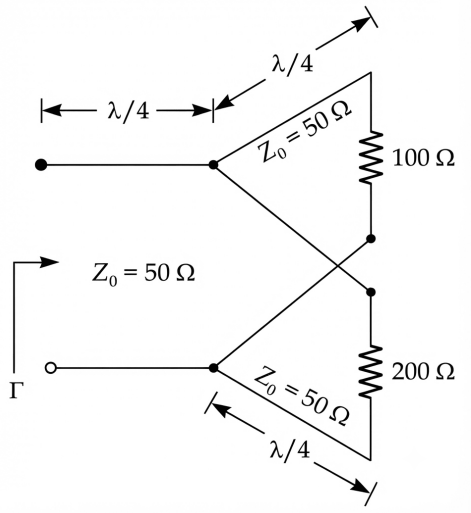


Fig. Q3.07.2

A. $-j\frac{7}{5}$ **B.** $\frac{-5}{7}$ **C.** $j\frac{5}{7}$ **D.** $\frac{5}{7}$

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions(Q3.06.1 and Q3.06.2) : A 30 Volts a battery with zero source resistance is connected to a co-axial line of characteristic impedance of 50 Ω at $t = 0$ seconds and terminated in an unknown resistive load. The line length is such that it takes 400 μs for an electromagnetic wave to travel from source end to load end and vice-versa. At $t = 400\ \mu$s, the voltage at the load end is found to be 40 Volts.

Q3.06.1 MCQ 2006 2M

The load resistance is

A. 25 Ω **B.** 50 Ω
C. 75 Ω **D.** 100 Ω

Q3.06.2 MCQ 2006 2M

The steady-state current through the load resistance is

A. 1.2 Amps. **B.** 0.3 Amps.
C. 0.6 Amps. **D.** 0.4 Amps.

Q3.05.1 MCQ 2005 2M

Many circles are drawn in a Smith chart used for transmission line calculations. The circles shown in the figure represent

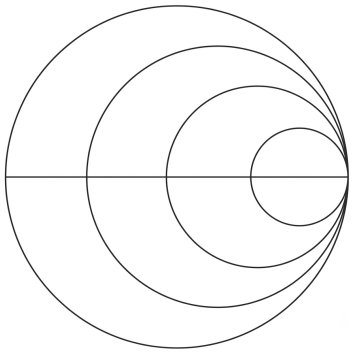


Fig. Q3.05.1

A. unit circles

B. constant resistance circles

C. constant reactance circles

D. constant reflection coefficient circles

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions (Q3.05.2 and Q3.05.3): Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50 Ω and a resistive load is shown in the figure.

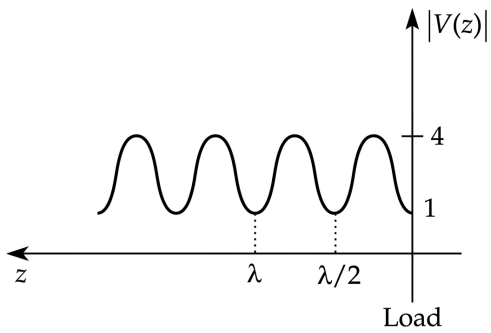


Fig. Q3.05.1

Q3.05.2 MCQ 2005 2M

The value of the load resistance is

A. 50 Ω **B.** 200 Ω
C. 12.5 Ω **D.** 0 Ω

Q3.05.3 MCQ 2005 2M

The reflection coefficient is given by

A. -0.6 **B.** -1
C. 0.6 **D.** 0

Q3.05.4 MCQ 2005 2M

Characteristic impedance of a transmission line is 50 Ω. Input impedance of the open-circuited line is $Z_{OC} = 100 + j150$ Ω. When the transmission line is short-circuited, then value of the input impedance will be

A. 50 Ω **B.** $100 + j150$ Ω
C. $7.69 + j11.54$ Ω **D.** $7.69 - j11.54$ Ω

Q3.04.1 MCQ 2004 2M

A lossless transmission line is terminated in a load which reflects a part of the incident power. The measured VSWR is 2. The percentage of the power that is reflected back is

A. 57.73 **B.** 33.33
C. 0.11 **D.** 11.11

Q3.04.2 MCQ 2004 2M

Consider an impedance $Z = R + jX$ marked with point P in an impedance Smith chart as shown in the figure. The movement from point P along a constant resistance circle in the clockwise direction by an angle 45° is equivalent to

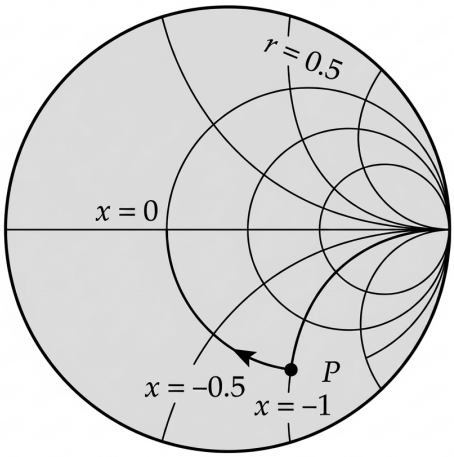


Fig. Q3.04.2

A. adding an inductance in series with Z

B. adding a capacitance in series with Z

C. adding an inductance in shunt across Z

D. adding a capacitance in shunt across Z

Q3.04.3 MCQ 2004 2M ○○○

Consider a 300 Ω, quarter wave long (at 1 GHz) transmission line as shown in figure. It is connected to a 10 V, 50 Ω source at one end and is left open-circuited at the other end. The magnitude of the voltage at the open-circuit end of the line is,

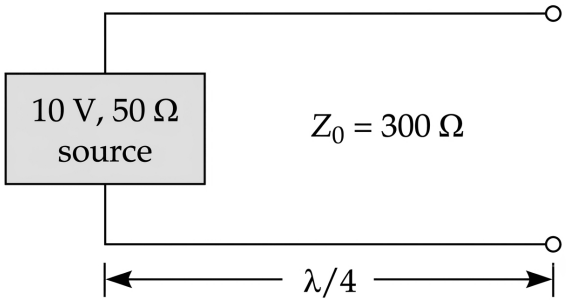


Fig. Q3.04.3

A. 10 V **B.** 5 V

C. 60 V **D.** 60/7 V

Q3.03.1 MCQ 2003 2M ○○○

A short-circuited stub is shunt connected to a transmission line as shown in the figure. If $Z_0 = 50\,\Omega$, the admittance Y seen at the junction of the stub and the transmission line is

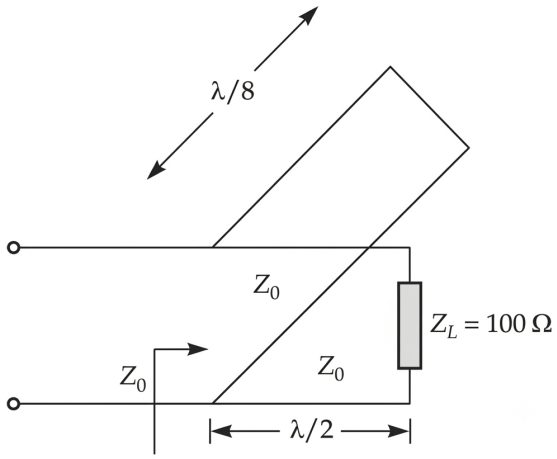


Fig. Q3.03.1

A. (0.01 - j0.02) ℧ **B.** (0.02 - j0.01) ℧

C. (0.04 - j0.02) ℧ **D.** (0.02 + j0) ℧

Q3.02.1 MCQ 2002 1M ○○○

In an impedance Smith chart, a clockwise movement along a constant resistance circle gives rise to

A. a decrease in the value of reactance.

B. an increase in the value of reactance.

C. no change in the reactance value.

D. no change in the impedance value.

Q3.02.2 MCQ 2002 1M ○○○

The VSWR can have any value between

A. 0 and 1 **B.** -1 and +1

C. 0 and ∞ **D.** 1 and ∞

Q3.01.1 MCQ 2001 2M ○○○

A uniform plane electromagnetic wave incident normally on a plane surface of a dielectric material is reflected with a VSWR of 3. What is the percentage of incident power that is reflected?

A. 10% **B.** 25%

C. 50% **D.** 75%

Q3.01.2 MCQ 2001 1M

A transmission line is distortionless if

A. $RL = \dfrac{1}{GC}$ **B.** $RL = GC$

C. $LG = RC$ **D.** $RG = LC$

Q3.00.1 MCQ 2000 1M

The magnitude of the open-circuit and short-circuit input impedance of a transmission line are 100 Ω and 25 Ω respectively. The characteristic impedance of the line is,

A. 25 Ω **B.** 50 Ω

C. 75 Ω **D.** 100 Ω

Q3.99.1 MCQ 1999 2M

In air, a lossless transmission line of length 50 cm with $L = 10\,\mu$H/m, $C = 40$ pF/m is operated at 25 MHz. Its electrical path length is

A. 0.5 meters **B.** λ meters

C. $\pi/2$ radians **D.** 180 degrees

Q3.99.2 MCQ 1999 2M

In a twin-wire transmission line in air, the adjacent voltage maxima are at 12.5 cm and 27.5 cm. The operating frequency is

A. 300 MHz **B.** 1 GHz

C. 2 GHz **D.** 6.28 GHz

Q3.98.1 MCQ 1998 1M

All transmission line section in figure, have a characteristic impedance $R_0 + j0$. The input impedance Z_{in} equals,

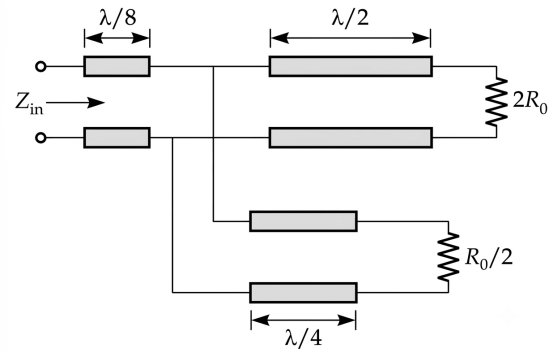


Fig. Q3.98.1

A. $\dfrac{2}{3}R_0$ **B.** R_0 **C.** $\dfrac{3}{2}R_0$ **D.** $2R_0$

Q3.97.1 MCQ 1997 1M

A transmission line of 50 Ω characteristic impedance is terminated with a 100 Ω resistance. The minimum impedance measured on the line is equal to

A. 0 Ω **B.** 25 Ω

C. 50 Ω **D.** 100 Ω

Q3.96.1 MCQ 1996 1M

The capacitance per unit length and the characteristic impedance of a lossless transmission line are C and Z_0 respectively. The velocity of a travelling wave on the transmission line is

A. Z_0C **B.** $\dfrac{1}{(Z_0C)}$

C. $\dfrac{Z_0}{C}$ **D.** $\dfrac{C}{Z_0}$

Q3.96.2 MCQ 1996 1M

A lossless transmission line having 50 Ω characteristic impedance and length $\lambda/4$ is short-circuited at one end and connected to an ideal voltage source of 1 V at the other end. The current drawn from the voltage sources is

A. 0 **B.** 0.02 A

C. ∞ **D.** None of these

Q3.94.1 MCQ 1994 2M

If a pure resistance load, when connected to a lossless 75 Ω line, produces a VSWR of 3 on the line, then the load impedance can only be 25 Ω.

A. True

B. False

Q3.94.2 NAT 1994 1M

A load impedance, $(200 + j0)$ Ω is to be matched to a 50 Ω lossless transmission line by using a quarter wave line transformer (QWT). The characteristic impedance of the QWT required is _____.

Q3.93.1 MCQ 1993 2M

Consider a transmission line of characteristic impedance 50 Ω. Let it be terminated at one end by $(+j50)$ Ω. The VSWR produced by it in the transmission line will be

A. +1 **B.** 0

C. ∞ **D.** $+j$

Q3.92.1 MSQ 1992 2M

A transmission line whose characteristic impedance is a real

- [ ] **A.** must be a lossless line.
- [ ] **B.** must be a distortionless line.
- [ ] **C.** may not be a lossless line.
- [ ] **D.** may not be a distortionless line.

Q3.91.1 MCQ 1991 2M

The input impedance of a short-circuited lossless transmission line quarter wave long is

A. purely reactive

B. purely resistive

C. infinite

D. dependent on the characteristic impedance of the line

Q3.89.1 MCQ 1989 2M

A 50 Ω lossless transmission line has a pure reactance of $(j100)$ Ω as its load. The VSWR in the line is

A. 1/2 (Half) **B.** 2 (Two)

C. 4 (Four) **D.** ∞ (Infinity)

Q3.88.1 MCQ 1988 2M

A two-wire transmission line of characteristic impedance Z_0 is connected to a load of impedance $Z_L (Z_L \neq Z_0)$. Impedance matching cannot be achieved with

A. a quarter-wavelength transformer

B. a half-wavelength transformer

C. an open-circuited parallel stub

D. a short-circuited parallel stub

Q3.87.1 MCQ 1987 2M

A transmission line of real characteristic impedance is terminated with an unknown load. The measured value of VSWR on the line is equal to 2 and a voltage minimum point is found to be at the load. The load impedance is then

A. complex **B.** purely capacitive

C. purely resistive **D.** purely inductive

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 B	Q3.26.2 20.38	Q3.25.1 6.4
Q3.24.1 200	Q3.24.2 D	Q3.23.1 0.13
Q3.23.2 B,C,D	Q3.23.3 B,C	Q3.23.4 A
Q3.21.1 D	Q3.20.1 D	Q3.20.2 6.25
Q3.18.1 0.001	Q3.18.2 C	Q3.17.1 0.375
Q3.17.2 49 to 50	Q3.17.3 0.85 to 0.88	Q3.16.1 32.40 to 32.80

Q3.16.2 0.1	Q3.16.3 *	Q3.16.4 B
Q3.15.1 30	Q3.15.2 120.22	Q3.15.3 D
Q3.15.4 B	Q3.14.1 B	Q3.14.2 C
Q3.14.3 33.33	Q3.14.4 70.71	Q3.14.5 40
Q3.14.6 B	Q3.13.1 A	Q3.12.1 C
Q3.12.2 MTA	Q3.11.1 A	Q3.11.2 C

Q3.10.1 C	Q3.10.2 B	Q3.10.3 D
Q3.09.1 D	Q3.08.1 D	Q3.08.2 D
Q3.07.1 C	Q3.07.2 D	Q3.06.1 D
Q3.06.2 B	Q3.05.1 B	Q3.05.2 C
Q3.05.3 A	Q3.05.4 D	Q3.04.1 D
Q3.04.2 A	Q3.04.3 C	Q3.03.1 A

Q3.02.1 B	Q3.02.2 D	Q3.01.1 B
Q3.01.2 C	Q3.00.1 B	Q3.99.1 C
Q3.99.2 B	Q3.98.1 B	Q3.97.1 B
Q3.96.1 B	Q3.96.2 A	Q3.94.1 B
Q3.94.2 100	Q3.93.1 C	Q3.92.1 B,C
Q3.91.1 C	Q3.89.1 D	Q3.88.1 B
Q3.87.1 C		

PrepFusion

UNIT

IV

Rectangular & Circular Waveguides

Topics

1. TE & TM Modes
2. Waveguide Parameters & Power Calculations
3. Field Patterns

Unit Snapshot*

Stream: EC	**Questions:** 35	**MCQ:** 21	**MSQ:** 5	**NAT:** 9
1M: 11	**2M:** 24	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 3M(2026)–9M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 1M

The cutoff frequency (in GHz) for the dominant TE_{10} mode of an air-filled rectangular waveguide of inner dimension 0.28 inch × 0.14 inch is _____ (rounded off to two decimal places).

Q4.23.1 MCQ 2023 1M

Consider a narrow band signal, propagating in a lossless dielectric medium ($\epsilon_r = 4, \mu_r = 1$), with phase velocity v_p and group velocity v_g. Which of the following statement is true? (c is the velocity of light in vacuum.)

A. $v_p > c, v_g > c$ **B.** $v_p < c, v_g > c$

C. $v_p > c, v_g < c$ **D.** $v_p < c, v_g < c$

Q4.22.1 MSQ 2022 2M

A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

☐ **A.** 6×10^{15} Hz

☐ **B.** 0.5×10^{12} Hz

☐ **C.** 8×10^{14} Hz

☐ **D.** 1×10^{13} Hz

Q4.22.2 MSQ 2022 2M

Consider the following wave equation:

$$\frac{\partial^2 f(x,t)}{\partial t^2} = 10000 \frac{\partial^2 f(x,t)}{\partial x^2}$$

Which of the given options is/are solution(s) to the given wave equation?

☐ **A.** $f(x,t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$

☐ **B.** $f(x,t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$

☐ **C.** $f(x,t) = e^{-(x-100t)} + \sin(x + 100t)$

☐ **D.** $f(x,t) = e^{-100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Q4.21.1 NAT 2021 2M

A standard air-filled rectangular waveguide with dimensions, $a = 8$ cm, $b = 4$ cm, operates at 3.4 GHz. For the dominant mode of wave propagation, the phase velocity of the signal is v_p. The value (Rounded off to two decimal places) of v_p/c, where 'c' denotes the velocity of light, is _____.

Q4.19.1 MCQ 2019 2M

The dispersion equation of a waveguide, which relates the wave number 'k' to the frequency ω, is $k(\omega) = \left(\frac{1}{c}\right)\sqrt{\omega^2 - \omega_0^2}$ where the speed of light $c = 3 \times 10^8$ m/s, and ω_0 is a constant. If the group velocity is 2×10^8 m/s, then the phase velocity is

A. 2×10^8 m/s **B.** 3×10^8 m/s

C. 1.5×10^8 m/s **D.** 4.5×10^8 m/s

Q4.19.2 NAT 2019 2M

A rectangular waveguide of width 'w' and height 'h' has cut-off frequencies for TE_{10} and TE_{11} modes in the ratio 1 : 2. The aspect ratio w/h, rounded off to two decimal places, is ____.

Q4.18.1 NAT 2018 2M

The cut-off frequency of TE_{01} mode of an air filled rectangular waveguide having inner dimensions a cm × b cm ($a > b$) is twice that of the dominant TE_{10} mode. When the waveguide is operated at a frequency which is 25% higher than the cut-off frequency of the dominant mode, the guide wavelength is found to be 4 cm. The value of b (in cm) correct to two decimal places, is ____.

Q4.17.1 MCQ 2017 2M

Standard air filled rectangular waveguides of dimensions $a = 2.29$ cm and $b = 1.02$ cm are designed for radar applications. It is desired that these waveguides operate only in the dominant TE_{10} mode with the operating frequency at least 25% above the cut-off frequency of the TE_{10} mode but not higher than 95% of the next higher cut-off frequency. The range of the allowable operating frequency, f is ____.

A. 8.19 GHz ≤ f ≤ 13.1 GHz

B. 8.19 GHz ≤ f ≤ 12.45 GHz

C. 6.55 GHz ≤ f ≤ 13.1 GHz

D. 1.64 GHz ≤ f ≤ 10.24 GHz

Q4.16.1 MCQ 2016 2M

Consider an air filled rectangular waveguide with dimensions, $a = 2.286$ and $b = 1.016$ cm. The increasing order of the cut-off frequencies for different modes is

A. $TE_{01} < TE_{10} < TE_{11} < TE_{20}$

B. $TE_{20} < TE_{11} < TE_{10} < TE_{01}$

C. $TE_{10} < TE_{20} < TE_{01} < TE_{11}$

D. $TE_{10} < TE_{11} < TE_{20} < TE_{01}$

Q4.16.2 NAT 2016 2M

Consider an air filled rectangular waveguide with dimensions, $a = 2.286$ cm and $b = 1.016$ cm. At 10 GHz operating frequency, the value of the propagation constant (per meter) of the corresponding propagating mode is ____.

Q4.15.1 NAT 2015 2M

An air filled rectangular waveguide of interval dimensions a cm $\times$ b cm ($a > b$) has a cut-off frequency of 6 GHz for the dominant TE_{10} mode. For the same waveguide, if the cut-off frequency of the TM_{11} mode is 15 GHz, the cut-off frequency of the TE_{01} mode (in GHz) is ____.

Q4.15.2 MCQ 2015 2M

The longitudinal component of the magnetic field inside an air filled rectangular waveguide made of a perfect electric conductor is given by the following expression: $H_z(x, y, z, t) = 0.1\cos(25\pi x)\cos(30.3\pi y)\cos(12\pi \times 10^9 t - \beta z)$ A/m The cross-sectional dimensions of the waveguide are given as $a = 0.08$ m and $b = 0.033$ m. The mode of propagation inside the waveguide is

A. TM_{12} **B.** TM_{21}
C. TE_{21} **D.** TE_{12}

Q4.14.1 NAT 2014 2M

Consider an air filled rectangular waveguide with a cross-section of 5 cm $\times$ 3 cm. For this waveguide, the cut-off frequency (in MHz) of TE_{21} mode is ____.

Q4.14.2 NAT 2014 2M

For a rectangular waveguide of internal dimensions $a \times b$ ($a > b$), the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If $a = \sqrt{5}$ cm, the value of b (in cm) is ____.

Q4.12.1 MCQ 2012 2M

The magnetic field along the propagation direction inside a rectangular waveguide with the cross-section shown in figure is

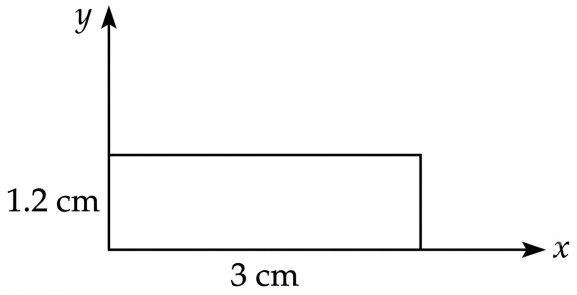


Fig. Q4.12.1

$H_z = 3\cos(2.094 \times 10^2 x)\cos(2.618 \times 10^2 y)\cos(6.283 \times 10^{10} t - \beta z)$ The phase velocity v_p of the wave inside the waveguide satisfies

A. $v_p > c$ **B.** $v_p = c$
C. $0 < v_p < c$ **D.** $v_p = 0$

Q4.11.1 MCQ 2011 1M

The modes in a rectangular waveguide are denoted by TE_{mn}/TM_{mn} where 'm' and 'n' are the eigen numbers along the larger and smaller dimensions of the waveguide respectively. Which one of the following statements is true?

A. The TM_{10} mode of the waveguide does not exist.

B. The TE_{10} mode of the waveguide does not exist.

C. The TM_{10} and the TE_{10} modes both exist and have the same cut-off frequencies.

D. The TM_{10} and the TE_{01} modes both exist and have the same cut-off frequencies.

Q4.09.1 MCQ 2009 1M

Which of the following statements is true regarding the fundamental mode of the metallic waveguides shown?

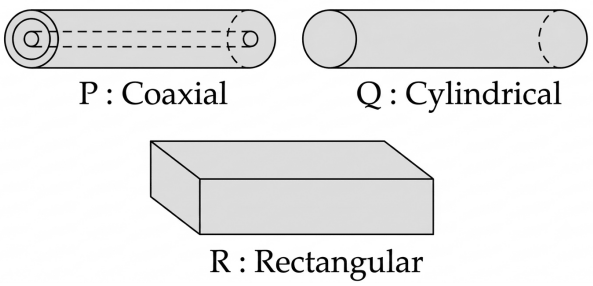


Fig. Q4.09.1

A. Only P has no cut-off frequency.

B. Only Q has no cut-off frequency.

C. Only R has no cut-off frequency.

D. All three have cut-off frequencies.

Q4.08.1 MCQ 2008 2M

A rectangular waveguide of internal dimensions (a = 4 cm and b = 3 cm) is to be operated in TE_{11} mode. The minimum operating frequency is

A. 6.25 GHz **B.** 6.0 GHz

C. 5.0 GHz **D.** 3.75 GHz

Q4.07.1 MCQ 2007 2M

The $\vec{E}$ field in a rectangular waveguide of inner dimensions $a \times b$ is given by

$$\vec{E} = \frac{\omega\mu}{h^2}\left(\frac{\pi}{a}\right) H_0 \sin\left(\frac{2\pi x}{a}\right) \sin(\omega t - \beta z)\,\hat{y}$$

where H_0 is a constant, and a and b are the dimensions along the x-axis and the y-axis respectively. The mode of propagation in the waveguide is.

A. TE_{20} **B.** TE_{11}

C. TM_{20} **D.** TE_{10}

Q4.07.2 MCQ 2007 2M

An air filled rectangular waveguide has inner dimensions of 3 cm × 2 cm. The wave impedance of the TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz is (free space impedance η_0 = 377 Ω)

A. 308 Ω **B.** 355 Ω

C. 400 Ω **D.** 461 Ω

Q4.06.1 MCQ 2006 2M

A rectangular waveguide having TE_{10} mode as dominant mode is having a cut-off frequency of 18 GHz for the TE_{30} mode. The inner broad wall dimension of the rectangular waveguide is

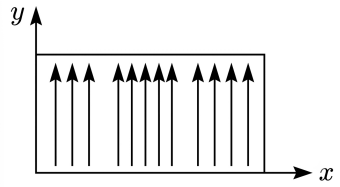
A. 5/3 cm **B.** 5 cm

C. 5/2 cm **D.** 10 cm

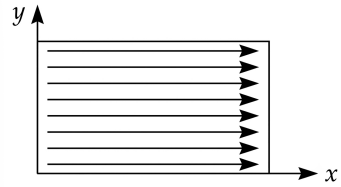
Q4.05.1 MCQ 2005 2M

Which one of the following does represent the electric field lines for the TE_{02} mode in the cross-section of a hollow rectangular metallic waveguide?

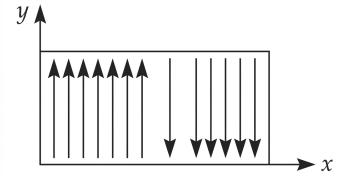
A.



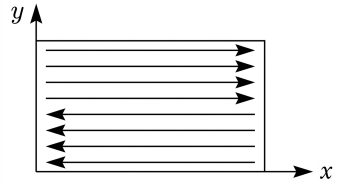
B.



C.



D.



Q4.04.1 MCQ 2004 1M

The phase velocity of an electromagnetic wave propagating in a hollow metallic rectangular waveguide in the TE_{10} mode is

A. equal to its group velocity.

B. less than the velocity of light in free space.

C. equal to the velocity of light in free space.

D. greater than the velocity of light in free space.

Q4.03.1 MCQ 2003 2M

A rectangular metal wave guide filled with a dielectric material of relative permittivity ϵ_r = 4 has the inside dimensions 3.0 cm × 1.2 cm. The cut-off frequency for the dominant mode is

A. 2.5 GHz **B.** 5.0 GHz

C. 10.0 GHz **D.** 12.5 GHz

Q4.02.1 MCQ 2002 1M

The phase velocity for the TE_{10} mode in an air filled rectangular waveguide is Note: c is the velocity of plane waves in free space.

A. less than c **B.** equal to c

C. greater than c **D.** none of the above

Q4.01.1 MCQ 2001 1M

The dominant mode in a rectangular waveguide is TE_{10}, because this mode has

A. no attenuation

B. no cut-off

C. no magnetic field component

D. the highest cut-off wavelength

Q4.01.2 MCQ 2001 1M

The phase velocity of waves propagating in a hollow metal waveguide is

A. greater than the velocity of light in free space.

B. less than the velocity of light in free space.

C. equal to the velocity of light in free space.

D. equal to the group velocity.

Q4.00.1 MCQ 2000 2M

A rectangular waveguide has dimensions 1 cm $\times$ 0.5 cm. Its cut-off frequency is

A. 5 GHz **B.** 10 GHz

C. 15 GHz **D.** 12 GHz

Q4.99.1 MCQ 1999 1M

Assuming perfect conductors of a transmission line, pure TEM propagation is not possible in

A. coaxial cable

B. air filled cylindrical waveguide

C. parallel twin-wire line in air

D. semi-infinite parallel to plate waveguide

Q4.97.1 MCQ 1997 1M

A rectangular air filled waveguide has a cross-section of 4 cm $\times$ 10 cm. The minimum frequency which can propagate in the waveguide is

A. 1.5 GHz **B.** 2.0 GHz

C. 2.5 GHz **D.** 3.0 GHz

Q4.94.1 NAT 1994 1M

The interior of a $\frac{20}{3}$ cm $\times$ $\frac{20}{4}$ cm rectangular waveguide is completely filled with a dielectric of $\epsilon_r = 4$. Waves of free space wave lengths shorter than _____ can be propagated in the TE_{11} mode.

Q4.90.1 MSQ 1990 2M

Choose the correct statements: For a wave propagating in an air filled rectangular waveguide

☐ **A.** guided wavelength is never less than the free space wavelength.

☐ **B.** wave impedance is never less than the free space impedance.

☐ **C.** phase velocity is never less than the free space velocity.

☐ **D.** TEM mode is possible if the dimensions of the wave guide are properly chosen.

Q4.88.1 MSQ 1988 2M

For a normal mode EM wave propagating in a hollow rectangular wave guide

☐ **A.** the phase velocity is greater than the group velocity.

☐ **B.** the phase velocity is greater than velocity of light in free space.

☐ **C.** the phase velocity is less than the velocity of light in free space.

☐ **D.** the phase velocity may be either greater than or less than the group velocity.

Q4.87.1 MSQ 1987 2M

The cut-off frequency of a waveguide depends upon

☐ **A.** the dimensions of waveguide.

☐ **B.** the dielectric property of the medium in the waveguide.

☐ **C.** the characteristic impedance of the waveguide.

☐ **D.** the transverse and axial components of the fields.

Answer Key

Q4.26.1 21.09	**Q4.23.1** D	**Q4.22.1** A,C
Q4.22.2 A,C	**Q4.21.1** 1.198	**Q4.19.1** D
Q4.19.2 1.72 to 1.74	**Q4.18.1** 0.75	**Q4.17.1** B
Q4.16.1 C	**Q4.16.2** 157.5 to 158.5	**Q4.15.1** 13.70 to 13.80
Q4.15.2 C	**Q4.14.1** 7800 to 7820	**Q4.14.2** 2
Q4.12.1 D	**Q4.11.1** A	**Q4.09.1** A
Q4.08.1 A	**Q4.07.1** A	**Q4.07.2** C
Q4.06.1 C	**Q4.05.1** D	**Q4.04.1** D
Q4.03.1 A	**Q4.02.1** C	**Q4.01.1** D
Q4.01.2 A	**Q4.00.1** C	**Q4.99.1** B
Q4.97.1 A	**Q4.94.1** 8	**Q4.90.1** A,C
Q4.88.1 A,B	**Q4.87.1** A,B	

PrepFusion

UNIT

V

Antenna & Antenna Array

Topics

1. Types of Antenna & Their Radiation Patterns
2. Antenna Parameters
3. Friis Equations
4. Antenna Arrays & Their Radiation Patterns

Unit Snapshot*

Stream: EC	**Questions:** 46	**MCQ:** 38	**MSQ:** 1	**NAT:** 7
1M: 19	**2M:** 27	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2015)–5M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 1M

Consider the Friis' transmission equation,

$$P_R = \frac{P_T G_T G_R \lambda^2}{(4\pi D)^2}$$

where P_R and P_T are the received and the transmitted powers, respectively. G_T and G_R are the gain of transmitting and receiving antennas, respectively, D is the distance between the transmitting and receiving antennas, and λ is the wavelength in free space. Given: $G_T = G_R = 1.0$, $\lambda = 0.30$ m and $P_T = +10$ dBm. Choose the distance (D), in km, from the following options at which the received power, $P_R = -90$ dBm?

A. $\frac{15}{4\pi}$ **B.** $\frac{15}{2\pi}$

C. $\frac{75}{2\pi}$ **D.** $\frac{3}{4\pi}$

Q5.21.1 NAT 2021 2M

An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain (Rounded off to 3 decimal places) is _____ V/m.

Q5.20.1 MCQ 2020 2M

For an infinitesimally small dipole in free space, the electric field E_θ in the far field is proportional to $(\frac{e^{-jkr}}{r})\sin\theta$, where $k = 2\pi/\lambda$. A vertical infinitesimally small electric dipole ($\delta I \ll \lambda$) is placed at a distance $h (h > 0)$ above an infinite ideal conducting plane, as shown in the figure. The minimum value of h, for which one of the maxima in the far field radiation pattern occurs at $\theta = 60°$, is

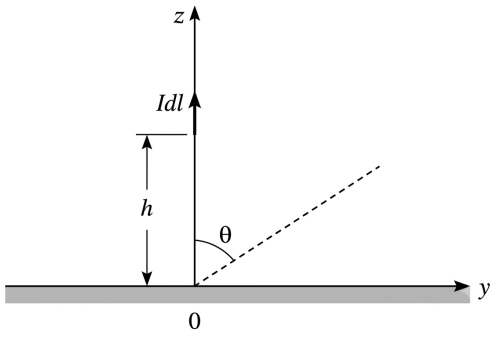


Fig. Q5.20.1

A. 0.75λ **B.** λ

C. 0.25λ **D.** 0.5λ

Q5.19.1 NAT 2019 1M

Radiation resistance of a small dipole current element of length l at a frequency of 3 GHz is 3 Ω. If the length is changed by 1%, then the percentage change in the radiation resistance, rounded off to two decimal places, is _____ %.

Q5.17.1 MCQ 2017 2M

A half wavelength dipole is kept in the x-y plane and oriented along 45° from the x-axis. Determine the direction of null in the radiation pattern for $0 \le \phi \le \pi$. Here the angle $\theta (0 \le \theta \le \pi)$ is measured from the z-axis, and the angle $\phi (0 \le \theta \le 2\pi)$ is measured from the x-axis in the x-y plane.

A. $\theta = 90°, \phi = 45°$ **B.** $\theta = 45°, \phi = 90°$

C. $\theta = 90°, \phi = 135°$ **D.** $\theta = 45°, \phi = 135°$

Q5.17.2 MCQ 2017 1M

Consider a wireless communication link between a transmitter and a receiver located in free space, with finite and strictly positive capacity. If the effective areas of the transmitter and the receiver antennas, and the distance between them are all doubled, and everything else remains unchanged, the maximum capacity of the wireless link

A. increased by a factor of 2

B. decreased by a factor of 2

C. remains unchanged

D. decreases by a factor of $\sqrt{2}$

Q5.16.1 MCQ 2016 2M

The far zone power density radiated by a helical antenna is approximated as,

$$\vec{W}_{\text{rad}} = \vec{W}_{\text{average}} \approx \hat{a}_r C_0 \frac{1}{r^2} \cos^4\theta$$

The radiated power density is symmetrical with respect to ϕ and exists only in the upper hemisphere $0 \le \theta \le \frac{\pi}{2}, 0 \le \phi \le 2\pi, C_0$ is a constant. The power radiated by the antenna (in Watts) and the maximum directivity of antenna, respectively, are

A. $15C_0$, 10 dB **B.** $1.256C_0$, 10 dB

C. $1.256C_0$, 12 dB **D.** $1.5C_0$, 12 dB

Q5.16.2 NAT 2016 2M ○○○

A radar operating at 5 GHz uses a common antenna for transmission and reception. The antenna has a gain of 150 and a aligned for maximum directional radiation and reception to a target 1 km away having radar cross-section of 3 m^2. If it transmits 100 kW, then the received power (in μW) is _____.

Q5.16.3 MCQ 2016 2M ○○○

The electric field of a uniform plane wave travelling along the negative z-direction is given by the following equation:

$$\vec{E}_w^i = (\hat{a}_x + j\hat{a}_y)E_0 e^{jkz}$$

This wave is incident upon a receiving antenna placed at the origin and whose radiated electric field towards the incident wave is given by the following equation:

$$\vec{E}_a = (\hat{a}_x + 2\hat{a}_y)E_1 \frac{1}{r} e^{-jkr}$$

The polarization of the incident wave, the polarization of the antenna and losses due to polarization mismatch are, respectively,

A. linear, circular (clockwise), −5 dB

B. circular (clockwise), linear, −5 dB

C. circular (clockwise), linear, −3 dB

D. circular (anticlockwise), linear, −3 dB

Q5.16.4 NAT 2016 2M ○○○

Two lossless X-band horn antennas are separated by a distance of 200λ. The amplitude reflection coefficients at the terminals of the transmitting and receiving antennas are 0.15 and 0.18, respectively. The maximum directivities of the transmitting and receiving antennas (over the isotropic antenna) are 18 dB and 22 dB, respectively. Assuming that the input power in the lossless transmission line connected to the antenna is 2 W, and that the antennas are perfectly aligned and polarization matched, the power (in mW) delivered to the load at the receiver is _____.

Q5.15.1 MCQ 2015 1M ○○○

The directivity of an antenna array can be increased by adding more antenna elements, as a larger number of elements

A. improves the radiation efficiency

B. increases the effective area of the antenna

C. results in a better impedance matching

D. allows more power to be transmitted by the antenna

Q5.15.2 NAT 2015 2M ○○○

Two half-wave dipole antennas placed as shown in the figure are excited with sinusoidally varying currents of frequency 3 MHz and phase shift of $\pi/2$ between them (the element at the origin leads in phase). If the maximum radiated E field at the point P in the x-y plane occurs at an azimuthal angle of 60°, the distance d (in meters) between the antenna is _____.

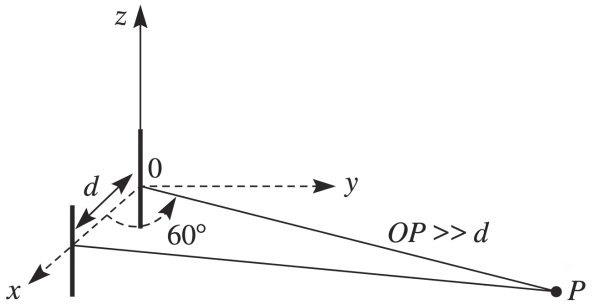


Fig. Q5.15.2

Q5.14.1 MCQ 2014 1M ○○○

Match Column-A with Column-B.

Column-A	Column-B
1. Point electromagnetic source	**P.** Highly directional
2. Dish antenna	**Q.** End fire
3. Yagi-Uda antenna	**R.** Isotropic

A. 1 - P, 2 - Q, 3 - R

B. 1 - R, 2 - P, 3 - Q

C. 1 - Q, 2 - P, 3 - R

D. 1 - R, 2 - Q, 3 - P

Q5.14.2 NAT 2014 1M ○○○

For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12 mV/m. Given the intrinsic impedance of the free space is 120$\pi\Omega$, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in nW/m^2) is _____.

Q5.14.3 NAT 2014 2M

In spherical coordinates, let $\hat{a}_\theta, \hat{a}_\phi$ denote unit vectors along the θ, ϕ-directions.

$$\vec{E} = \frac{100}{r} \sin\theta \cos(\omega t - \beta r)\hat{a}_\theta \text{ V/m}$$

$$\vec{H} = \frac{0.265}{r} \sin\theta \cos(\omega t - \beta r)\hat{a}_\phi \text{ V/m}$$

represent the electric and magnetic field components of the EM wave at large distances r from a dipole antenna, in free space. The average power (W) crossing the hemispherical shell located at $r = 1$ km, $0 \le \theta \le \pi/2$ is _____.

Q5.12.1 MCQ 2012 1M

The radiation pattern of an antenna in spherical co-ordinates is given by

$$F(\theta) = \cos^4\theta, \quad 0 \le \theta \le \frac{\pi}{2}$$

The directivity of the antenna is

A. 10 dB **B.** 12.6 dB

C. 11.5 dB **D.** 18 dB

Q5.11.1 MCQ 2011 1M

Consider the following statements regarding the complex pointing vector $\vec{P}$ for the power radiated by a point source in an infinite homogenous and lossless medium. $\text{Re}(\vec{P})$ denotes the real part of $\vec{P}$, S denotes a spherical surface whose center is at the point source, and $\hat{n}$ denotes the unit surface normal on S. Which of the following statements is true?

A. $\text{Re}(\vec{P})$ remains constant at any radial distance from the source.

B. $\text{Re}(\vec{P})$ increases with increasing radial distance from the source.

C. $\iint_s \text{Re}(\vec{P}) \cdot \hat{n}\, dS$ remains constant at any radial distance from the source.

D. $\iint_s \text{Re}(\vec{P}) \cdot \hat{n}\, dS$ decreases with increasing radial distance from the source.

Q5.08.1 MCQ 2008 2M

At 20 GHz, the gain of a parabolic dish antenna of 1 meter diameter and 70% efficiency is

A. 15 dB **B.** 25 dB

C. 35 dB **D.** 45 dB

Q5.08.2 MCQ 2008 1M

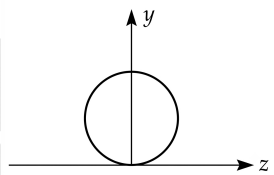
For a Hertz dipole antenna, the Half Power Beam Width (HPBW) in the E-plane is

A. 360° **B.** 180° **C.** 90° **D.** 45°

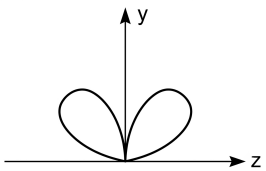
Q5.07.1 MCQ 2007 2M

A $\lambda/2$ dipole is kept horizontally at a height of $\lambda_0/2$ above a perfectly conducting infinite ground plane. The radiation pattern in the plane of the dipole ($\vec{E}$ plane) looks approximately as,

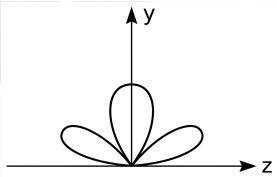
A.



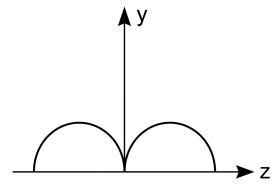
B.



C.



D.



Q5.06.1 MCQ 2006 2M

A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base fed at a frequency of 600 kHz. The radiation resistance of the antenna (in Ω) is

A. $\frac{2\pi^2}{5}$ **B.** $\frac{\pi^2}{5}$

C. $\frac{4\pi^2}{5}$ **D.** $20\pi^2$

Q5.06.2 MCQ 2006 1M

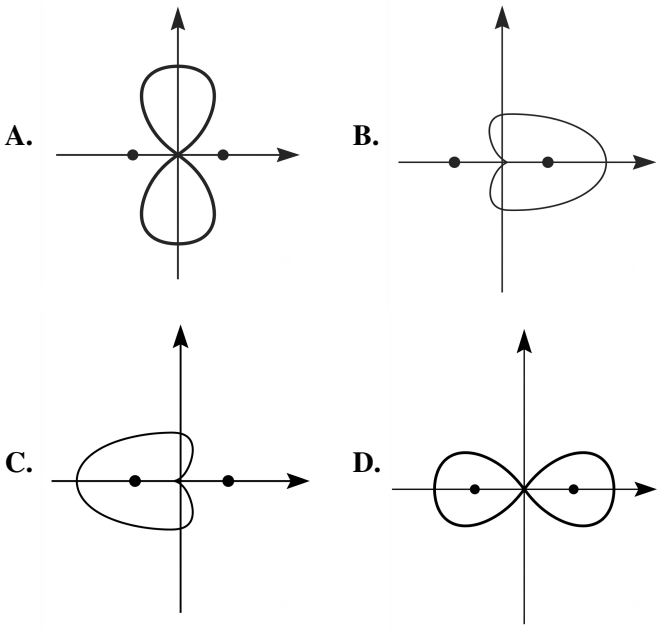
A transmission line is feeding 1 W of power to a horn antenna having gain of 10 dB. The antenna is matched to the transmission line. The total power radiated

A. 10 Watts **B.** 1 Watt

C. 0.1 Watt **D.** 0.01 Watt

Q5.05.1 MCQ 2005 2M

Two identical and parallel dipole antennas are kept apart by a distance of $\lambda/4$ in the H-plane. They are fed with equal currents but the right most antenna has a phase shift of +90°. The radiation pattern is given as,



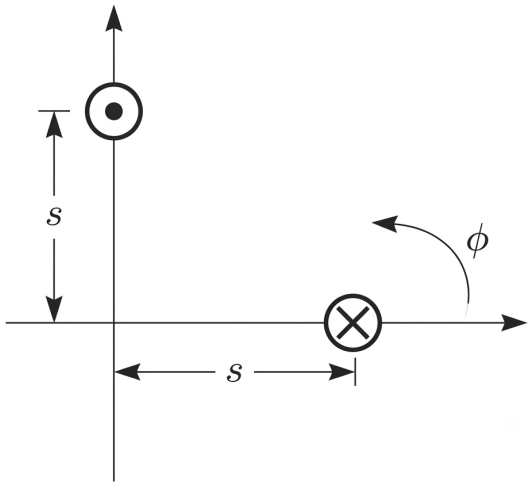


Fig. Q5.03.1

A. $2\cos\left(\frac{2\pi s}{\lambda}\right)$ **B.** $2\sin\left(\frac{2\pi s}{\lambda}\right)$

C. $2\cos\left(\frac{\pi s}{\lambda}\right)$ **D.** $2\sin\left(\frac{\pi s}{\lambda}\right)$

Q5.02.1 MCQ 2002 2M ○○○

A person with a receiver is 5 km away from the transmitter. What is the distance that this person must move further to detect a 3 dB decrease in signal strength?

A. 942 m **B.** 2070 m

C. 4978 m **D.** 5320 m

Q5.02.2 MCQ 2002 1M ○○○

The line of sight communication requires the transmit and receive antennas to face each other. If the transmit antenna is vertically polarized, for best reception the receiver antenna should be

A. horizontally polarized

B. vertically polarized

C. at 45° with respect to horizontal polarized

D. at 45° with respect to vertical polarized

Q5.01.1 MCQ 2001 2M ○○○

In a uniform linear array, four isotropic radiating elements are spaced $\lambda/4$ apart. The progressive phase shift between the elements required for forming the main beam at 60° off the end fire is

A. $-\pi$ radians **B.** $-\pi/2$ radians

C. $-\pi/4$ radians **D.** $-\pi/8$ radians

Q5.04.1 MCQ 2004 1M ○○○

Consider a lossless antenna with a directive gain of +6 dB. If 1 mW of power is fed to it the total power radiated by the antenna will be

A. 4 mW **B.** 1 mW

C. 7 mW **D.** $\frac{1}{4}$ mW

Q5.03.1 MCQ 2003 2M ○○○

Two identical antennas are placed in the $\theta = \pi/2$ plane as shown in the figure. The elements have equal amplitude excitation with 180° polarity difference, operating at wavelength λ. The correct values of the magnitude of the far zone resultant electric field strength normalized with that of a single element, both computed for $\phi = 0$, is

Q5.01.2 MCQ 2001 2M

A medium wave radio transmitter operating at a wavelength of 492 m has a tower antenna of height 124 m. What is the radiation resistance of the antenna?

A. 25Ω **B.** 36.5Ω

C. 50Ω **D.** 73Ω

Q5.00.1 MCQ 2000 2M

For an 8 feet (2.4 m) parabolic dish antenna operating at 4 GHz, the minimum distance required for far field measurement is closest to

A. 7.5 cm **B.** 15 cm

C. 15 m **D.** 150 m

Q5.00.2 MCQ 2000 1M

If the diameter of a $\lambda/2$ dipole antenna is increased from $\frac{\lambda}{100}$ to $\frac{\lambda}{50}$, then its

A. bandwidth increases **B.** bandwidth decreases

C. gain increases **D.** gain decreases

Q5.99.1 MCQ 1999 2M

A transmitting antenna radiates 251 W isotropically. A receiving antenna, located 100 m away from the transmitting antenna, has an effective aperture of 500 cm^2. The total power received by the antenna is

A. 10 μW **B.** 1 μW

C. 20 μW **D.** 100 μW

Q5.98.1 MCQ 1998 1M

The far field on an antenna varies with distance 'r' as,

A. $\frac{1}{r}$ **B.** $\frac{1}{r^2}$

C. $\frac{1}{r^3}$ **D.** $\frac{1}{\sqrt{r}}$

Q5.98.2 MCQ 1998 1M

An antenna in free space receives 2 μW of power when the incident electric field is 20 mV/m rms. The effective aperture of the antenna is

A. 0.005 m^2 **B.** 0.05 m^2

C. 1.885 m^2 **D.** 3.77 m^2

Q5.98.3 MCQ 1998 1M

The radiation resistance of a circular loop of one turn is 0.01 Ω. The radiation resistance of five turns of such a loop will be

A. 0.002 Ω **B.** 0.01 Ω

C. 0.05 Ω **D.** 0.25 Ω

Q5.98.4 MCQ 1998 1M

The vector $\vec{H}$ in the far field on an antenna satisfies

A. $\nabla \cdot \vec{H} = 0$ and $\nabla \times \vec{H} = 0$

B. $\nabla \cdot \vec{H} \neq 0$ and $\nabla \times \vec{H} \neq 0$

C. $\nabla \cdot \vec{H} = 0$ and $\nabla \times \vec{H} \neq 0$

D. $\nabla \cdot \vec{H} \neq 0$ and $\nabla \times \vec{H} = 0$

Q5.97.1 MCQ 1997 2M

A parabolic dish antenna has conical beam 2° wide. The directivity of the antenna is approximately

A. 20 dB **B.** 30 dB

C. 40 dB **D.** 50 dB

Q5.96.1 MCQ 1996 2M

At 1 km long microwave link use two antennas each having 30 dB gain. If the power transmitted by one antenna is 1 W at 3 GHz, the power received by the other antenna is approximately

A. 98.6 μW **B.** 76.8 μW

C. 63.4 μW **D.** 55.2 μW

Q5.96.2 MCQ 1996 2M

A transverse electromagnetic wave with circular polarization is received by a dipole antenna. Due to polarization mismatch, the power transfer efficiency from the wave to the antenna is reduced to about

A. 50% **B.** 35.3%

C. 25% **D.** 0%

Q5.95.1 MCQ 1995 1M

An antenna, when radiating has a highly directional radiation pattern. When the antenna is receiving, its radiation pattern

A. is more directive **B.** is less directive

C. is the same **D.** exhibits no directivity at all

Q5.94.1 MCQ 1994 1M

For a half-wave dipole antenna, which of the following statements are right?

A. The radiation intensity is maximum along the normal to the dipole axis.

B. The current distribution along its length is uniform irrespective of the length.

C. The effective length equals its physical length.

D. The input impedance is independent of the location of the feed point.

Q5.92.1 MCQ 1992 2M

The beamwidth between first nulls of a uniform linear array of N equally spaced (element spacing = d), equal excited antennas, is determine by

A. N alone and not by d **B.** d alone and not by N

C. the ratio, (N/d) **D.** the product, (Nd)

Q5.92.2 MCQ 1992 2M

Two dissimilar antennas having their maximum directivities equal, which of the following statements are right?

A. must have their beam widths also equal.

B. cannot have their beam widths equal because they are dissimilar antenna.

C. may not necessarily have their maximum power gains equal.

D. must have their effective aperture areas (capture areas) also equal.

Q5.91.1 MCQ 1991 2M

In a broad side array of 20 isotropic radiators, equally spaced at distance of $\lambda/2$, the beam width between first nulls is

A. 51.3 degrees **B.** 11.46 degrees

C. 22.9 degrees **D.** 102.6 degrees

Q5.90.1 MCQ 1990 2M

Two isotropic antennas are separated by a distance of two wavelengths. If both the antennas are fed with currents of equal phase and magnitude, the number of lobes in the radiation pattern in the horizontal plane are

A. 2 **B.** 4

C. 6 **D.** 8

Q5.88.1 MSQ 1988 2M

The electric field E and the magnetic field H of a short dipole antenna satisfy the condition

- ☐ **A.** the r component of E is equal to zero.
- ☐ **B.** both r and θ components of H are equal to zero.
- ☐ **C.** the θ component of E dominates the r component in the far field region.
- ☐ **D.** the θ and ϕ components of H are of the same order of magnitude in the near field region.

Answer Key

Q5.26.1 B	Q5.21.1 0.244	Q5.20.1 B
Q5.19.1 2.01	Q5.17.1 A	Q5.17.2 C
Q5.16.1 B	Q5.16.2 0.012	Q5.16.3 C
Q5.16.4 3	Q5.15.1 B	Q5.15.2 50
Q5.14.1 B	Q5.14.2 47.5 to 48.0	Q5.14.3 55.5
Q5.12.1 A	Q5.11.1 C	Q5.08.1 D
Q5.08.2 C	Q5.07.1 B	Q5.06.1 A
Q5.06.2 B	Q5.05.1 C	Q5.04.1 B
Q5.03.1 D	Q5.02.1 B	Q5.02.2 B
Q5.01.1 C	Q5.01.2 B	Q5.00.1 D
Q5.00.2 C	Q5.99.1 D	Q5.98.1 A
Q5.98.2 C	Q5.98.3 D	Q5.98.4 C
Q5.97.1 C	Q5.96.1 C	Q5.96.2 A
Q5.95.1 C	Q5.94.1 A	Q5.92.1 D
Q5.92.2 C	Q5.91.1 B	Q5.90.1 D
Q5.88.1 B,C		

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